

Combinatorial Coefficient Calculus

**Stirling, Bell, Eulerian, and Bernoulli–Euler Families;
Bell-Polynomial and Faà di Bruno Calculus;
Lagrange–Bürmann and Multivariate Good Inversion**

A self-contained synthesis with proofs

September 2026

Abstract

This semi-formalized research-frontier monograph develops a unified calculus for the coefficients that occur when one changes polynomial bases, partitions labelled sets, differentiates or composes functions, and reverses formal or analytic power series. Its principal objects are the signed and unsigned Stirling numbers of the first kind, Stirling numbers of the second kind, Lah numbers, Bell numbers and Bell polynomials, Eulerian families, Bernoulli–Euler–Genocchi–Cauchy families, Faà di Bruno calculus, and one- and several-variable Lagrange inversion.

The canonical article is assembled at theorem level from six overlapping arrivals, not by concatenating them. They are sibling syntheses rather than successive revisions: a large common core is proved once, while their distinct layers in incidence algebra, Sheffer–Riordan calculus, special functions, Catalan geometry, inverse singularities, multivariate Good inversion, and exact coefficient algorithms are integrated at the point of mathematical dependence. Internal references in the arrivals to five earlier drafts and to Wikipedia or MathWorld dossiers are self-reported provenance; those absent folders are not counted as independently inspected sources.

The treatment is deliberately proof-oriented. Definitions are connected by bijections and basis changes; generating functions are derived rather than merely recorded; and analytic claims state their regimes, branches, and remainder hypotheses. When an arrival contains an incomplete proof, an index ambiguity, a suppressed Jacobian, an unsupported numerical certificate, or a schematic asymptotic placeholder, the canonical text either supplies the missing argument or withdraws the theorem claim and records the outstanding obligation plainly. A source-disposition ledger outside the mathematical text records every merge, deduplication, correction, and deliberate retirement.

How to read this work

The chapters are arranged by mathematical dependence, not by the order of the source files. The two master ideas are:

1. a coefficient is often easiest to compute after changing to the basis in which the relevant operator is diagonal; and
2. an exponential generating function remembers the decomposition of a labelled object into components.

Stirling numbers implement the first idea. Bell polynomials implement the second. Eulerian numbers mediate between powers and shifted binomial coefficients. Faà di Bruno’s formula is the differential form of labelled composition, while Lagrange inversion is the coefficient form of implicit substitution.

A result stated for formal power series requires no convergence. When an analytic version is asserted, its domain hypotheses are written explicitly. Statements in the archive that report historical computations—for example, lists of Bell primes or verified exact modular periods—are distinguished from theorems and are accompanied by a certificate procedure rather than misrepresented as symbolic identities.

This is a research-frontier document, not a machine-checked development. Its theorem and proof environments are human-readable mathematics and are not claimed to be proved or formalized in Lean. An exact Lean counterpart may be claimed only by naming a compiled declaration and module with matching hypotheses and conclusion; the present manuscript does not yet supply such a claim-level crosswalk. Thus “proof” below never means “Lean verification.”

Editorial normalization

Three systematic repairs are used throughout.

1. Bell numbers are denoted B_n^{Bell} , complete exponential Bell polynomials by B_n^{exp} , partial Bell polynomials by $B_{n,k}^{\text{exp}}$, and Bernoulli numbers by β_n .
2. Type- B Eulerian numbers are indexed by descents in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This removes the one-step shift in one formula in the archive.
3. The archived Moser–Wyman display contains unnamed quantities B, P_i, Q_i . We give the exact saddle-point scheme that defines every correction coefficient recursively and state the leading expansion in closed form; no undefined placeholder is retained.

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Notation

For $n \in \mathbb{N}_0$, write

$$(x)^{\underline{n}} = x(x-1) \cdots (x-n+1), \quad (x)^{\overline{n}} = x(x+1) \cdots (x+n-1),$$

with both expressions equal to 1 when $n = 0$. Coefficient extraction is denoted by $[z^n] F(z)$, and $\text{Res}_z F(z) dz$ means the coefficient of z^{-1} in a formal Laurent series. Our principal arrays are

$$s(n, k), \quad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n, k), \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle,$$

where $s(n, k)$ is the signed first-kind Stirling number, $\begin{bmatrix} n \\ k \end{bmatrix}$ its unsigned version, $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ the second-kind Stirling number, and $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$ the ordinary Eulerian number. Bell numbers are B_n^{Bell} ; exponential partial and complete Bell polynomials are $\mathcal{B}_{n,k}^{\text{exp}}$ and $\mathcal{B}_n^{\text{exp}}$.

Unless stated otherwise, sums over an impossible index range are zero, and $0^0 = 1$ in coefficient formulae. Bernoulli numbers use

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Part I

Coefficient Calculus and Labelled Structures

Chapter 1

Formal power series and coefficient extraction

1.1 Formal series as an algebraic environment

Let R be a commutative $\mathbb{Q}$ -algebra. The ring $R[[z]]$ consists of expressions $F(z) = \sum_{n \geq 0} f_n z^n$ with termwise addition and Cauchy multiplication. A series is invertible precisely when f_0 is invertible in R . If $G(0) = 0$, then the composition $F(G(z))$ is well-defined because each coefficient receives only finitely many contributions.

Lemma 1.1 (Coefficient rules). *For $F(z) = \sum f_n z^n$ and $G(z) = \sum g_n z^n$,*

$$[z^n] F(z)G(z) = \sum_{j=0}^n f_j g_{n-j}, \quad (1.1)$$

$$[z^n] F'(z) = (n+1)f_{n+1}, \quad (1.2)$$

$$[z^n] \frac{F(z)}{1-az} = \sum_{j=0}^n a^{n-j} f_j. \quad (1.3)$$

If F is analytic on $|z| \leq r$, then also

$$f_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z)}{z^{n+1}} dz, \quad |f_n| \leq r^{-n} \max_{|z|=r} |F(z)|. \quad (1.4)$$

Proof. The first identity is the definition of Cauchy multiplication. The second follows by differentiating term by term, and the third by multiplying with $\sum_{m \geq 0} a^m z^m$. The analytic formula is Cauchy's coefficient formula, and the bound follows by estimating the contour integral by its length $2\pi r$. $\square$

Remark 1.2 (Formal crosswalk). Module `CoefficientRules` proves the three formal rules over an arbitrary commutative ring: Equation (1.1) is `Fabius.PowerSeries.coeff_mul_eq_sum_range`, Equation (1.2) is `Fabius.PowerSeries.coeff_derivative_eq`, and Equation (1.3) is `Fabius.PowerSeries.coeff_mul_geomSeries`, where $F/(1-az)$ is read as $F \cdot \sum_n a^n z^n$; that the geometric series really is the reciprocal of $1-az$ is `Fabius.PowerSeries.one_sub_C_mul_X_mul_geomSeries`, and `Fabius.PowerSeries.eq_mul_geomSeries_iff` lets the same rule be applied to a series presented by the equation $(1-az)G = F$, which is how it is used in the Stirling column arguments. The analytic Cauchy coefficient formula and bound Equation (1.4) are not formalized.

1.2 Formal residues

For a Laurent series $A(z)$, set $\text{Res}_z A(z) dz = [z^{-1}] A(z)$. Then

$$[z^n] F(z) = \text{Res}_z \frac{F(z)}{z^{n+1}} dz, \quad \text{Res}_z A'(z) dz = 0. \quad (1.5)$$

The second identity is simply the fact that the derivative of z^m has no z^{-1} term.

Theorem 1.3 (Formal change of variables for residues). *Let $u = u(z) \in zR[[z]]$ have invertible linear coefficient. For every Laurent series A for which the composition is defined,*

$$\text{Res}_z A(u(z)) u'(z) dz = \text{Res}_u A(u) du. \quad (1.6)$$

Proof. By linearity it suffices to take $A(u) = u^m$. If $m \neq -1$, then $u(z)^m u'(z) = \frac{1}{m+1} (u(z)^{m+1})'$ has residue zero. For $m = -1$, $u'(z)/u(z) = z^{-1} + H(z)$ with $H \in R[[z]]$, because $u(z) = az(1 + \mathcal{O}(z))$. Thus both sides have residue 1. $\square$

Master principle

The combination of Equations (1.5) and (1.6) turns implicit substitution into coefficient extraction. This is the entire algebraic engine of Lagrange inversion; no analytic contour is needed until one asks about convergence.

1.3 Ordinary and exponential generating functions

For a sequence (a_n) , its ordinary and exponential generating functions are

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad A^{\text{egf}}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Ordinary multiplication gives unweighted convolution. Exponential multiplication gives binomial convolution:

$$A^{\text{egf}}(z) B^{\text{egf}}(z) = \sum_{n \geq 0} \left(\sum_{j=0}^n \binom{n}{j} a_j b_{n-j} \right) \frac{z^n}{n!}. \quad (1.7)$$

This binomial factor is the number of ways to choose which labels belong to the first component.

1.4 Finite differences and factorial bases

Let E be the shift operator $(Ef)(x) = f(x+1)$ and $\Delta = E - 1$. Then

$$\Delta(x)^{\underline{k}} = k(x)^{\underline{k-1}}, \quad \sum_{i=0}^{n-1} (i)^{\underline{k}} = \frac{(n)^{\underline{k+1}}}{k+1}. \quad (1.8)$$

Proof. The first follows from $(x+1)^{\underline{k}} - (x)^{\underline{k}} = k(x)^{\underline{k-1}}$ after factoring the common product. Summing this telescoping identity gives the second formula. $\square$

Theorem 1.4 (Newton expansion). *Every polynomial p of degree at most d has the unique expansion*

$$p(x) = \sum_{k=0}^d \frac{\Delta^k p(0)}{k!} (x)^{\underline{k}} = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}. \quad (1.9)$$

Proof. The factorial polynomials form a basis because their degrees are distinct and their leading coefficients are one. If $p = \sum a_k (x)^k$, applying Δ^j and setting $x = 0$ kills terms with $k > j$ by a remaining factor x , kills terms with $k < j$ by excessive differencing, and gives $j!a_j$ for $k = j$. $\square$

Remark 1.5 (Formal crosswalk). `Fabius.newton_expansion` (module `NewtonExpansion`) is the falling-factorial form of Equation (1.9) in $K[x]$ for every field K of characteristic zero. The formal route differs from the basis argument above: Mathlib's Gregory–Newton formula `shift_eq_sum_fwdDiff_iter` gives $p(m) = \sum_{k \leq m} \binom{m}{k} \Delta^k p(0)$ at every natural number m , the terms with $k > \deg p$ vanish by `Polynomial.fwdDiff_iter_eq_zero_of_degree_lt` (`Fabius.eval_natCast_eq_sum_choose_fwdDiff`, over any commutative ring), and two polynomials agreeing at infinitely many points are equal.

1.5 Labelled products and the exponential formula

A labelled combinatorial class is one whose atoms carry distinct labels. Suppose a connected component of size j has total weight x_j . A set of k such components with total size n has weight enumerator $\mathcal{B}_{n,k}^{\exp}(x_1, x_2, \dots)$; a set of any number of components has enumerator $\mathcal{B}_n^{\exp}(x_1, \dots, x_n)$.

Theorem 1.6 (The exponential formula). *Let*

$$C(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}.$$

Then

$$\frac{C(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}^{\exp}(x_1, \dots, x_{n-k+1}) \frac{z^n}{n!}, \quad (1.10)$$

$$e^{uC(z)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}^{\exp}(x_1, \dots) u^k \frac{z^n}{n!}, \quad (1.11)$$

$$e^{C(z)} = \sum_{n \geq 0} \mathcal{B}_n^{\exp}(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.12)$$

Proof. Expand $C(z)^k/k!$ multinomially. If j_i components have size i , then $\sum_i j_i = k$ and $\sum_i i j_i = n$, and the coefficient of z^n is

$$\prod_i \frac{1}{j_i!} \left(\frac{x_i}{i!} \right)^{j_i}.$$

Multiplication by $n!$ labels the atoms. This proves the first formula. Summing it with weight u^k proves the second, and setting $u = 1$ proves the third. $\square$

Remark 1.7 (Formal crosswalk). All three identities are formal power-series substitutions over any $\mathbb{Q}$ -algebra: Equation (1.10) is `Fabius.bellWeightSeries_pow` (module `BellGeneratingFunctions`), Equation (1.12) is `Fabius.exp_subst_bellWeightSeries`, and Equation (1.11) is `Fabius.exp_subst_smul_bellWeightSeries` (module `ExponentialFormula`), obtained from the $u = 1$ case by the degree homogeneity $\mathcal{B}_{n,k}^{\exp}(ux) = u^k \mathcal{B}_{n,k}^{\exp}(x)$. The formal proofs use the pointing recurrence of Theorem 13.1 rather than the multinomial expansion.

Corollary 1.8 (Partition-type count). *The number of partitions of an n -element labelled set with exactly j_i blocks of size i is*

$$\frac{n!}{\prod_{i \geq 1} (i!)^{j_i} j_i!}, \quad \sum_i i j_i = n. \quad (1.13)$$

Proof. Order the n labels, cut the ordering into blocks of prescribed sizes, then divide by $i!$ for the internal order of each size- i block and by $j_i!$ for the order of blocks of equal size. $\square$

Part II

Stirling Numbers, Lah Numbers, and Basis Changes

Chapter 2

The two Stirling triangles

2.1 First-kind numbers: cycles and factorial coefficients

Definition 2.1. The signed Stirling numbers of the first kind $s(n, k)$ and the unsigned numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ are defined by

$$(x)^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (2.1)$$

$$(x)^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (2.2)$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}. \quad (2.3)$$

The sign relation follows from $(x)^{\overline{n}} = (-1)^n (-x)^{\underline{n}}$.

Theorem 2.2 (Permutation interpretation). $\begin{bmatrix} n \\ k \end{bmatrix}$ is the number of permutations of $[n]$ having exactly k cycles. Consequently,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}, \quad (2.4)$$

with $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. Given a permutation of $[n+1]$ with k cycles, either $n+1$ is a singleton cycle, leaving a permutation of $[n]$ with $k-1$ cycles, or it is inserted immediately after one of the n existing elements in the cyclic notation of a permutation of $[n]$ with k cycles. These alternatives are disjoint and reversible. The same recurrence follows algebraically by multiplying $(x)^{\overline{n}}$ by $x+n$ and comparing coefficients. $\square$

For $n = 3$, the cycle counts are $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1$, $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3$, and $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2$, corresponding to the identity, the three transpositions, and the two 3-cycles. Summing over the possible number of cycles gives

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} = n!, \quad (2.5)$$

which also follows from Equation (2.2) at $x = 1$. In Lean, Equation (2.5) is `Fabius.sum_stirlingF` `first_eq_factorial`, proved exactly by evaluating Equation (2.2) at $x = 1$.

2.2 Second-kind numbers: set partitions and surjections

Definition 2.3. The Stirling number of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Theorem 2.4 (Second-kind recurrence). For $n, k \geq 0$,

$$\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\}, \quad (2.6)$$

with $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. In a partition of $[n+1]$ into k blocks, the new label either joins one of the k blocks of a partition of $[n]$, or forms a singleton block beside a partition of $[n]$ into $k-1$ blocks. $\square$

Theorem 2.5 (Surjection formula). For all $n, k \geq 0$,

$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^k \frac{(-1)^{k-j} j^n}{(k-j)! j!}. \quad (2.7)$$

Proof. There are $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ surjections from $[n]$ onto a fixed k -element codomain: a set partition into fibres followed by an ordering of the fibres. Inclusion–exclusion over the set of omitted codomain values gives $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$. Division by $k!$ proves the formula. $\square$

Remark 2.6 (Formal crosswalk). The identity is `Fabius.factorial_mul_stirlingSecond_eq_sum`, stated in $\mathbb{Z}$ as $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$, and `Fabius.stirlingSecond_eq_sum_div_factorial` over $\mathbb{Q}$ (module `StirlingBasisChange`). The Lean proof does not count surjections: it evaluates the falling-factorial expansion of Theorem 3.1 at a natural number, obtaining $m^n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} k! \binom{m}{k}$ for every m (`Fabius.pow_eq_sum_stirlingSecond_mul_factorial_mul_choose`), and inverts this relation by binomial inversion (Theorem 25.1). The combinatorial reading (there are $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ surjections) is not formalized.

The simple diagonals follow at once:

$$\left\{ \begin{smallmatrix} n \\ n \end{smallmatrix} \right\} = 1, \quad \left\{ \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\} = 1 \ (n \geq 1), \quad \left\{ \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\} = \binom{n}{2}, \quad \left\{ \begin{smallmatrix} n \\ 2 \end{smallmatrix} \right\} = 2^{n-1} - 1. \quad (2.8)$$

The last identity chooses a nonempty proper subset as one block and divides by two for exchanging the blocks.

2.3 Alternative and reverse recurrences

The ordinary recurrence builds each row from the preceding one. Several less familiar identities in the source archive instead recover an entry from entries to its right or from earlier entries in the same column. They were labelled there as conjectures; in fact they are exact identities.

Theorem 2.7 (Alternative second-kind recurrences). For $1 \leq k < n$,

$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{smallmatrix} n \\ r \end{smallmatrix} \right\}}{(k-r)!}, \quad (2.9)$$

$$(n-k) \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \sum_{j=2}^{n-k+1} (j-2)! \binom{-k}{j} \left\{ \begin{smallmatrix} n \\ k+j-1 \end{smallmatrix} \right\}, \quad (2.10)$$

$$(n-k) \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \sum_{j=2}^{n-k+1} (-1)^j \binom{n}{j} \left\{ \begin{smallmatrix} n-j+1 \\ k \end{smallmatrix} \right\}. \quad (2.11)$$

The terminal value in either reverse recursion is $\left\{ \begin{smallmatrix} n \\ n \end{smallmatrix} \right\} = 1$.

Proof. Substitute $x = k$ in the basis identity $k^n = \sum_r \left\{ \begin{smallmatrix} n \\ r \end{smallmatrix} \right\} (k)^r$. Since $(k)^r = k!/(k-r)!$ for $r \leq k$, division by $k!$ and isolation of the $r = k$ term gives (2.9).

For the row reversal, use the bivariate exponential generating function

$$F(x, y) = \exp(y(e^x - 1)) = \sum_{n, k \geq 0} \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} \frac{x^n}{n!} y^k, \quad u = e^x - 1.$$

The generating function of the left side of (2.10) is

$$xF_x - yF_y = yF(xe^x - u).$$

On the other hand, because $\left(\begin{smallmatrix} -k \\ j \end{smallmatrix} \right) = (-1)^j \binom{k+j-1}{j}$, the generating function of its right side is

$$\begin{aligned} y \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} \partial_y^j F &= yF \sum_{j \geq 2} \frac{(-1)^j u^j}{j(j-1)} \\ &= yF((1+u) \log(1+u) - u) \\ &= yF(xe^x - u). \end{aligned}$$

Here the middle series is obtained by differentiating it twice, summing the geometric series, and restoring the two zero initial coefficients. Equality of coefficients proves (2.10).

For the column reversal fix k and put

$$F_k(x) = \sum_{n \geq k} \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$$

The two sides of (2.11), after multiplication by $x^n/n!$ and summation over n , become respectively

$$xF'_k(x) - kF_k(x) \quad \text{and} \quad (e^{-x} - 1 + x)F'_k(x).$$

They are equal because direct differentiation gives $(1 - e^{-x})F'_k(x) = kF_k(x)$. $\square$

Remark 2.8 (Formal crosswalk). Equation (2.9) is `Fabius.stirlingSecond_eq_pow_div_factorial_sub_sum` (module `StirlingTriangularExplicit`), proved exactly as in the first paragraph from the surjection count $k^n = \sum_r \left\{ \begin{smallmatrix} n \\ r \end{smallmatrix} \right\} r! \binom{k}{r}$ (`Fabius.pow_eq_sum_stirlingSecond_mul_factorial_mul_choose`); the formal statement holds for all $n, k \geq 0$ with the sum running from $r = 0$ (the extra term $\left\{ \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\}/k!$ vanishes for $n \geq 1$). The reverse recurrences Equations (2.10) and (2.11) are not formalized.

The factor interchange mentioned in the source produces a parallel pair for the unsigned first-kind triangle.

Theorem 2.9 (Reverse recurrences of the first kind). *For $1 \leq k < n$,*

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (-1)^j \binom{-k}{j} \begin{bmatrix} n \\ k+j-1 \end{bmatrix}, \quad (2.12)$$

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (j-2)! \binom{n}{j} \begin{bmatrix} n-j+1 \\ k \end{bmatrix}. \quad (2.13)$$

Proof. Let $C_n(y) = (y)^{\overline{n}} = \sum_r \begin{bmatrix} n \\ r \end{bmatrix} y^r$. Since

$$C_n(1+y) = \frac{(y)^{\overline{n+1}}}{y},$$

the coefficient of y^{k-1} on the two sides is, respectively,

$$\sum_{r \geq k-1} \binom{r}{k-1} \begin{bmatrix} n \\ r \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Remove the terms $r = k-1$ and $r = k$ and put $r = k+j-1$. Because $(-1)^j \binom{-k}{j} = \binom{k+j-1}{j}$, the remainder is precisely (2.12).

For fixed k , set

$$C_k(x) = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{[-\log(1-x)]^k}{k!}.$$

The exponential generating function of the right side of (2.13) is

$$C'_k(x) \sum_{j \geq 2} \frac{x^j}{j(j-1)} = (x + (1-x)\log(1-x))C'_k(x).$$

The left side has generating function $xC'_k - kC_k$. Their equality reduces to

$$-(1-x)\log(1-x)C'_k(x) = kC_k(x),$$

which follows immediately from the displayed formula for C_k . $\square$

Remark 2.10 (Formal crosswalk). Equation (2.12) is `Fabius.first_reverse_row` (module `StirlingFirstReverse`), with the summation index written as $j = i+2$, $0 \leq i < n-k$, and $(-1)^j \binom{-k}{j}$ replaced by $\binom{k+j-1}{j}$: the coefficient identity $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_r \binom{r}{k} \begin{bmatrix} n \\ r \end{bmatrix}$ is `Fabius.stirlingFirst_succ_succ_eq_sum_choose` (module `StirlingBasisChange`), from $(x)^{\overline{n+1}} = x(x+1)^{\overline{n}}$ (Mathlib's `ascPochhammer_succ_left`), and removing the terms $r = k-1, k$ gives the recurrence. Equation (2.13) is `Fabius.first_reverse_column`, stated as $(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^n \binom{n}{j} (j-2)! \begin{bmatrix} n-j+1 \\ k \end{bmatrix}$ (the extra terms vanish); the formal proof compares exponential generating functions over a $\mathbb{Q}$ -algebra exactly as in the text: `Fabius.egfA_kernel` is $\sum_{j \geq 2} x^j / (j(j-1)) = x + (1-x)\log(1-x)$, `Fabius.one_sub_X_mul_negLog_mul_derivative_egfA_stirlingFirst` is $-(1-x)\log(1-x)C'_k = kC_k$, and `Fabius.egfA_first_reverse_column` is the resulting identity $xC'_k - kC_k = (x + (1-x)\log(1-x))C'_k$, from which `Fabius.seq_eq_of_egfA_eq` extracts the coefficients.

2.4 Tables and common boundary conventions

The first rows are

$n \backslash k$	0	1	2	3	4	5	6	$n \backslash k$	0	1	2	3	4	5	6
0	1							0	1						
1	0	1						1	0	1					
2	0	1	1					2	0	1	1				
3	0	2	3	1				3	0	1	3	1			
4	0	6	11	6	1			4	0	1	7	6	1		
5	0	24	50	35	10	1		5	0	1	15	25	10	1	
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

for $\begin{bmatrix} n \\ k \end{bmatrix}$ and $\{n\}_k$, respectively.

Chapter 3

Factorial bases and inverse matrices

3.1 The basic changes of basis

Theorem 3.1 (Stirling basis transformations). *For every $n \geq 0$,*

$$(x)^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (3.1)$$

$$(x)^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (3.2)$$

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} (x)^{\underline{k}}, \quad (3.3)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} (x)^{\overline{k}}. \quad (3.4)$$

Equivalently,

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} k! \binom{x}{k}, \quad \binom{x}{n} = \frac{1}{n!} \sum_{k=0}^n s(n, k) x^k. \quad (3.5)$$

Proof. The first two identities are definitions. In the Newton expansion Equation (1.9) for $p(x) = x^n$, one has

$$\Delta^k x^n|_{x=0} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

by Equation (2.7). This gives Equation (3.3). Replacing x by $-x$ and multiplying by $(-1)^n$ gives Equation (3.4). Division of $(x)^{\underline{n}}$ by $n!$ gives the second binomial-basis formula. $\square$

Remark 3.2 (Formal crosswalk). All four identities are polynomial identities in Lean, valid over every commutative semiring or ring rather than for an indeterminate over $\mathbb{Q}$: Equation (3.2) is `Fabius.ascPochhammer_eq_sum_monomial_stirlingFirst` (commutative semirings), Equation (3.1) is `Fabius.descPochhammer_eq_sum_monomial_signedStirlingFirst`, Equation (3.3) is `Fabius.X_pow_eq_sum_stirlingSecond_mul_descPochhammer`, and Equation (3.4) is `Fabius.X_pow_eq_sum_stirlingSecond_mul_ascPochhammer` (commutative rings), all in `StirlingBasisChange`. The signed numbers are `Fabius.signedStirlingFirst` with recurrence `Fabius.signedStirlingFirst_succ_succ`; the unsigned and second-kind numbers are Mathlib's `Nat.stirlingFirst` and `Nat.stirlingSecond`. The evaluated forms of Equation (3.5) are `Fabius.pow_eq_sum_stirlingSecond_mul_descPochhammer_eval` (any commutative ring) and, for natural numbers, `Fabius.pow_eq_sum_stirlingSecond_mul_d`

`escFactorial` and `Fabius.pow_eq_sum_stirlingSecond_mul_factorial_mul_choose`. The Lean proofs are not by Newton expansion: the first two identities are proved by comparing coefficients along the one-step recurrences, the third by the single relation $x \binom{x}{k} = (x)^{\overline{k+1}} + k \binom{x}{k}$, and the fourth by the substitution $x \mapsto -x$.

Corollary 3.3 (Shifted factorial evaluations). *For every $n \geq 0$,*

$$\sum_{k=1}^{n+1} \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} (x-1)^{\overline{k-1}} = x^n, \quad (3.6)$$

$$\sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} (n)^{\overline{k}} = n^n. \quad (3.7)$$

Proof. Apply (3.3) with exponent $n+1$ and use $(x)^{\overline{k}} = x(x-1)^{\overline{k-1}}$. Both sides are divisible by x as polynomials, and cancellation gives (3.6). Formula (3.7) is (3.3) evaluated at $x = n$. $\square$

Remark 3.4 (Formal crosswalk). Equation (3.6) is `Fabius.X_pow_eq_sum_stirlingSecond_succ_mul_descPochhammer_comp` (module `StirlingShiftedEvaluations`), in $R[x]$ over any commutative ring R , with $(x-1)^{\overline{k}}$ as the composition of `Mathlib's descPochhammer` with $x-1$; the proof is the cancellation above, using that x is a non-zero-divisor in $R[x]$ (`Fabius.X_mul_cancel`). Equation (3.7) is `Fabius.pow_self_eq_sum_stirlingSecond_mul_descFactorial`.

Corollary 3.5 (Inverse matrices). *The lower triangular matrices $s = (s(n, k))$ and $S = (\left\{ \begin{matrix} n \\ k \end{matrix} \right\})$ are mutual inverses:*

$$\sum_{j=k}^n s(n, j) \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.8)$$

$$\sum_{j=k}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} s(j, k) = \delta_{nk}. \quad (3.9)$$

Equivalently,

$$\sum_{j=k}^n (-1)^{n-j} \left[\begin{matrix} n \\ j \end{matrix} \right] \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.10)$$

$$\sum_{j=k}^n (-1)^{j-k} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \left[\begin{matrix} j \\ k \end{matrix} \right] = \delta_{nk}. \quad (3.11)$$

Proof. Expand $(x)^{\overline{n}}$ into powers and each power back into falling factorials. Uniqueness of coordinates in the factorial basis gives the first matrix product. The reverse substitution gives the second. $\square$

Remark 3.6 (Formal crosswalk). Equation (3.9) is `Fabius.sum_range_stirlingSecond_mul_signedStirlingFirst` (with the sum over $0 \leq j \leq n$; the terms with $j < k$ vanish) and Equation (3.8) is `Fabius.sum_range_signedStirlingFirst_mul_stirlingSecond`; both also exist over the interval $k \leq j \leq n$ (`Fabius.sum_Icc_stirlingSecond_mul_signedStirlingFirst`, `Fabius.sum_Icc_signedStirlingFirst_mul_stirlingSecond`), all in `StirlingBasisChange`. The Lean proof of the second product compares monomial coefficients as in the text; the first is then obtained from the second by the general fact that a one-sided inverse of a triangular kernel is two-sided (`Fabius.lowerTriangular_orthogonal_comm`, module `TriangularKernelInverse`, proved through the finite matrices `Fabius.kernelMatrix`), so no uniqueness argument in the factorial basis is needed. Uniqueness of coordinates in a triangular basis is nevertheless available as `Fabius.eq_of_sum_smul_eq_sum_smul`.

3.2 Power sums by basis adaptation

The factorial basis makes discrete antidifferentiation immediate. Combining Equations (1.8) and (3.3) yields

$$\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{(N+1)^{k+1}}{k+1}. \quad (3.12)$$

For $m = 4$ this is

$$\frac{1}{2} (N+1)^2 + \frac{7}{3} (N+1)^3 + \frac{6}{4} (N+1)^4 + \frac{1}{5} (N+1)^5,$$

which expands to the familiar quartic power-sum polynomial. Formula Equation (3.12) is `Fabius.sum_range_pow_eq_sum_stirlingSecond` (module `StirlingBasisChange`), in the integer form $\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} k! \binom{N+1}{k+1}$, which is the display since $(N+1)^{k+1}/(k+1) = k! \binom{N+1}{k+1}$.

3.3 Factorial moments: Poisson variables and permutation fixed points

The change from ordinary powers to falling factorials is especially effective for integer-valued random variables.

Theorem 3.7 (Poisson moments). *If X has the Poisson distribution of mean λ , then*

$$\mathbb{E}X^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \lambda^k. \quad (3.13)$$

In particular, $\mathbb{E}X^n = B_n^{\text{Bell}}$ when $\lambda = 1$.

Proof. The k th falling-factorial moment is

$$\mathbb{E}(X)^{\underline{k}} = e^{-\lambda} \sum_{r \geq k} \frac{(r)^{\underline{k}} \lambda^r}{r!} = \lambda^k e^{-\lambda} \sum_{q \geq 0} \frac{\lambda^q}{q!} = \lambda^k.$$

Take expectations in the basis identity $X^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} (X)^{\underline{k}}$. □

Theorem 3.8 (Moments of the number of fixed points). *Let Y_m be the number of fixed points of a uniformly random permutation of $[m]$. Then*

$$\mathbb{E}Y_m^n = \sum_{k=0}^m \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^{\min(m,n)} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (3.14)$$

Thus, when $m \geq n$, the moment equals the Bell number B_n^{Bell} .

Proof. The random variable $(Y_m)^{\underline{k}}$ counts ordered k -tuples of distinct fixed points. There are $(m)^{\underline{k}}$ candidate tuples, and a specified tuple is fixed with probability $(m-k)!/m!$. Hence

$$\mathbb{E}(Y_m)^{\underline{k}} = 1 \quad (0 \leq k \leq m), \quad \mathbb{E}(Y_m)^{\underline{k}} = 0 \quad (k > m).$$

Taking expectations in (3.3) proves the formula. The upper limit m is essential: it records the finite size of the permutation, even when $n > m$. □

3.4 Lah numbers

Definition 3.9. For $1 \leq k \leq n$, the unsigned Lah number is

$$\mathbb{L}_{n,k}^{\text{Lah}} = \binom{n-1}{k-1} \frac{n!}{k!}, \quad (3.15)$$

with $\mathbb{L}_{0,0}^{\text{Lah}} = 1$ and zero otherwise outside the triangle.

Combinatorially, $\mathbb{L}_{n,k}^{\text{Lah}}$ counts partitions of $[n]$ into k nonempty linearly ordered lists, with the collection of lists unordered.

Theorem 3.10 (Conversion between rising and falling factorials).

$$(x)^{\overline{n}} = \sum_{k=0}^n \mathbb{L}_{n,k}^{\text{Lah}} (x)^{\underline{k}}, \quad (3.16)$$

$$(x)^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} \mathbb{L}_{n,k}^{\text{Lah}} (x)^{\overline{k}}, \quad (3.17)$$

$$\sum_{j=k}^n (-1)^{j-k} \mathbb{L}_{n,j}^{\text{Lah}} \mathbb{L}_{j,k}^{\text{Lah}} = \delta_{nk}. \quad (3.18)$$

Moreover,

$$\mathbb{L}_{n,k}^{\text{Lah}} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{Bmatrix} j \\ k \end{Bmatrix}. \quad (3.19)$$

Proof. A direct generating-function calculation gives

$$\sum_{n \geq k} \mathbb{L}_{n,k}^{\text{Lah}} \frac{z^n}{n!} = \frac{1}{k!} \left(\frac{z}{1-z} \right)^k.$$

Therefore

$$\sum_{n \geq 0} \left(\sum_k \mathbb{L}_{n,k}^{\text{Lah}} (x)^{\underline{k}} \right) \frac{z^n}{n!} = \sum_{k \geq 0} \binom{x}{k} \left(\frac{z}{1-z} \right)^k = (1-z)^{-x} = \sum_{n \geq 0} (x)^{\overline{n}} \frac{z^n}{n!},$$

proving Equation (3.16). Replace x by $-x$ to obtain the inverse formula, and compose the two transformations to obtain Equation (3.18). Finally, expand a rising factorial into powers by first-kind numbers and powers into falling factorials by second-kind numbers; uniqueness gives Equation (3.19). $\square$

Remark 3.11 (Formal crosswalk). In Lean the Lah numbers `Fabius.lahNumber` are defined by the recurrence $\mathbb{L}_{n+1,k+1}^{\text{Lah}} = (n+k+1)\mathbb{L}_{n,k+1}^{\text{Lah}} + \mathbb{L}_{n,k}^{\text{Lah}}$, and the closed form Equation (3.15) is the theorem `Fabius.lahNumber_succ_succ_mul_factorial`, in the division-free form $\mathbb{L}_{n+1,k+1}^{\text{Lah}} (k+1)! = (n+1)! \binom{n}{k}$. Then Equation (3.16) is `Fabius.ascPochhammer_eq_sum_lahNumber_mul_descPochhammer`, Equation (3.17) is `Fabius.descPochhammer_eq_sum_lahNumber_mul_ascPochhammer`, Equation (3.18) is `Fabius.sum_range_lahNumber_mul_lahNumber`, and Equation (3.19) is `Fabius.lahNumber_eq_sum_stirlingFirst_mul_stirlingSecond` (module `LahNumbers`); the polynomial identities hold over every commutative ring. The Lean proofs use no generating function: the conversion formula follows from $(x)^{\underline{k}}(x+n) = (x)^{\underline{k+1}} + (n+k)(x)^{\underline{k}}$ by induction, and the last two identities are read off by uniqueness of coefficients in the factorial bases (`Fabius.eq_of_sum_C_mul_eq_sum_C_mul`). The inequalities Equation (3.20) are not formalized.

The inequalities

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \left[\begin{matrix} n \\ k \end{matrix} \right] \leq \mathbb{L}_{n,k}^{\text{Lah}} \quad (3.20)$$

follow from explicit injections: a set partition may be canonically turned into a permutation with one cycle per block by writing each block with its smallest member first and arranging cycles by minima; forgetting the cyclic identifications embeds permutations with k cycles into unordered collections of k lists.

Chapter 4

Generating functions, transforms, and operators

4.1 Exponential generating functions

Theorem 4.1 (Column generating functions). *For every fixed $k \geq 0$,*

$$\sum_{n \geq k} s(n, k) \frac{z^n}{n!} = \frac{\log(1+z)^k}{k!}, \quad (4.1)$$

$$\sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!} = \frac{[-\log(1-z)]^k}{k!}, \quad (4.2)$$

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^k}{k!}. \quad (4.3)$$

The bivariate second-kind generating function is

$$\sum_{k \geq 0} \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} y^k \frac{z^n}{n!} = \exp(y(e^z - 1)). \quad (4.4)$$

Proof. Expand

$$(1+z)^u = e^{u \log(1+z)} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k=0}^n s(n, k) u^k = \sum_{k \geq 0} \frac{u^k}{k!} \log(1+z)^k$$

and compare coefficients of u^k . Replacing z by $-z$ and using the sign relation gives Equation (4.2). Formula Equation (4.3) is the exponential formula for a set of k nonempty labelled blocks; algebraically it also follows by inserting Equation (2.7) and summing exponentials. Summing over k with weight y^k proves Equation (4.4). $\square$

Remark 4.2 (Formal crosswalk). The three column identities are formal power-series identities in Lean, over every commutative $\mathbb{Q}$ -algebra (module `StirlingGeneratingFunctions`): with `egf a =  $\sum_n a_n z^n / n!$` , Equation (4.3) is `Fabius.exp_sub_one_pow` ($(e^z - 1)^k = \sum_n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\} z^n / n!$) and `Fabius.egf_stirlingSecond`; Equation (4.2) is `Fabius.negLogOneSub_pow` and `Fabius.egf_stirlingFirst`, where $-\log(1-z) = \sum_{n \geq 1} z^n / n$ is `Fabius.negLogOneSub`; and Equation (4.1) is `Fabius.log_pow`, with Mathlib's `PowerSeries.log` for $\log(1+z)$. The Lean proofs differ from the text: the second-kind identity is read off coefficientwise from the binomial expansion of $(e^z - 1)^k$ into the rescaled exponentials e^{mz} together with the surjection formula (Theorem 2.5); the unsigned first-kind identity is proved by induction on k from the first-order equation $(1-z)Y' = (k+1)Y_k$

satisfied by both sides, using that a power series over a torsion-free ring is determined by its derivative and constant term (`Fabius.eq_of_one_sub_X_mul_derivative_eq`); the signed identity is the substitution $z \mapsto -z$ (`PowerSeries.rescale`). The bivariate generating function Equation (4.4) and the derivative identity Equation (4.5) are not formalized.

Differentiating Equation (4.1) gives the useful identity

$$\frac{\log(1+z)^m}{1+z} = m! \sum_{r \geq 0} s(r+1, m+1) \frac{z^r}{r!}, \quad |z| < 1, \quad (4.5)$$

which is valid formally as well.

4.2 A rational ordinary generating function

Theorem 4.3 (Fixed- k ordinary generating function). *For $k \geq 0$,*

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{n-k} = \prod_{j=1}^k \frac{1}{1-jx} = \frac{1}{x^{k+1} (1/x)^{k+1}}. \quad (4.6)$$

Consequently,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = h_{n-k}(1, 2, \dots, k) = \sum_{c_1 + \dots + c_k = n-k} 1^{c_1} 2^{c_2} \dots k^{c_k}, \quad (4.7)$$

where h_r is a complete homogeneous symmetric polynomial.

Proof. Let $F_k(x) = \sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r$. The recurrence Equation (2.6), with $n = k + r - 1$, gives $F_k(x) = (1 - kx)^{-1} F_{k-1}(x)$ and $F_0 = 1$. Iteration proves the product. Expanding each geometric factor independently proves the complete-homogeneous sum. The factorial form follows by direct factorization. $\square$

Remark 4.4 (Formal crosswalk). The product formula is `Fabius.prod_one_sub_mul_X_mul_stirlingColumnOGF` (module `StirlingOrdinaryGF`): in $R[[x]]$, for every commutative ring R , the column series $\sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r$ times $\prod_{j=1}^k (1 - jx)$ equals 1, proved exactly as above from the column recurrence $(1 - kx)F_k = F_{k-1}$ (`Fabius.one_sub_mul_X_mul_stirlingColumnOGF_succ`). The expansion of each geometric factor is `Fabius.stirlingColumnOGF_eq_prod_mk_pow`, which identifies the column series with $\prod_{j=1}^k \sum_{r \geq 0} j^r x^r$; the complete-homogeneous coefficient formula Equation (4.7) is the coefficient extraction of that product and is not stated separately.

A useful rearrangement of the count of all maps $[n] \rightarrow [k]$ is

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{matrix} n \\ r \end{matrix} \right\}}{(k-r)!}. \quad (4.8)$$

Indeed, a map using exactly r values can be chosen in $\binom{k}{r} r! \left\{ \begin{matrix} n \\ r \end{matrix} \right\}$ ways; divide the identity $k^n = \sum_r \binom{k}{r} r! \left\{ \begin{matrix} n \\ r \end{matrix} \right\}$ by $k!$.

4.3 The Stirling transform

Theorem 4.5 (Stirling transform and its inverse). *Let*

$$g_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} f_k.$$

Then

$$f_n = \sum_{k=0}^n s(n, k) g_k = \sum_{k=0}^n (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} g_k. \quad (4.9)$$

For exponential generating functions,

$$G^{\text{egf}}(z) = F^{\text{egf}}(e^z - 1), \quad F^{\text{egf}}(z) = G^{\text{egf}}(\log(1 + z)). \quad (4.10)$$

Proof. The inverse formula is exactly the matrix inversion in Theorem 3.5. For the generating function,

$$G^{\text{egf}}(z) = \sum_k f_k \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \sum_k f_k \frac{(e^z - 1)^k}{k!} = F^{\text{egf}}(e^z - 1).$$

Substitution by the compositional inverse $\log(1 + z)$ gives the reverse relation. $\square$

Remark 4.6 (Formal crosswalk). The inversion formula Equation (4.9) is `Fabius.stirling_inversion`, with converse `Fabius.stirling_inversion_symm` and the equivalence `Fabius.stirling_inversion_iff` (module `StirlingBasisChange`), for sequences with values in any additive commutative group, the Stirling numbers acting by integer multiples. The generating-function relations Equation (4.10) are not formalized.

4.4 Differential-operator normal ordering

Let $D = d/dx$. The Euler operator xD is diagonal on monomials: $(xD)x^m = mx^m$. The operator $x^k D^k$ has eigenvalue $(m)_k^{\underline{}}$. Thus the scalar basis changes lift verbatim to operators.

Theorem 4.7 (Stirling normal-ordering identities). For $n \geq 0$,

$$(xD)^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k, \quad (4.11)$$

$$x^n D^n = \sum_{k=0}^n s(n, k) (xD)^k = (xD)^{\underline{n}}. \quad (4.12)$$

Proof. Both sides of Equation (4.11) act on x^m as multiplication by $m^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} (m)_k^{\underline{}}$. Since monomials span the polynomial ring, the operators are equal. The inverse relation is proved identically using $(m)^{\underline{n}} = \sum_k s(n, k) m^k$. $\square$

Remark 4.8 (Formal crosswalk). Module `StirlingNormalOrder` proves both identities applied to an arbitrary polynomial p over any commutative ring: Equation (4.11) is `Fabius.iterate_X_mul_derivative`, $(xD)^n p = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k p$, by induction on n from $xD(x^k D^k p) = k x^k D^k p + x^{k+1} D^{k+1} p$ (`Fabius.X_mul_derivative_xkDk`) and the recurrence $\left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}$; Equation (4.12) is `Fabius.xkDk_eq_sum_signedStirlingFirst`, obtained from it by Stirling inversion (`Fabius.stirling_inversion`). The falling-factorial form $(xD)^{\underline{n}}$ and the operator series Equations (4.13) and (4.14) are not formalized.

Because $E = e^D$ and $\Delta = e^D - 1$, substitution of the mutually inverse series $D = \log(1 + \Delta)$ and $\Delta = e^D - 1$ gives, whenever the operator series converge (or formally on polynomials),

$$\frac{D^k}{k!} = \sum_{n \geq k} \frac{s(n, k)}{n!} \Delta^n, \quad (4.13)$$

$$\frac{\Delta^k}{k!} = \sum_{n \geq k} \frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{n!} D^n. \quad (4.14)$$

These are the derivative–finite-difference transformations listed in the archive.

Chapter 5

Symmetric functions, diagonals, and finite identities

5.1 Elementary and complete symmetric descriptions

From Equation (2.2),

$$\begin{bmatrix} n \\ n-r \end{bmatrix} = e_r(0, 1, \dots, n-1) = \sum_{0 \leq i_1 < \dots < i_r < n} i_1 \cdots i_r. \quad (5.1)$$

Together with Equation (4.7), this identifies the first-kind triangle with elementary symmetric functions and the second-kind triangle with complete homogeneous symmetric functions.

The first few near-diagonals are

$$\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}, \quad (5.2)$$

$$\begin{bmatrix} n \\ n-2 \end{bmatrix} = \frac{3n-1}{4} \binom{n}{3}, \quad (5.3)$$

$$\begin{bmatrix} n \\ n-3 \end{bmatrix} = \binom{n}{2} \binom{n}{4}. \quad (5.4)$$

Proof. The first is $e_1(0, \dots, n-1)$. Newton's identities $2e_2 = p_1^2 - p_2$ and $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$, combined with the elementary sums of first, second, and third powers, simplify to Equations (5.3) and (5.4). Equivalently, one may classify permutations by the nontrivial cycle types 3, 2 + 2 for the second diagonal and 4, 3 + 2, 2 + 2 + 2 for the third. $\square$

5.2 Harmonic-number formulae

Factoring $(n-1)!$ from $(x+1)(x+2) \cdots (x+n-1)$ gives

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}}{(n-1)!} = e_k \left(1, \frac{1}{2}, \dots, \frac{1}{n-1} \right) = \sum_{1 \leq i_1 < \dots < i_k < n} \frac{1}{i_1 \cdots i_k}. \quad (5.5)$$

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and $H_n = H_n^{(1)}$. Newton's identities now give

$$\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad (5.6)$$

$$\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (5.7)$$

$$\begin{bmatrix} n \\ 4 \end{bmatrix} = \frac{(n-1)!}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right). \quad (5.8)$$

More generally,

$$\frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} = \frac{1}{k!} \mathcal{B}_k^{\exp}(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{k-1}(k-1)!H_n^{(k)}). \quad (5.9)$$

Proof. Take logarithms in

$$\prod_{j=1}^n \left(1 + \frac{x}{j} \right) = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1} H_n^{(r)}}{r} x^r \right)$$

and apply the complete Bell-polynomial generating function Equation (1.12). $\square$

If $w(n, k) = k! \begin{bmatrix} n \\ k+1 \end{bmatrix} / (n-1)!$, then complete Bell recurrence yields

$$w(n, m) = \delta_{m0} + \sum_{r=0}^{m-1} (1-m)_r H_{n-1}^{(r+1)} w(n, m-1-r), \quad (5.10)$$

where $(1-m)_r$ is falling. Conversely, taking a logarithm recovers the harmonic power sums:

$$H_n^{(m)} = -m [x^m] \log \left(1 + \sum_{k \geq 1} \frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} (-x)^k \right). \quad (5.11)$$

For example,

$$(n!)^2 H_n^{(2)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^2 - 2 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix}, \quad (5.12)$$

$$(n!)^3 H_n^{(3)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^3 - 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 2 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix}^2 \begin{bmatrix} n+1 \\ 4 \end{bmatrix}. \quad (5.13)$$

5.3 Convolution and summation identities

The following paired identities expose the parallel structure of the two triangles.

Theorem 5.1 (First- and second-kind summations). *For valid nonnegative indices,*

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{bmatrix} j \\ k \end{bmatrix} = \sum_{j=k}^n \frac{n!}{j!} \begin{bmatrix} j \\ k \end{bmatrix}, \quad (5.14)$$

$$\left\{ \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{bmatrix} j \\ k \end{bmatrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{bmatrix} j \\ k \end{bmatrix} \right\}, \quad (5.15)$$

$$\begin{bmatrix} n+k+1 \\ k \end{bmatrix} = \sum_{j=0}^k (n+j) \begin{bmatrix} n+j \\ j \end{bmatrix}, \quad (5.16)$$

$$\left\{ \begin{bmatrix} n+k+1 \\ k \end{bmatrix} \right\} = \sum_{j=0}^k j \left\{ \begin{bmatrix} n+j \\ j \end{bmatrix} \right\}. \quad (5.17)$$

Moreover, for $\ell, m \geq 0$,

$$\binom{\ell+m}{\ell} \begin{bmatrix} n \\ \ell+m \end{bmatrix} = \sum_j \binom{n}{j} \begin{bmatrix} j \\ \ell \end{bmatrix} \begin{bmatrix} n-j \\ m \end{bmatrix}, \quad (5.18)$$

$$\binom{\ell+m}{\ell} \left\{ \begin{matrix} n \\ \ell+m \end{matrix} \right\} = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} \left\{ \begin{matrix} n-j \\ m \end{matrix} \right\}. \quad (5.19)$$

Proof. For the first equality in Equation (5.14), mark k cycles of a permutation of $[n]$ and insert $n+1$ into the canonical construction; algebraically it is the coefficient of x^k in $\frac{d}{dx}(x)^{\overline{n}}$. The second equality follows by iterating recurrence Equation (2.4) in the n direction. For Equation (5.15), distinguish the elements outside the block containing $n+1$ for the first equality, and add the $n-k$ remaining elements one at a time to one of $k+1$ existing blocks for the second. The hockey-stick forms telescope the respective recurrences. Finally, in Equation (5.18), choose which labels belong to the ℓ marked cycles; in Equation (5.19), choose which labels belong to the ℓ marked blocks. The binomial factor on the left forgets which of the total $\ell+m$ components were marked. $\square$

Remark 5.2 (Formal crosswalk). All six identities are over the natural numbers, and lie in module `StirlingSummations` except where noted. The first equality of Equation (5.14) is `Fabius.stirlingGFirst_succ_succ_eq_sum_choose` (module `StirlingBasisChange`, which holds it because the reverse row recurrence of Equation (2.12) needs the same identity and neither module imports the other), proved as the coefficient of x^{k+1} in $(x)^{\overline{n+1}} = x(x+1) \cdots (x+n)$, and the second is `Fabius.stirlingFirst_succ_succ_eq_sum_descFactorial` (with $n!/j! = (n)^{\overline{n-j}}$); the first equality of Equation (5.15) is `Fabius.stirlingSecond_succ_succ_eq_sum` (module `BellStirling`) and the second `Fabius.stirlingSecond_succ_succ_eq_sum_pow`, both iterated recurrences. The hockey sticks Equations (5.16) and (5.17) are `Fabius.stirlingFirst_add_succ_eq_sum` and `Fabius.stirlingSecond_add_succ_eq_sum`, telescoping the recurrences by induction on k . The convolutions Equations (5.18) and (5.19) are `Fabius.choose_mul_stirlingFirst_add` and `Fabius.choose_mul_stirlingSecond_add`, specializations of the block-colour convolution of partial Bell polynomials (Theorem 13.2, `Fabius.factorial_mul_partialBell_add`) to the weights $(j-1)!$ and 1.

Binomial inversion of the first identity in Equation (5.14) gives

$$\begin{bmatrix} n \\ m \end{bmatrix} = \sum_{k=m}^n (-1)^{k-m} \binom{k}{m} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}. \quad (5.20)$$

The alternative form

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix} = \sum_{k=m}^n \frac{n!}{k!} \begin{bmatrix} k \\ m \end{bmatrix} \quad (5.21)$$

is the second equality in Equation (5.14). All finite sums displayed in the source follow from these and the two convolution identities by relabelling indices.

5.4 Symmetric cross-formulae

The archive records a striking symmetric pair. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\}, \quad (5.22)$$

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \begin{bmatrix} j-k \\ j-n \end{bmatrix}. \quad (5.23)$$

Proof. Set $r = n - k$ and use the symmetric-function descriptions $\begin{bmatrix} n \\ k \end{bmatrix} = e_r(0, 1, \dots, n-1)$ and $\{n\}_k = h_r(1, 2, \dots, k)$. The involution ω on the ring of symmetric functions sends e_r to h_r and is implemented on generating functions by $E(t)H(-t) = 1$. Expand the finite alphabets after adjoining and removing the common arithmetic progression, then apply the binomial translation formula $h_r(X + c) = \sum_j \binom{m+r-1}{r-j} c^{r-j} h_j(X)$ and its elementary analogue. Collecting coefficients gives Equation (5.22); applying ω gives Equation (5.23). Equivalently, direct substitution of the inclusion-exclusion formula Equation (2.7) and two applications of Vandermonde's identity produce the same finite sum. $\square$

5.5 An explicit two-sum formula of the first kind

Unlike the second-kind inclusion-exclusion formula, the first-kind triangle has no comparably short single finite sum. Substitution into the symmetric cross-formula nevertheless gives a completely explicit double sum.

Theorem 5.3 (Explicit first-kind double sum). *For $1 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} \binom{j-1}{k-1} \binom{2n-k}{j} \sum_{m=0}^{j-n} \frac{(-1)^{m+n-k} m^{j-k}}{m! (j-n-m)!}. \quad (5.24)$$

Proof. Start from (5.22) and apply (2.7) with upper index $j - k$ and lower index $j - n$:

$$\left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\} = \sum_{m=0}^{j-n} \frac{(-1)^{j-n-m} m^{j-k}}{(j-n-m)! m!}.$$

Multiplication by the outer sign $(-1)^{j-k}$ gives $(-1)^{2j-k-n-m} = (-1)^{m+n-k}$, because the exponents differ by the even integer $2(j-n-m)$. Substitution proves (5.24). $\square$

5.6 A partition formula for signed near-diagonal coefficients

The near-diagonal coefficient $s(n, n-p)$ is a polynomial in n of degree $2p$. The following finite formula makes every monomial contribution explicit.

Theorem 5.4 (Near-diagonal partition formula). *Let $n \geq 1$ and $0 \leq p \leq n-1$. Then*

$$s(n, n-p) = \frac{1}{(n-p-1)!} \sum_{\substack{k_1, \dots, k_p \geq 0 \\ \sum_{r=1}^p r k_r = p}} (-1)^K \frac{(n+K-1)!}{\prod_{r=1}^p k_r! (r+1)!^{k_r}}, \quad K = \sum_{r=1}^p k_r. \quad (5.25)$$

For $p = 0$ the sum and product are empty and the formula reads $s(n, n) = 1$.

Proof. The signed first-kind exponential generating function gives

$$s(n, n-p) = \frac{n!}{(n-p)!} [z^p] \left(\frac{\log(1+z)}{z} \right)^{n-p}. \quad (5.26)$$

Put $t = \log(1+z)$ and

$$\phi(t) = \frac{t}{e^t - 1}.$$

Then $t = z\phi(t)$ and $t/z = \phi(t)$. For $p \geq 1$, the Lagrange–Bürmann formula applied to $H(t) = \phi(t)^{n-p}$ gives

$$\begin{aligned} [z^p] H(t(z)) &= \frac{1}{p} [t^{p-1}] H'(t) \phi(t)^p \\ &= \frac{n-p}{p} [t^{p-1}] \phi'(t) \phi(t)^{n-1} \\ &= \frac{n-p}{n} [t^p] \phi(t)^n. \end{aligned}$$

The same conclusion is immediate for $p = 0$. Hence

$$s(n, n-p) = \frac{(n-1)!}{(n-p-1)!} [t^p] \left(\frac{t}{e^t - 1} \right)^n. \quad (5.27)$$

Now write

$$A(t) = \sum_{r \geq 1} \frac{t^r}{(r+1)!} = \frac{e^t - 1 - t}{t}, \quad \frac{t}{e^t - 1} = (1 + A(t))^{-1}.$$

The negative-binomial expansion yields

$$(n-1)! [t^p] (1 + A)^{-n} = \sum_{K \geq 0} (-1)^K \frac{(n+K-1)!}{K!} [t^p] A(t)^K.$$

Finally, the multinomial theorem gives

$$[t^p] A(t)^K = \sum_{\substack{k_1 + \dots + k_p = K \\ \sum r k_r = p}} \frac{K!}{\prod_{r=1}^p k_r!} \prod_{r=1}^p \frac{1}{(r+1)!^{k_r}}.$$

Insert this in (5.27) to obtain (5.25). □

The unsigned value is $\left[\begin{smallmatrix} n \\ n-p \end{smallmatrix} \right] = (-1)^p s(n, n-p)$. For $p = 1, 2, 3$ the theorem reduces to the closed forms in Equations (5.2) to (5.4); its advantage is that it works at arbitrary fixed distance from the diagonal without a new case analysis.

Chapter 6

Congruences and extensions to negative indices

6.1 Congruences for first-kind numbers

For a prime p ,

$$p \mid \begin{bmatrix} p \\ k \end{bmatrix} \quad (1 < k < p). \quad (6.1)$$

Indeed, in $\mathbb{F}_p[x]$,

$$(x)^{\overline{p}} = \prod_{a \in \mathbb{F}_p} (x + a) = x^p - x,$$

so every intermediate coefficient vanishes.

More generally, reducing each factor modulo 2 or 3 gives exact coefficient rules:

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/2 \rceil} (x+1)^{\lfloor n/2 \rfloor} = \begin{pmatrix} \lfloor n/2 \rfloor \\ m - \lceil n/2 \rceil \end{pmatrix} \pmod{2}, \quad (6.2)$$

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/3 \rceil} (x+1)^{\lceil (n-1)/3 \rceil} (x+2)^{\lfloor n/3 \rfloor} \pmod{3}. \quad (6.3)$$

Proof. Use $(x)^{\overline{n}} = \prod_{j=0}^{n-1} (x+j)$ and count how often each residue class occurs among $0, 1, \dots, n-1$. The displayed binomial form modulo 2 is immediate; expanding the last two factors gives the finite binomial sum written in the source for modulus 3. $\square$

For fixed m , expansion of Equation (6.2) yields the source's first four parity formulae. They are most cleanly read as integer-valued expressions multiplied by Iverson brackets; for example

$$\begin{bmatrix} n \\ 1 \end{bmatrix} \equiv 2^{n-2} [n \geq 2] + [n = 1] \pmod{2},$$

and similarly the quoted expressions for $m = 2, 3, 4$. Formula Equation (6.2) is both shorter and proves all of them simultaneously.

Theorem 6.1 (Rigorous finite-modulus replacement for the archived spectral form). *Fix $h \geq 2$ and $m \geq 0$. The sequence $n \mapsto \begin{bmatrix} n \\ m \end{bmatrix} \pmod{h}$ is generated by the coefficient of x^m in*

$$\left(\prod_{r=0}^{h-1} (x+r) \right)^q \prod_{r=0}^{s-1} (x+r), \quad n = qh + s, \quad (6.4)$$

and therefore satisfies a linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and is eventually an exponential-polynomial sequence over any finite splitting extension.

Proof. Group the factors of $(x)^{\bar{n}}$ into blocks of h consecutive integers and reduce each block modulo h . For fixed m , coefficient extraction takes place in a finite free module of polynomials of degree at most m , on which multiplication by the fixed block polynomial is a linear endomorphism. Cayley–Hamilton supplies a linear recurrence. Over a splitting extension, the usual Jordan decomposition writes its powers as sums $p_i(q)\omega_i^q$. This is the precise content intended by the archived formula involving unspecified roots $\omega_{h,i}$ and polynomials $p_{h,i}^{[m]}$. $\square$

Remark 6.2 (Formal crosswalk). `Fabius.stirlingFirst_cast_eq_coeff_block` (module `StirlingFirstModH`) is Equation (6.4): with $n = qh + s$, $\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] \left(\prod_{r < h} (x + r) \right)^q \prod_{r < s} (x + r)$ modulo h . It follows from the product form of the rising factorial, $(x)^{\bar{n}} = \prod_{j < n} (x + j)$ (`Fabius.ascPochhammer_eq_prod_range`, which the corpus previously had only in evaluated form), by splitting the product into q blocks of length h and reducing each index modulo h . The editorial remark below is also a theorem: `Fabius.stirlingFirst_cast_eq_zero_of_lt` proves that $\begin{bmatrix} qh+s \\ m \end{bmatrix} \equiv 0 \pmod{h}$ whenever $m < q$, because the block polynomial has constant term $\prod_{r < h} r = 0$ in $\mathbb{Z}/h\mathbb{Z}$, so x^q divides its q th power. The linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and the Jordan decomposition are not formalized.

Remark 6.3 (Editorial: the spectral form is vacuous here). For fixed m the sequence $q \mapsto \begin{bmatrix} qh+s \\ m \end{bmatrix} \pmod{h}$ is not merely eventually exponential-polynomial: it is eventually *identically zero*. The block polynomial $\prod_{r=0}^{h-1} (x+r)$ has constant term $\prod_{r=0}^{h-1} r = 0$ modulo h , so x^q divides its q th power and every coefficient below x^q vanishes; hence $\begin{bmatrix} qh+s \\ m \end{bmatrix} \equiv 0 \pmod{h}$ as soon as $q > m$. The recurrence and the Jordan decomposition are therefore correct but carry no information in this regime, and the archived spectral formula should be read as describing only the finitely many $q \leq m$.

6.2 Parity of second-kind numbers

Theorem 6.4 (Parity criterion). *Let $1 \leq k \leq n$ and*

$$z = n - \left\lceil \frac{k+1}{2} \right\rceil, \quad w = \left\lfloor \frac{k-1}{2} \right\rfloor.$$

Then

$$\begin{Bmatrix} n \\ k \end{Bmatrix} \equiv \begin{pmatrix} z \\ w \end{pmatrix} \pmod{2}. \quad (6.5)$$

Equivalently,

$$\begin{Bmatrix} n \\ k \end{Bmatrix} \equiv 1 \pmod{2} \iff (n-k) \& \left\lfloor \frac{k-1}{2} \right\rfloor = 0. \quad (6.6)$$

In particular, $\begin{Bmatrix} 2n \\ n \end{Bmatrix}$ is odd precisely when n is a power of two.

Proof. Reduce the ordinary generating function Equation (4.6) modulo 2. The even factors become 1, while every odd j contributes $(1-x)^{-1}$. There are $\lceil k/2 \rceil$ odd integers from 1 to k , so

$$\sum_{r \geq 0} \begin{Bmatrix} k+r \\ k \end{Bmatrix} x^r \equiv (1-x)^{-\lceil k/2 \rceil}.$$

The coefficient of x^{n-k} is $\begin{pmatrix} n-k+\lceil k/2 \rceil-1 \\ \lceil k/2 \rceil-1 \end{pmatrix} = \begin{pmatrix} z \\ w \end{pmatrix}$. Lucas's theorem modulo 2 says $\begin{pmatrix} z \\ w \end{pmatrix}$ is odd exactly when every 1-bit of w is also a 1-bit of z , equivalently when $w \& (z-w) = 0$; here $z-w = n-k$. Putting $k = n$ in Equation (6.6) gives $n \& \lfloor (n-1)/2 \rfloor = 0$, which is equivalent to n having one 1-bit. $\square$

Remark 6.5 (Formal crosswalk). `Fabius.stirlingSecond_modEq_choose_two` (module `StirlingParity`) is Equation (6.5) for $1 \leq k \leq n$, as the congruence $\{n\}_k \equiv \binom{n - \lfloor (k+2)/2 \rfloor}{\lfloor (k-1)/2 \rfloor} \pmod{2}$ of natural numbers (note $\lceil (k+1)/2 \rceil = \lfloor (k+2)/2 \rfloor$). The formal proof is the one above: over $\mathbb{F}_2$ the product $\prod_{j=1}^k (1 - jt)$ equals $(1 - t)^{\lceil k/2 \rceil}$ (`Fabius.prod_one_sub_mul_X_zmod_two`, by induction on k with the parity of the new factor), so the column series of Equation (4.6) becomes $(1 - t)^{-\lceil k/2 \rceil}$ (`Fabius.stirlingColumnOGF_zmod_two`) and its coefficients are the binomial coefficients (`Fabius.stirlingSecond_add_zmod_two`). The bitwise form Equation (6.6) and the power-of-two criterion for $\{n\}_k$ rest on Lucas's theorem and are not formalized.

6.3 Roman extension and negative Bell values

There are several inequivalent extensions of Stirling numbers outside the nonnegative triangle. The one used in the archive is the *Roman* extension: continue the two triangular recurrences wherever they determine a value and impose the duality

$$\begin{bmatrix} n \\ k \end{bmatrix} = \begin{Bmatrix} -k \\ -n \end{Bmatrix}, \quad \begin{Bmatrix} n \\ k \end{Bmatrix} = \begin{bmatrix} -k \\ -n \end{bmatrix}. \quad (6.7)$$

These identities are definitions of a coherent reciprocal array, not assertions about the original finite counting problems.

For $n \geq 1$ and $k \geq 0$, the mixed quadrant admits the finite expression

$$\begin{bmatrix} -n \\ k \end{bmatrix} = \frac{1}{n!} \sum_{j=1}^n (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}. \quad (6.8)$$

For example,

$$\begin{bmatrix} -5 \\ k \end{bmatrix} = \frac{1}{120} \left(5 - \frac{10}{2^k} + \frac{10}{3^k} - \frac{5}{4^k} + \frac{1}{5^k} \right). \quad (6.9)$$

Proof. The ordinary generating function of the right-hand side in k is a partial-fraction expansion of

$$\frac{(-1)^{n-1}}{(1-x)(2-x) \cdots (n-x)}.$$

Multiplication by the denominator shows that its coefficients satisfy the continued first-kind recurrence and the required boundary value. Uniqueness proves Equation (6.8); setting $n = 5$ gives the example. $\square$

Define negative Bell values by

$$B_{-k}^{\text{Bell}} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} \quad (k \geq 1). \quad (6.10)$$

Then absolute convergence and Equation (6.8) give

$$B_{-k}^{\text{Bell}} = e^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}. \quad (6.11)$$

Proof. Interchange the absolutely convergent sums and write $n = j + r$:

$$\sum_{n \geq j} \frac{(-1)^{n-j}}{n!} \binom{n}{j} = \frac{1}{j!} \sum_{r \geq 0} \frac{(-1)^r}{r!} = \frac{e^{-1}}{j!}.$$

For $k = 1$ this also yields

$$\mathbf{B}_{-1}^{\text{Bell}} = e^{-1} \int_0^1 \frac{e^t - 1}{t} dt = 0.4848291 \dots,$$

since $(e^t - 1)/t = \sum_{j \geq 1} t^{j-1}/j!$.

□

Chapter 7

Asymptotics, bounds, and special-function sums

7.1 Elementary asymptotic regimes

Theorem 7.1 (Fixed column and fixed diagonal). *For fixed $k \geq 1$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \sim \frac{k^n}{k!} \quad (n \rightarrow \infty). \quad (7.1)$$

For fixed $r \geq 0$,

$$\left[\begin{matrix} n+r \\ n \end{matrix} \right] \sim \left\{ \begin{matrix} n+r \\ n \end{matrix} \right\} \sim \frac{n^{2r}}{2^r r!} \quad (n \rightarrow \infty). \quad (7.2)$$

Finally, uniformly for $k = o(\log n)$,

$$\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right] \sim \frac{n!}{k!} (\log n + \gamma)^k. \quad (7.3)$$

Proof. In Equation (2.7), the term $j = k$ has size $k^n/k!$, and every other term is exponentially smaller. For the near diagonal, use Equations (4.7) and (5.1). In either e_r or h_r , the leading contribution is obtained by treating repetitions/collisions as lower order and equals $\frac{1}{r!}(1+2+\dots+n)^r \sim n^{2r}/(2^r r!)$; terms with a collision lose at least one power of n . For Equation (7.3), apply Cauchy's coefficient formula to

$$\frac{\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]}{n!} = \left[t^k \right] \exp \left(H_n t - \frac{H_n^{(2)}}{2} t^2 + \frac{H_n^{(3)}}{3} t^3 - \dots \right).$$

The saddle for the first exponential is $t = k/H_n = o(1)$. Uniformly on the saddle circle, the remaining exponent is $\mathcal{O}(k^2/H_n^2) = o(1)$, while $H_n = \log n + \gamma + o(1)$. Hence the coefficient is asymptotic to $H_n^k/k!$. $\square$

7.2 Bounds for second-kind numbers

For $n \geq 2$ and $1 \leq k \leq n-1$, the archive states

$$\frac{k^2 + k + 2}{2} k^{n-k-1} - 1 \leq \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \frac{1}{2} \binom{n}{k} k^{n-k}. \quad (7.4)$$

Proof. For the upper bound, in every partition choose the k least elements of its blocks. After ordering these representatives increasingly, each of the remaining $n-k$ elements chooses one of k blocks. This gives at most $\binom{n}{k} k^{n-k}$ encodings. Except at $k=1$, each partition is encoded at least twice by

reversing which of the two largest block minima is declared first; the boundary case is immediate. This yields the factor $1/2$.

For the lower bound, restrict to encodings in which the first k labels are block minima. There are k^{n-k} assignments. Count separately the assignments in which every block receives a prescribed witness among the next one or two labels; inclusion–exclusion through two terms gives $\frac{1}{2}(k^2 + k + 2)k^{n-k-1} - 1$ distinct partitions. The omitted higher inclusion–exclusion terms are nonnegative after pairing by the least missing block, so the truncated expression is a lower bound. This is the standard elementary proof of the displayed estimate. $\square$

7.3 Real zeros, log-concavity, and the location of the maximum

For fixed n , package a row of the second-kind triangle into the Touchard polynomial

$$\mathsf{T}_n^{\text{Tot}}(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

The recurrence (2.6) is equivalent to

$$\mathsf{T}_{n+1}^{\text{Tot}}(x) = x(\mathsf{T}_n^{\text{Tot}}(x) + (\mathsf{T}_n^{\text{Tot}})'(x)), \quad \mathsf{T}_0^{\text{Tot}}(x) = 1. \quad (7.5)$$

Theorem 7.2 (Real-rootedness and strict log-concavity). *For every $n \geq 1$, all zeros of $\mathsf{T}_n^{\text{Tot}}$ are real, simple, and nonpositive. Hence*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}^2 > \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\} \left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} \quad (1 < k < n), \quad (7.6)$$

and the row $\left\{ \begin{matrix} n \\ 1 \end{matrix} \right\}, \dots, \left\{ \begin{matrix} n \\ n \end{matrix} \right\}$ is unimodal. Its maximum is attained at either one index or two consecutive indices.

Proof. The assertion is clear for $\mathsf{T}_1^{\text{Tot}}(x) = x$. Suppose $f = \mathsf{T}_n^{\text{Tot}}$ has simple zeros

$$r_1 < r_2 < \dots < r_n = 0.$$

Away from these zeros,

$$\frac{f(x) + f'(x)}{f(x)} = 1 + \frac{f'(x)}{f(x)} = 1 + \sum_{i=1}^n \frac{1}{x - r_i}.$$

This function is strictly decreasing on every complementary interval. On $(-\infty, r_1)$ it decreases from 1 to $-\infty$, and on each (r_i, r_{i+1}) it decreases from $+\infty$ to $-\infty$. Therefore $f + f'$ has exactly one simple zero in each of these n intervals and none to the right of 0. Formula (7.5) multiplies this polynomial by x ; since $(f + f')(0) = f'(0) = 1$, it adds a new simple zero at 0. Induction proves real-rootedness and interlacing.

Newton's inequalities for a degree- n polynomial with real nonpositive zeros say that its normalized coefficients are log-concave:

$$\left(\frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{\binom{n}{k}} \right)^2 \geq \frac{\left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}}{\binom{n}{k-1}} \frac{\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\}}{\binom{n}{k+1}}.$$

The binomial-factor quotient is strictly larger than 1, and all interior coefficients are positive, giving (7.6). Consequently the successive ratios $\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} / \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ strictly decrease. They can cross 1 only once; an equality at the crossing permits exactly two adjacent equal maxima and no longer plateau. $\square$

Let k_n denote the smaller index at which the maximum in row n is attained. The source records both its scale and the logarithm of the maximal entry.

Theorem 7.3 (Asymptotic mode and maximal entry). *As $n \rightarrow \infty$,*

$$k_n \sim \frac{n}{\log n}, \quad (7.7)$$

$$\log \left\{ \frac{n}{k_n} \right\} = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.8)$$

Proof. Two elementary estimates suffice. Label the k blocks of a partition to obtain at most k^n surjections, whence

$$\left\{ \frac{n}{k} \right\} \leq \frac{k^n}{k!}. \quad (7.9)$$

Conversely, require the first k labels to lie in distinct blocks and assign each of the remaining labels to one of them. This injection gives

$$\left\{ \frac{n}{k} \right\} \geq k^{n-k}. \quad (7.10)$$

Take $k_0 = \lfloor n/\log n \rfloor$. The lower estimate and $\log k_0 = \log n - \log \log n + \mathcal{O}(\log n/n)$ give

$$\log \left\{ \frac{n}{k_0} \right\} \geq n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.11)$$

For the upper estimate, Stirling's formula, uniformly for $1 \leq k \leq n$, gives

$$\log \left\{ \frac{n}{k} \right\} \leq n \log k - k \log k + k + \mathcal{O}(\log n).$$

The continuous majorant on the right is maximized when $k \log k = n$, that is, at $k = n/W(n)$. Since

$$W(n) = \log n - \log \log n + O\left(\frac{\log \log n}{\log n}\right),$$

its maximum is the right side of (7.8). Together with (7.11) this proves the maximal-value estimate.

It remains to localize the maximizing index. Put $k = cn/\log n$, with c bounded away from 0 and ∞ . Comparing the last upper bound with (7.11) gives

$$\log \left\{ \frac{n}{k} \right\} - \log \left\{ \frac{n}{k_0} \right\} \leq n(\log c - c + 1) + o(n).$$

The function $\log c - c + 1$ is strictly negative for $c \neq 1$. Values outside every fixed compact subinterval of $(0, \infty)$ are excluded directly by the same majorant, which increases up to $n/W(n)$ and decreases afterwards. Therefore any maximizing k satisfies $k/(n/\log n) \rightarrow 1$, proving (7.7). $\square$

7.4 A uniform saddle-point approximation

The phrase “away from the two extreme edges” is too vague for a uniform asymptotic statement. The following compact-ratio formulation makes both the limiting regime and the dependence of the error constant explicit.

Theorem 7.4 (Uniform Stirling saddle estimate). *Fix $\delta > 0$ and $M > 1 + \delta$. Uniformly for integers $1 \leq k < n$ such that*

$$1 + \delta \leq v := \frac{n}{k} \leq M$$

as $k \rightarrow \infty$, let $r = r(v) > 0$ be the unique solution of $r/(1 - e^{-r}) = v$ and put $G = v - r$. Equivalently, $G \in (0, 1)$ is the unique solution

$$G = ve^{G-v}. \quad (7.12)$$

Then

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sqrt{\frac{v-1}{v(1-G)}} \left(\frac{v-1}{v-G} \right)^{n-k} \frac{k^n}{n^k} e^{k(1-G)} \binom{n}{k} (1 + \mathcal{O}_{\delta,M}(k^{-1})). \quad (7.13)$$

Since n/k remains in a compact subinterval of $(1, \infty)$, the remainder may equivalently be written $\mathcal{O}_{\delta,M}(n^{-1})$.

Proof. The function $a(r) = r/(1 - e^{-r})$ is continuous and strictly increasing from 1 to ∞ : differentiation gives

$$a'(r) = \frac{1 - e^{-r} - r e^{-r}}{(1 - e^{-r})^2} > 0,$$

because $e^r > 1 + r$. Thus $r(v)$ exists uniquely and ranges over a compact subinterval of $(0, \infty)$. Since $G = v e^{-r}$, we also have $0 < G < 1$.

By Equation (4.3),

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{2\pi k! r^n} \int_{-\pi}^{\pi} (e^{r e^{i\theta}} - 1)^k e^{-i n \theta} d\theta.$$

Near zero choose the analytic logarithm and write the normalized phase as

$$\Phi_v(\theta) = \log \frac{e^{r e^{i\theta}} - 1}{e^r - 1} - i v \theta.$$

The saddle equation gives $\Phi'_v(0) = 0$, while direct differentiation gives

$$\Phi''_v(0) = -b(v), \quad b(v) = r a'(r) = v(1 - G) > 0.$$

Both $b(v)$ and the first five derivatives of Φ_v are bounded uniformly, with $b(v)$ bounded away from zero, because $v \in [1 + \delta, M]$.

For completeness, the needed off-saddle estimate follows directly from aperiodicity. For $0 < |\theta| \leq \pi$, strict triangle inequality in

$$e^{r e^{i\theta}} - 1 = \sum_{m \geq 1} \frac{r^m e^{i m \theta}}{m!}$$

shows that its modulus is smaller than $e^r - 1$: equality would require the phases for $m = 1$ and $m = 2$ to agree. Compactness supplies a uniform geometric drop when $|\theta|$ is bounded away from zero. Taylor's theorem and the lower bound for $b(v)$ supply constants $c, \eta > 0$ such that $\Re \Phi_v(\theta) \leq -c\theta^2$ for $|\theta| \leq \eta$, uniformly in v .

On $|\theta| \leq k^{-2/5}$, expand Φ_v through degree four and put $u = \sqrt{k b(v)} \theta$. The cubic term is odd and cancels between the two halves of the conjugate-symmetric contour. The quartic term, the square of the cubic term, and the Taylor remainder contribute $\mathcal{O}_{\delta,M}(k^{-1})$ relative to the Gaussian integral. The preceding quadratic and geometric drops make the rest of the contour exponentially smaller. Hence

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{k!} \frac{(e^r - 1)^k}{r^n} \frac{1}{\sqrt{2\pi k v (1 - G)}} (1 + \mathcal{O}_{\delta,M}(k^{-1})).$$

Finally $G = v e^{-r}$, $r = v - G$, and the uniform Stirling expansion of $(n - k)!$ (valid because $(n - k)/k = v - 1 \in [\delta, M - 1]$) transforms this display algebraically into Equation (7.13). $\square$

7.5 Zeta and Hurwitz-zeta sums

Theorem 7.5 (Stirling–zeta identities). *For $i \geq 1$,*

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n n!} = \zeta(i+1), \quad (7.14)$$

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n (v)^{\overline{n}}} = \zeta(i+1, v), \quad \Re v > 0. \quad (7.15)$$

Proof. Integrate Equation (4.2) after division by x :

$$\sum_{n \geq i} \frac{[n]_i}{n n!} = \frac{1}{i!} \int_0^1 \frac{[-\log(1-x)]^i}{x} dx.$$

With $u = -\log(1-x)$ this becomes $\frac{1}{i!} \int_0^\infty u^i / (e^u - 1) du = \zeta(i+1)$. For the Hurwitz form use $1/(v)^{\overline{n}} = \Gamma(v)/\Gamma(v+n)$ and the beta integral

$$\frac{1}{(v)^{\overline{n}}} = \frac{1}{(n-1)!} \int_0^1 t^{v-1} (1-t)^{n-1} dt,$$

then sum Equation (4.2) under the integral and use the standard Mellin integral for $\zeta(i+1, v)$. $\square$

A repeated partial-fraction expansion of x^{-k} around $1-x$ gives, for $k \geq 3$ and $\Re z > k-1$,

$$\begin{aligned} \int_0^1 \frac{\log^z(1-x)}{x^k} dx &= \frac{(-1)^z \Gamma(z+1)}{(k-1)!} \sum_{r=1}^{k-1} s(k-1, r) \\ &\quad \times \sum_{m=0}^r \binom{r}{m} (k-2)^{r-m} \zeta(z+1-m), \end{aligned} \quad (7.16)$$

with the principal interpretation of $(-1)^z$ when z is not integral.

Proof. Substitute $u = -\log(1-x)$, expand $(1-e^{-u})^{-k}$ after $k-1$ integrations by parts, and express the resulting polynomial in the Euler operator uD in the ordinary-power basis by $s(k-1, r)$. Each term is a Mellin integral $\int_0^\infty u^{z-m} / (e^u - 1) du = \Gamma(z+1-m) \zeta(z+1-m)$; simplifying gamma ratios yields Equation (7.16). $\square$

The more elaborate Hurwitz expansion in the archive,

$$\begin{aligned} \zeta(s, v) &= \frac{k!}{(s-k)^{\overline{k}}} \sum_{n \geq 0} \frac{[n+k]}{(n+k)!} \\ &\quad \times \sum_{\ell=0}^{n+k-1} (-1)^\ell \binom{n+k-1}{\ell} (\ell+v)^{k-s}, \end{aligned} \quad (7.17)$$

for $k \geq 1$ (initially in the common domain of absolute convergence, then by analytic continuation), follows by applying Equation (4.13) to the k -fold antiderivative of $(x+v)^{-s}$ and summing over $x \in \mathbb{N}_0$. The inner sum is a finite difference, while the outer coefficient is the first-kind conversion coefficient. This derivation also supplies the convergence conditions suppressed in the compact archived display.

7.6 Convergent Stirling series for $\log \Gamma$, ψ , and the Stieltjes constants

The source archive lists three nested Stirling series without their convergence mechanism. All three arise from the same substitution

$$u = 1 - e^{-2\pi x}, \quad x = \frac{-\log(1-u)}{2\pi}, \quad \frac{dx}{e^{2\pi x} - 1} = \frac{du}{2\pi u}. \quad (7.18)$$

We first record the coefficient estimate that justifies integration at the endpoint $u = 1$.

Lemma 7.6 (Logarithmic transfer estimate). *Suppose h is analytic in the unit disk, continues to a dented neighborhood of $u = 1$, and, there,*

$$h(u) = h_0 + O\left(\frac{(1 + \log \log \frac{1}{|1-u|})^M}{\log \frac{1}{|1-u|}}\right)$$

for some fixed $M \geq 0$. If $h(u) = \sum_{n \geq 0} h_n u^n$, then

$$h_n = O\left(\frac{(1 + \log \log n)^M}{n \log^2 n}\right). \quad (7.19)$$

Consequently $\sum_{n \geq 1} |h_n|/n$ converges.

Proof. Apply Cauchy's coefficient formula on a keyhole contour that follows the two sides of the cut $[1, \infty)$. The part separated from 1 is exponentially small. On the local part put $u = 1 - w/n$. The constant h_0 has no positive-index coefficients, while the jump of $1/\log(1/(1-u))$ across the cut is $\mathcal{O}(\log^{-2} n)$; multiplication by a power of $\log \log(1/(1-u))$ contributes at most $(1 + \log \log n)^M$. The Cauchy kernel contributes $n^{-1}e^{w/n}$, and the two rays of the keyhole have an integrable exponential majorant. This gives (7.19). Division by another factor n makes the resulting majorant summable. $\square$

7.6.1 The logarithm and logarithmic derivative of the gamma function

Binet's second formula, valid for $z > 0$, is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx. \quad (7.20)$$

It follows, for example, by applying the Abel–Plana summation formula to $w \mapsto \log(z + w)$, subtracting the elementary integral and endpoint terms, and normalizing at $z = 1$ by the Wallis product. Differentiation under the integral recovers the usual Binet integral for the digamma function, so the normalization also proves (7.20) rather than merely determining it up to a constant.

Theorem 7.7 (Convergent Stirling series for $\log \Gamma$ and ψ). *For real $z > 1/6$,*

$$\begin{aligned} \log \Gamma(z) = & \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) \\ & + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \end{aligned} \quad (7.21)$$

The series is locally uniformly convergent in $z > 1/6$, and termwise differentiation gives

$$\begin{aligned} \psi(z) = & \frac{\Gamma'(z)}{\Gamma(z)} = \log z - \frac{1}{2z} \\ & - \frac{1}{\pi z} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell+1)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \end{aligned} \quad (7.22)$$

Proof. Put $L(u) = -\log(1 - u)$. The formal Taylor series of the composite function is

$$\begin{aligned} \arctan \frac{L(u)}{2\pi z} &= \sum_{\ell \geq 0} \frac{(-1)^\ell}{2\ell + 1} \frac{L(u)^{2\ell+1}}{(2\pi z)^{2\ell+1}} \\ &= \sum_{n \geq 1} \frac{u^n}{n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \begin{bmatrix} n \\ 2\ell + 1 \end{bmatrix}, \end{aligned} \quad (7.23)$$

where the second line uses $L(u)^r/r! = \sum_{n \geq r} \begin{bmatrix} n \\ r \end{bmatrix} u^n/n!$. For $z > 1/6$, the analytic continuation of this composite from $u = 0$ has no singularity in $|u| < 1$. Indeed, the possible arctangent branch points would require $\log(1 - u) = \pm 2\pi iz$; when $1/6 < z < 1/4$ their solutions have modulus $2 \sin(\pi z) > 1$, and for $z \geq 1/4$ the principal logarithm on the disk cannot have imaginary part $\pm 2\pi z$. The only boundary singularity relevant to the coefficient estimate is therefore $u = 1$.

As $u \rightarrow 1$ in a dented domain,

$$\arctan \frac{L(u)}{2\pi z} = \frac{\pi}{2} + O\left(\frac{1}{L(u)}\right).$$

Thus Theorem 7.6 applies, uniformly when z ranges over a compact subinterval of $(1/6, \infty)$. We may divide the Taylor coefficients by n and sum at $u = 1$. Substituting (7.18) in (7.20) now gives

$$2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx = \frac{1}{\pi} \int_0^1 \arctan \frac{L(u)}{2\pi z} \frac{du}{u},$$

and termwise integration of (7.23) proves (7.21). The same uniform estimate after one z -derivative justifies termwise differentiation; differentiating $(2\pi z)^{-2\ell-1}$ produces the factor $-(2\ell + 1)/z$ and yields (7.22). $\square$

For complex z with $\Re z > 0$, the same proof works whenever the two points $1 - e^{2\pi iz}$ and $1 - e^{-2\pi iz}$ lie outside the closed unit disk; the stated real half-line is the simplest connected domain containing all commonly used positive arguments.

7.6.2 Stieltjes constants

Define the Stieltjes constants by the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m}{m!} (s-1)^m. \quad (7.24)$$

Theorem 7.8 (Stirling series for the Stieltjes constants). *For every integer $m \geq 0$,*

$$\begin{aligned} \gamma_m &= \frac{1}{2} \delta_{m0} + \frac{(-1)^m m!}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \\ &\quad \times \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k}{(2\pi)^{2k+1}} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix} \begin{bmatrix} n \\ 2k+1 \end{bmatrix}. \end{aligned} \quad (7.25)$$

The outer series converges absolutely.

Proof. The Abel–Plana formula applied to $(1+w)^{-s}$ gives Hermite’s representation

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{1}{i} \int_0^\infty \frac{(1-ix)^{-s} - (1+ix)^{-s}}{e^{2\pi x} - 1} dx, \quad \Re s > 0, \quad (7.26)$$

with the pole omitted at $s = 1$. Expand the two powers at $s = 1$ and compare with (7.24). Differentiation under the exponentially convergent integral yields

$$\gamma_m = \frac{1}{2}\delta_{m0} + \frac{1}{i} \int_0^\infty \frac{1}{e^{2\pi x} - 1} \left\{ \frac{\log^m(1 - ix)}{1 - ix} - \frac{\log^m(1 + ix)}{1 + ix} \right\} dx. \quad (7.27)$$

Differentiate the signed first-kind generating function to obtain, for $|w| < 1$,

$$\frac{\log^m(1 + w)}{1 + w} = m! \sum_{r \geq m} s(r + 1, m + 1) \frac{w^r}{r!}. \quad (7.28)$$

Only odd powers survive after replacing w by $-ix$ and ix and taking the difference. Since $s(2k + 2, m + 1) = (-1)^{m+1} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$, one finds

$$\begin{aligned} \frac{1}{i} \left\{ \frac{\log^m(1 - ix)}{1 - ix} - \frac{\log^m(1 + ix)}{1 + ix} \right\} \\ = 2(-1)^m m! \sum_{k \geq 0} \frac{(-1)^k \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}}{(2k + 1)!} x^{2k+1}. \end{aligned} \quad (7.29)$$

Now make the substitution (7.18). For every k ,

$$\frac{L(u)^{2k+1}}{(2k + 1)!} = \sum_{n \geq 2k+1} \begin{bmatrix} n \\ 2k + 1 \end{bmatrix} \frac{u^n}{n!}.$$

The composite integrand in (7.27) is analytic in the unit disk and, as $u \rightarrow 1$, is

$$O\left(\frac{(1 + \log L(u))^m}{L(u)}\right).$$

Hence Theorem 7.6 permits absolute termwise integration after division by u . The Jacobian contributes $1/(2\pi)$, the factor 2 in (7.29) cancels it to $1/\pi$, and $\int_0^1 u^{n-1} du = 1/n$. Collecting the finite contributions to each coefficient u^n gives exactly (7.25). $\square$

For $m = 0$, the theorem is a convergent Stirling-number expansion of the Euler–Mascheroni constant. The factors $\begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$ show that, at every fixed outer index n , only finitely many orders of logarithmic differentiation contribute; this is the same triangularity that underlies Faà di Bruno's formula and series reversion.

7.7 Gregory coefficients and derivatives of logarithmic powers

The Gregory coefficients G_n^{Greg} are defined by

$$\frac{z}{\log(1 + z)} = 1 + \sum_{n \geq 1} G_n^{\text{Greg}} z^n.$$

Multiplying by the first-kind expansion of powers of $\log(1 + z)$ gives

$$n! G_n^{\text{Greg}} = \sum_{\ell=0}^n \frac{s(n, \ell)}{\ell + 1}. \quad (7.30)$$

Indeed, $z/\log(1 + z) = \int_0^1 (1 + z)^t dt$ and comparison with $(1 + z)^t = \sum_n \sum_\ell s(n, \ell) t^\ell z^n / n!$ proves the claim.

For arbitrary μ and $n \geq 1$,

$$\frac{d^n}{dx^n} (\log x)^\mu = x^{-n} \sum_{k=1}^n (\mu)^k s(n, n - k + 1) (\log x)^{\mu-k}. \quad (7.31)$$

Proof. Write $D_x = x^{-1}D_t$ with $t = \log x$. Then $D_x^n = x^{-n}(D_t)^n$, as follows inductively from $x^{-1}D_t x^{-m} = x^{-m-1}(D_t - m)$. Expand $(D_t)^n$ by signed first-kind numbers and apply powers of D_t to t^μ ; reindexing gives the displayed form. $\square$

7.8 Generalized factorial coefficients

Given $f : \mathbb{N}_0 \rightarrow \mathbb{C}$ and $t \neq 0$, define

$$(x)_{n,f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} := \left[x^{k-1} \right] (x)_{n,f,t}. \quad (7.32)$$

Multiplication by the last factor immediately yields

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} = f(n-1)t^{1-n} \left[\begin{matrix} n-1 \\ k \end{matrix} \right]_{f,t} + \left[\begin{matrix} n-1 \\ k-1 \end{matrix} \right]_{f,t}, \quad (7.33)$$

with the usual delta boundary convention. The generalized harmonic sums $F_n^{(r)}(t) = \sum_{j \leq n} t^j / f(j)^r$ enter the coefficients through the same Newton/Bell-polynomial calculation as Equation (5.9).

More broadly, any triangular array satisfying

$$a(n, k) = (\alpha n + \beta k + \gamma)a(n-1, k) + (\alpha' n + \beta' k + \gamma')a(n-1, k-1) + \delta_{n0}\delta_{k0} \quad (7.34)$$

is governed by a first-order differential equation for its bivariate generating function. This umbrella contains both Stirling kinds, Lah numbers, Eulerian variants, and many generalized factorial coefficients.

Chapter 8

Restricted variants of set partitions

The second-kind recurrence survives many natural restrictions, but its boundary term changes according to how the block containing the newest label is allowed to look.

8.1 r -Stirling numbers of the second kind

For fixed $r \geq 0$, let

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$$

denote the number of partitions of $[n]$ into k nonempty blocks in which the labels $1, \dots, r$ lie in distinct blocks. The definition is meaningful for $n \geq r$ and forces $k \geq r$.

Theorem 8.1 (r -Stirling recurrence). *For $n > r$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}_r + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}_r, \quad (8.1)$$

with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the natural zero boundary conditions.

Proof. Because $n > r$, the new label n is not one of the distinguished labels. It either joins one of the k blocks of an admissible partition of $[n-1]$, or forms a singleton beside an admissible partition into $k-1$ blocks. The distinction of $1, \dots, r$ is preserved in both constructions, and deletion of n reverses them. $\square$

Remark 8.2 (Formal crosswalk). Module `RStirling` defines $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$ by the recurrence Equation (8.1) with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the zero boundary (`Fabius.rStirling`, in the shifted indices $T_r(n, k) = \left\{ \begin{matrix} n+r \\ k+r \end{matrix} \right\}_r$ as `Fabius.rStirlingShift`); the count of partitions with $1, \dots, r$ in distinct blocks is not formalized. The recurrence is `Fabius.rStirling_succ_succ` (for $n \geq r$). Since the recurrence says that $n \mapsto T_r(n, k)$ is the binomial convolution of $j \mapsto r^j$ with $j \mapsto \left\{ \begin{matrix} j \\ k \end{matrix} \right\}$ (`Fabius.rStirlingShift_eq_binomialConv`, by induction with the shift rule `Bell.binomialConv_succ`), one gets the explicit formula $\left\{ \begin{matrix} n+r \\ k+r \end{matrix} \right\}_r = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\} r^{n-j}$ (`Fabius.rStirlingShift_eq_sum`, `Fabius.rStirling_eq_sum`), and the mixed generating function Equation (8.2) follows from the product rule for exponential generating functions: `Fabius.egfA_rStirlingPoly` is $\sum_n (\sum_k \left\{ \begin{matrix} n+r \\ k+r \end{matrix} \right\}_r y^k) z^n / n! = e^{rz} \exp(y(e^z - 1))$ in $\mathbb{Q}[y][[z]]$, which is Equation (8.2) after multiplication by y^r .

The same labelled-component argument gives the mixed exponential generating function

$$\sum_{n \geq r} \sum_{k \geq r} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r e^{rz} \exp(y(e^z - 1)). \quad (8.2)$$

Indeed, the r distinguished blocks each consist of one distinguished atom and an arbitrary set of ordinary atoms, contributing $y^r e^{rz}$, while the remaining blocks form an ordinary set partition.

8.2 Associated Stirling numbers

Let $S_r(n, k)$ count partitions of $[n]$ into k blocks, every block having size at least r .

Theorem 8.3 (Associated recurrence). *For $r \geq 1$,*

$$S_r(n+1, k) = kS_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1). \quad (8.3)$$

Moreover,

$$\sum_{n \geq 0} S_r(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(e^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k. \quad (8.4)$$

Proof. Consider the block containing $n+1$. If deleting $n+1$ leaves a block of size at least r , start from a partition counted by $S_r(n, k)$ and choose one of its k blocks for insertion. Otherwise the block containing $n+1$ has exactly r elements: choose its $r-1$ companions and partition the remaining $n-r+1$ labels into $k-1$ admissible blocks. These cases are disjoint and exhaustive, proving the recurrence. For the generating function, one admissible block is a labelled set of size at least r , with EGF $e^z - \sum_{j < r} z^j/j!$; an unordered set of k such blocks contributes its k th power divided by $k!$. $\square$

Remark 8.4 (Formal crosswalk). Module `AssociatedStirling` defines $S_r(n, k)$ as the partial Bell polynomial $\mathcal{B}_{n,k}^{\text{exp}}$ with the weights $x_j = [j \geq r]$ (`Fabius.associatedStirling`); the count of partitions with all blocks of size at least r is not formalized. With this definition Equation (8.4) is `Fabius.egfA_associatedStirling`, immediate from the column theorem `Fabius.bellWeightSeries_pow` once the weight series is identified with $e^z - \sum_{j < r} z^j/j!$ (`Fabius.bellWeightSeries_assocWeight`), and Equation (8.3) is `Fabius.associatedStirling_succ_succ`, obtained by differentiating the generating function: $W' = W + z^{r-1}/(r-1)!$ (`Fabius.derivative_bellWeightSeries_assocWeight`) gives $F'_k = kF_k + F_{k-1}z^{r-1}/(r-1)!$ for $F_k = W^k/k!$, and the coefficient of $z^n/n!$ is the recurrence.

For $r = 2$ these are often called Ward numbers. Their occurrence as coefficients of other polynomial families comes from the same exclusion of singleton components expressed by (8.4).

8.3 Reduced Stirling numbers and graph coloring

For $d \geq 1$, let $S^{[d]}(n, k)$ count partitions of $[n]$ into k blocks such that two labels in the same block always differ by at least d .

Theorem 8.5 (Reduction formula). *For $n \geq k \geq d$,*

$$S^{[d]}(n, k) = \left\{ \begin{matrix} n-d+1 \\ k-d+1 \end{matrix} \right\}. \quad (8.5)$$

In particular $S^{[1]}(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Proof. Join two vertices $i, j \in [n]$ when $0 < |i-j| < d$, and call the resulting graph $G_{n,d}$. Its independent-set partitions are exactly the partitions counted by $S^{[d]}(n, k)$. A proper coloring can therefore be obtained by choosing such a partition and assigning distinct colors to its blocks, so

$$\chi_{G_{n,d}}(q) = \sum_k S^{[d]}(n, k) (q)^k. \quad (8.6)$$

Color the vertices in their natural order. The first $d-1$ vertices must receive distinct colors; every subsequent vertex sees the preceding $d-1$ vertices, which form a clique, and hence has $q-d+1$ choices. Thus

$$\chi_{G_{n,d}}(q) = (q)^{d-1} (q-d+1)^{n-d+1}.$$

Apply the ordinary Stirling basis expansion to the last power:

$$\begin{aligned}\chi_{G_{n,d}}(q) &= (q)^{\overline{d-1}} \sum_{r \geq 0} \left\{ \begin{matrix} n-d+1 \\ r \end{matrix} \right\} (q-d+1)^r \\ &= \sum_{r \geq 0} \left\{ \begin{matrix} n-d+1 \\ r \end{matrix} \right\} (q)^{\overline{r+d-1}}.\end{aligned}$$

Comparison with (8.6) gives $k = r + d - 1$ and proves (8.5). □

Part III

Bell Numbers and Set Partitions

Chapter 9

Set partitions and the many meanings of the Bell numbers

9.1 Definition, initial values, and equivalence relations

Definition 9.1. For $n \geq 0$, the n th Bell number B_n^{Bell} is the number of partitions of an n -element labelled set into nonempty, pairwise disjoint blocks. Thus

$$B_0^{\text{Bell}} = B_1^{\text{Bell}} = 1, \quad (B_n^{\text{Bell}})_{n \geq 0} = 1, 1, 2, 5, 15, 52, 203, 877, 4140, \dots$$

The empty set has one partition, namely the empty family of blocks.

For example, the five partitions of $\{a, b, c\}$ are

$$abc, \quad a \mid bc, \quad b \mid ac, \quad c \mid ab, \quad a \mid b \mid c,$$

where vertical bars separate blocks.

Proposition 9.2. Partitions of a set S are in natural bijection with equivalence relations on S . Consequently B_n^{Bell} also counts equivalence relations on an n -element set.

Proof. Given a partition π , declare $x \sim_\pi y$ when x and y lie in the same block. Reflexivity, symmetry, and transitivity follow immediately. Conversely, the equivalence classes of an equivalence relation are nonempty, disjoint, cover S , and hence form a partition. The two constructions undo one another. $\square$

9.2 Ordered set partitions and the Fubini numbers

If the blocks themselves are linearly ordered, the resulting numbers are usually called the *ordered Bell numbers* or *Fubini numbers*. To prevent a collision with the Bell notation, write them as F_n .

Theorem 9.3 (Ordered Bell numbers). For $n \geq 0$,

$$F_n = \sum_{k=0}^n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (9.1)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (9.2)$$

$$F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (9.3)$$

with $F_0 = 1$.

Proof. A partition into k blocks may be ordered in $k!$ ways, proving (9.1). By the second-kind exponential generating function,

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{k \geq 0} k! \frac{(e^z - 1)^k}{k!} = \sum_{k \geq 0} (e^z - 1)^k = \frac{1}{2 - e^z},$$

first as a formal identity and then analytically near the origin. Finally, in a nonempty ordered partition choose the elements of its first block, which may be any nonempty subset, and order-partition the complement. Summing over the first-block size gives (9.3). $\square$

Remark 9.4 (Formal crosswalk). Module `OrderedBell` defines `Fabius.fubini` by Equation (9.1). Equation (9.2) is `Fabius.two_sub_exp_mul_egfA_fubini`, in the form $(2 - e^t) \sum_n F_n t^n / n! = 1$ over any $\mathbb{Q}$ -algebra, proved as in the text: by the exponential composition theorem with block weights $k!$ the generating function is $1/(1 - u)$ at $u = e^t - 1$ (`Fabius.egfA_fubini`), and $(1 - u) \cdot 1/(1 - u) = 1$ is substituted. Reading $1/(1 - u) = 1 + u/(1 - u)$ at $u = e^t - 1$ coefficientwise gives Equation (9.3) as `Fabius.fubini_succ`, written $F_{n+1} = \sum_{i \leq n} \binom{n+1}{i+1} F_{n-i}$; the ordered-partition count is not formalized.

Thus

$$(F_n)_{n \geq 0} = 1, 1, 3, 13, 75, 541, 4683, \dots,$$

which contrasts with the unordered Bell sequence by recording the order of the blocks rather than only their underlying family.

9.3 Squarefree factorizations, rhyme schemes, and restricted-growth words

Proposition 9.5. *If $N = p_1 \cdots p_n$ is squarefree, then B_n^{Bell} is the number of unordered factorizations of N into integers greater than 1.*

Proof. To a set partition π of $[n]$ associate the multiset of factors $\{\prod_{i \in B} p_i : B \in \pi\}$. Unique factorization recovers each block from its factor, so this is a bijection. For $30 = 2 \cdot 3 \cdot 5$ the five factorizations are

$$30, \quad 2 \cdot 15, \quad 3 \cdot 10, \quad 5 \cdot 6, \quad 2 \cdot 3 \cdot 5.$$

$\square$

A rhyme scheme of an n -line stanza is likewise a partition of the set of lines: two indices lie in the same block exactly when the corresponding lines rhyme. The customary lettering convention is the following canonical encoding.

Definition 9.6. A *restricted-growth word* of length n is a word $a_1 \cdots a_n$ of nonnegative integers such that $a_1 = 0$ and

$$a_i \leq 1 + \max(a_1, \dots, a_{i-1}) \quad (i > 1).$$

Proposition 9.7. *Set partitions of $[n]$, restricted-growth words of length n , and canonical rhyme schemes of length n are mutually bijective.*

Proof. Order the blocks by increasing least element, number them $0, 1, \dots$, and put $a_i = j$ when i belongs to block j . The displayed restriction says precisely that a new block receives the next unused number. Conversely, equal letters form the blocks. Replacing $0, 1, 2, \dots$ by $A, B, C, \dots$ gives the rhyme scheme. $\square$

9.4 A card-shuffling bijection

Consider the operation that removes the top card from a deck of n labelled cards and reinserts it in any of the n positions. There are n^n sequences of n such operations.

Theorem 9.8 (Return-to-order shuffles). *Exactly B_n^{Bell} of those n^n operation sequences return an initially ordered deck to its original order. Hence the return probability is B_n^{Bell}/n^n .*

Proof. Reverse time. A reversed operation chooses an arbitrary card and moves it to the top. Thus a reversed history is a word $c_1 \cdots c_n$ in the card labels. After all moves, the cards that occurred in the word appear first, in decreasing order of their last occurrence times; cards never chosen follow in their original relative order.

Partition the time set $[n]$ according to equality of the letters c_i . For the final deck to be $1, 2, \dots, n$, the labels that occur must be an initial interval $1, \dots, m$, and their blocks of occurrence times must be labelled in decreasing order of their maxima: the block with largest maximum receives label 1, the next receives 2, and so on. Therefore every set partition of $[n]$ determines exactly one successful word, and every successful word determines its equality partition. This is the required bijection. $\square$

9.5 Bell-enumerated vincular pattern classes

A dash in a permutation pattern means that the corresponding entries need not be adjacent; an undashed pair must be adjacent. For instance, an occurrence of $23-1$ consists of an adjacent ascent followed later by a smaller entry.

Theorem 9.9. *The permutations of $[n]$ avoiding $23-1$ are counted by B_n^{Bell} . The same is true for each pattern in*

$$1-23, 12-3, 32-1, 3-21, 1-32, 3-12, 21-3,$$

and for the equivalent barred-pattern class in which every 321 occurrence extends to a 3241 occurrence.

Proof. For the pattern $23-1$, order the blocks of a partition by increasing least element and write the entries of each block in decreasing order. Concatenate the resulting words. In the resulting permutation, each block is a decreasing run, and every ascent occurs between two blocks. Its lower endpoint is a right-to-left minimum, so no later entry is smaller; hence $23-1$ is avoided.

Conversely, cut a $23-1$ -avoiding permutation immediately after each right-to-left minimum. Every resulting segment is decreasing: otherwise an ascent inside a segment, together with its terminal right-to-left minimum, would form $23-1$. The segment minima increase from left to right, so the segments, viewed as sets, are blocks already in canonical order. The two maps are inverse.

Reversal, complement, inverse, and their compositions are bijections on permutations and carry vincular occurrences to occurrences of the seven listed patterns; therefore all eight classes are equinumerous. Finally, cutting at right-to-left minima shows that failure of the decreasing-run condition is simultaneously an occurrence of $23-1$ and a 321 that cannot be extended by an intervening entry to 3241 . Thus the barred-pattern formulation describes the same class. $\square$

Remark 9.10. Classical avoidance classes have at most exponential growth in n , whereas Bell numbers grow faster than C^n for every fixed C . The adjacency requirement is therefore essential: these Bell-enumerated classes cannot be classical single-pattern avoidance classes.

9.6 The Bell–Aitken triangle

Define a triangular array $(a_{n,k})_{0 \leq k \leq n}$ by

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1} \quad (1 \leq k \leq n).$$

Its first rows are

$$\begin{array}{cccccc}
 1 & & & & & \\
 1 & 2 & & & & \\
 2 & 3 & 5 & & & \\
 5 & 7 & 10 & 15 & & \\
 15 & 20 & 27 & 37 & 52 &
 \end{array}$$

Theorem 9.11. For $0 \leq k \leq n$,

$$a_{n,k} = \sum_{j=0}^k \binom{k}{j} B_{n-j}^{\text{Bell}}. \quad (9.4)$$

Consequently the left edge is $a_{n,0} = B_n^{\text{Bell}}$ and the right edge is $a_{n,n} = B_{n+1}^{\text{Bell}}$.

Proof. The formula is immediate for $k = 0$. Pascal's identity gives

$$a_{n,k-1} + a_{n-1,k-1} = \sum_j \left(\binom{k-1}{j} + \binom{k-1}{j-1} \right) B_{n-j}^{\text{Bell}},$$

which is (9.4). For $k = n$, reverse the index and use the Bell recurrence proved in Theorem 10.1 below. $\square$

Chapter 10

The enumerative calculus of Bell numbers

10.1 Two fundamental sums

Theorem 10.1 (Binomial recurrence). *For $n \geq 0$,*

$$B_{n+1}^{\text{Bell}} = \sum_{k=0}^n \binom{n}{k} B_k^{\text{Bell}}. \quad (10.1)$$

Proof. In a partition of $\{0, 1, \dots, n\}$, let k be the number of elements outside the block containing 0. Choose those k elements and partition them arbitrarily; the block containing 0 is then forced. $\square$

Theorem 10.2 (Stirling decomposition).

$$B_n^{\text{Bell}} = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^n \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n. \quad (10.2)$$

Proof. The first equality groups partitions by their number of blocks. Substitute the inclusion-exclusion formula for $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ from Equation (2.7) to obtain the second equality. $\square$

Remark 10.3 (Formal crosswalk). The first equality is `Fabius.bell_eq_sum_stirlingSecond` and the second `Fabius.bell_eq_sum_sum_div_factorial` (module `BellStirling`). Mathlib's `Nat.bell` is defined by the binomial recurrence of Theorem 10.1 (`Nat.bell_succ`), so the Lean proof shows that $\sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ satisfies the same recurrence, through the vertical recurrence $\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{i \leq n} \binom{n}{i} \left\{ \begin{matrix} i \\ k \end{matrix} \right\}$ (`Fabius.stirlingSecond_succ_succ_eq_sum`). The same recurrence gives the Touchard polynomials $T_n(x) = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k$ (`Fabius.touchardPolynomial`) the binomial-type relation $T_{n+1}(x) = x \sum_k \binom{n}{k} T_k(x)$ (`Fabius.touchardPolynomial_succ`) and $T_n(1) = B_n^{\text{Bell}}$ (`Fabius.touchardPolynomial_eval_one`).

10.2 Spivey's identity and binomial inversions

Theorem 10.4 (Spivey's identity). *For $m, n \geq 0$,*

$$B_{m+n}^{\text{Bell}} = \sum_{k=0}^n \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{n}{k} j^{n-k} B_k^{\text{Bell}}. \quad (10.3)$$

Proof. Partition the first m labels into j blocks. Of the final n labels, choose k whose blocks contain no old label and partition them in B_k^{Bell} ways. Each of the remaining $n - k$ labels independently joins one of the j old blocks. This classification is unique and gives the summand. $\square$

Remark 10.5 (Formal crosswalk). `Fabius.spivey` (module `BellShiftEGF`) proves Equation (10.3) from the shifted generating function `Fabius.egfA_bell_add`, $\sum_{n \geq 0} B_{m+n}^{\text{Bell}} \frac{t^n}{n!} = T_m^{\text{Tou}}(e^t) e^{e^t-1}$, which is the m -th derivative of the Bell generating function. The induction on m uses $(\sum_n B_{n+1}^{\text{Bell}} t^n/n!) = e^t \sum_n B_n^{\text{Bell}} t^n/n!$ (`Fabius.egfA_bell_succ`, the binomial recurrence) and the Touchard recurrence at $x = e^t$, $T_{m+1}^{\text{Tou}}(e^t) = (T_m^{\text{Tou}}(e^t))' + T_m^{\text{Tou}}(e^t)e^t$ (`Fabius.derivative_touchardExp`); the coefficient of $t^n/n!$ in $e^{jt}e^{e^t-1}$ is the binomial convolution $\sum_k \binom{n}{k} B_k^{\text{Bell}} j^{n-k}$.

Binomial inversion of (10.1) gives the first identity below. The second is a useful shifted form.

Theorem 10.6 (Bell inversion identities). *For $n, k \geq 0$,*

$$B_n^{\text{Bell}} = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} B_{j+1}^{\text{Bell}}, \quad (10.4)$$

$$\sum_{j=0}^n \binom{n}{j} B_{k+j}^{\text{Bell}} = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} B_{n+i+1}^{\text{Bell}}. \quad (10.5)$$

Proof. The first formula is ordinary binomial inversion. For the second let X be a Poisson random variable of mean 1; Theorem 10.12 below gives $\mathbb{E}X^r = B_r^{\text{Bell}}$. The left side is $\mathbb{E}[X^k(X+1)^n]$. Poisson summation by parts,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[h(X+1)], \quad (10.6)$$

iterated k times in the equivalent form $\mathbb{E}[X^{n+1}(X-1)^k] = \mathbb{E}[(X+1)^n X^k]$, turns it into the right side after expanding $(X-1)^k$. $\square$

Remark 10.7 (Formal crosswalk). Equation (10.4) is `Fabius.bell_eq_sum_neg_one_pow_choose_bell_succ` (module `BellShiftEGF`): binomial inversion (`Fabius.binomial_inversion_ring`) of the recurrence Equation (10.1) in the form `Fabius.bell_succ_eq_sum_choose`. Equation (10.5) is not formalized.

The first-kind Stirling numbers package all repeated shifts at once.

Theorem 10.8 (Weighted Bell shift). *For $a, b \in \mathbb{C}$ and $k, n \geq 0$,*

$$\begin{aligned} \sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j^{\text{Bell}} \\ = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} \sum_{j=0}^n \binom{n}{j} a^j (b - ak)^{n-j} B_{j+i}^{\text{Bell}}. \end{aligned} \quad (10.7)$$

In particular,

$$\sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j^{\text{Bell}} = \sum_{j=0}^n \binom{n}{j} a^j (b - a)^{n-j} B_{j+1}^{\text{Bell}}, \quad (10.8)$$

$$\sum_{j=0}^n \binom{n}{j} k^{n-j} B_j^{\text{Bell}} = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} B_{n+i}^{\text{Bell}}. \quad (10.9)$$

Proof. By Dobiński's formula, the left side of (10.7) is

$$e^{-1} \sum_{m \geq 0} \frac{(am + b)^n}{m!}.$$

On the right, use $\sum_i (-1)^{k-i} \binom{k}{i} m^i = (m)^k$ inside the same Poisson sum. It becomes

$$e^{-1} \sum_{m \geq 0} \frac{(m)^k [am + b - ak]^n}{m!} = e^{-1} \sum_{r \geq 0} \frac{(ar + b)^n}{r!},$$

after $m = r + k$. The special cases follow by setting $k = 1$, and then $a = 1, b = k$. $\square$

Remark 10.9 (Formal crosswalk). `Fabius.weighted_bell_shift` (module `BellUmbra`) is Equation (10.7), `Fabius.weighted_bell_shift_one` is Equation (10.8) and `Fabius.signedStirlingFirst_mul_bell_eq` is Equation (10.9), with a, b in an arbitrary commutative ring and $(-1)^{k-i} \binom{k}{i}$ as the signed number `Fabius.signedStirlingFirst`. The formal proof replaces the Poisson sum by the Bell umbra, the linear functional L on $R[x]$ with $L(x^n) = B_n^{\text{Bell}}$ (`Fabius.bellUmbra`): the binomial recurrence Equation (10.1) says $L(xf(x)) = L(f(x+1))$ (`Fabius.bellUmbra_X_mul`), induction on k with $(x)^{k+1} = x(x-1)^k$ gives the umbral shift $L((x)^k f(x)) = L(f(x+k))$ (`Fabius.bellUmbra_descPochhammer_mul`), and applying it to $f(x) = (ax+b-ak)^n$ gives $L((ax+b)^n)$ on the left, which is the binomial expansion of Equation (10.7). In particular all factorial moments of the Bell umbra are 1 (`Fabius.bellUmbra_descPochhammer`).

10.3 Exponential generating function and differential equation

Theorem 10.10. *The Bell exponential generating function is*

$$\mathcal{B}(z) := \sum_{n \geq 0} B_n^{\text{Bell}} \frac{z^n}{n!} = \exp(e^z - 1), \quad \mathcal{B}'(z) = e^z \mathcal{B}(z). \quad (10.10)$$

Proof. A set partition is a labelled set of nonempty labelled sets. The EGF of a nonempty set is $e^z - 1$, and the exponential formula therefore gives $\exp(e^z - 1)$. Differentiation gives the differential equation; conversely, coefficient comparison in that equation gives (10.1) and $\mathcal{B}(0) = 1$, so it uniquely determines the series. $\square$

Remark 10.11 (Formal crosswalk). The identity $\mathcal{B}(z) = \exp(e^z - 1)$ is `Fabius.exp_subst_exp_sub_one` (module `BellGeneratingFunctions`), stated as the substitution of Mathlib's exponential series into itself minus one, over any commutative $\mathbb{Q}$ -algebra; it is the case $x = (1, 1, \dots)$ of the complete Bell generating function `Fabius.exp_subst_bellWeightSeries`, whose Lean proof is the uniqueness argument of this proof: both sides satisfy $Y' = e^z Y$ with $Y(0) = 1$.

10.4 Dobiński's formula and Poisson moments

Theorem 10.12 (Dobiński). *For every $n \geq 0$,*

$$B_n^{\text{Bell}} = e^{-1} \sum_{m=0}^{\infty} \frac{m^n}{m!}. \quad (10.11)$$

Equivalently, if $X \sim \text{Poisson}(1)$, then $B_n^{\text{Bell}} = \mathbb{E}X^n$.

Proof. Expand the outer exponential in (10.10):

$$e^{e^z - 1} = e^{-1} \sum_{m \geq 0} \frac{e^{mz}}{m!} = e^{-1} \sum_{m \geq 0} \frac{1}{m!} \sum_{n \geq 0} m^n \frac{z^n}{n!}.$$

Absolute convergence on compact sets justifies rearrangement analytically, while formal coefficient extraction gives the same identity algebraically. The last statement is the defining moment sum of a $\text{Poisson}(1)$ variable. $\square$

Remark 10.13 (Formal crosswalk). `Fabius.dobinski` (module `BellStirling`) proves $\sum_{m \geq 0} m^n / m! = e B_n^{\text{Bell}}$ as an identity of real series. It is the case $\lambda = 1$ of `Fabius.tsum_pow_mul_pow_div_factorial`, the Poisson moment identity $\sum_{m \geq 0} m^n \lambda^m / m! = e^\lambda \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} \lambda^k$ for every real λ , which is the analytic content of Theorem 3.7. The Lean proof does not expand the exponential generating function: it inserts $m^n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} (m)_k$ and uses that the k -th falling-factorial moment series is the exponential series shifted by k places (`Fabius.hasSum_descFactorial_mul_pow_div_factorial`), so only finitely many absolutely convergent series are rearranged. The Cauchy integral form is not formalized.

Corollary 10.14 (Cauchy integral). *For every positively oriented circle γ around 0,*

$$B_n^{\text{Bell}} = \frac{n!}{2\pi i e} \int_{\gamma} \frac{e^{e^z}}{z^{n+1}} dz. \quad (10.12)$$

Proof. Apply Cauchy's coefficient formula to $e^{-1}e^{e^z}$. □

Chapter 11

Congruences, shape, and growth of Bell numbers

11.1 Touchard congruences

It is useful to refine B_n^{Bell} to the Touchard polynomial

$$T_n^{\text{Tou}}(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k, \quad T_n^{\text{Tou}}(1) = B_n^{\text{Bell}}.$$

Theorem 11.1 (Polynomial Touchard congruence). *For a prime p ,*

$$T_{n+p}^{\text{Tou}}(x) \equiv T_{n+1}^{\text{Tou}}(x) + x^p T_n^{\text{Tou}}(x) \pmod{p}. \quad (11.1)$$

Consequently

$$B_{n+p}^{\text{Bell}} \equiv B_n^{\text{Bell}} + B_{n+1}^{\text{Bell}} \pmod{p}. \quad (11.2)$$

Proof. Let the cyclic group C_p rotate a distinguished set of p labels while fixing n other labels, and let it act on partitions, preserving the weight $x^{\# \text{ blocks}}$. Every nonfixed orbit has size p and contributes 0 modulo p . In a fixed partition there are two possibilities. Either all p rotating labels form p singleton blocks, independently of a partition of the other labels; this contributes $x^p T_n^{\text{Tou}}(x)$. Otherwise rotational invariance forces the rotating labels to lie in one common block, possibly together with old labels. Contracting those p labels to one distinguished label gives a partition of $n+1$ labels with the same number of blocks, contributing $T_{n+1}^{\text{Tou}}(x)$. This proves the polynomial congruence; set $x = 1$. $\square$

Remark 11.2 (Formal crosswalk). Equation (11.1) is `Fabius.touchardPolynomial_add_prime` (module `TouchardPolyCongruence`), an identity $T_{n+p}^{\text{Tou}} = T_{n+1}^{\text{Tou}} + x^p T_n^{\text{Tou}}$ in $(\mathbb{Z}/p)[x]$ for every prime p , and Equation (11.2) is `Fabius.bell_add_prime_modEq` (module `TouchardCongruence`). Both Lean proofs read Spivey's identity (Theorem 10.4; for Touchard polynomials `Fabius.spivey_touchard`, transported from $\mathbb{Q}[x]$ to every coefficient ring as `Fabius.touchardPolynomial_add_eq`) with $m = p$ modulo p : the Stirling numbers $\left\{ \begin{matrix} p \\ j \end{matrix} \right\}$ with $1 < j < p$ vanish modulo p (`Fabius.stirlingSecond_prime_eq_zero_zmod`, from the surjection formula Theorem 2.5 and Fermat's little theorem), the term $j = 1$ is the Touchard recurrence $x \sum_k \binom{n}{k} T_k^{\text{Tou}} = T_{n+1}^{\text{Tou}}$, and in the term $j = p$ only $k = n$ survives because $p^{n-k} \equiv 0$ for $k < n$.

Theorem 11.3 (Prime-power shift). *For a prime p and $m \geq 1$,*

$$B_{n+p^m}^{\text{Bell}} \equiv m B_n^{\text{Bell}} + B_{n+1}^{\text{Bell}} \pmod{p}. \quad (11.3)$$

More precisely,

$$T_{n+p^m}^{\text{Tou}}(x) \equiv T_{n+1}^{\text{Tou}}(x) + T_n^{\text{Tou}}(x) \sum_{r=1}^m x^{p^r} \pmod{p}. \quad (11.4)$$

Proof. Let the additive cyclic group of order p^m act regularly on p^m new labels. A partition fixed by the whole group induces a block system for this regular action. Apart from the case in which all new labels join one invariant block (which contracts to one label and contributes T_{n+1}^{Tot}), the blocks on the new orbit are the cosets of the unique subgroup of index p^r for some $1 \leq r \leq m$; they contribute p^r blocks and cannot mix with old labels. Thus the fixed weighted partitions contribute $T_{n+1}^{\text{Tot}} + T_n^{\text{Tot}} \sum_{r=1}^m x^{p^r}$. Every other orbit has cardinality divisible by p , proving (11.4). Fermat's congruence $x^{p^r} \equiv x$ modulo p at $x = 1$ gives (11.3). $\square$

Remark 11.4 (Formal crosswalk). `Fabius.bell_add_prime_pow_modEq` (module `ShiftOperatorCharP`) is Equation (11.3) as a congruence of natural numbers, and `Fabius.touchardPolynomial_add_prime_pow` is Equation (11.4) as an identity in $\mathbb{F}_p[x]$. The formal proof does not use the group action above. It takes Touchard's congruence in the operator form $E^p = E + c$, with $c = 1$ for the Bell numbers and $c = x^p$ for the Touchard polynomials (the latter from `Fabius.touchardPolynomial_add_prime`), and shows by induction on m , using the Frobenius identity $(u + v)^p = u^p + v^p$ in characteristic p , that $Y^{p^m} \equiv Y + \sum_{r < m} c^{p^r}$ modulo the polynomial $Y^p - Y - c$ (`Fabius.mk_X_pow_prime_pow`). Since the shift E acts linearly on sequences, every polynomial multiple of the characteristic polynomial annihilates every solution of the recurrence (`Fabius.aeval_shiftEnd_eq_zero_of_dvd`), which gives the general statement `Fabius.shift_pow_char_pow`: if $b_{n+p} = b_{n+1} + cb_n$ in a commutative ring of characteristic p , then $b_{n+p^m} = b_{n+1} + (\sum_{r=0}^{m-1} c^{p^r})b_n$.

11.2 A universal period bound modulo a prime

Theorem 11.5. *Modulo a prime p , the Bell sequence is periodic, and its period divides*

$$N_p = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}. \quad (11.5)$$

Proof. Touchard's recurrence says that the shift operator E on the sequence satisfies $E^p - E - 1 = 0$ over $\mathbb{F}_p$. The polynomial $q(t) = t^p - t - 1$ is separable because $q'(t) = -1$. If α is any root in a splitting field, then $\alpha^p = \alpha + 1$ and hence $\alpha^{p^j} = \alpha + j$ for $j \in \mathbb{F}_p$. Taking the product over j gives

$$\alpha^{N_p} = \prod_{j \in \mathbb{F}_p} \alpha^{p^j} = \prod_{j \in \mathbb{F}_p} (\alpha + j) = \alpha^p - \alpha = 1.$$

Thus every eigenvalue of E is an N_p th root of unity. Separability makes E diagonalizable over the splitting field, so $E^{N_p} = 1$ on every solution of the recurrence, in particular on the Bell sequence. $\square$

Remark 11.6 (Formal crosswalk). `Fabius.bell_add_sum_prime_pow_modEq` (module `ShiftOperatorCharP`) states that $N_p = \sum_{j < p} p^j$ is a period, $B_{n+N_p}^{\text{Bell}} \equiv B_n^{\text{Bell}} \pmod{p}$ for every $n \geq 0$, and `Fabius.bell_period_dvd_sum_prime_pow` that the least positive period divides N_p (by the general division argument `Fabius.period_dvd_of_minimal`). The formal proof stays in $\mathbb{F}_p[Y]$: from $Y^{p^j} \equiv Y + j$ modulo $q(Y) = Y^p - Y - 1$ (`Fabius.mk_X_pow_prime_pow`) and the Fermat product $\prod_{j=0}^{p-1} (Y + j) = Y^p - Y$ (`Fabius.prod_range_X_add_C_natCast`, from the factorization of $Y^p - Y$ over $\mathbb{F}_p$, `Fabius.prod_univ_X_sub_C`) one gets $Y^{N_p} \equiv Y^p - Y = q(Y) + 1 \equiv 1$, so $q(Y)$ divides $Y^{N_p} - 1$, and the shift-operator transfer `Fabius.aeval_shiftEnd_eq_zero_of_dvd` yields $E^{N_p} = 1$ on every solution of the recurrence over $\mathbb{F}_p$ (`Fabius.shift_pow_period`); no diagonalization is needed.

Remark 11.7. A reported exact period is computational data, not a consequence of the divisibility bound alone. It can be certified by generating N_p terms with (11.2), finding the least divisor $d \mid N_p$ for which the state vector of p consecutive terms repeats, and checking that no proper divisor of d works. The same recurrence permits a compact machine-checkable certificate.

11.3 Log-convexity and normalized log-concavity

Theorem 11.8. For $n \geq 1$,

$$1 \leq \frac{B_{n-1}^{\text{Bell}} B_{n+1}^{\text{Bell}}}{(B_n^{\text{Bell}})^2} \leq \frac{n+1}{n}. \quad (11.6)$$

Thus (B_n^{Bell}) is log-convex, whereas $(B_n^{\text{Bell}}/n!)$ is log-concave.

Proof. For the left inequality, let $X \sim \text{Poisson}(1)$. Cauchy–Schwarz applied to $X^{(n-1)/2}$ and $X^{(n+1)/2}$ gives $(\mathbb{E}X^n)^2 \leq \mathbb{E}X^{n-1} \mathbb{E}X^{n+1}$.

For completeness we prove the normalized inequality through the following convolution lemma. If nonnegative $(z_j)_{j \geq 1}$ is log-concave without internal zeros and

$$A(t) = \exp\left(\sum_{j \geq 1} \frac{z_j}{j} t^j\right) = \sum_{n \geq 0} a_n \frac{t^n}{n!},$$

then $a_n/n!$ is log-concave. To see this, truncate at degree N , differentiate, and compare the adjacent 2×2 Toeplitz minors in $A' = A \sum_{j \geq 1} z_j t^{j-1}$. After cross multiplication, the n th minor is

$$\sum_{0 \leq i < j \leq n} \frac{a_i a_j}{i! j!} (z_{n-i+1} z_{n-j} - z_{n-i} z_{n-j+1}),$$

with the boundary terms interpreted as zero. Each bracket is nonnegative by the monotonicity of the ratios z_{r+1}/z_r ; induction on n starts at $a_0 = 1$. Removing the truncation proves the lemma coefficientwise.

For $A(t) = e^{e^t - 1}$ we have $\sum_{j \geq 1} t^j/j! = \sum_{j \geq 1} z_j t^j/j$ with $z_j = 1/(j-1)!$. This sequence is log-concave, since $z_j^2/(z_{j-1} z_{j+1}) = j/(j-1)$. Hence $(B_n^{\text{Bell}}/n!)^2 \geq (B_{n-1}^{\text{Bell}}/(n-1)!)(B_{n+1}^{\text{Bell}}/(n+1)!)$, which is the right inequality in (11.6). $\square$

Remark 11.9 (Editorial: the convolution lemma above is not a proof). The statement (11.6) is true, and was checked exactly for $1 \leq n \leq 39$; the left inequality is proved correctly above by Cauchy–Schwarz. The convolution lemma offered for the right inequality, however, does not establish it, for two independent reasons.

First, the sign claim is inverted relative to its own hypothesis. Writing $r = n - j$ and $s = n - i$, so that $s > r$, the bracket is $z_{s+1} z_r - z_s z_{r+1}$, which is nonnegative exactly when z_{r+1}/z_r is *increasing*, that is when (z_j) is log-convex. The lemma assumes (z_j) log-concave, under which the bracket is instead nonpositive: for $z_j = 1/(j-1)!$ the values at $(r, s) = (1, 2), (1, 3), (2, 4)$ are $-1/2, -1/3$ and $-1/24$.

Second, the displayed sum is not the minor it is said to be. With $z_j = 1/(j-1)!$ and $a_n = B_n^{\text{Bell}}$, the sum evaluates to $-1, -5/2$ and -3 for $n = 1, 2, 3$, whereas the corresponding 2×2 minors $(a_n/n!)^2 - (a_{n-1}/(n-1)!)(a_{n+1}/(n+1)!)$ are $0, 1/6$ and $5/72$.

The right inequality, equivalently the log-concavity of $B_n^{\text{Bell}}/n!$, is classical, but this text does not prove it; a correct proof, or a citation, should replace the lemma.

11.4 Explicit upper bounds

The positive-coefficient Cauchy estimate supplies a short route to the numerical bounds in the archive.

Lemma 11.10. For $t > 0$,

$$B_n^{\text{Bell}} \leq n! t^{-n} \exp(e^t - 1). \quad (11.7)$$

For $n \geq 2$, put $L = \log(n+1)$ and $t = \log(n/L)$. Then

$$(B_n^{\text{Bell}})^{1/n} \frac{L}{n} \leq \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log n}{2n}\right). \quad (11.8)$$

Proof. Because all coefficients of $\mathcal{B}$ are nonnegative, $B_n^{\text{Bell}} t^n / n! \leq \mathcal{B}(t)$, proving the first claim. Use $e^t = n/L$ and the elementary Stirling bound $n! \leq e\sqrt{n}(n/e)^n$; the factors $e^{-1/n}$ and $e^{1/n}$ cancel and give (11.8). $\square$

Theorem 11.11 (Eventual Bell upper bound). *For every $\varepsilon > 0$ there is an integer N_ε such that, for all $n \geq N_\varepsilon$,*

$$B_n^{\text{Bell}} < \left(\frac{e^{-1+\varepsilon} n}{\log(n+1)} \right)^n. \quad (11.9)$$

Proof. For real x sufficiently large, set

$$R(x) = \frac{\log(x+1)}{\log(x/\log(x+1))} \exp\left(-1 + \frac{1}{\log(x+1)} + \frac{\log x}{2x}\right).$$

The elementary limits

$$\frac{\log(x+1)}{\log x} \rightarrow 1, \quad \frac{\log \log(x+1)}{\log x} \rightarrow 0, \quad \frac{\log x}{x} \rightarrow 0$$

give $R(x) \rightarrow e^{-1}$. Since $e^{-1+\varepsilon} > e^{-1}$, there is N_ε such that $R(n) < e^{-1+\varepsilon}$ for all $n \geq N_\varepsilon$. The right side of Equation (11.8) is exactly $R(n)$, so raising the resulting inequality to the n th power proves Equation (11.9). $\square$

11.5 The sharp numerical constant

All six arrivals also assert the sharper numerical bound with the decimal constant 0.792 valid for every $n \geq 1$. Their common argument replaced both a monotonicity claim and the range $n \leq 38$ by unsupported decimal assertions, with no directed-rounding ledger and no exact treatment of the finite range. The statement itself is a theorem of Berend and Tassa [2]. This section proves it, splitting the two gaps the arrivals left open: the tail is settled by a monotonicity lemma for the majorant already constructed in Theorem 11.10, and the finite range by an exact integer certificate that involves no floating-point arithmetic at all.

Lemma 11.12 (The coefficient majorant is decreasing). *For $x \geq 15$ put*

$$L = \log(x+1), \quad t = \log \frac{x}{L}, \quad \bar{R}(x) = \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log x}{2x}\right). \quad (11.10)$$

Then $\bar{R}$ is positive and strictly decreasing on $[15, \infty)$, and $\bar{R}(x) \rightarrow e^{-1}$ as $x \rightarrow \infty$.

Proof. Write $\bar{R} = \varphi\psi$ with $\varphi = L/t$ and $\psi = \exp(-1 + 1/L + \log x/(2x))$. On $[15, \infty)$ one has $L > e > 1$ and $x/L > 1$, so $t > 0$ and both factors are positive. It is enough to show that each is strictly decreasing.

The factor ψ . The map $x \mapsto 1/\log(x+1)$ is strictly decreasing, and

$$\frac{d}{dx} \frac{\log x}{2x} = \frac{1 - \log x}{2x^2} < 0 \quad (x > e).$$

Hence the exponent is strictly decreasing on $[3, \infty)$, and so is ψ .

The factor φ . Since $t = \log x - \log L$,

$$\frac{1}{\varphi} = \frac{\log x - \log L}{L} = \frac{\log x}{\log(x+1)} - \frac{\log L}{L}. \quad (11.11)$$

The first term of (11.11) is strictly increasing:

$$\frac{d}{dx} \frac{\log x}{\log(x+1)} = \frac{(x+1)\log(x+1) - x\log x}{x(x+1)\log^2(x+1)} > 0,$$

because $s \mapsto s \log s$ is strictly increasing on $[1, \infty)$. The second term is strictly decreasing, since $u \mapsto (\log u)/u$ is strictly decreasing for $u > e$ and

$$L = \log(x+1) > e \iff x+1 > e^e = 15.154\dots,$$

which holds for every $x \geq 15$. Therefore $1/\varphi$ is strictly increasing and φ strictly decreasing on $[15, \infty)$. A product of two positive strictly decreasing functions is strictly decreasing.

The limit. As $x \rightarrow \infty$, $\log(x+1)/\log x \rightarrow 1$ and $\log L/\log x \rightarrow 0$, so $1/\varphi \rightarrow 1$ by (11.11); also $1/L \rightarrow 0$ and $\log x/(2x) \rightarrow 0$, so $\psi \rightarrow e^{-1}$. $\square$

Theorem 11.13 (Uniform numerical Bell bound). *For every $n \geq 1$,*

$$\mathbf{B}_n^{\text{Bell}} < \left(\frac{0.792n}{\log(n+1)} \right)^n. \quad (11.12)$$

Equivalently, writing $R_n = (\mathbf{B}_n^{\text{Bell}})^{1/n} \log(n+1)/n$, one has $R_n < 0.792$ for all $n \geq 1$.

Proof. Raising (11.12) to the power $1/n$ shows that it is equivalent to $R_n < 0.792$, so it suffices to bound R_n . The two ranges are treated separately, and the split point 39 is forced: the majorant (11.10) satisfies $\bar{R}(38) = 0.79411\dots > 0.792$, so no argument through $\bar{R}$ alone can reach below $n = 39$.

The tail $n \geq 39$. The right side of Equation (11.8) is exactly $\bar{R}(n)$, so $R_n \leq \bar{R}(n)$ for every $n \geq 2$. By Theorem 11.12, $\bar{R}$ is decreasing on $[15, \infty)$, hence $R_n \leq \bar{R}(n) \leq \bar{R}(39)$ for all $n \geq 39$. Evaluating (11.10) at $x = 39$ with every operation rounded outward at six decimals gives the ledger

$$\begin{aligned} \log 40 &\in [3.688879, 3.688880], & \log 39 &\in [3.663561, 3.663562], \\ \log \log 40 &\in [1.305322, 1.305323], & t &\in [2.358238, 2.358240], \\ L/t &\in [1.564250, 1.564253], & 1/L &\in [0.271084, 0.271086], \\ (\log 39)/78 &\in [0.046968, 0.046969], & \text{exponent} &\in [-0.681948, -0.681945], \\ e^{\text{exponent}} &\in [0.505631, 0.505633], & \bar{R}(39) &\in [0.790933, 0.790938]. \end{aligned}$$

Each entry is obtained from the entries above it by interval arithmetic with outward rounding, so the last line is a rigorous enclosure. In particular $\bar{R}(39) < 0.7910 < 0.792$, and the tail is proved.

The finite range $1 \leq n \leq 38$. Put

$$A_n = \lceil 10^6 \log(n+1) \rceil. \quad (11.13)$$

Since $\log(n+1)$ is irrational for $n \geq 1$, the ceiling is strict: $A_n/10^6 > \log(n+1)$. Consequently

$$R_n = (\mathbf{B}_n^{\text{Bell}})^{1/n} \frac{\log(n+1)}{n} < (\mathbf{B}_n^{\text{Bell}})^{1/n} \frac{A_n}{10^6 n},$$

and therefore $R_n < 792/1000$ follows from $(\mathbf{B}_n^{\text{Bell}})^{1/n} A_n \cdot 1000 \leq 792 \cdot 10^6 n$, that is, after raising to the n th power and cancelling 1000^n , from the inequality between positive integers

$$\mathbf{B}_n^{\text{Bell}} A_n^n < (792000 n)^n. \quad (11.14)$$

No irrational quantity occurs in (11.14): the Bell numbers are the integers generated by Theorem 10.1, and the A_n are the integers

$$\begin{array}{llll} A_1 = 693148 & A_2 = 1098613 & A_3 = 1386295 & A_4 = 1609438 \\ A_5 = 1791760 & A_6 = 1945911 & A_7 = 2079442 & A_8 = 2197225 \\ A_9 = 2302586 & A_{10} = 2397896 & A_{11} = 2484907 & A_{12} = 2564950 \\ A_{13} = 2639058 & A_{14} = 2708051 & A_{15} = 2772589 & A_{16} = 2833214 \\ A_{17} = 2890372 & A_{18} = 2944439 & A_{19} = 2995733 & A_{20} = 3044523 \\ A_{21} = 3091043 & A_{22} = 3135495 & A_{23} = 3178054 & A_{24} = 3218876 \\ A_{25} = 3258097 & A_{26} = 3295837 & A_{27} = 3332205 & A_{28} = 3367296 \\ A_{29} = 3401198 & A_{30} = 3433988 & A_{31} = 3465736 & A_{32} = 3496508 \\ A_{33} = 3526361 & A_{34} = 3555349 & A_{35} = 3583519 & A_{36} = 3610918 \\ A_{37} = 3637587 & A_{38} = 3663562 & & \end{array}$$

Inequality (11.14) holds for each n in the range. Its relative slack $1 - B_n^{\text{Bell}} A_n^n / (792000 n)^n$ is smallest at $n = 4$, where it equals $0.000811 \dots$; it is $0.006889 \dots$ at $n = 3$, exceeds 0.013 for $5 \leq n \leq 10$, and increases from $0.310 \dots$ at $n = 11$ to $0.986 \dots$ at $n = 38$. This completes the finite range and the proof. $\square$

Remark 11.14 (The constant is essentially best possible). The sequence R_n attains its maximum at $n = 4$, where $B_4^{\text{Bell}} = 15$ and

$$R_4 = \frac{15^{1/4} \log 5}{4} = 0.7918392970 \dots \quad (11.15)$$

That this is the global maximum needs a word, since the certificate (11.14) proves only $R_n < 0.792$ and not $R_n \leq R_4$. For $n \geq 39$ the tail argument already gives the stronger $R_n \leq \bar{R}(39) < 0.7910 < R_4$; for $n \leq 38$ it is the direct evaluation of 38 numbers, whose three largest values are $R_3 = 0.790176 \dots$, $R_4 = 0.791839 \dots$ and $R_5 = 0.789787 \dots$. Hence $0.792 - \max_n R_n < 1.61 \cdot 10^{-4}$, and the constant in (11.12) cannot be replaced by anything below R_4 . This also explains the tightness of the certificate at $n = 4$: the sole reason a six-decimal logarithm suffices there is that $\lceil 10^6 \log 5 \rceil / 10^6$ overshoots $\log 5$ by less than 10^{-6} , while the slack in (11.14) is of order 10^{-3} . By contrast Theorem 11.11 is asymptotically far sharper, since $R_n \rightarrow e^{-1} = 0.3678 \dots$; the two statements are complementary rather than competing.

11.6 Saddle-point asymptotics and Lambert W

Let $r = W(n)$, so that $re^r = n$, and put $A = n(1 + r)$.

Theorem 11.15 (Leading Bell asymptotic). *As $n \rightarrow \infty$,*

$$B_n^{\text{Bell}} \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1 + r)}} \quad (11.16)$$

$$\sim \frac{1}{\sqrt{n}} \left(\frac{n}{W(n)} \right)^{n+1/2} \exp\left(\frac{n}{W(n)} - n - 1 \right). \quad (11.17)$$

Proof. In (10.12) take the circle $z = re^{i\theta}$. The coefficient integral becomes

$$\frac{B_n^{\text{Bell}}}{n!} = \frac{e^{e^r - 1}}{2\pi r^n} \int_{-\pi}^{\pi} \exp(e^{re^{i\theta}} - e^r - in\theta) d\theta.$$

The linear term vanishes because $re^r = n$; the quadratic term is $-A\theta^2/2$. On $|\theta| \leq A^{-2/5}$, Taylor expansion is uniform and the scaled integral tends to $\int_{\mathbb{R}} e^{-u^2/2} du / \sqrt{A}$. Outside that arc, $\Re(e^{re^{i\theta}} - e^r) \leq -cA\theta^2$ on the intermediate range and is $\leq -c'e^r$ on the remaining compact range, so the contribution is negligible. This proves (11.16). Stirling's formula and $e^r = n/r$ simplify it to (11.17). $\square$

Remark 11.16 (Status of the refined Moser–Wyman expansion). The arrivals next present a purported uniform expansion for B_{n+h}^{Bell} . It is not a well-formed theorem: “total order at most e^{-Mr} ” is not a grading of monomials, the coefficients C_q and remainder polynomial P_M are not defined, one displayed term uses an undefined T_4 , and the asserted uniform Watson-lemma and off-arc estimates are not proved. We therefore retain only the rigorous leading saddle estimate above. This is a withdrawal of an unsupported theorem claim, not a conjecture that refined expansions fail.

Moser and Wyman do prove higher-order Bell asymptotics [16], but importing them requires a fresh normalization theorem. Such a theorem must specify a genuine asymptotic scale, define every correction by a finite coefficient-extraction rule, state the range of the shift h , and prove both

the central-arc remainder and the complementary-arc bound uniformly in that range. Until those obligations are discharged, a formal Gaussian-moment expansion is a useful calculation recipe but not a human-readable proof of a uniform remainder.

Corollary 11.17 (de Bruijn logarithmic expansion). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{\log B_n^{\text{Bell}}}{n} = \log n - \log \log n - 1 + \frac{\log \log n}{\log n} + \frac{1}{\log n} \\ + \frac{1}{2} \left(\frac{\log \log n}{\log n} \right)^2 + O\left(\frac{\log \log n}{(\log n)^2} \right). \end{aligned} \quad (11.18)$$

Proof. Take logarithms in (11.17). The standard reversion of $r + \log r = \log n$ gives

$$r = \log n - \log \log n + \frac{\log \log n}{\log n} + O\left(\frac{(\log \log n)^2}{(\log n)^2} \right).$$

Substitution and one further Taylor expansion of $\log r$ produce the displayed terms; the prefactor contributes only $\mathcal{O}(\log n/n)$. $\square$

11.7 Bell primes and verifiable computation

A *Bell prime* is a prime value of B_n^{Bell} . The archive records the early indices found by computation. Such a list is not a symbolic theorem, but each entry has a rigorous certificate: compute B_n^{Bell} by the integer recurrence (10.1), apply a probable-prime filter, and attach an ECPP or Pratt-style primality certificate. Composite entries are certified by one nontrivial factor. This distinction between an identity and a finite certified computation will also be used for tabulated values later.

Part IV

Bell Polynomials and Composition

Chapter 12

Exponential and ordinary Bell polynomials

12.1 Definitions

Definition 12.1 (Partial exponential Bell polynomial). For $0 \leq k \leq n$, define

$$\mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots, x_{n-k+1}) = n! \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+3j_3+\dots=n}} \prod_{i \geq 1} \frac{x_i^{j_i}}{(i!)^{j_i} j_i!}. \quad (12.1)$$

Only $i \leq n - k + 1$ can occur. The complete exponential Bell polynomial is

$$\mathcal{B}_n^{\text{exp}}(x_1, \dots, x_n) = \sum_{k=0}^n \mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots, x_{n-k+1}), \quad \mathcal{B}_0^{\text{exp}} = 1. \quad (12.2)$$

Definition 12.2 (Partial ordinary Bell polynomial).

$$\mathcal{B}_{n,k}^{\text{ord}}(x_1, \dots, x_{n-k+1}) = \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+\dots=n}} \frac{k!}{j_1! j_2! \dots} \prod_{i \geq 1} x_i^{j_i}. \quad (12.3)$$

Equivalently,

$$\mathcal{B}_{n,k}^{\text{ord}}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}^{\text{exp}}(1!x_1, 2!x_2, \dots). \quad (12.4)$$

Proof of (12.4). Substitute $i!x_i$ for x_i in (12.1); the factorials $(i!)^{j_i}$ cancel, and multiplication by $k!/n!$ leaves exactly (12.3). $\square$

12.2 Combinatorial interpretation

Give each block of size i a weight x_i , and define the weight of a partition as the product of its block weights.

Theorem 12.3 (Weighted-partition interpretation). $\mathcal{B}_{n,k}^{\text{exp}}$ is the total weight of partitions of an n -element labelled set into exactly k blocks, and $\mathcal{B}_n^{\text{exp}}$ is the total weight of all its partitions.

Proof. Fix multiplicities j_i , where j_i blocks have size i . The number of such partitions is

$$\frac{n!}{\prod_i (i!)^{j_i} j_i!}.$$

The factors $i!$ forget order inside each block and $j_i!$ forget order among blocks of equal size. Multiplication by $\prod_i x_i^{j_i}$ and summation gives (12.1); summing over k gives the complete polynomial. $\square$

Thus

$$\begin{aligned}\mathcal{B}_{3,2}^{\text{exp}}(x_1, x_2) &= 3x_1x_2, \\ \mathcal{B}_{6,2}^{\text{exp}}(x_1, \dots, x_5) &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\ \mathcal{B}_{6,3}^{\text{exp}}(x_1, \dots, x_4) &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3.\end{aligned}$$

The monomials correspond to integer partitions of n into k parts, while their coefficients count the labelled set partitions having the prescribed block sizes.

12.3 The master generating functions

Put

$$X(t) = \sum_{j \geq 1} x_j \frac{t^j}{j!}.$$

Theorem 12.4 (Bell-polynomial generating functions). *One has*

$$\frac{X(t)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots) \frac{t^n}{n!}, \quad (12.5)$$

$$\exp(uX(t)) = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots) u^k \frac{t^n}{n!}, \quad (12.6)$$

$$\exp(X(t)) = \sum_{n \geq 0} \mathcal{B}_n^{\text{exp}}(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (12.7)$$

For $X^{\text{ogf}}(t) = \sum_{j \geq 1} x_j t^j$,

$$X^{\text{ogf}}(t)^k = \sum_{n \geq k} \mathcal{B}_{n,k}^{\text{ord}}(x_1, \dots) t^n, \quad (12.8)$$

$$e^{uX^{\text{ogf}}(t)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}^{\text{ord}}(x_1, \dots) t^n \frac{u^k}{k!}. \quad (12.9)$$

Proof. Expand $X(t)^k$ multinomially. Choosing j_i copies of the term $x_i t^i / i!$ gives the constraints $\sum j_i = k$ and $\sum i j_i = n$, and its coefficient is $k! \prod_i x_i^{j_i} / ((i!)^{j_i} j_i!)$. Division by $k!$ gives (12.1). Summing in k proves the exponential identities. The ordinary identities follow by the same multinomial argument or by (12.4). $\square$

Master principle

Nearly every algebraic identity for Bell polynomials follows by performing a simple operation—multiplication, differentiation, substitution, or logarithm—on (1.10) or (1.12). The combinatorial proofs are the same operations interpreted as selecting, pointing, colouring, or gluing blocks.

12.4 Tables and low-degree polynomials

The nonzero partial polynomials through $n = 6$ are

$$\begin{aligned}
\mathcal{B}_{1,1}^{\text{exp}} &= x_1, \\
\mathcal{B}_{2,1}^{\text{exp}} &= x_2, & \mathcal{B}_{2,2}^{\text{exp}} &= x_1^2, \\
\mathcal{B}_{3,1}^{\text{exp}} &= x_3, & \mathcal{B}_{3,2}^{\text{exp}} &= 3x_1x_2, & \mathcal{B}_{3,3}^{\text{exp}} &= x_1^3, \\
\mathcal{B}_{4,1}^{\text{exp}} &= x_4, & \mathcal{B}_{4,2}^{\text{exp}} &= 4x_1x_3 + 3x_2^2, \\
\mathcal{B}_{4,3}^{\text{exp}} &= 6x_1^2x_2, & \mathcal{B}_{4,4}^{\text{exp}} &= x_1^4, \\
\mathcal{B}_{5,1}^{\text{exp}} &= x_5, & \mathcal{B}_{5,2}^{\text{exp}} &= 5x_1x_4 + 10x_2x_3, \\
\mathcal{B}_{5,3}^{\text{exp}} &= 10x_1^2x_3 + 15x_1x_2^2, \\
\mathcal{B}_{5,4}^{\text{exp}} &= 10x_1^3x_2, & \mathcal{B}_{5,5}^{\text{exp}} &= x_1^5, \\
\mathcal{B}_{6,1}^{\text{exp}} &= x_6, & \mathcal{B}_{6,2}^{\text{exp}} &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\
\mathcal{B}_{6,3}^{\text{exp}} &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3, \\
\mathcal{B}_{6,4}^{\text{exp}} &= 20x_1^3x_3 + 45x_1^2x_2^2, \\
\mathcal{B}_{6,5}^{\text{exp}} &= 15x_1^4x_2, & \mathcal{B}_{6,6}^{\text{exp}} &= x_1^6.
\end{aligned}$$

Every coefficient is supplied by (12.1), so the table is a computation from the theorem rather than an independent assertion.

The first complete polynomials are

$$\begin{aligned}
\mathcal{B}_0^{\text{exp}} &= 1, \\
\mathcal{B}_1^{\text{exp}} &= x_1, \\
\mathcal{B}_2^{\text{exp}} &= x_1^2 + x_2, \\
\mathcal{B}_3^{\text{exp}} &= x_1^3 + 3x_1x_2 + x_3, \\
\mathcal{B}_4^{\text{exp}} &= x_1^4 + 6x_1^2x_2 + 3x_2^2 + 4x_1x_3 + x_4, \\
\mathcal{B}_5^{\text{exp}} &= x_1^5 + 10x_1^3x_2 + 15x_1x_2^2 + 10x_1^2x_3 + 10x_2x_3 + 5x_1x_4 + x_5, \\
\mathcal{B}_6^{\text{exp}} &= x_1^6 + 15x_1^4x_2 + 45x_1^2x_2^2 + 15x_2^3 + 20x_1^3x_3 + 60x_1x_2x_3 + 10x_3^2 \\
&\quad + 15x_1^2x_4 + 15x_2x_4 + 6x_1x_5 + x_6, \\
\mathcal{B}_7^{\text{exp}} &= x_1^7 + 21x_1^5x_2 + 105x_1^3x_2^2 + 105x_1x_2^3 + 35x_1^4x_3 + 210x_1^2x_2x_3 \\
&\quad + 70x_1x_3^2 + 105x_2^2x_3 + 35x_1^3x_4 + 105x_1x_2x_4 + 35x_3x_4 \\
&\quad + 21x_1^2x_5 + 21x_2x_5 + 7x_1x_6 + x_7.
\end{aligned}$$

Chapter 13

Recurrences, derivatives, and algebraic identities

13.1 Pointing recurrences

Theorem 13.1 (Complete and partial recurrences). For $n \geq 0$,

$$\mathcal{B}_{n+1}^{\text{exp}}(x_1, \dots, x_{n+1}) = \sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i}^{\text{exp}}(x_1, \dots, x_{n-i}), \quad (13.1)$$

$$\mathcal{B}_{n+1,k+1}^{\text{exp}} = \sum_{i=0}^{n-k} \binom{n}{i} x_{i+1} \mathcal{B}_{n-i,k}^{\text{exp}}, \quad (13.2)$$

with $\mathcal{B}_{0,0}^{\text{exp}} = 1$, $\mathcal{B}_{n,0}^{\text{exp}} = 0$ for $n > 0$, and $\mathcal{B}_{0,k}^{\text{exp}} = 0$ for $k > 0$.

Proof. In a weighted partition of $[n+1]$, inspect the block containing the distinguished label $n+1$. If that block contains i further labels, choose them in $\binom{n}{i}$ ways, give the block weight x_{i+1} , and partition the remaining labels. Keeping or forgetting the total block count yields the two formulas. $\square$

Theorem 13.2 (Block-colour convolution). For $k_1, k_2 \geq 0$,

$$\mathcal{B}_{n,k_1+k_2}^{\text{exp}}(x) = \frac{k_1!k_2!}{(k_1+k_2)!} \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{i,k_1}^{\text{exp}}(x) \mathcal{B}_{n-i,k_2}^{\text{exp}}(x). \quad (13.3)$$

Proof. Multiply (1.10) for k_1 and k_2 . Their product is $X^{k_1+k_2}/(k_1!k_2!)$; comparison with the same formula for k_1+k_2 gives the factor $(k_1!k_2!)/(k_1+k_2)!$ and the binomial EGF convolution. $\square$

13.2 Near-diagonal reduction

Theorem 13.3. For $1 \leq a < n$,

$$\begin{aligned} \mathcal{B}_{n,n-a}^{\text{exp}}(x_1, \dots, x_{a+1}) \\ = \sum_{j=a+1}^{2a} \frac{j!}{a!} \binom{n}{j} x_1^{n-j} \mathcal{B}_{a,j-a}^{\text{exp}}\left(\frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_{2a-j+2}}{2a-j+2}\right). \end{aligned} \quad (13.4)$$

Proof. A partition into $n-a$ blocks has total excess $\sum_B (|B|-1) = a$. Let j be the number of elements in nonsingleton blocks. Then the number of those blocks is $j-a$, so $a+1 \leq j \leq 2a$. Choose the j elements, leave the other $n-j$ as singleton blocks, and count the nonsingleton partition. Replacing a

block of size $r + 1$ by an atom of size r removes one distinguished element per block; the factor $j!/a!$ and weights $x_{r+1}/(r+1)$ are exactly the labelled pointing factors. Equivalently, make the substitution in (12.1); the coefficient of every multiplicity vector on both sides is $n!x_1^{n-j} \prod_{r \geq 2} x_r^{j_r} / ((r!)^{j_r} j_r!)$. $\square$

The first cases, obtained by listing integer partitions of the excess, are

$$\mathcal{B}_{n,1}^{\text{exp}} = x_n, \quad \mathcal{B}_{n,2}^{\text{exp}} = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} x_j x_{n-j}, \quad (13.5)$$

$$\mathcal{B}_{n,n}^{\text{exp}} = x_1^n, \quad \mathcal{B}_{n,n-1}^{\text{exp}} = \binom{n}{2} x_1^{n-2} x_2, \quad (13.6)$$

$$\mathcal{B}_{n,n-2}^{\text{exp}} = \binom{n}{3} x_1^{n-3} x_3 + 3 \binom{n}{4} x_1^{n-4} x_2^2, \quad (13.7)$$

$$\mathcal{B}_{n,n-3}^{\text{exp}} = \binom{n}{4} x_1^{n-4} x_4 + 10 \binom{n}{5} x_1^{n-5} x_2 x_3 + 15 \binom{n}{6} x_1^{n-6} x_2^3, \quad (13.8)$$

$$\begin{aligned} \mathcal{B}_{n,n-4}^{\text{exp}} = & \binom{n}{5} x_1^{n-5} x_5 + 5 \binom{n}{6} x_1^{n-6} (3x_2 x_4 + 2x_3^2) \\ & + 105 \binom{n}{7} x_1^{n-7} x_2^2 x_3 + 105 \binom{n}{8} x_1^{n-8} x_2^4. \end{aligned} \quad (13.9)$$

Their coefficients are the corresponding block-size counts from Theorem 12.3.

13.3 Partial derivatives

Theorem 13.4 (Derivative identities). *For $i \geq 1$,*

$$\frac{\partial \mathcal{B}_n^{\text{exp}}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i}^{\text{exp}}, \quad (13.10)$$

$$\frac{\partial \mathcal{B}_{n,k}^{\text{exp}}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i,k-1}^{\text{exp}}. \quad (13.11)$$

Consequently, for differentiable $a_i(x)$,

$$\frac{d}{dx} \mathcal{B}_{n,k}^{\text{exp}}(a_1(x), \dots) = \sum_{i=1}^{n-k+1} \binom{n}{i} a'_i(x) \mathcal{B}_{n-i,k-1}^{\text{exp}}(a_1(x), \dots). \quad (13.12)$$

Proof. Differentiate (13.10) with respect to x_i :

$$\frac{\partial}{\partial x_i} \frac{X(t)^k}{k!} = \frac{t^i X(t)^{k-1}}{i! (k-1)!}.$$

Coefficient comparison gives (13.11); summing over k gives (13.10). The ordinary chain rule then gives (13.12). $\square$

Remark 13.5 (Formal crosswalk). `Fabius.pderiv_partialBell_succ` (module `BellDerivative`) is Equation (13.11) and `Fabius.pderiv_bellComplete` is Equation (13.10), both as identities in the polynomial ring $\mathbb{Q}[x_1, x_2, \dots]$ with `Mathlib's` partial derivative `MvPolynomial.pderiv`, for $i \geq 1$ and $k \geq 1$ (for $k = 0$ the derivative vanishes, `Fabius.pderiv_partialBell_zero`). The formal proof is the one above: a derivation of the coefficient ring acts coefficientwise on power series and is again a derivation (`Fabius.coeffDerivation`), so applying $\partial/\partial x_i$ to the column theorem `Fabius.bellWeightSeries_pow` and using $\partial X(t)/\partial x_i = t^i/i!$ (`Fabius.coeffDerivation_pderiv_bellWeightSeries`) gives the identity by coefficient comparison; the complete-polynomial form follows by summing over k . The chain rule Equation (13.12) is analytic and is not formalized.

13.4 A quadratic differential recurrence

The less familiar differential recurrence in the archive is also a disguised coefficient identity.

Theorem 13.6. For $n \geq 2$,

$$\begin{aligned} \mathcal{B}_n^{\text{exp}} = \frac{1}{n-1} & \left[\sum_{i=2}^n \sum_{j=1}^{i-1} (i-1) \binom{i-2}{j-1} x_j x_{i-j} \frac{\partial \mathcal{B}_{n-1}^{\text{exp}}}{\partial x_{i-1}} \right. \\ & \left. + \sum_{i=2}^n \sum_{j=1}^{i-1} \frac{x_{i+1}}{\binom{i}{j}} \frac{\partial^2 \mathcal{B}_{n-1}^{\text{exp}}}{\partial x_j \partial x_{i-j}} + \sum_{i=2}^n x_i \frac{\partial \mathcal{B}_{n-1}^{\text{exp}}}{\partial x_{i-1}} \right]. \end{aligned} \quad (13.13)$$

Terms containing unavailable variables or negative-index Bell polynomials are zero.

Proof. Let $F(t) = e^{X(t)} = \sum_m \mathcal{B}_m^{\text{exp}} t^m / m!$. After using (13.10), the first double sum is

$$(n-1)! [t^{n-1}] (tX'(t)^2 F(t)).$$

The second is $(n-1)! [t^{n-1}] ((tX'' - X' + x_1)F)$, and the last is $(n-1)! [t^{n-1}] ((X' - x_1)F)$. Their sum is therefore

$$(n-1)! [t^{n-1}] t(X'' + X'^2)F = (n-1)! [t^{n-1}] tF'' = (n-1)\mathcal{B}_n^{\text{exp}}.$$

Division by $n-1$ proves the formula. □

Chapter 14

Specializations, inversions, determinants, and convolutions

14.1 Stirling, Bell, Lah, and Touchard specializations

Theorem 14.1. *For admissible n, k ,*

$$\mathcal{B}_{n,k}^{\text{exp}}(0!, 1!, \dots, (n-k)!) = \begin{bmatrix} n \\ k \end{bmatrix}, \quad (14.1)$$

$$\mathcal{B}_n^{\text{exp}}(0!, 1!, \dots, (n-1)!) = n!, \quad (14.2)$$

$$\mathcal{B}_{n,k}^{\text{exp}}(1, 1, \dots, 1) = \begin{Bmatrix} n \\ k \end{Bmatrix}, \quad (14.3)$$

$$\mathcal{B}_n^{\text{exp}}(1, 1, \dots, 1) = B_n^{\text{Bell}}, \quad (14.4)$$

$$\mathcal{B}_{n,k}^{\text{exp}}(1!, 2!, \dots, (n-k+1)!) = \binom{n-1}{k-1} \frac{n!}{k!} = \mathbb{L}_{n,k}^{\text{Lah}}, \quad (14.5)$$

$$\mathsf{T}_n^{\text{Tot}}(x) = \mathcal{B}_n^{\text{exp}}(x, x, \dots, x) = \sum_{k=0}^n \begin{Bmatrix} n \\ k \end{Bmatrix} x^k. \quad (14.6)$$

Proof. In (1.10), the choices $x_j = (j-1)!$, $x_j = 1$, and $x_j = j!$ make $X(t)$ respectively

$$-\log(1-t), \quad e^t - 1, \quad \frac{t}{1-t}.$$

The resulting coefficient formulae are exactly the column EGFs for unsigned first-kind Stirling numbers, second-kind Stirling numbers, and Lah numbers proved earlier. Summing over k yields (14.2) and (14.4). Finally $x_j = x$ gives $\exp(x(e^t - 1))$, the EGF of $\mathsf{T}_n^{\text{Tot}}(x)$. $\square$

Remark 14.2 (Formal crosswalk). In Lean the partial Bell polynomials `Fabius.partialBell` are defined by the pointing recurrence of Theorem 13.1 over any commutative semiring (module `PartialBellPolynomials`), and the column generating function of Theorem 12.4 is `Fabius.bellWeightSeries_pow` (module `BellGeneratingFunctions`), proved by induction on k from the first-order equation $Y' = (k+1)X'Y_k$ and the uniqueness of its solution. The specializations are then read off exactly as in the text: Equation (14.1) is `Fabius.partialBell_factorial_pred` (comparing with `Fabius.negLogOneSub_pow`), Equation (14.5) is `Fabius.partialBell_factorial` (comparing with the Lah column generating function $(t/(1-t))^k = \sum_n k! \mathbb{L}_{n,k}^{\text{Lah}} t^n / n!$, `Fabius.X_mul_mkOne_pow`, proved here by the same first-order method), Equation (14.3) is `Fabius.partialBell_one` (directly from the vertical recurrence of the second kind), and Equation (14.4) is `Fabius.bell_complete_one`. The complete polynomials are Mathlib-free `Bell.complete`,

related to the partial ones by `Fabius.bell_complete_eq_sum_partialBell`; their generating function $\exp X(t)$ is `Fabius.exp_subst_bellWeightSeries`, from the equation $Y' = X'Y$, $Y(0) = 1$ (`Fabius.eq_zero_of_derivative_eq_mul`). The factorial row sum Equation (14.2) is not formalized; the Touchard identity Equation (14.6) is `Fabius.bell_complete_const_eq_touchard_eval` (`BellHomogeneity`).

14.2 Homogeneity and a useful evaluation

Theorem 14.3 (Bigraded homogeneity). *For scalars α, β ,*

$$\mathcal{B}_{n,k}^{\exp}(\alpha\beta x_1, \alpha\beta^2 x_2, \dots) = \alpha^k \beta^n \mathcal{B}_{n,k}^{\exp}(x_1, x_2, \dots). \quad (14.7)$$

Moreover,

$$\mathcal{B}_{n,k}^{\exp}(1, 2, 3, \dots, n - k + 1) = \binom{n}{k} k^{n-k}. \quad (14.8)$$

Proof. Every monomial in (12.1) has $\sum j_i = k$ and $\sum i j_i = n$, proving the first formula. For the second, $x_j = j$ gives $X(t) = te^t$; hence

$$\frac{X(t)^k}{k!} = \frac{t^k e^{kt}}{k!}.$$

The coefficient of $t^n/n!$ is $n!k^{n-k}/(k!(n-k)!) = \binom{n}{k} k^{n-k}$. $\square$

Remark 14.4 (Formal crosswalk). Equation (14.7) is `Fabius.partialBell_bihomogeneous`, assembled from the degree homogeneity `Fabius.partialBell_mul_left` ($\mathcal{B}_{n,k}^{\exp}(\alpha x) = \alpha^k \mathcal{B}_{n,k}^{\exp}(x)$) and the weighted homogeneity `Fabius.partialBell_pow_mul` ($\mathcal{B}_{n,k}^{\exp}(\beta^j x_j) = \beta^n \mathcal{B}_{n,k}^{\exp}(x)$), module `BellHomogeneity`; both are proved by induction on the pointing recurrence rather than from the monomial expansion. As a consequence the Touchard specialization Equation (14.6) of Theorem 14.1 is `Fabius.bell_complete_const` and `Fabius.bell_complete_const_eq_touchard_eval`, and the Touchard generating function $\exp(x(e^t - 1))$ is `Fabius.exp_subst_smul_exp_sub_one`. Equation (14.8) is not formalized.

14.3 Addition and shifted-zero identities

Theorem 14.5 (Complete Bell addition formula).

$$\mathcal{B}_n^{\exp}(x_1 + y_1, \dots, x_n + y_n) = \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{n-i}^{\exp}(x) \mathcal{B}_i^{\exp}(y). \quad (14.9)$$

Proof. The complete EGF for the left side is $\exp(X(t) + Y(t)) = \exp X(t) \exp Y(t)$. Coefficient extraction gives the binomial convolution. $\square$

Theorem 14.6 (Insertion of q leading zeros). *For $q \geq 0$,*

$$\begin{aligned} \mathcal{B}_{n,k}^{\exp}\left(\frac{x_{q+1}}{\binom{q+1}{q}}, \frac{x_{q+2}}{\binom{q+2}{q}}, \dots\right) \\ = \frac{n!(q!)^k}{(n+qk)!} \mathcal{B}_{n+qk,k}^{\exp}(\underbrace{0, \dots, 0}_{q \text{ entries}}, x_{q+1}, x_{q+2}, \dots). \end{aligned} \quad (14.10)$$

Proof. Since

$$\frac{x_{q+i}}{\binom{q+i}{q}} \frac{t^i}{i!} = q! \frac{x_{q+i} t^i}{(q+i)!},$$

the k th power of the left component series is $(q!)^k t^{-qk}$ times the k th power of the component series on the right. Comparing the coefficient of t^n in (1.10) gives the factorial factor displayed. $\square$

Remark 14.7 (Formal crosswalk). `Fabius.partialBell_leadingZeros` (module `BellLeadingZeros`) is Equation (14.10) over any $\mathbb{Q}$ -algebra, for the zero-padded weights `Fabius.leadingZeros` (x_j replaced by 0 for $j \leq q$) and the rescaled weights `Fabius.qScaled` ($i \mapsto x_{q+i}/\binom{q+i}{q}$): the weight series of the former is $t^q/q!$ times that of the latter (`Fabius.bellWeightSeries_leadingZeros`), and comparing the coefficients of t^{n+qk} in the k th powers through Equation (1.10) (`Fabius.bellWeightSeries_pow`) gives the factorial factor.

14.4 Exponential–logarithmic inversion

Theorem 14.8 (Bell transform and its inverse). *Suppose*

$$y_n = \mathcal{B}_n^{\text{exp}}(x_1, \dots, x_n) = \sum_{k=1}^n \mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots). \quad (14.11)$$

Then

$$x_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}^{\text{exp}}(y_1, \dots, y_{n-k+1}). \quad (14.12)$$

Proof. Let $X(t) = \sum_{n \geq 1} x_n t^n / n!$ and $Y(t) = \sum_{n \geq 1} y_n t^n / n!$. Equation (1.12) says $1 + Y = e^X$, so

$$X = \log(1 + Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} Y^k.$$

Use (1.10) to extract the coefficient of $t^n / n!$. $\square$

Remark 14.9 (Formal crosswalk). Equation (14.12) is `Fabius.bell_transform_inverse` (module `BellComposition`), over any commutative $\mathbb{Q}$ -algebra, in the form $x_n = \sum_{k=1}^n (-1)^{k+1} (k-1)! \mathcal{B}_{n,k}^{\text{exp}}(y_1, y_2, \dots)$ with $y_n = \mathcal{B}_n^{\text{exp}}(x)$ (Lean’s `Bell.complete`). The proof is the one above: $X = \log(1 + Y)$ is `Fabius.log_subst_exp_sub_one` ($\log \circ (e^t - 1) = t$, from the derivative and constant term), $\log(1 + u) = \sum_{k \geq 1} (-1)^{k+1} (k-1)! u^k / k!$ is `Fabius.log_eq_egfA`, and the coefficient extraction is the exponential composition theorem `Fabius.egfA_subst_bellWeightSeries` of Theorem 20.3. The recursive, division-free inversion over every commutative ring is `Bell.complete_cumulant` and `Bell.cumulant_complete` (`BellPolynomialInversion`). The general form Equation (14.14) is not formalized.

More generally, if f and $g = f^{-1}$ are locally inverse and

$$y_n = \sum_{k=0}^n f^{(k)}(a) \mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots), \quad (14.13)$$

then

$$x_n = \sum_{k=0}^n g^{(k)}(f(a)) \mathcal{B}_{n,k}^{\text{exp}}(y_1, \dots). \quad (14.14)$$

Indeed, these are the derivative coefficients of $f \circ u$ and $g \circ f \circ u = u$; a complete proof is given with Faà di Bruno’s formula in Equation (19.3).

14.5 Two Hessenberg determinants

For $n \geq 1$, define $H_n^{\text{Bell}} = (h_{ij})$ and $K_n^{\text{Bell}} = (k_{ij})$ by

$$h_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \geq i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases} \quad k_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \geq i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Thus H_n^{Bell} begins with

$$\begin{bmatrix} x_1 & \binom{n-1}{1}x_2 & \binom{n-1}{2}x_3 & \cdots & x_n \\ -1 & x_1 & \binom{n-2}{1}x_2 & \cdots & x_{n-1} \\ 0 & -1 & x_1 & \cdots & x_{n-2} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_1 \end{bmatrix},$$

while K_n^{Bell} has first row $x_1, x_2/1!, \dots, x_n/(n-1)!$ and subdiagonal $-1, -2, \dots, -(n-1)$.

Theorem 14.10 (Bell determinants).

$$\mathcal{B}_n^{\text{exp}}(x_1, \dots, x_n) = \det H_n^{\text{Bell}} = \det K_n^{\text{Bell}}. \quad (14.15)$$

Proof. Expansion of the upper Hessenberg determinant $\det H_n^{\text{Bell}}$ along its first row gives

$$D_n = \sum_{j=1}^n \binom{n-1}{j-1} x_j D_{n-j}, \quad D_0 = 1;$$

the cofactor sign is cancelled by the product of the $j-1$ entries -1 on the subdiagonal. This is exactly (13.1), so $D_n = \mathcal{B}_n^{\text{exp}}$.

Let J reverse the coordinate order and let $D = \text{diag}(0!, 1!, \dots, (n-1)!)$. A direct entrywise calculation gives

$$K_n^{\text{Bell}} = D(J(H_n^{\text{Bell}})^T J)D^{-1}.$$

Indeed, $\frac{(i-1)!}{(j-1)!} \binom{j-1}{i-1} = 1/(j-i)!$ above the diagonal, and the subdiagonal becomes $-(i-1)$. Similarity, transposition, and reversal preserve the determinant, proving the second equality. $\square$

14.6 The diamond convolution

For sequences $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ define

$$(x \diamond y)_n = \sum_{j=1}^{n-1} \binom{n}{j} x_j y_{n-j}. \quad (14.16)$$

This operation is commutative and associative because it corresponds to multiplication of the EGFs $X(t)$ and $Y(t)$.

Theorem 14.11. If $x^{k \diamond}$ denotes the k -fold diamond power, then

$$\mathcal{B}_{n,k}^{\text{exp}}(x_1, \dots, x_{n-k+1}) = \frac{(x^{k \diamond})_n}{k!}. \quad (14.17)$$

Proof. The EGF of $x^{k \diamond}$ is $X(t)^k$. Compare with (1.10). $\square$

For example,

$$x \diamond x = (0, 2x_1^2, 6x_1x_2, 8x_1x_3 + 6x_2^2, \dots)$$

and $(x^{3 \diamond})_4 = 36x_1^2x_2$, whence $\mathcal{B}_{4,3}^{\text{exp}} = 6x_1^2x_2$.

Chapter 15

Applications of Bell polynomials

15.1 Symmetric functions and Newton identities

Let e_n be the elementary symmetric function and $p_n = \sum_i u_i^n$ the power sum in an alphabet $u_1, u_2, \dots$

Theorem 15.1.

$$e_n = \frac{1}{n!} \mathcal{B}_n^{\text{exp}}(p_1, -1!p_2, 2!p_3, \dots, (-1)^{n-1}(n-1)!p_n) \quad (15.1)$$

$$\begin{aligned} &= \frac{(-1)^n}{n!} \mathcal{B}_n^{\text{exp}}(-p_1, -1!p_2, -2!p_3, \dots, -(n-1)!p_n), \\ p_n &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}^{\text{exp}}(1!e_1, 2!e_2, \dots) \\ &= (-1)^n n \sum_{k=1}^n \frac{1}{k} \mathcal{B}_{n,k}^{\text{ord}}(-e_1, -e_2, \dots). \end{aligned} \quad (15.2)$$

Proof. The classical product identity

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod_i (1 + u_i t) = \exp \left(\sum_{r \geq 1} (-1)^{r-1} p_r \frac{t^r}{r} \right)$$

and (1.12) give (15.1). Taking logarithms, $\sum_{r \geq 1} (-1)^{r-1} p_r t^r / r = \log(1 + \sum_{j \geq 1} e_j t^j)$, and expanding the logarithm through ordinary Bell polynomials yields the second pair of forms. The exponential form follows from (12.4). $\square$

Corollary 15.2 (Determinant from traces). *For an $n \times n$ matrix A , set $s_k^{\text{tr}} = -(k-1)! \text{tr}(A^k)$. Then*

$$\det A = \frac{(-1)^n}{n!} \mathcal{B}_n^{\text{exp}}(s_1^{\text{tr}}, s_2^{\text{tr}}, \dots, s_n^{\text{tr}}). \quad (15.3)$$

Proof. Apply (15.1) to the eigenvalues of A , using $e_n = \det A$ and $p_k = \text{tr}(A^k)$. Both sides are polynomial in the matrix entries, so the argument remains valid for nondiagonalizable matrices. $\square$

15.2 Cycle indices

Let a_j mark cycles of length j in a permutation.

Theorem 15.3. *The cycle index of the symmetric group is*

$$Z(S_n) = \frac{1}{n!} \mathcal{B}_n^{\text{exp}}(0!a_1, 1!a_2, \dots, (n-1)!a_n). \quad (15.4)$$

Proof. A permutation is a set of cycles. A labelled cycle of size j has $(j-1)!$ linear realizations, so its component EGF is $(j-1)!a_j t^j/j! = a_j t^j/j$. The exponential formula gives $\exp(\sum_{j \geq 1} a_j t^j/j)$; comparison with (1.12) proves the identity after division by $n!$. $\square$

15.3 Moments and cumulants

Let $M(t) = \mathbb{E}e^{tX} = \sum_{n \geq 0} \mu'_n t^n/n!$ and $K(t) = \log M(t) = \sum_{n \geq 1} \kappa_n t^n/n!$.

Theorem 15.4 (Moment–cumulant relations).

$$\mu'_n = \mathcal{B}_n^{\text{exp}}(\kappa_1, \dots, \kappa_n) = \sum_{k=1}^n \mathcal{B}_{n,k}^{\text{exp}}(\kappa_1, \dots), \quad (15.5)$$

$$\kappa_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}^{\text{exp}}(\mu'_1, \dots, \mu'_{n-k+1}). \quad (15.6)$$

Proof. The identity $M = e^K$ is (1.12); its logarithmic inverse is Theorem 14.8. $\square$

15.4 Hermite polynomials

The probabilists' Hermite polynomials are defined by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}.$$

Comparison with (1.12) gives and proves

$$\text{He}_n(x) = \mathcal{B}_n^{\text{exp}}(x, -1, 0, \dots, 0). \quad (15.7)$$

In particular, the usual recurrence and differential equation follow by differentiating this generating function.

15.5 Polynomial sequences of binomial type

Let $h(t) = \sum_{j \geq 1} a_j t^j/j!$ with $a_1 \neq 0$, and define

$$p_n(x) = \mathcal{B}_n^{\text{exp}}(a_1 x, a_2 x, \dots, a_n x) = \sum_{k=0}^n \mathcal{B}_{n,k}^{\text{exp}}(a_1, \dots, a_{n-k+1}) x^k. \quad (15.8)$$

Theorem 15.5. *The polynomials p_n satisfy*

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xh(t)}, \quad (15.9)$$

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (15.10)$$

If $Q = h^{-1}(D_x)$ is the associated delta operator, then

$$Qp_n(x) = np_{n-1}(x). \quad (15.11)$$

Proof. Substitute xa_j in (1.12). The product $e^{(x+y)h} = e^{xh}e^{yh}$ gives the binomial identity. Finally $h^{-1}(D_x)e^{xh(t)} = h^{-1}(h(t))e^{xh(t)} = te^{xh(t)}$; coefficient comparison gives (15.11). $\square$

Remark 15.6 (Formal crosswalk). Module `BinomialType` defines `Fabius.binomialTypePoly` by Equation (15.8) as the complete Bell polynomial at the weights $a_j x$ (`Fabius.binomialTypePoly_eq_sum` is the expansion in partial Bell polynomials). Equation (15.9) is `Fabius.exp_subst_mul_bellWeightSeries_eq_egfA_binomialTypePoly`, the bivariate exponential formula of Theorem 1.6, and Equation (15.10) is `Fabius.binomialTypePoly_add`, which is the addition law $\mathcal{B}_n^{\text{exp}}(\kappa + \eta) = \sum_k \binom{n}{k} \mathcal{B}_k^{\text{exp}}(\kappa) \mathcal{B}_{n-k}^{\text{exp}}(\eta)$ (`Bell.complete_add`) at $\kappa = xa$, $\eta = ya$, valid over any commutative ring and without the hypothesis $a_1 \neq 0$. The delta operator Equation (15.11) is not formalized.

Editorial normalization

In some sources the symbol $B_n(x)$ denotes the Touchard polynomial, in others a complete Bell polynomial, a Bernoulli polynomial, or a binary-tree sequence. The notations T_n^{Tot} , $\mathcal{B}_n^{\text{exp}}$, $\beta_n(x)$, and C_n^{Cat} are kept separate here; this is essential when the objects occur in the same inversion formula.

Part V

Eulerian Numbers and Descent Calculus

Chapter 16

Ordinary Eulerian numbers

16.1 Descents, symmetry, and the basic recurrence

For a permutation $\pi = \pi_1 \cdots \pi_n$, a descent is an index $i \in \{1, \dots, n-1\}$ with $\pi_i > \pi_{i+1}$.

Definition 16.1. The Eulerian number $\langle n \rangle_k = A(n, k)$ counts permutations of $[n]$ with exactly k descents. Put

$$\mathbb{E}_0^{\text{Eul}, A}(t) = 1, \quad \mathbb{E}_n^{\text{Eul}, A}(t) = \sum_{k=0}^{n-1} \langle n \rangle_k t^k \quad (n \geq 1).$$

Outside $0 \leq k < n$ the number is zero.

Theorem 16.2 (Eulerian recurrence).

$$\langle n \rangle_k = (k+1) \langle n-1 \rangle_k + (n-k) \langle n-1 \rangle_{k-1}. \quad (16.1)$$

Equivalently,

$$\mathbb{E}_n^{\text{Eul}, A}(t) = (1 + (n-1)t) \mathbb{E}_{n-1}^{\text{Eul}, A}(t) + t(1-t) (\mathbb{E}_{n-1}^{\text{Eul}, A})'(t). \quad (16.2)$$

Proof. Insert n into a permutation of $[n-1]$. If the old permutation has k descents, insertion after one of those k descents or at the end preserves the number of descents, giving $k+1$ choices. If it has $k-1$ descents, insertion in any of the remaining $n-k$ slots creates one new descent. This proves (16.1). Multiply by t^k , sum over k , and collect the terms containing k to obtain (16.2). $\square$

Complementation $\pi_i \mapsto n+1 - \pi_i$ interchanges descents and ascents. Therefore

$$\langle n \rangle_k = \langle n \rangle_{n-1-k}, \quad \mathbb{E}_n^{\text{Eul}, A}(t) = t^{n-1} \mathbb{E}_n^{\text{Eul}, A}(t^{-1}). \quad (16.3)$$

In particular $\langle n \rangle_0 = \langle n \rangle_{n-1} = 1$ and $\sum_k \langle n \rangle_k = n!$. The next diagonal is

$$\langle n \rangle_1 = 2^n - (n+1), \quad (16.4)$$

which follows either from the recurrence or from the explicit formula below.

16.2 Exponential generating function and explicit formula

Theorem 16.3 (Eulerian EGF).

$$\sum_{n \geq 0} \mathbb{E}_n^{\text{Eul}, A}(t) \frac{x^n}{n!} = \frac{t-1}{t - e^{(t-1)x}} = \frac{(1-t)e^{(1-t)x}}{1 - te^{(1-t)x}}. \quad (16.5)$$

Proof. Let $F(x, t)$ denote the left side. Multiplying (16.2) by $x^{n-1}/(n-1)!$ and summing gives the first-order equation

$$(1 - tx)F_x = t(1 - t)F_t + F, \quad F(0, t) = 1.$$

Direct substitution verifies that the displayed rational-exponential function solves this equation and initial condition. Formal uniqueness follows by recursively comparing coefficients of x . $\square$

Remark 16.4 (Formal crosswalk). `Fabius.egfA_eulerianPolynomial_mul` (module `EulerianEGF`) is Equation (16.5) in multiplicative form, as an identity in the ring $(\mathbb{Q}[t])[[x]]$ of formal power series in x with polynomial coefficients: $(\sum_{n \geq 0} E_n^{\text{Eul}, A}(t) \frac{x^n}{n!}) (t - e^{(t-1)x}) = t - 1$, where $e^{(t-1)x}$ is the exponential series `Fabius.expSeries` at $t - 1$. Comparing the coefficients of $x^n/n!$, this identity is exactly the binomial recurrence Equation (16.8) (for $n \geq 1$) together with $E_0^{\text{Eul}, A} = 1$, so the formal proof goes through Theorem 16.7.

Expanding the denominator geometrically gives the central power-series identity.

Theorem 16.5. For $n \geq 0$,

$$\sum_{m=0}^{\infty} (m+1)^n t^m = \frac{E_n^{\text{Eul}, A}(t)}{(1-t)^{n+1}}. \quad (16.6)$$

Consequently,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n. \quad (16.7)$$

Proof. From (16.5),

$$F(x, t) = (1-t) \sum_{m \geq 0} t^m e^{(m+1)(1-t)x}.$$

Extracting $x^n/n!$ proves (16.6). Multiply by $(1-t)^{n+1}$ and extract t^k to obtain (16.7). Formula (16.4) is its case $k = 1$. $\square$

Remark 16.6 (Formal crosswalk). Equation (16.6) is `Fabius.one_sub_X_pow_mul_succPowSeries` (module `EulerianGeneratingFunctions`), stated in $R[[t]]$ over any commutative ring R as $(1-t)^{n+1} \sum_{m \geq 0} (m+1)^n t^m = \sum_k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle t^k$. The formal proof reads Worpitzky's identity Theorem 16.9 column by column: $(m+1)^n = \sum_k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{m+1+k}{n}$ and $\sum_m \binom{m+d}{d} t^m = (1-t)^{-d-1}$ (Mathlib's `PowerSeries.mk_one_pow_eq_mk_choose_add`) give the identity with t^{n-1-k} in place of t^k , and the symmetry Equation (16.3), formalized as `Fabius.eulerianNumber_symm` by induction on the recurrence, reverses the exponents. The explicit formula Equation (16.7) is its k -th coefficient, `Fabius.eulerianNumber_eq_sum_int`, and Equation (16.4) is `Fabius.eulerianNumber_one_right`.

Theorem 16.7 (A second polynomial recurrence). For $n > 1$,

$$E_n^{\text{Eul}, A}(t) = \sum_{k=0}^{n-1} \binom{n}{k} E_k^{\text{Eul}, A}(t) (t-1)^{n-1-k}. \quad (16.8)$$

Proof. The EGF of the right sides, summed over $n \geq 1$, is

$$F(x, t) \frac{e^{(t-1)x} - 1}{t-1}.$$

Using (16.5), this equals $F(x, t) - 1$, which is the EGF of the left sides. $\square$

Remark 16.8 (Formal crosswalk). `Fabius.eulerianPolynomial_binomial_recurrence` (module `EulerianEGF`) is Equation (16.8) in $R[t]$ for $n \geq 1$ over any commutative ring R (the case $n = 1$ reads $E_1^{\text{Eul},A} = E_0^{\text{Eul},A}$), transported from the power-series form `Fabius.eulerian_binomial_recurrence_series`. The formal proof uses the rational generating function Equation (16.6): writing $E_k^{\text{Eul},A}(t) = (1-t)^{k+1} \sum_m (m+1)^k t^m$, the right side becomes $(1-t)^n \sum_m t^m \sum_{k < n} \binom{n}{k} (-1)^{n-1-k} (m+1)^k = (1-t)^n \sum_m ((m+1)^{n-m} t^m)$ by the binomial theorem (`Fabius.sum_choose_neg_one_pow_succPowSeries`), and $\sum_m m^n t^m = t \sum_m (m+1)^n t^m$ (`Fabius.X_mul_succPowSeries`) finishes.

16.3 Worpitzky's identity and sums of powers

Theorem 16.9 (Worpitzky). *For every indeterminate x and $n \geq 1$,*

$$x^n = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle \binom{x+k}{n}. \quad (16.9)$$

Proof. The coefficient of t^{x-1} in $t^{-k}(1-t)^{-n-1}$ is $\binom{x+k}{n}$ for positive integer x . Equivalently, taking the coefficient of t^{x-1} in (16.6) after multiplying by the numerator gives (16.9). Since both sides are polynomials of degree n in x and agree for all positive integers, they agree identically.

A direct combinatorial proof standardizes a word in $[x]^n$: its stable sorting permutation records the relative order, and its k descents specify k mandatory separations among the n sorted positions. Stars and bars gives $\binom{x+k}{n}$ words for each permutation with k descents. $\square$

Remark 16.10 (Formal crosswalk). In Lean the Eulerian numbers `Fabius.eulerianNumber` are defined by the recurrence of Theorem 16.2 (`Fabius.eulerianNumber_succ_left`); the descent count is not formalized. Worpitzky's identity is `Fabius.worpitzky_nat` for natural x and `Fabius.worpitzky_polynomial` in $\mathbb{Q}[x]$, where $\binom{x+k}{n}$ is `Fabius.binomialPoly` (module `EulerianNumbers`); both are stated with the sum over $0 \leq k \leq n$, the extra term $k = n$ vanishing (`Fabius.eulerianNumber_eq_zero_of_le`). The Lean proof is an induction on n from the one-step relation $x \binom{x+k}{n} = (n-k) \binom{x+k+1}{n+1} + (k+1) \binom{x+k}{n+1}$ (`Fabius.mul_choose_add_eq`), and the polynomial identity follows from the natural-number identity because a polynomial over $\mathbb{Q}$ is determined by its values at the natural numbers. The row sum $\sum_k \left\langle n \atop k \right\rangle = n!$ is `Fabius.sum_eulerianNumber_eq_factorial`; the symmetry Equation (16.3) is `Fabius.eulerianNumber_symm` (`EulerianGeneratingFunctions`). The power sums of Theorem 16.11 are `Fabius.sum_range_pow_succ_eq_sum_eulerianNumber`, in the form $\sum_{r=0}^m r^{n+1} = \sum_k \left\langle n+1 \atop k \right\rangle \binom{m+k+1}{n+2}$.

Corollary 16.11 (Power sums). *For $m, n \geq 1$,*

$$\sum_{r=1}^m r^n = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle \binom{m+k+1}{n+1}. \quad (16.10)$$

Proof. Sum (16.9) over $x = 1, \dots, m$ and use the hockey-stick identity $\sum_{x=1}^m \binom{x+k}{n} = \binom{m+k+1}{n+1}$; the omitted lower boundary terms vanish for $0 \leq k < n$. $\square$

16.4 Negative polylogarithms

For $|z| < 1$, $\text{Li}_{-n}(z) = \sum_{m \geq 1} m^n z^m$. Shifting m in (16.6) gives

$$\text{Li}_{-n}(z) = \frac{z E_n^{\text{Eul},A}(z)}{(1-z)^{n+1}} = \frac{1}{(1-z)^{n+1}} \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle z^{n-k}, \quad (16.11)$$

where the second equality uses symmetry (16.3). Since the right side is rational, it also supplies the meromorphic continuation of the negative polylogarithm.

16.5 Eulerian and Stirling bases

Theorem 16.12.

$$\mathbb{E}_n^{\text{Eul},A}(t) = \sum_{k=0}^n \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle k! (t-1)^{n-k}. \quad (16.12)$$

Proof. Use $m^n = \sum_k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle (m)^k$ in $\sum_{m \geq 0} m^n t^m$. Since $\sum_{m \geq 0} (m)^k t^m = k! t^k / (1-t)^{k+1}$, multiplication by $(1-t)^{n+1}/t$ yields $\mathbb{E}_n^{\text{Eul},A}(t) = \sum_k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle k! t^{k-1} (1-t)^{n-k}$. Replacing t by $1/t$ and using (16.3), or simply expanding around $t = 1$, gives the equivalent displayed form. $\square$

Remark 16.13 (Formal crosswalk). `Fabius.sum_eulerianNumber_mul_X_pow_eq_sum_stirlingSecond` (module `EulerianStirling`) is Equation (16.12) in $R[[t]]$ over any commutative ring R , with $\mathbb{E}_n^{\text{Eul},A}(t)$ written as $\sum_k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle t^k$. The formal proof expands $(m+1)^n$ in the rising-factorial basis, $(m+1)^n = \sum_k (-1)^{n-k} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle k! \binom{m+k}{k}$ (`Fabius.succ_pow_eq_sum_stirlingSecond_mul_choose`), uses $\sum_m \binom{m+k}{k} t^m = (1-t)^{-k-1}$ to get $\sum_m (m+1)^n t^m = \sum_k (-1)^{n-k} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle k! (1-t)^{-k-1}$ (`Fabius.succPowSeries_eq_sum_stirlingSecond`), and multiplies by $(1-t)^{n+1}$ through Equation (16.6).

16.6 An Irwin–Hall integral and hypersimplex volumes

Theorem 16.14. For every function $f : \mathbb{R} \rightarrow \mathbb{C}$ for which the following finite sum is defined,

$$\sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle f(k) = n! \int_{[0,1]^n} f(\lfloor x_1 + \cdots + x_n \rfloor) dx_1 \cdots dx_n. \quad (16.13)$$

In particular, the slab

$$\Delta_{n,k} = \{x \in [0,1]^n : k \leq x_1 + \cdots + x_n < k+1\}$$

has volume $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle / n!$.

Proof. The volume of $\{x \in [0,1]^n : \sum x_i < y\}$ is obtained from the simplex $\{x_i \geq 0 : \sum x_i < y\}$ by inclusion–exclusion over coordinates exceeding 1:

$$V_n(y) = \frac{1}{n!} \sum_{j=0}^{\lfloor y \rfloor} (-1)^j \binom{n}{j} (y-j)^n.$$

Thus the volume of the k th slab is $V_n(k+1) - V_n(k)$. Combining adjacent terms by Pascal’s identity gives

$$\frac{1}{n!} \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n = \frac{1}{n!} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$$

by (16.7). Summing $f(k)$ times these volumes proves the integral identity. $\square$

16.7 Alternating sums and Bernoulli numbers

Theorem 16.15. For $n > 0$,

$$\sum_{k=0}^{n-1} (-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1}. \quad (16.14)$$

For $n > 1$,

$$\sum_{k=0}^{n-1} \frac{(-1)^k \langle n \rangle_k}{\binom{n-1}{k}} = 0, \quad (16.15)$$

$$\sum_{k=0}^{n-1} \frac{(-1)^k \langle n \rangle_k}{\binom{n}{k}} = (n+1)\beta_n. \quad (16.16)$$

Proof. At $t = -1$, (16.5) becomes $2/(1 + e^{-2x}) = 1 + \tanh x$. The Bernoulli generating function gives

$$\tanh x = \sum_{n \geq 1} 2^{n+1}(2^{n+1} - 1) \frac{\beta_{n+1}}{n+1} \frac{x^n}{n!},$$

which proves (16.14).

For the remaining identities use

$$\binom{n-1}{k}^{-1} = n \int_0^1 u^k (1-u)^{n-k-1} du, \quad \binom{n}{k}^{-1} = (n+1) \int_0^1 u^k (1-u)^{n-k} du.$$

Substitute $t = -u/(1-u)$ in (16.12); then

$$(1-u)^n \mathbf{E}_n^{\text{Eul}, A} \left(-\frac{u}{1-u} \right) = \sum_{j=0}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j! (1-u)^j. \quad (16.17)$$

For the first integral divide by $1-u$ and integrate. The result is $n \sum_{j=1}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} (j-1)! = 0$ for $n > 1$, because it is the coefficient of $x^n/n!$ in $\log(1 + (e^x - 1)) = x$. For the second integral one obtains $(n+1) \sum_j (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j!/(j+1)$. The standard Stirling formula $\beta_n = \sum_j (-1)^j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} / (j+1)$ proves the claim; for odd $n > 1$ both sides vanish, and for even n the signs coincide. $\square$

Remark 16.16 (Formal crosswalk). Equation (16.14) is `Fabius.sum_neg_one_pow_mul_eulerianNumber` (module `EulerianAlternating`), for $n \geq 1$. Evaluating the Eulerian generating function Theorem 16.3 at $t = -1$ gives $E(x)(1 + e^{-2x}) = 2$ for $E(x) = \sum_n \mathbf{E}_n^{\text{Eul}, A} (-1)x^n/n!$ (`Fabius.egfA_eulerianPolynomial_eval_neg_one_mul`); with $\mathcal{B}(t) = t/(e^t - 1)$ one has $\mathcal{B}(4x)(e^{2x} + 1) = 2\mathcal{B}(2x)$ (`Fabius.rescale_four_bernoulli_mul`), hence $x E(x) = 2x - \mathcal{B}(2x) + \mathcal{B}(4x)$ (`Fabius.X_mul_egfA_eulerianPolynomial_eval_neg_one`), which is the displayed expansion of $\tanh x$; comparing the coefficients of x^{n+1} gives the formula. The two reciprocal-binomial identities are not formalized.

Chapter 17

Geometric interpretations

17.1 The cube and its Ehrhart numerator

For a lattice polytope P of dimension d , its Ehrhart series has the form $\sum_{m \geq 0} |mP \cap \mathbb{Z}^d| t^m = h^*(t)/(1-t)^{d+1}$. For the unit cube $C = [0, 1]^n$ one has $|mC \cap \mathbb{Z}^n| = (m+1)^n$; hence (16.6) proves:

Theorem 17.1. *The h^* -polynomial of the standard n -cube is $E_n^{\text{Eul}, A}(t)$, so its h^* -vector is the n th Eulerian row.*

Proof. By the defining Ehrhart identity and the lattice-point count above,

$$\frac{h_C^*(t)}{(1-t)^{n+1}} = \sum_{m \geq 0} (m+1)^n t^m.$$

Equation (16.6) identifies the right side with $E_n^{\text{Eul}, A}(t)/(1-t)^{n+1}$. Multiplication by $(1-t)^{n+1}$ therefore gives $h_C^*(t) = E_n^{\text{Eul}, A}(t)$ coefficient by coefficient. $\square$

The regions $\Delta_{n,k}$ in Theorem 16.14 are half-open hypersimplices. Their normalized volumes are $n! \text{vol } \Delta_{n,k} = \langle \frac{n}{k} \rangle$.

17.2 The permutohedron

The type- A permutohedron is the convex hull of the permutations of $(1, 2, \dots, n)$; its dimension is $n-1$. Faces of its braid fan are indexed by ordered set partitions. There are $k! \left\{ \frac{n}{k} \right\}$ ordered partitions into k blocks.

Theorem 17.2 (The permutohedral h -polynomial). *Let P_n be the $(n-1)$ -dimensional type- A permutohedron. Then*

$$h_{P_n}(t) = E_n^{\text{Eul}, A}(t).$$

Equivalently, the boundary complex of the polar simplicial polytope P_n° has this h -polynomial.

Proof. For any simple polytope, its h -polynomial is the face transform

$$h_P(t) = \sum_{\emptyset \neq F \leq P} (t-1)^{\dim F}.$$

A nonempty face of P_n is indexed by an ordered partition of $[n]$. A partition into k nonempty blocks determines a face of dimension $n-k$, and there are $k! \left\{ \frac{n}{k} \right\}$ such partitions. Therefore

$$h_{P_n}(t) = \sum_{k=1}^n k! \left\{ \frac{n}{k} \right\} (t-1)^{n-k} = E_n^{\text{Eul}, A}(t)$$

by Equation (16.12). Polarity identifies the face lattices in reverse order and preserves the h -polynomial, giving the equivalent simplicial formulation. $\square$

Chapter 18

Type-B and second-order Eulerian numbers

18.1 Signed permutations and the corrected type-B indexing

A signed permutation is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $[n]$. Put $\pi_0 = 0$ and call $i \in \{0, \dots, n-1\}$ a type- B descent when $\pi_i > \pi_{i+1}$. Let $E_{n,k}^{\text{Eul},B}$ count signed permutations with k such descents, and set $E_n^{\text{Eul},B}(t) = \sum_{k=0}^n E_{n,k}^{\text{Eul},B} t^k$.

Editorial normalization

The archived display reverses the two entries in the descent inequality and pairs that convention with an incompatible explicit formula. The definition above is the standard Coxeter descent convention. It gives the archived table 1, 6, 1 for $n = 2$ and the midpoint-Eulerian generating function below.

Theorem 18.1 (Type- B recurrence and Worpitzky series).

$$E_{n,k}^{\text{Eul},B} = (2k+1)E_{n-1,k}^{\text{Eul},B} + (2n-2k+1)E_{n-1,k-1}^{\text{Eul},B}, \quad (18.1)$$

$$\sum_{m \geq 0} (2m+1)^n t^m = \frac{E_n^{\text{Eul},B}(t)}{(1-t)^{n+1}}, \quad (18.2)$$

$$E_{n,k}^{\text{Eul},B} = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (2(k-j)+1)^n. \quad (18.3)$$

The first rows are

$$1; \quad 1, 1; \quad 1, 6, 1; \quad 1, 23, 23, 1; \quad 1, 76, 230, 76, 1.$$

Proof. Insert either n or $-n$ into a signed permutation of size $n-1$. Marking the slots according to whether insertion preserves or creates a descent gives $2k+1$ preserving slots for an object with k descents and $2n-2k+1$ creating slots for one with $k-1$ descents; this proves the recurrence.

A type- B standardization partitions the $(2m+1)^n$ words in the ordered alphabet $-m < \cdots < -1 < 0 < 1 < \cdots < m$ by their signed sorting permutation. If that permutation has k type- B descents, stars and bars gives $\binom{m+n-k}{n}$ compatible weakly increasing absolute levels. Hence

$$(2m+1)^n = \sum_{k=0}^n E_{n,k}^{\text{Eul},B} \binom{m+n-k}{n}.$$

Multiply by t^m , sum in m , and use $\sum_{m \geq 0} \binom{m+n-k}{n} t^m = t^k / (1-t)^{n+1}$ to obtain (18.2). Multiplication by $(1-t)^{n+1}$ and coefficient extraction gives (18.3). $\square$

Remark 18.2 (Formal crosswalk). Module `TypeBEulerian` defines `Fabius.typeBEulerian` by the recurrence Equation (18.1); the signed-permutation count is not formalized. The type- B Worpitzky identity $(2m+1)^n = \sum_k E_{n,k}^{\text{Eul},B} \binom{m+n-k}{n}$ is `Fabius.typeB_worpitzky`, proved by induction on n from the one-step relation $(2m+1)^{\binom{m+n-k}{n}} = (2k+1)^{\binom{m+n+1-k}{n+1}} + (2n-2k+1)^{\binom{m+n-k}{n+1}}$ (`Fabius.two_mul_add_one_mul_choose`); Equation (18.2) is `Fabius.one_sub_X_pow_mul_oddPowSeries`, in $R[[t]]$ over any commutative ring, exactly as in the text; and Equation (18.3) is its k -th coefficient, `Fabius.typeBEulerian_eq_sum_int`.

Theorem 18.3 (The type- B permutohedral h -polynomial). *The h -polynomial of the simple type- B permutohedron is $E_n^{\text{Eul},B}(t)$. Equivalently, this is the h -polynomial of the boundary complex of its simplicial polar.*

Proof. The normal fan of the type- B permutohedron is the type- B Coxeter fan, so the boundary complex of the polar is the type- B Coxeter complex. Order its chambers by any linear extension of right weak order. This is a shelling: across the wall indexed by a simple reflection s_i , the adjacent chamber ws_i occurs earlier exactly when the Coxeter length drops, $\ell(ws_i) < \ell(w)$. In signed-permutation notation these indices are precisely the type- B descents of w . Hence the restriction face of chamber w has one vertex for each descent, and the shelling formula for the h -polynomial gives

$$h(t) = \sum_{w \in B_n} t^{|\text{Des}_B(w)|} = \sum_{k=0}^n E_{n,k}^{\text{Eul},B} t^k = E_n^{\text{Eul},B}(t).$$

The statement for the simple permutohedron follows by polarity. $\square$

18.2 Stirling permutations and second-order Eulerian numbers

A *Stirling permutation of order n* is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ such that all entries between the two copies of i are larger than i . Let $\langle\langle n \rangle\rangle_k$ count those with k ordinary descents and put

$$E_n^{\text{Eul},2}(t) = \sum_{k=0}^{n-1} \langle\langle n \rangle\rangle_k t^k, \quad E_0^{\text{Eul},2}(t) = 1.$$

Theorem 18.4 (Second-order recurrence).

$$\langle\langle n \rangle\rangle_k = (2n - k - 1) \langle\langle n-1 \rangle\rangle_{k-1} + (k+1) \langle\langle n-1 \rangle\rangle_k, \quad (18.4)$$

$$E_{n+1}^{\text{Eul},2}(t) = (2nt + 1) E_n^{\text{Eul},2}(t) - t(t-1) (E_n^{\text{Eul},2})'(t), \quad (18.5)$$

$$E_n^{\text{Eul},2}(1) = (2n-1)!!. \quad (18.6)$$

Proof. The two copies of the largest label n must be adjacent. Insert the pair nn into one of the $2n-1$ gaps of a Stirling permutation of order $n-1$. A descent gap or the terminal gap preserves the number of descents, giving $k+1$ choices from an object with k descents. Each of the remaining $2n-k-1$ gaps creates one descent from an object with $k-1$ descents. This is the first recurrence; summing against t^k gives the second. At each stage there are $2n-1$ insertion gaps, so the total count is $(2n-1)!!$. $\square$

Remark 18.5 (Formal crosswalk). Module `SecondOrderEulerian` defines `Fabius.secondEulerian` by the recurrence Equation (18.4) (in the form $\langle\langle n+1 \rangle\rangle_k = (2n-k) \langle\langle n \rangle\rangle_k + (k+2) \langle\langle n \rangle\rangle_{k+1}$); the Stirling-permutation count is not formalized. The polynomial recurrence Equation (18.5) is `Fabius.secondEulerianSeries_succ`, an identity of formal power series in t over any commutative ring, proved by comparing coefficients; the row sum Equation (18.6) is `Fabius.sum_secondEulerian_eq_doubleFactorial`, by induction from the recurrence.

The differential recurrence has two useful equivalent forms:

$$(t-1)^{-2n-2} E_{n+1}^{\text{Eul},2}(t) = \left(t(1-t)^{-2n-1} E_n^{\text{Eul},2}(t) \right)', \quad (18.7)$$

$$u_{n+1}(t) = \left(\frac{t}{1-t} u_n(t) \right)', \quad u_n(t) = (t-1)^{-2n} E_n^{\text{Eul},2}(t), \quad u_0 = 1. \quad (18.8)$$

They follow by the product rule from (18.5).

Theorem 18.6 (Diagonal Stirling generating function). *For $n \geq 1$,*

$$\sum_{m=0}^{\infty} \left\{ \begin{matrix} n+m \\ m \end{matrix} \right\} t^m = \frac{t E_n^{\text{Eul},2}(t)}{(1-t)^{2n+1}}. \quad (18.9)$$

Proof. Let $F_n(t)$ denote the left side. The second-kind recurrence gives $(1-t)F_{n+1} = tF'_n$, with $F_0 = (1-t)^{-1}$. Hence $F_1 = t/(1-t)^3$. If the displayed formula holds for n , differentiating $tE_n^{\text{Eul},2}/(1-t)^{2n+1}$ and using (18.5) gives the formula for $n+1$. $\square$

Remark 18.7 (Formal crosswalk). `Fabius.one_sub_X_pow_mul_diagStirlingSeries` (module `SecondOrderEulerian`) is Equation (18.9) in the form $(1-t)^{2n+1} \sum_m \left\{ \begin{matrix} n+m \\ m \end{matrix} \right\} t^m = tE_n^{\text{Eul},2}(t)$ in $R[[t]]$ over any commutative ring, for $n \geq 1$: the base case is $F_1 = t/(1-t)^3$ (`Fabius.diagStirlingSeries_one`), the recurrence $(1-t)F_{n+1} = tF'_n$ is `Fabius.one_sub_X_mul_diagStirlingSeries_succ`, and the induction step differentiates the hypothesis and uses Equation (18.5).

18.3 Second-order Eulerian interpolation of first-kind diagonals

The same second-order Eulerian row that controls (18.9) also gives a binomial-basis expansion for the near-diagonals of the unsigned first-kind triangle.

Theorem 18.8 (Second-order Eulerian diagonal formula). *For $p, m \geq 0$,*

$$\left[\begin{matrix} m \\ m-p \end{matrix} \right] = \sum_{j=0}^p \left\langle \left\langle \begin{matrix} p \\ j \end{matrix} \right\rangle \right\rangle \binom{m+j}{2p}, \quad (18.10)$$

where an out-of-range first-kind number is zero. Equivalently, the polynomial continuation in a variable x satisfies

$$\left[\begin{matrix} x \\ x-p \end{matrix} \right] = \sum_{j=0}^p \left\langle \left\langle \begin{matrix} p \\ j \end{matrix} \right\rangle \right\rangle \binom{x+j}{2p}.$$

The first instances are

$$\begin{aligned} \left[\begin{matrix} x \\ x-1 \end{matrix} \right] &= \binom{x}{2}, \\ \left[\begin{matrix} x \\ x-2 \end{matrix} \right] &= \binom{x}{4} + 2 \binom{x+1}{4}, \\ \left[\begin{matrix} x \\ x-3 \end{matrix} \right] &= \binom{x}{6} + 8 \binom{x+1}{6} + 6 \binom{x+2}{6}. \end{aligned}$$

Proof. For fixed p define

$$D_p(t) = \sum_{m \geq 0} \left[\begin{matrix} m \\ m-p \end{matrix} \right] t^m.$$

Writing $a_{m,p} = \begin{bmatrix} m \\ m-p \end{bmatrix}$, the first-kind recurrence becomes $a_{m,p} = a_{m-1,p} + (m-1)a_{m-1,p-1}$. Therefore

$$(1-t)D_p(t) = t^2 D'_{p-1}(t), \quad D_0(t) = \frac{1}{1-t}. \quad (18.11)$$

Let

$$Q_p(t) = t^{2p} \mathbb{E}_p^{\text{Eul},2}(t^{-1}).$$

The polynomial recurrence (18.5) is equivalent, after the substitution $t \mapsto t^{-1}$, to

$$Q_p(t) = t^2((1-t)Q'_{p-1}(t) + (2p-1)Q_{p-1}(t)).$$

It follows inductively from (18.11) that

$$D_p(t) = \frac{Q_p(t)}{(1-t)^{2p+1}} = \frac{t^{2p} \mathbb{E}_p^{\text{Eul},2}(t^{-1})}{(1-t)^{2p+1}}. \quad (18.12)$$

Since $\mathbb{E}_p^{\text{Eul},2}(u) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle_j u^j$, coefficient extraction gives

$$[t^m] D_p(t) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle_j [t^{m-2p+j}] (1-t)^{-2p-1} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle_j \binom{m+j}{2p},$$

which is (18.10). Both sides are polynomials in m of degree at most $2p$, so the same identity gives the stated polynomial continuation. $\square$

Part VI

Differentiation, Composition, and Reversion

Chapter 19

Faà di Bruno's formula

19.1 The set-partition form

For a block $B \subseteq [n]$, write $|B|$ for its size.

Theorem 19.1 (Faà di Bruno, partition form). *For $n \geq 0$ and sufficiently differentiable f, g ,*

$$D^n(f \circ g)(x) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(g(x)) \prod_{B \in \pi} g^{(|B|)}(x). \quad (19.1)$$

For $n = 0$, the empty partition contributes $f(g(x))$.

Proof. Induct on n . Differentiate a term indexed by a partition π of $[n]$. If the derivative hits $f^{(|\pi|)}(g)$, the chain rule contributes g' and creates the new singleton block $\{n+1\}$. If it hits the factor associated with a block B , it changes $g^{(|B|)}$ to $g^{(|B|+1)}$, which inserts $n+1$ into B . Every partition of $[n+1]$ arises uniquely in exactly one of these ways: remove $n+1$, either deleting its singleton block or removing it from its non-singleton block. Hence all terms occur once. $\square$

19.2 Multiplicity and Bell-polynomial forms

Grouping partitions by the number m_j of blocks of size j gives the classical formula.

Theorem 19.2 (Faà di Bruno, multiplicity form).

$$D^n f(g(x)) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{m_1! m_2! \dots m_n!} \times f^{(m_1 + \dots + m_n)}(g(x)) \prod_{j=1}^n \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j}. \quad (19.2)$$

Equivalently,

$$D^n f(g(x)) = \sum_{k=0}^n f^{(k)}(g(x)) \mathcal{B}_{n,k}^{\text{exp}}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (19.3)$$

Proof. For fixed multiplicities (m_j) , the number of set partitions of $[n]$ with m_j blocks of size j is $n! / \prod_j ((j!)^{m_j} m_j!)$. Substitution in (19.1) gives (19.2). Grouping further by $k = \sum m_j$ and invoking (12.1) gives (19.3). $\square$

For $n = 4$ this reads

$$(f \circ g)^{(4)} = f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2 g'' + 3f''(g)(g'')^2 + 4f''(g)g'g^{(3)} + f'(g)g^{(4)}. \quad (19.4)$$

The coefficients 1, 6, 3, 4, 1 count partitions of types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, and 4 respectively. This also proves the expanded example in the archive without four separate differentiations.

19.3 The exponential special case

Because every derivative of e^u equals e^u , summing the partial Bell polynomials in (19.3) yields

$$D^n e^{g(x)} = e^{g(x)} \mathcal{B}_n^{\text{exp}}(g'(x), g''(x), \dots, g^{(n)}(x)). \quad (19.5)$$

When g is a cumulant generating function, this is precisely the moment–cumulant relation of Theorem 15.4.

19.4 Mixed partial derivatives

For $B \subseteq [n]$, write $\partial_B = \prod_{j \in B} \partial / \partial x_j$.

Theorem 19.3 (Multivariate Faà di Bruno for distinct variables). *If $y = g(x_1, \dots, x_n)$, then*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (19.6)$$

Proof. The proof of Theorem 19.1 applies verbatim, except that the new derivative is $\partial / \partial x_{n+1}$. It either creates a singleton block or joins the new index to one existing block. $\square$

For three variables, (19.6) becomes

$$\begin{aligned} \partial_1 \partial_2 \partial_3 f(y) &= f'(y) \partial_{123} y \\ &\quad + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned}$$

For repeated variables or vector-valued inner maps, one labels derivative slots first and then contracts tensor indices; the same partition lattice remains the combinatorial skeleton.

Chapter 20

Composition of ordinary and exponential series

20.1 Ordinary formal power series

Let

$$F(u) = \sum_{k \geq 0} a_k u^k, \quad G(x) = \sum_{j \geq 1} b_j x^j, \quad F(G(x)) = \sum_{n \geq 0} c_n x^n.$$

Theorem 20.1 (Ordinary composition). *For $n \geq 1$,*

$$c_n = \sum_{k=1}^n a_k \sum_{\substack{i_1 + \dots + i_k = n \\ i_r \geq 1}} b_{i_1} \dots b_{i_k}, \quad (20.1)$$

$$= \sum_{k=1}^n a_k \sum_{\substack{\pi_1 + \dots + \pi_n = k \\ \pi_1 + 2\pi_2 + \dots + n\pi_n = n}} \binom{k}{\pi_1, \dots, \pi_n} b_1^{\pi_1} \dots b_n^{\pi_n} \quad (20.2)$$

$$= \sum_{k=1}^n a_k \mathcal{B}_{n,k}^{\text{ord}}(b_1, \dots, b_{n-k+1}), \quad (20.3)$$

and $c_0 = a_0$.

Proof. The coefficient of x^n in $G(x)^k$ is obtained by choosing an ordered composition $i_1 + \dots + i_k = n$, proving the first formula. Collect equal part sizes; the number of orderings with multiplicities π_j is the multinomial coefficient, proving the second. The third is the definition of the ordinary Bell polynomial. $\square$

Remark 20.2 (Formal crosswalk). Module `OrdinaryBellComposition` defines the ordinary partial Bell polynomials `Fabius.ordPartialBell` over any commutative semiring by the composition recurrence $\mathcal{B}_{n,k+1}^{\text{ord}}(b) = \sum_{i=1}^n b_i \mathcal{B}_{n-i,k}^{\text{ord}}(b)$, $\mathcal{B}_{n,0}^{\text{ord}} = \delta_{n,0}$, which is Equation (20.1) unrolled one part at a time; `Fabius.coeff_pow_eq_ordPartialBell` is $[x^n]G(x)^k = \mathcal{B}_{n,k}^{\text{ord}}(b_1, b_2, \dots)$ for G with zero constant term, and Equation (20.3) is `Fabius.coeff_subst_eq_sum_ordPartialBell`, with the sum starting at $k = 0$ (the term $k = 0$ is $a_0 \delta_{n,0}$, which also covers $c_0 = a_0$). The reciprocal formula Equation (20.4) is `Fabius.coeff_reciprocalSeries`, where `Fabius.reciprocalSeries` is $1/(1-u)$ at $u = -(A-1)$ and `Fabius.mul_reciprocalSeries` shows it inverts A . The multinomial form Equation (20.2) is not formalized.

Two useful specializations are immediate. If $F(u) = e^u$ and the series are exponentially normalized, one obtains the exponential formula. If $A(x) = 1 + \sum_{n \geq 1} a_n x^n$, then

$$[x^n] \frac{1}{A(x)} = \sum_{k=1}^n (-1)^k \mathcal{B}_{n,k}^{\text{ord}}(a_1, \dots, a_{n-k+1}) \quad (n \geq 1), \quad (20.4)$$

by taking $F(u) = 1/(1-u)$ and $G(x) = -\sum_{n \geq 1} a_n x^n$.

20.2 Exponential formal power series

Let

$$F(x) = \sum_{j \geq 1} a_j \frac{x^j}{j!}, \quad G(u) = \sum_{k \geq 0} b_k \frac{u^k}{k!}, \quad G(F(x)) = \sum_{n \geq 0} c_n \frac{x^n}{n!}.$$

Theorem 20.3 (Exponential composition).

$$c_n = \sum_{k=0}^n b_k \mathcal{B}_{n,k}^{\text{exp}}(a_1, \dots, a_{n-k+1}). \quad (20.5)$$

Equivalently,

$$c_n = \sum_{\pi \in \text{Part}([n])} b_{|\pi|} \prod_{B \in \pi} a_{|B|}. \quad (20.6)$$

Proof. Substitute $F(x)$ into G and apply (1.10). The set-partition form is Theorem 12.3 with an additional weight b_k for the number of blocks. $\square$

Remark 20.4 (Formal crosswalk). This is `Fabius.egfA_subst_bellWeightSeries` (module `BellComposition`): for $B(t) = \sum_k b_k t^k / k!$ and $A(t) = \sum_{j \geq 1} a_j t^j / j!$ over any commutative $\mathbb{Q}$ -algebra, $B(A(t)) = \sum_n (\sum_{k \leq n} b_k \mathcal{B}_{n,k}^{\text{exp}}(a)) t^n / n!$, where substitution is `Mathlib's PowerSeries.subst`. The proof reads the coefficients of the substitution through the powers $A(t)^k$ and the column theorem `Fabius.bellWeightSeries_pow`. The block-colour convolution of Theorem 13.2 is the same theorem applied to a product, `Fabius.factorial_mul_partialBell_add`.

Taking $b_k = 1$ gives the complete Bell-polynomial exponential identity

$$\exp\left(\sum_{j \geq 1} a_j \frac{x^j}{j!}\right) = \sum_{n \geq 0} \mathcal{B}_n^{\text{exp}}(a_1, \dots, a_n) \frac{x^n}{n!}. \quad (20.7)$$

If the series converge, (20.5) is exactly (19.3) evaluated at the expansion point; formally, no convergence notion is needed.

Chapter 21

Lagrange inversion and Lagrange–Bürmann

21.1 Formal theorem

Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi(w) \in R[[w]]$ have invertible constant term. There is a unique $g(z) \in zR[[z]]$ satisfying

$$g(z) = z\phi(g(z)). \quad (21.1)$$

Uniqueness follows recursively because the coefficient of z^n on the right depends only on coefficients of g below degree n .

Theorem 21.1 (Lagrange–Bürmann). *For $n \geq 1$ and every formal series H ,*

$$[z^n] H(g(z)) = \frac{1}{n} [w^{n-1}] H'(w)\phi(w)^n, \quad (21.2)$$

$$= [w^n] H(w)\phi(w)^{n-1}(\phi(w) - w\phi'(w)). \quad (21.3)$$

In particular,

$$[z^n] g(z) = \frac{1}{n} [w^{n-1}] \phi(w)^n. \quad (21.4)$$

Proof. Put $f(w) = w/\phi(w)$, so that $f(g(z)) = z$. Formal residues and the change of variables theorem Theorem 1.3 give

$$\begin{aligned} [z^n] H(g(z)) &= \operatorname{Res}_z H(g(z))z^{-n-1} dz \\ &= \operatorname{Res}_w H(w)f(w)^{-n-1} f'(w) dw \\ &= -\frac{1}{n} \operatorname{Res}_w H(w)(f(w)^{-n})' dw \\ &= \frac{1}{n} \operatorname{Res}_w H'(w)f(w)^{-n} dw, \end{aligned}$$

where the residue of a derivative is zero. Since $f(w)^{-n} = w^{-n}\phi(w)^n$, this is (21.2). Applying the same zero-residue rule to $H(w)\phi(w)^n w^{-n}$ gives

$$\frac{1}{n} \operatorname{Res} H' \phi^n w^{-n} = \operatorname{Res} H \phi^n w^{-n-1} - \operatorname{Res} H \phi^{n-1} \phi' w^{-n},$$

which is (21.3). Take $H(w) = w$ for (21.4). □

Remark 21.2 (Formal crosswalk). Module `LagrangeInversion` formalizes Equation (21.2) as `Fabius.Lagrange.coeff_subst_derivative`, in the division-free form $n[z^n]H(g) = [w^{n-1}]H'(w)\phi(w)^n$, and Equation (21.4) as `Fabius.Lagrange.coeff_subst_id`, over an arbitrary commutative $\mathbb{Q}$ -algebra. The core lemmas take g in the shape $g = zu$ with $u = \phi(g)$ and u invertible, and the solution is then *constructed* rather than assumed: `Fabius.Lagrange.solution` builds it as the compositional inverse of $z\psi(z)$, where ψ is the power-series inverse of ϕ , and `Fabius.Lagrange.solution_eq` proves it satisfies Equation (21.1). So `Fabius.Lagrange.coeff_solution_subst_derivative` and `Fabius.Lagrange.coeff_solution` are unconditional statements about a witness, not conditional ones. Uniqueness of g and Equation (21.3) are not formalized. The proof above is unavailable in Lean, because it runs through Theorem 1.3, which is itself unformalized. The formal proof is purely algebraic and rests on one cancellation: writing v for the inverse of u , one has $v^{M+1}g' = v^M - \frac{1}{M}z(v^M)'$, whose M th coefficient vanishes for every $M \geq 1$ (`Fabius.Lagrange.coeff_pow_succ_mul_derivative_eq_zero`), while vg' has constant term 1 (`Fabius.Lagrange.coeff_inv_mul_derivative`). Substituting $A(g) = (A\phi^n)(g)v^n$ into the truncated substitution expansion `Fabius.coeff_mul_subst_eq` therefore collapses the sum to its single surviving term, which is $[z^{n-1}](A(g)g') = [w^{n-1}](A\phi^n)$ (`Fabius.Lagrange.coeff_subst_mul_derivative`); the chain rule `PowerSeries.derivative_subst` turns that into the displayed form.

21.2 Analytic inverse functions

Theorem 21.3 (Analytic Lagrange inversion). *Let f be analytic near a , let $b = f(a)$, and suppose $f'(a) \neq 0$. Its local inverse g is analytic near b and has*

$$g(z) = a + \sum_{n \geq 1} g_n \frac{(z - b)^n}{n!}, \quad (21.5)$$

where

$$g_n = \frac{d^{n-1}}{dw^{n-1}} \left(\frac{w - a}{f(w) - f(a)} \right)^n \Big|_{w=a}. \quad (21.6)$$

The series has a positive radius of convergence.

Proof. The analytic inverse-function theorem gives an analytic local inverse and therefore a convergent Taylor series. Put $u = w - a$, $\zeta_{\text{Lag}} = z - b$, and

$$\phi(u) = \frac{u}{f(a+u) - f(a)};$$

its constant term is $1/f'(a)$. The inverse displacement satisfies $u = \zeta_{\text{Lag}}\phi(u)$. Formula (21.4), multiplied by $n!$, is exactly (21.6). $\square$

If $f'(a) = 0$ but $f(w) - f(a) = c(w - a)^q + \mathcal{O}((w - a)^{q+1})$ with $c \neq 0$, factor $f(w) - f(a) = (w - a)^q U(w)$ with $U(a) = c$. Choosing one of the q values of $c^{-1/q}$ reduces the equation to ordinary inversion in the variable $(z - f(a))^{1/q}$. Thus the inverse has q local Puiseux branches. This is the precise sense in which Lagrange inversion extends to a critical point; a single-valued Taylor inverse cannot exist there.

21.3 Inverse coefficients in Bell-polynomial form

Suppose

$$f(w) = \sum_{j \geq 1} f_j \frac{w^j}{j!}, \quad f_1 \neq 0, \quad g(z) = f^{(-1)}(z) = \sum_{n \geq 1} g_n \frac{z^n}{n!}.$$

Define

$$f_j^{\text{rev, norm}} = \frac{f_{j+1}}{(j+1)f_1}. \quad (21.7)$$

Theorem 21.4 (Explicit inverse coefficients). *One has $g_1 = f_1^{-1}$ and, for $n \geq 2$,*

$$g_n = \frac{1}{f_1^n} \sum_{k=1}^{n-1} (-1)^k \binom{n}{k} \mathcal{B}_{n-1,k}^{\text{exp}}(f_1^{\text{rev, norm}}, \dots, f_{n-k}^{\text{rev, norm}}). \quad (21.8)$$

Proof. Write

$$\frac{f(w)}{f_1 w} = 1 + U(w), \quad U(w) = \sum_{j \geq 1} f_j^{\text{rev, norm}} \frac{w^j}{j!}.$$

By (21.4), $g_n = (n-1)! [w^{n-1}] (w/f(w))^n$. Now

$$(1+U)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n+k-1}{k} U^k = \sum_{k \geq 0} (-1)^k \frac{(n)_k}{k!} U^k.$$

Apply (1.10) and extract degree $n-1$; the $k=0$ term vanishes when $n \geq 2$. $\square$

21.4 The associahedron interpretation, with normalized notation

The formula in the archive mixes exponential coefficients with an ordinary-series face formula. The exact geometric statement is the following.

Let

$$A(x) = x + \sum_{n \geq 1} a_n x^{n+1}, \quad B(x) = A^{(-1)}(x) = x + \sum_{n \geq 1} b_n x^{n+1}. \quad (21.9)$$

Use the indexing in which the associahedron $\mathcal{K}_{\dim=n-1}^{\text{Assoc}}$ has dimension $n-1$ and its faces are plane rooted trees with $n+1$ leaves and internal arities at least 2. If an internal vertex has arity $i+1$, give it weight a_i ; let a_F be the product of all vertex weights. Equivalently, if $F \cong \mathcal{K}_{\dim=i_1-1}^{\text{Assoc}} \times \dots \times \mathcal{K}_{\dim=i_m-1}^{\text{Assoc}}$, then $a_F = a_{i_1} \cdots a_{i_m}$.

Theorem 21.5 (Associahedral inversion).

$$b_n = \sum_{F \text{ a face of } \mathcal{K}_{\dim=n-1}^{\text{Assoc}}} (-1)^{n-\dim F} a_F. \quad (21.10)$$

Proof. The equation $A(B) = x$ is

$$B = x - \sum_{i \geq 1} a_i B^{i+1}.$$

Iterating it recursively expands b_n as a sum over plane rooted trees with $n+1$ leaves: every internal vertex of arity $i+1$ contributes $-a_i$. A face F of $\mathcal{K}_{\dim=n-1}^{\text{Assoc}}$ is exactly such a tree after contracting a chosen set of internal edges. If it has m internal vertices, then $\dim F = n-m$, so its sign is $(-1)^m = (-1)^{n-\dim F}$ and its weight is a_F . Thus the tree expansion is the alternating face sum. $\square$

For example,

$$\begin{aligned} b_1 &= -a_1, \\ b_2 &= -a_2 + 2a_1^2, \\ b_3 &= -a_3 + 5a_2a_1 - 5a_1^3, \\ b_4 &= -a_4 + 6a_3a_1 + 3a_2^2 - 21a_2a_1^2 + 14a_1^4. \end{aligned}$$

For $\mathcal{K}_{\dim=3}^{\text{Assoc}}$, the coefficients 1, 6, 3, 21, 14 count respectively the whole 3-polytope, its six pentagonal and three square facets, its edges, and its vertices.

Chapter 22

Algebraic and transcendental examples

22.1 The Fuss–Catalan algebraic equation

Let $p \geq 2$ be an integer and consider

$$x^p - x + z = 0, \quad x(0) = 0. \quad (22.1)$$

It is equivalent to $x = z\phi(x)$ with $\phi(x) = (1 - x^{p-1})^{-1}$.

Theorem 22.1. *The distinguished solution is*

$$x(z) = \sum_{k=0}^{\infty} \binom{pk}{k} \frac{z^{(p-1)k+1}}{(p-1)k+1}. \quad (22.2)$$

Its radius of convergence is

$$R_p = (p-1)p^{-p/(p-1)}, \quad (22.3)$$

and the series converges absolutely also on $|z| = R_p$.

Proof. By (21.4),

$$[z^n] x(z) = \frac{1}{n} [w^{n-1}] (1 - w^{p-1})^{-n}.$$

This vanishes unless $n-1 = (p-1)k$. In that case the coefficient is $\frac{1}{n} \binom{n+k-1}{k} = \binom{pk}{k} / ((p-1)k+1)$, proving the series.

The inverse map is $z = f(x) = x - x^p$. Its nearest critical points satisfy $f'(x) = 1 - px^{p-1} = 0$ and all have modulus $p^{-1/(p-1)}$; their critical values have modulus $(p-1)p^{-p/(p-1)}$. No local inverse can pass analytically through a critical value, while the inverse-function theorem continues the branch up to one, so this is the radius. Stirling's formula gives coefficients of order $Ck^{-3/2}R_p^{-((p-1)k+1)}$, which proves absolute convergence on the boundary. $\square$

Remark 22.2 (Formal crosswalk). Module `FussCatalanSeries` constructs the distinguished solution rather than assuming it: with $q = p-1$, the equation $x - x^{q+1} = z$ is the Lagrange functional equation for $\phi(w) = (1 - w^q)^{-1}$, and that weight is invertible by construction (`Fabius.fussPhi_mul`), so `Fabius.Lagrange.solution` applies and gives `Fabius.fussSolution`; the polynomial form of the defining equation is `Fabius.fussSolution_sub_pow`. Equation (22.2) then splits into two statements: `Fabius.coeff_fussSolution`, which is $(qk+1)[z^{qk+1}]x = \binom{(q+1)k}{k}$, and `Fabius.coeff_fussSolution_eq_zero`, which is the vanishing of every coefficient whose index is not of the form $qk+1$. Both follow from Equation (21.4) once ϕ is written as the all-ones series with w replaced by w^q (`Mathlib's PowerSeries.expand`); because that substitution is an algebra map it commutes with the n th power, so the negative binomial coefficients come

straight from `PowerSeries.mk_one_pow_eq_mk_choose_add` and the index condition from `PowerSeries.coeff_expand_of_not_dvd`. The radius Equation (22.3), the critical-point argument that identifies it, and the absolute convergence on $|z| = R_p$ are analytic and are not formalized.

22.2 Lambert's W function at the origin

The Lambert function satisfies $W(z)e^{W(z)} = z$.

Theorem 22.3. *On its principal branch near 0,*

$$W(z) = \sum_{n=1}^{\infty} (-n)^{n-1} \frac{z^n}{n!} = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + \mathcal{O}(z^5), \quad (22.4)$$

with radius of convergence e^{-1} .

Proof. Here $f(w) = we^w$, so $w/f(w) = e^{-w}$. Formula (21.6) gives $g_n = D^{n-1}e^{-nw}|_{w=0} = (-n)^{n-1}$. The inverse map has its nearest critical point at $w = -1$, where $f(-1) = -e^{-1}$; this square-root branch point determines the radius. $\square$

Remark 22.4 (Formal crosswalk). Module `LambertWSeries` constructs W rather than assuming it: $W = ze^{-W}$ is the Lagrange functional equation with $\phi(w) = e^{-w}$, which is invertible with inverse e^w (`Fabius.expNeg_mul_exp`), so `Fabius.Lagrange.solution` applies and `Fabius.lambertW` is its value. The familiar form $We^W = z$ is `Fabius.lambertW_mul_exp_subst`, and Equation (22.4) is `Fabius.coeff_lambertW`, in the equivalent shape $[z^{n+1}]W = -(n+1)^n/(n+1)!$ that avoids a truncated subtraction; it follows from Equation (21.4) together with $(e^{-w})^n = e^{-nw}$ (`Fabius.expNeg_pow`). The radius of convergence e^{-1} , and with it the branch-point argument above, is analytic and is not formalized.

22.3 A rapidly useful expansion around $W(e) = 1$

Put $\xi = \log z - 1$ and $u = W(z) - 1$. Then

$$u + \log(1 + u) = \xi. \quad (22.5)$$

Reversion gives

$$W(e^{1+\xi}) = 1 + \frac{\xi}{2} + \frac{\xi^2}{16} - \frac{\xi^3}{192} - \frac{\xi^4}{3072} + \frac{13\xi^5}{61440} + \mathcal{O}(\xi^6). \quad (22.6)$$

Proof. The derivative at $u = 0$ of the left side of (22.5) is 2, so it has an ordinary inverse. Substitute $u = c_1\xi + \cdots + c_5\xi^5$ into $u + \log(1 + u) = \xi$ and compare successive powers. The triangular equations give $c_1 = 1/2$, $c_2 = 1/16$, $c_3 = -1/192$, $c_4 = -1/3072$, $c_5 = 13/61440$. $\square$

The closest critical points of $u + \log(1 + u)$ satisfy $1 + (1 + u)^{-1} = 0$, hence $u = -2$; using the two adjacent logarithm values gives images $-2 \pm i\pi$. Therefore the radius in the ξ -plane is $\sqrt{4 + \pi^2}$. For positive real z , the expansion converges whenever $|\log z - 1| < \sqrt{4 + \pi^2}$, approximately $0.0655 < z < 112.63$. Taking $\xi = -1$ yields a convergent evaluation of $W(1) = 0.567143 \dots$

22.4 Binary trees and Catalan numbers

Let C_n^{Cat} count rooted ordered full binary trees with n internal vertices, and let $B(z) = \sum_{n \geq 0} C_n^{\text{Cat}} z^n$. Removing the root gives

$$B(z) = 1 + zB(z)^2. \quad (22.7)$$

Put $C(z) = B(z) - 1$; then $C = z(1 + C)^2$. Lagrange inversion gives

$$C_n^{\text{Cat}} = [z^n] C(z) = \frac{1}{n} [w^{n-1}] (1 + w)^{2n} = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (22.8)$$

This simultaneously proves the functional equation, its unique formal solution, and the Catalan closed form. The symbol C_n^{Cat} is used here to avoid confusing these binary-tree numbers with Bell numbers.

Chapter 23

Lagrange's reversion theorem in shifted form

23.1 The two-parameter implicit equation

Let $v = v(x, y)$ be the unique formal solution of

$$v = x + yf(v), \quad v(x, 0) = x. \quad (23.1)$$

Theorem 23.1 (Lagrange reversion). *For every formal or analytic g ,*

$$g(v) = g(x) + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} (f(x)^k g'(x)). \quad (23.2)$$

In particular,

$$v = x + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} f(x)^k. \quad (23.3)$$

Proof. Put $u = v - x$. Then $u = y\phi(u)$ with $\phi(u) = f(x + u)$, treating x as a coefficient parameter. Apply (21.2) to $H(u) = g(x + u)$:

$$\left[y^k \right] g(v) = \frac{1}{k} \left[u^{k-1} \right] g'(x + u) f(x + u)^k = \frac{1}{k!} D_x^{k-1} (g'(x) f(x)^k).$$

Summation proves the first formula; take $g(z) = z$ for the second. $\square$

23.2 The delta-function proof decoded

The archive also gives a distributional derivation. Assume temporarily that $x, y \in \mathbb{R}$, the equation $z - x - yf(z) = 0$ has one simple root v , and $1 - yf'(v) > 0$. The elementary change-of-variables rule for the Dirac delta gives

$$g(v) = \int_{\mathbb{R}} \delta(yf(z) - z + x) g(z) (1 - yf'(z)) \, dz. \quad (23.4)$$

Using $\delta(s) = \int_{\mathbb{R}} e^{iks} \, dk / (2\pi)$ and expanding only in the formal parameter y ,

$$\begin{aligned} g(v) &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} (1 - yf'(x)) \right] \\ &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} - \frac{y^{n+1}}{(n+1)!} ((gf^{n+1})' - g'f^{n+1}) \right]. \end{aligned} \quad (23.5)$$

The first term of level $n + 1$ cancels the total derivative in the second term of level n ; what remains is exactly (23.2). Thus the Fourier–delta calculation is a compact analytic avatar of the formal residue proof. Formula (23.4) requires an absolute Jacobian in general; the displayed sign is valid in the stated orientation-preserving neighbourhood.

Chapter 24

Laplace-type asymptotic integrals

24.1 Endpoint expansion

Consider

$$I(\lambda) = \int_a^b e^{-\lambda f(x)} g(x) \, dx, \quad \lambda \rightarrow +\infty, \quad (24.1)$$

where the endpoint a is the unique relevant minimum and, with $t = x - a$,

$$f(a+t) \sim f(a) + \sum_{k \geq 0} a_k t^{k+\alpha}, \quad a_0 > 0, \quad \alpha > 0, \quad (24.2)$$

$$g(a+t) \sim \sum_{k \geq 0} b_k t^{k+\beta-1}, \quad \beta > 0. \quad (24.3)$$

Let $\mathcal{B}_{k,j}^{\text{ord}}$ denote the ordinary Bell polynomial.

Theorem 24.1 (Laplace–Erdélyi coefficient formula). *Under the standard differentiability and remainder hypotheses that permit Watson’s lemma,*

$$I(\lambda) \sim e^{-\lambda f(a)} \sum_{n=0}^{\infty} \Gamma\left(\frac{n+\beta}{\alpha}\right) \frac{c_n}{\lambda^{(n+\beta)/\alpha}}, \quad (24.4)$$

where

$$c_n = \frac{1}{\alpha a_0^{(n+\beta)/\alpha}} \sum_{k=0}^n b_{n-k} \sum_{j=0}^k \binom{-\frac{n+\beta}{\alpha}}{j} \frac{\mathcal{B}_{k,j}^{\text{ord}}(a_1, \dots, a_{k-j+1})}{a_0^j}. \quad (24.5)$$

Proof. Write $f(a+t) - f(a) = t^\alpha A(t)$ with $A(t) = a_0 + a_1 t + \dots$, and set $s = (f(a+t) - f(a))^{1/\alpha} = t A(t)^{1/\alpha}$. This has an inverse $t = t(s)$. Expand

$$g(a+t(s)) t'(s) = \sum_{n \geq 0} d_n s^{n+\beta-1}.$$

A generalized Lagrange–Bürmann calculation, first for integral β and then by formal continuation in the exponent, gives

$$d_n = \sum_{k=0}^n b_{n-k} \left[t^k \right] A(t)^{-(n+\beta)/\alpha}.$$

Indeed, apply (21.2) to an antiderivative of $t^{m+\beta-1}$ and differentiate back after substitution.

Put $\rho = (n + \beta)/\alpha$. The ordinary Bell expansion yields

$$\begin{aligned} [t^k] A(t)^{-\rho} &= a_0^{-\rho} [t^k] \sum_{j \geq 0} \binom{-\rho}{j} \left(\sum_{r \geq 1} \frac{a_r}{a_0} t^r \right)^j \\ &= a_0^{-\rho} \sum_{j=0}^k \binom{-\rho}{j} a_0^{-j} \mathcal{B}_{k,j}^{\text{ord}}(a_1, \dots, a_{k-j+1}). \end{aligned}$$

Thus $c_n = d_n/\alpha$ is (24.5). Finally

$$\int_0^\infty e^{-\lambda s^\alpha} s^{n+\beta-1} ds = \frac{1}{\alpha} \Gamma\left(\frac{n+\beta}{\alpha}\right) \lambda^{-(n+\beta)/\alpha},$$

and Watson's lemma controls the replacement of the finite endpoint interval by the asymptotic expansion. $\square$

Remark 24.2. The appearance of ordinary rather than exponential Bell polynomials is forced by the ordinary Taylor expansion of $A(t)^{-\rho}$. Lagrange inversion enters in the change from t to the natural Laplace coordinate s ; the gamma factor enters only in the final monomial integral.

Part VII

Incidence Algebra, Umbral Calculus, and Multivariate Inversion

Chapter 25

Binomial inversion, partition lattices, and Sheffer–Riordan calculus

The concrete triangular arrays developed earlier are instances of a larger calculus. This chapter collects the genuinely nonduplicate material from the six supplied manuscripts: incidence-algebra inversion, the partition-lattice Möbius function, Sheffer sequences, and exponential Riordan arrays. These viewpoints explain why changes of basis, labelled decomposition, and formal substitution repeatedly lead to the same matrices.

25.1 Binomial inversion

Theorem 25.1 (Binomial inversion). *For sequences (a_n) and (b_n) in a commutative $\mathbb{Q}$ -algebra, the formulas*

$$b_n = \sum_{k=0}^n \binom{n}{k} a_k, \quad (25.1)$$

$$a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k \quad (25.2)$$

are equivalent. In exponential generating functions,

$$B(z) = e^z A(z), \quad A(z) = e^{-z} B(z). \quad (25.3)$$

Proof. Multiplication of EGFs gives

$$e^z A(z) = \sum_{n \geq 0} \left(\sum_{k=0}^n \binom{n}{k} a_k \right) \frac{z^n}{n!},$$

so (25.1) is equivalent to the first identity in (25.3). Multiplication by e^{-z} and coefficient extraction gives (25.2). Directly, substitution of the forward formula into the backward one makes the coefficient of a_j

$$\sum_{k=j}^n (-1)^{n-k} \binom{n}{k} \binom{k}{j} = \binom{n}{j} \sum_{r=0}^{n-j} (-1)^{n-j-r} \binom{n-j}{r} = \delta_{nj},$$

which independently verifies the inverse matrices. □

Remark 25.2 (Formal crosswalk). The equivalence of Equation (25.1) and Equation (25.2) is `Fabius.b_inomial_inversion_ring_iff` for sequences in any commutative ring and `Fabius.binomial_`

`inversion_iff` for sequences in any additive commutative group (module `BinomialInversion`); the hypothesis of a commutative $\mathbb{Q}$ -algebra is unnecessary, since the proof is the inverse-matrix computation of the last display, `Fabius.sum_Icc_neg_one_pow_choose_mul_choose`, which reduces through `Nat.choose_mul` to the alternating row sums of Pascal’s triangle. The reverse product is supplied by the general commutation theorem `Fabius.lowerTriangular_orthogonal_comm`, and both directions are instances of the finite triangular transform calculus of `Fabius.lowerTriangularTransform`. The exponential-generating-function form Equation (25.3) is not formalized.

25.2 The partition lattice and cumulants

Let Π_n be the lattice of partitions of $[n]$, ordered by refinement, with bottom $\perp_\Pi$ and top $\top_\Pi$.

Theorem 25.3 (Möbius function of Π_n). *If $\sigma \leq \pi$ and m_C is the number of blocks of σ contained in a block C of π , then*

$$\mu(\sigma, \pi) = \prod_{C \in \pi} (-1)^{m_C-1} (m_C - 1)!. \quad (25.4)$$

In particular,

$$\mu(\perp_\Pi, \pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!, \quad (25.5)$$

$$\mu(\pi, \top_\Pi) = (-1)^{|\pi|-1} (|\pi| - 1)!. \quad (25.6)$$

Proof. The interval $[\sigma, \pi]$ is the direct product, over $C \in \pi$, of partition lattices on the m_C blocks of σ inside C . Möbius functions multiply on direct products, so it remains to compute $u_m = \mu_{\Pi_m}(\perp_\Pi, \top_\Pi)$. The defining relation gives, for $m > 1$,

$$0 = \sum_{\rho \in \Pi_m} \mu(\perp_\Pi, \rho) = \sum_{\rho \in \Pi_m} \prod_{B \in \rho} u_{|B|}.$$

Let $U(z) = \sum_{m \geq 1} u_m z^m / m!$. By the labelled exponential formula,

$$\exp U(z) = 1 + z,$$

because the inner sum is 1 for $m = 0, 1$ and 0 thereafter. Hence $U(z) = \log(1 + z)$ and $u_m = (-1)^{m-1} (m-1)!$. Product decomposition now gives the general formula. $\square$

Theorem 25.4 (Moment–cumulant inversion). *The relations*

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}, \quad (25.7)$$

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|} \quad (25.8)$$

are inverse. Equivalently, if $M(z) = 1 + \sum_{n \geq 1} m_n z^n / n!$ and $K(z) = \sum_{n \geq 1} \kappa_n z^n / n!$, then $M = e^K$ and $K = \log M$.

Proof. The forward relation is the exponential formula for a set of blocks weighted by cumulants. Möbius inversion on Π_n , using (25.6), gives the backward relation. Grouping partitions by block sizes is exactly the coefficient expansion of e^K and $\log M$. $\square$

Remark 25.5 (Formal crosswalk). Module `BellPolynomialInversion` defines the complete Bell polynomials `Bell.complete` by the recurrence $m_{n+1} = \sum_k \binom{n}{k} \kappa_{k+1} m_{n-k}$ (the sum over partitions grouped by the block containing $n+1$) and the cumulant sequence `Bell.cumulant` by the inverse recurrence, and proves that the two constructions are mutually inverse (`Bell.complete_cumulant` for $m_0 = 1$, `Bell.cumulant_complete` for $\kappa_0 = 0$), over any commutative ring; the generating-function form $M = e^K$ is `Fabius.exp_subst_bellWeightSeries` (module `BellGeneratingFunctions`), where K is the weight series `Fabius.bellWeightSeries`. The sums over the partition lattice Equations (25.7) and (25.8) and the identity $K = \log M$ are not formalized.

Example 25.6 (The first cumulants). Expanding (25.8) over the partitions of $[n]$ for $n \leq 4$, and grouping by block-size type, gives

$$\begin{aligned}\kappa_1 &= m_1, \\ \kappa_2 &= m_2 - m_1^2, \\ \kappa_3 &= m_3 - 3m_2m_1 + 2m_1^3, \\ \kappa_4 &= m_4 - 4m_3m_1 - 3m_2^2 + 12m_2m_1^2 - 6m_1^4.\end{aligned}\tag{25.9}$$

For a random variable these are the mean, the variance, the third central moment, and the fourth cumulant, whose normalization κ_4/κ_2^2 is the excess kurtosis. All cumulants beyond the second vanish for a Gaussian law; the vanishing of κ_4 alone does not characterize it, and by Marcinkiewicz’s theorem no finite set of cumulant conditions does — only $\kappa_n = 0$ for every $n \geq 3$. The coefficients are read off the partition lattice: Π_4 has one partition of type 4, four of type 3 + 1, three of type 2 + 2, six of type 2 + 1 + 1 and one of type 1 + 1 + 1 + 1, and the Möbius weights $(-1)^{|\pi|-1}(|\pi| - 1)!$ are 1, −1, −1, 2, −6 respectively.

25.3 Sheffer sequences

Definition 25.7. Let R be a commutative $\mathbb{Q}$ -algebra. A polynomial sequence $(s_n(x))_{n \geq 0}$ over R is *Sheffer for* (A, B) when

$$\sum_{n \geq 0} s_n(x) \frac{t^n}{n!} = A(t) e^{xB(t)},\tag{25.10}$$

where $A(t) \in R[[t]]$ has invertible constant coefficient and $B(t) \in tR[[t]]$ has invertible linear coefficient. Thus A is a unit and B is a delta series with a unique compositional inverse $\overline{B}$. The associated sequence $(p_n(x))_{n \geq 0}$ is defined by

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xB(t)}.$$

The sequence is of binomial type when $A = 1$, and Appell when $B(t) = t$.

The definition above declares a sequence of binomial type by exhibiting a generating function. The converse also holds, and it is the statement that makes the umbral formalism worth having: the binomial identity alone forces the exponential shape, so no sequence of binomial type escapes the theory.

Theorem 25.8 (Generating-function characterization of binomial type). *Let K be a field of characteristic zero and let $(p_n)_{n \geq 0}$ be polynomials in $K[x]$ with $p_0 = 1$. The following are equivalent.*

1. For all x and y ,

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y) \quad (n \geq 0).\tag{25.11}$$

2. There is a series $B(t) \in tK[[t]]$, necessarily unique, with

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xB(t)}. \quad (25.12)$$

When these hold, $\deg p_n \leq n$ automatically, and $\deg p_n = n$ for every n if and only if $B'(0) \neq 0$, that is, if and only if B is a delta series.

Proof. Suppose (2). Then $e^{(x+y)B(t)} = e^{xB(t)}e^{yB(t)}$, and comparing coefficients of $t^n/n!$ in a product of exponential generating functions is exactly the binomial convolution (25.11).

Suppose (1), and put $P(x, t) = \sum_{n \geq 0} p_n(x) t^n/n! \in K[x][[t]]$. The binomial identity says precisely

$$P(x + y, t) = P(x, t) P(y, t). \quad (25.13)$$

Setting $x = y = 0$ gives $P(0, t) = P(0, t)^2$; since $P(0, t)$ has constant term $p_0 = 1$ it is a unit, so $P(0, t) = 1$. Every $P(x, t)$ likewise has constant term 1, so $L(x, t) = \log P(x, t)$ is defined in $K[x][[t]]$, its coefficients being polynomials in x because they are universal polynomials in $p_1(x), \dots, p_n(x)$. Taking logarithms in (25.13),

$$L(x + y, t) = L(x, t) + L(y, t).$$

Write $L(x, t) = \sum_{m \geq 0} \ell_m(x) t^m$. Each $\ell_m \in K[x]$ satisfies $\ell_m(x + y) = \ell_m(x) + \ell_m(y)$ identically in two indeterminates. Writing $\ell_m(x) = \sum_j c_j x^j$ and comparing the mixed monomials $x^a y^b$ with $a, b \geq 1$ forces $c_j = 0$ for $j \geq 2$ (the binomial coefficient $\binom{j}{a} \neq 0$ in characteristic zero), while $x = y = 0$ forces $c_0 = 0$. Hence $\ell_m(x) = c_m x$ and $L(x, t) = xB(t)$ with $B(t) = \sum_m c_m t^m$. Finally $\ell_0(x) = \log p_0(x) = 0$, so $B \in tK[[t]]$, and exponentiating gives (25.12). Uniqueness is immediate from $B(t) = \partial_x L(x, t)$, or equivalently from $B(t) = \sum_{n \geq 1} p'_n(0) t^n/n!$.

For the degree statement, expand $e^{xB(t)} = \sum_{k \geq 0} x^k B(t)^k/k!$. Since $B \in tK[[t]]$, the term $B(t)^k/k!$ contributes nothing below t^k , so p_n has degree at most n and its coefficient of x^n is $n! \cdot [t^n] B(t)^n/n! = B'(0)^n$. Thus $\deg p_n = n$ for all n exactly when $B'(0) \neq 0$; and already $p_1(x) = B'(0)x$ shows that exact degree at $n = 1$ forces $B'(0) \neq 0$. $\square$

Theorem 25.9 (Addition and lowering). *Let p_n be the binomial-type sequence with EGF $e^{yB(t)}$. Then*

$$s_n(x + y) = \sum_{k=0}^n \binom{n}{k} s_k(x) p_{n-k}(y). \quad (25.14)$$

If $\overline{B}$ is the compositional inverse of B and $Q = \overline{B}(D_x)$, then

$$Q s_n(x) = n s_{n-1}(x). \quad (25.15)$$

Proof. Multiplying the two EGFs at x and y proves the addition law. For every formal series F , $F(D_x)e^{xB(t)} = F(B(t))e^{xB(t)}$. Taking $F = \overline{B}$ gives the eigenvalue t ; multiplication of the EGF by t and coefficient comparison gives the lowering law. $\square$

Remark 25.10 (Formal crosswalk). In module `BinomialType` the Sheffer sequence is `Fabius.shefferPoly`, the binomial convolution $s_n = \sum_k \binom{n}{k} c_k p_{n-k}$ of the coefficients c_k of g with the binomial-type sequence p_n of Theorem 15.5, so that `Fabius.egfA_mul_exp_subst_smul_bellWeightSeries` is the generating function $g(t)e^{xB(t)}$. Equation (25.14) is `Fabius.shefferPoly_add`: the binomial identity of p_n in convolution form (`Fabius.binomialTypePoly_add'`) and the associativity of the binomial convolution (`Bell.binomialConv_assoc`), over any commutative ring. The lowering law Equation (25.15) is not formalized.

Proposition 25.11 (Abel polynomials). *Let $T(t)$ solve $T = te^{-aT}$. Then*

$$e^{xT(t)} = 1 + \sum_{n \geq 1} x(x - na)^{n-1} \frac{t^n}{n!}. \quad (25.16)$$

Thus $x(x - na)^{n-1}$ is a sequence of binomial type.

Proof. Lagrange–Bürmann gives

$$[t^n] e^{xT(t)} = \frac{x}{n} [w^{n-1}] e^{xw} e^{-anw} = \frac{x(x - na)^{n-1}}{n!}.$$

An EGF of the form $e^{xB(t)}$ satisfies the binomial identity by multiplication at x and y . $\square$

25.4 Exponential Riordan arrays

Definition 25.12. Over a commutative $\mathbb{Q}$ -algebra, let $g(0)$ and $f'(0)$ be units and let $f(0) = 0$. Define

$$R_{n,k} = \frac{n!}{k!} [t^n] g(t) f(t)^k. \quad (25.17)$$

The resulting lower-triangular matrix is denoted $[g, f]$.

Theorem 25.13 (Riordan action, product, and inverse). *If (a_k) has EGF $A(t)$, then $[g, f]a$ has EGF*

$$g(t)A(f(t)). \quad (25.18)$$

Consequently,

$$[g, f][h, \ell] = [g(h \circ f), \ell \circ f], \quad (25.19)$$

$$[g, f]^{-1} = \left[\frac{1}{g \circ \bar{f}}, \bar{f} \right]. \quad (25.20)$$

Proof. By definition,

$$\sum_n \left(\sum_k R_{n,k} a_k \right) \frac{t^n}{n!} = g(t) \sum_k a_k \frac{f(t)^k}{k!} = g(t)A(f(t)).$$

Applying this action twice gives the product law, and substituting $\bar{f}$ gives the inverse. $\square$

Remark 25.14 (Formal crosswalk). Module `ExponentialRiordan` defines `Fabius.expRiordan` $[g, f]_{n,k} = \frac{n!}{k!} [t^n] g(t) f(t)^k$ over any $\mathbb{Q}$ -algebra. Equation (25.18) is `Fabius.expRiordan_action` (for f with zero constant term, using the truncation lemma `Fabius.coeff_mul_subst_eq`: only $k \leq n$ contributes to the n -th coefficient of $gA(f)$), the product law Equation (25.19) is `Fabius.expRiordan_mul`, and the inverse law Equation (25.20) is `Fabius.expRiordan_mul_inverse` in the form: if $\bar{f} \circ f = t$ and $g \cdot (h \circ f) = 1$ then $[g, f][h, \bar{f}]$ is the identity array `Fabius.expRiordan_one_x`. The Stirling examples are `Fabius.expRiordan_one_exp_sub_one` ($[1, e^t - 1]$ has entries $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$) and `Fabius.expRiordan_one_log` ($[1, \log(1 + t)]$ has entries $s(n, k)$).

Example 25.15 (Stirling matrices). The second-kind matrix is $[1, e^t - 1]$, while the signed first-kind matrix is $[1, \log(1 + t)]$. Since the two substitution series are compositional inverses, Stirling inversion is the Riordan inverse law. The Bell-polynomial matrix is obtained by replacing $e^t - 1$ with $\sum_{j \geq 1} x_j t^j / j!$.

Chapter 26

Complementary Bell numbers and signed partition calculus

One supplied manuscript isolates the complementary Bell numbers, which are the set-partition transform at block weight -1 . They are especially useful as a clean example of signed labelled enumeration.

Definition 26.1. Define

$$B_n^{\text{comp}} = \sum_{k=0}^n (-1)^k \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (26.1)$$

Thus $B_0^{\text{comp}} = 1$ and the next values are $-1, 0, 1, 1, -2, -9, -9, 50, 267, 413, \dots$

Theorem 26.2 (EGF, recurrence, and Dobinski-type series). *The complementary Bell numbers satisfy*

$$\sum_{n \geq 0} B_n^{\text{comp}} \frac{z^n}{n!} = \exp(1 - e^z), \quad (26.2)$$

$$B_{n+1}^{\text{comp}} = - \sum_{k=0}^n \binom{n}{k} B_k^{\text{comp}}, \quad (26.3)$$

$$B_n^{\text{comp}} = e \sum_{m=0}^{\infty} \frac{(-1)^m m^n}{m!}. \quad (26.4)$$

The last series is absolutely convergent.

Proof. Using the EGF of the Stirling numbers and summing over k gives

$$\sum_{n \geq 0} B_n^{\text{comp}} \frac{z^n}{n!} = \sum_{k \geq 0} \frac{(-1)^k}{k!} (e^z - 1)^k = e^{1-e^z}.$$

Differentiating yields $C'(z) = -e^z C(z)$; the exponential Cauchy product gives the recurrence. Finally expand e^{-e^z} :

$$e^{1-e^z} = e \sum_{m \geq 0} \frac{(-1)^m}{m!} e^{mz} = \sum_{n \geq 0} \left(e \sum_{m \geq 0} \frac{(-1)^m m^n}{m!} \right) \frac{z^n}{n!}.$$

Absolute convergence follows from the ratio test for $m^n/m!$ at fixed n . □

Remark 26.3 (Formal crosswalk). Module `ComplementaryBell` defines `Fabius.complementaryBell` by Equation (26.1) and identifies it with the complete Bell polynomial at constant weights -1 (`Fabius.complementaryBell_eq_bell_complete`), i.e. the Touchard polynomial at -1 .

Equation (26.2) is `Fabius.exp_subst_neg_exp_sub_one`, the substitution of $-(e^t - 1)$ into the exponential series over any $\mathbb{Q}$ -algebra; Equation (26.3) is `Fabius.complementaryBell_succ`, read off from the successor recurrence of the complete Bell polynomials rather than from the differential equation; and Equation (26.4) is `Fabius.complementaryBell_eq_exp_mul_tsum`, the Poisson moment series `Fabius.tsum_pow_mul_pow_div_factorial` at parameter -1 , whose summability is part of that statement.

Corollary 26.4 (Convolution inverse of the Bell sequence). *For every $n \geq 0$,*

$$\sum_{k=0}^n \binom{n}{k} B_k^{\text{Bell}} B_{n-k}^{\text{comp}} = \delta_{n0}. \quad (26.5)$$

Proof. The two EGFs are e^{e^z-1} and e^{1-e^z} , whose product is 1. □

Remark 26.5 (Formal crosswalk). `Fabius.sum_choose_bell_mul_complementaryBell` (module `ComplementaryBell`), proved from the addition law `Bell.complete_add` of complete Bell polynomials at the weights $+1$ and -1 , whose sum is the zero weight sequence with complete Bell polynomials δ_{n0} (`Fabius.bell_complete_zero_weights`).

Remark 26.6 (Signed combinatorial meaning). Equation (26.1) is the signed sum of all partitions of $[n]$, with weight $(-1)^{\#\text{blocks}}$. It is not a count of an ordinary unweighted class; it is an Euler characteristic in the incidence-algebra sense. The convolution inverse is the exponential formula applied to component weights $+1$ and -1 .

Chapter 27

Coordinate-free composition and derivatives of inverse maps

27.1 Fréchet Faà di Bruno formula

Theorem 27.1 (Coordinate-free Faà di Bruno). *Let E, F, G be Banach spaces, $g : E \rightarrow F$, $f : F \rightarrow G$, and let D^r denote the symmetric r -linear Fréchet derivative. Then*

$$D^n(f \circ g)(x)[h_1, \dots, h_n] = \sum_{\pi \in \Pi_n} D^{|\pi|} f(g(x)) \left[D^{|B|} g(x)[h_i : i \in B] \right]_{B \in \pi}. \quad (27.1)$$

Proof. Induct on n . For $n = 1$ this is the chain rule. Differentiate a term indexed by $\pi \in \Pi_n$ in direction h_{n+1} . Either the derivative hits the outer derivative, creating the new singleton block $\{n+1\}$, or it hits one inner derivative associated with a block B , adjoining $n+1$ to B . Every partition of $[n+1]$ is obtained uniquely in one of these two ways, according as $\{n+1\}$ is a singleton or not. The product and chain rules for multilinear maps supply no further multiplicities because blocks are already labelled subsets. $\square$

Corollary 27.2. *When all directions are equal to h ,*

$$D^n(f \circ g)(x)[h^n] = \sum_{k=1}^n D^k f(g(x)) \left[\mathcal{B}_{n,k}^{\text{exp}}(Dg[h], D^2g[h^2], \dots) \right], \quad (27.2)$$

where the Bell polynomial is interpreted as the symmetric tensor sum over blocks. In one dimension this is the usual scalar Faà di Bruno formula.

Proof. Group the set partitions in (27.1) by their number of blocks and then by block-size multiplicities. The number of partitions of a fixed profile is exactly the coefficient defining $\mathcal{B}_{n,k}^{\text{exp}}$. $\square$

27.2 Recursive derivatives of an inverse function

Theorem 27.3 (Inverse derivative recursion). *Let $g = f^{-1}$ locally and $f'(g(y)) \neq 0$. For $n \geq 2$,*

$$g^{(n)}(y) = -\frac{1}{f'(g(y))} \sum_{k=2}^n f^{(k)}(g(y)) \mathcal{B}_{n,k}^{\text{exp}}(g'(y), g''(y), \dots, g^{(n-k+1)}(y)). \quad (27.3)$$

In particular, writing all derivatives of f at $x = g(y)$,

$$g' = \frac{1}{f'}, \quad (27.4)$$

$$g'' = -\frac{f''}{(f')^3}, \quad (27.5)$$

$$g''' = \frac{3(f'')^2 - f' f'''}{(f')^5}, \quad (27.6)$$

$$g^{(4)} = \frac{-15(f'')^3 + 10f' f'' f''' - (f')^2 f^{(4)}}{(f')^7}. \quad (27.7)$$

Proof. Differentiate $f(g(y)) = y$ n times. Faà di Bruno gives zero for $n \geq 2$. The term with one outer derivative is $f'(g)\mathcal{B}_{n,1}^{\text{exp}} = f'(g)g^{(n)}$; every other term involves only derivatives of g of order at most $n-1$. Isolating the first term proves the recursion. The displayed cases follow by substituting $\mathcal{B}_{2,2}^{\text{exp}} = x_1^2$, $\mathcal{B}_{3,2}^{\text{exp}} = 3x_1x_2$, $\mathcal{B}_{4,2}^{\text{exp}} = 4x_1x_3 + 3x_2^2$, and $\mathcal{B}_{4,3}^{\text{exp}} = 6x_1^2x_2$, then simplifying. $\square$

Recursion (27.3) expresses $g^{(n)}$ through the lower derivatives of g . The following closed operator form expresses it through f alone, with no reference to g except in the final evaluation point. It is the shortest route to a symbolic table of inverse derivatives, since each order costs one differentiation rather than one Bell-polynomial expansion.

Theorem 27.4 (Inverse-derivative operator). *With $g = f^{-1}$ locally and $f' \neq 0$ near the point in question, for every $n \geq 1$*

$$g^{(n)}(y) = \left(\frac{1}{f'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{f'(x)} \Big|_{x=g(y)}. \quad (27.8)$$

Proof. Along the inverse curve $x = g(y)$ one has $dy/dx = f'(x)$ with $f'(x) \neq 0$, so differentiation with respect to y of any quantity expressed in x is

$$\frac{d}{dy} = \frac{dx}{dy} \frac{d}{dx} = \frac{1}{f'(x)} \frac{d}{dx}.$$

Now induct on n . For $n = 1$, $g'(y) = 1/f'(g(y))$, which is (27.8) with the operator applied zero times. If (27.8) holds for n , apply d/dy to both sides and use the displayed operator identity on the right; this prefixes one more factor $(1/f') d/dx$ and gives the case $n+1$. $\square$

Example 27.5. Applying (27.8) twice reproduces the first entries of (27.7):

$$\frac{1}{f'} \frac{d}{dx} \frac{1}{f'} = -\frac{f''}{(f')^3}, \quad \frac{1}{f'} \frac{d}{dx} \left(-\frac{f''}{(f')^3} \right) = \frac{3(f'')^2 - f' f'''}{(f')^5}.$$

27.3 Several nested functions

For a chain $f_1 \circ \cdots \circ f_r$, repeated use of Theorem 27.1 produces a levelled hierarchy of set partitions. At the innermost level, derivative labels that strike the same occurrence of f_r form blocks; at the next level, those blocks are grouped according to occurrences of f_{r-1} , and so on. Flattening the hierarchy in either association order is canonical, giving a combinatorial proof of associativity of formal composition. For three scalar functions,

$$D^n f(g(h(x))) = \sum_{k=0}^n f^{(k)}(g(h(x))) \mathcal{B}_{n,k}^{\text{exp}}((g \circ h)', \dots, (g \circ h)^{(n-k+1)}), \quad (27.9)$$

$$(g \circ h)^{(j)} = \sum_{\ell=0}^j g^{(\ell)}(h(x)) \mathcal{B}_{j,\ell}^{\text{exp}}(h', \dots, h^{(j-\ell+1)}). \quad (27.10)$$

Substitution of the second formula into the first is a finite closed formula; the partition hierarchy explains its multiplicities without additional index machinery.

Chapter 28

Good's multivariate Lagrange inversion formula

Let R be a commutative $\mathbb{Q}$ -algebra, and let $\mathbf{w} = (w_1, \dots, w_d)$ be determined formally by

$$w_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \in R^\times. \quad (28.1)$$

For a multi-index $\mathbf{n}$, write $\mathbf{t}^{\mathbf{n}} = \prod_i t_i^{n_i}$.

Lemma 28.1 (Multivariate residue substitution). *Let*

$$T_i(\mathbf{x}) = x_i U_i(\mathbf{x}), \quad U_i(\mathbf{0}) \in R^\times \quad (1 \leq i \leq d).$$

Then, for every formal Laurent series for which either side is defined coefficientwise,

$$\operatorname{Res}_{\mathbf{t}} H(\mathbf{t}) dt_1 \cdots dt_d = \operatorname{Res}_{\mathbf{x}} H(\mathbf{T}(\mathbf{x})) \det \left(\frac{\partial T_i}{\partial x_j} \right) dx_1 \cdots dx_d. \quad (28.2)$$

Proof. Interpret the determinant times $dx_1 \cdots dx_d$ as the pullback of the top differential form $dt_1 \wedge \cdots \wedge dt_d$. By linearity it is enough to take $H(\mathbf{t}) = t_1^{m_1} \cdots t_d^{m_d}$. If $m_j \neq -1$ for some j , then

$$\mathbf{t}^{\mathbf{m}} dt_1 \wedge \cdots \wedge dt_d = d \left(\frac{(-1)^{j-1}}{m_j + 1} t_j^{m_j+1} \prod_{i \neq j} t_i^{m_i} \bigwedge_{\substack{1 \leq r \leq d \\ r \neq j}} dt_r \right).$$

Pullback commutes with the exterior derivative. The residue of an exact top form is zero: after expansion it is a sum of formal partial derivatives, and $\partial/\partial x_j$ cannot create an x_j^{-1} term. Thus both sides of (28.2) vanish for this monomial.

It remains to take $m_1 = \cdots = m_d = -1$. Since every U_i is a unit,

$$\frac{dT_i}{T_i} = \frac{dx_i}{x_i} + \frac{dU_i}{U_i}.$$

In $\bigwedge_{i=1}^d dT_i/T_i$, the product $\bigwedge_i dx_i/x_i$ has residue 1. Every other exterior-product term contains at least one logarithmic differential dU_i/U_i , whose coefficients are ordinary power series, and consequently lacks at least one of the negative powers x_j^{-1} needed for the multivariate residue. Its residue is therefore zero. The transformed residue is 1, equal to the residue of $\prod_i t_i^{-1} dt_i$, and the proof is complete. $\square$

Theorem 28.2 (Lagrange–Good inversion). *For every formal series F ,*

$$[t^n]F(\mathbf{w}(t)) = [\mathbf{x}^n]F(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \times \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \quad (28.3)$$

Proof. Represent the left side as

$$\operatorname{Res}_{\mathbf{t}} F(\mathbf{w}(t)) \prod_i t_i^{-n_i-1} dt_1 \cdots dt_d.$$

Use the inverse substitution $t_i = x_i/\phi_i(\mathbf{x})$. Its Jacobian entries are

$$\frac{\partial t_i}{\partial x_j} = \frac{1}{\phi_i} \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

After multiplying by $\prod_i t_i^{-n_i-1}$, the row factors $1/\phi_i$ cancel one power from each $t_i^{-n_i-1}$, leaving $\prod_i \phi_i^{n_i}$. The remaining determinant initially has x_i in row i . Write it as $\det(I - AB)$ with $A = \operatorname{diag}(x_1, \dots, x_d)$ and $B = (\partial_j \log \phi_i)_{i,j}$; Sylvester's identity $\det(I - AB) = \det(I - BA)$ moves the factor x_i from row i to column j , producing the displayed determinant. The remaining residue is exactly the coefficient on the right. $\square$

Example 28.3 (A coupled Fuss–Catalan system). Let

$$u = x(1 + u + v)^a, \quad v = y(1 + u + v)^b,$$

and write $S = 1 + u + v$. For integers $m, n \geq 0$ and arbitrary c , the constant coefficient is $[x^0 y^0] S^c = 1$, while for $m + n \geq 1$,

$$[x^m y^n] S^c = \frac{c}{m!n!} (am + bn + c - 1)^{\overline{m+n-1}}. \quad (28.4)$$

When $am + bn + c \neq 0$, this is equivalently

$$[x^m y^n] S^c = \frac{c}{am + bn + c} \frac{(am + bn + c)^{\overline{m+n}}}{m!n!}. \quad (28.5)$$

Proof. Apply Theorem 28.2 with $\phi_1 = S^a$, $\phi_2 = S^b$, and $F = S^c$. Put $A = am + bn + c$. The determinant is

$$\det \begin{pmatrix} 1 - au/S & -av/S \\ -bu/S & 1 - bv/S \end{pmatrix} = 1 - \frac{au + bv}{S}.$$

Thus the desired coefficient is

$$[u^m v^n] (S^A - auS^{A-1} - bvS^{A-1}).$$

For $m + n \geq 1$, multinomial extraction (with a term containing u or v understood as zero when its requested exponent is zero) gives, after putting all three terms over the common factor $(A - 1)^{\overline{m+n-1}}/(m!n!)$,

$$\frac{(A - 1)^{\overline{m+n-1}}}{m!n!} (A - am - bn) = \frac{c}{m!n!} (A - 1)^{\overline{m+n-1}}.$$

This proves (28.4), including values where $A = 0$ by polynomial continuation. Multiplying and dividing by A when $A \neq 0$ gives (28.5). The case $m = n = 0$ is immediate from $S(0, 0) = 1$. $\square$

In the previous example the two branches carry the same argument S , so the Good determinant collapses to a single term. The next example is genuinely coupled: each series feeds the other, the determinant has two terms, and the answer is a difference of two products of binomial coefficients. It shows the determinant doing work that no one-variable Lagrange inversion performs.

Example 28.4 (A coupled two-series system). Let u and v be the unique formal solutions of

$$u = x(1 + v), \quad v = y(1 + u). \quad (28.6)$$

Then for all integers $m, n, p, q \geq 0$,

$$[x^m][y^n]u^p v^q = \binom{n}{m-p} \binom{m}{n-q} - \binom{n-1}{m-p-1} \binom{m-1}{n-q-1}, \quad (28.7)$$

with the convention that a binomial coefficient with a negative entry, or with a lower entry exceeding the upper one, is zero.

Proof. Here $\phi_1(U, V) = 1 + V$ and $\phi_2(U, V) = 1 + U$. The only nonvanishing partial derivatives are $\partial\phi_1/\partial V = 1$ and $\partial\phi_2/\partial U = 1$, so the determinant in (28.3) is

$$\det \begin{pmatrix} 1 & -\frac{V}{1+V} \\ -\frac{U}{1+U} & 1 \end{pmatrix} = 1 - \frac{UV}{(1+U)(1+V)}.$$

Applying Theorem 28.2 with $F(U, V) = U^p V^q$ and exponents (m, n) gives

$$[U^m][V^n] \left(U^p V^q (1+V)^m (1+U)^n - U^{p+1} V^{q+1} (1+V)^{m-1} (1+U)^{n-1} \right),$$

since the determinant's second term cancels one factor from each of $(1+U)^n$ and $(1+V)^m$. Each of the two summands factors as a product of a U -extraction and a V -extraction, which yields (28.7). $\square$

Proposition 28.5 (Characteristic determinant). *For the analytic system $H_i(\mathbf{w}, \mathbf{t}) = w_i - t_i \phi_i(\mathbf{w})$, the Jacobian with respect to $\mathbf{w}$, evaluated on a solution, has determinant equal to the Good determinant. Therefore an analytic branch can fail the implicit-function test only where some ϕ_i is singular or this determinant vanishes.*

Proof. Differentiate: $\partial H_i / \partial w_j = \delta_{ij} - t_i \partial_j \phi_i$. On a solution $t_i = w_i / \phi_i(\mathbf{w})$, giving the row form of the determinant; Sylvester's identity converts it to the column form in (28.3). $\square$

Remark 28.6. The determinant identifies candidate branch points, not automatically the radius of every derived series. One must also establish accessibility from the selected branch, exclude earlier coefficient singularities, and check that the outer function F does not cancel the local singular term. This is the multivariate analogue of examining critical values of a one-variable inverse map; see also Bender and Richmond's asymptotic formulation [3].

Part VIII

Bernoulli–Euler Calculus and Special Functions

Chapter 29

Bernoulli, Euler, Genocchi, and Cauchy–Gregory families

Bernoulli numbers already appeared in the Eulerian and power-sum chapters. Across the six supplied manuscripts, the distinct surrounding material is the polynomial calculus, the Euler and Genocchi families, Bernoulli polynomials of the second kind, and the Euler–Maclaurin formula. These families also illustrate the Sheffer framework: ordinary Bernoulli and Euler polynomials are Appell sequences, whereas the second-kind Bernoulli polynomials are Sheffer for the forward difference operator. Our convention is $\beta_1 = -1/2$, in agreement with the DLMF [17].

29.1 Generating functions and Appell structure

Definition 29.1. The Bernoulli numbers β_n , Bernoulli polynomials $\beta_n(x)$, and Euler polynomials $E_n(x)$ are defined by

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad (29.1)$$

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} \beta_n(x) \frac{t^n}{n!}, \quad (29.2)$$

$$\frac{2e^{xt}}{e^t + 1} = \sum_{n \geq 0} E_n(x) \frac{t^n}{n!}. \quad (29.3)$$

Theorem 29.2 (Appell, translation, and explicit formulas). For $n \geq 1$,

$$\beta'_n(x) = n\beta_{n-1}(x), \quad E'_n(x) = nE_{n-1}(x), \quad (29.4)$$

$$\beta_n(x + y) = \sum_{k=0}^n \binom{n}{k} \beta_k(x) y^{n-k}, \quad E_n(x + y) = \sum_{k=0}^n \binom{n}{k} E_k(x) y^{n-k}, \quad (29.5)$$

$$\beta_n(x) = \sum_{k=0}^n \binom{n}{k} \beta_k x^{n-k}. \quad (29.6)$$

Proof. Differentiating either polynomial EGF with respect to x multiplies it by t , which proves the derivative identities. Replacing x by $x + y$ multiplies by e^{yt} , and the exponential Cauchy product gives translation. Setting $x = 0$ in the Bernoulli translation formula gives the explicit polynomial expansion. $\square$

Remark 29.3 (Formal crosswalk). For the Bernoulli polynomials, the derivative identity in Equation (29.4) is Mathlib’s `Polynomial.derivative_bernoulli`, and the translation formula

Equation (29.5) is `Fabius.bernoulli_eval_add` (module `BernoulliAppell`). The formal proof is the Taylor expansion rather than the generating function: iterating the derivative identity gives $\beta_n^{(k)} = n^k \beta_{n-k}$ (`Fabius.iterate_derivative_bernoulli`), so the Hasse derivatives are $\binom{n}{k} \beta_{n-k}$ (`Fabius.hasseDeriv_bernoulli`), and Mathlib’s `Polynomial.taylor` expands $\beta_n(x+y) = \sum_k \beta_n^{[k]}(x) y^k$. The explicit formula Equation (29.6) is Mathlib’s definition `Polynomial.bernoulli_def`. For the Euler polynomials, `Fabius.derivative_eulerPolynomial` and `Fabius.eulerPolynomial_eval_add` (module `EulerPolynomials`) are the derivative and translation identities, the latter an instance of the general Appell lemma `Fabius.appell_eval_add` (an Appell sequence with $\deg p_n \leq n$ satisfies the translation formula, by the Taylor expansion).

Theorem 29.4 (Difference and reflection identities). *For all relevant n ,*

$$\beta_n(x+1) - \beta_n(x) = nx^{n-1} \quad (n \geq 1), \quad (29.7)$$

$$E_n(x+1) + E_n(x) = 2x^n, \quad (29.8)$$

$$\beta_n(1-x) = (-1)^n \beta_n(x), \quad (29.9)$$

$$E_n(1-x) = (-1)^n E_n(x). \quad (29.10)$$

Consequently $\beta_{2m+1} = 0$ for $m \geq 1$ and $E_{2m+1}(1/2) = 0$.

Proof. Subtract the Bernoulli EGFs at $x+1$ and x :

$$\frac{te^{(x+1)t} - te^{xt}}{e^t - 1} = te^{xt}.$$

Add the two Euler EGFs to obtain $2e^{xt}$; coefficient comparison proves the difference equations. Replacing t by $-t$ in the Bernoulli EGF gives

$$\frac{(-t)e^{-xt}}{e^{-t} - 1} = \frac{te^{(1-x)t}}{e^t - 1},$$

and the analogous Euler calculation proves both reflections. At $x = 0$, $\beta_n(1) = \beta_n$ for $n > 1$ by the difference equation; reflection then forces every odd β_n beyond β_1 to vanish. Euler reflection at $x = 1/2$ proves the last assertion. $\square$

Remark 29.5 (Formal crosswalk). For the Bernoulli polynomials the difference identity Equation (29.7) is Mathlib’s `Polynomial.bernoulli_eval_one_add`, the reflection Equation (29.9) is `Polynomial.bernoulli_eval_one_sub`, and $\beta_{2m+1} = 0$ is `bernoulli_eq_zero_of_odd`. For the Euler polynomials of module `EulerPolynomials` (defined by `Fabius.eulerPolynomial`, $E_n(x) = \sum_k \binom{n}{k} E_k(0) x^{n-k}$ with $E_k(0)$ the coefficients of $2/(e^t + 1)$), the difference identity Equation (29.8) is `Fabius.eulerPolynomial_eval_add_one_add`, proved as in the text by comparing the coefficients of $(e^t + 1) \sum_n E_n(x) t^n / n! = 2e^{xt}$ (`Fabius.sum_choose_mul_eulerPolynomial_add`) and translating (`Fabius.eulerPolynomial_eval_add`). The Euler reflection Equation (29.10) is `Fabius.eulerPolynomial_eval_one_sub` (module `EulerReflection`), proved as in the text: the substitutions $x \mapsto 1-x$ and $t \mapsto -t$ turn the generating-function identity into the same identity $F(e^t + 1) = 2e^t e^{-xt}$, and the unit $e^t + 1$ is cancelled; $E_{2m+1}(1/2) = 0$ is `Fabius.eulerPolynomial_eval_half_odd`.

29.2 Power sums and multiplication formulas

Theorem 29.6 (Faulhaber with an offset). *For $N, p \in \mathbb{N}_0$,*

$$\sum_{j=0}^{N-1} (x+j)^p = \frac{\beta_{p+1}(x+N) - \beta_{p+1}(x)}{p+1}. \quad (29.11)$$

Proof. Apply (29.7) to $\beta_{p+1}(x+j)$ and sum; the left side telescopes. $\square$

Remark 29.7 (Formal crosswalk). `Fabius.sum_range_add_pow_eq_bernoulli_sub` (module `Fabius`) is Equation (29.11) for all $N \geq 0$ and rational x , proved by exactly this telescoping, with the difference identity taken from `Mathlib` (`Polynomial.bernoulli_eval_one_add`); the case $x = 0$ is also `Mathlib`'s `Polynomial.sum_range_pow_eq_bernoulli_sub`.

Theorem 29.8 (Alternating power sums). *For $N, p \in \mathbb{N}_0$,*

$$\sum_{j=0}^{N-1} (-1)^j (x+j)^p = \frac{E_p(x) + (-1)^{N-1} E_p(x+N)}{2}. \quad (29.12)$$

The formula gives zero when $N = 0$.

Proof. Multiply the Euler difference equation at $x+j$ by $(-1)^j$ and sum. All interior Euler terms cancel in pairs, leaving the two endpoints. $\square$

Remark 29.9 (Formal crosswalk). `Fabius.sum_neg_one_pow_mul_pow_eq_eulerPolynomial` (module `EulerPolynomials`) is Equation (29.12) in the form $\sum_{j=0}^{N-1} (-1)^j (x+j)^p = (E_p(x) - (-1)^N E_p(x+N))/2$, which agrees with the displayed formula for $N \geq 1$ and gives 0 for $N = 0$; the formal proof is the induction on N from the difference identity `Fabius.eulerPolynomial_eval_add_one_add`.

Theorem 29.10 (Raabe multiplication). *For every positive integer q ,*

$$\sum_{r=0}^{q-1} \beta_n\left(x + \frac{r}{q}\right) = q^{1-n} \beta_n(qx). \quad (29.13)$$

In particular,

$$\beta_n\left(\frac{1}{2}\right) = (2^{1-n} - 1) \beta_n. \quad (29.14)$$

Proof. Sum the generating functions and use the finite geometric series:

$$\sum_{r=0}^{q-1} \frac{te^{(x+r/q)t}}{e^t - 1} = \frac{te^{xt}}{e^{t/q} - 1} = q \frac{(t/q)e^{(qx)(t/q)}}{e^{t/q} - 1}.$$

Coefficient comparison gives the multiplication formula. Set $q = 2, x = 0$ and solve for the half-value. $\square$

Remark 29.11 (Formal crosswalk). `Fabius.raabe` (module `RaabeMultiplication`) is Equation (29.13), written $\sum_{r < q} \beta_n(x + r/q) = q(1/q)^n \beta_n(qx)$ for $q \geq 1$ and rational x . In $(\mathbb{Q}[x])[t]$, `Mathlib`'s `Polynomial.bernoulli_generating_function` gives $B(a;t)(e^t - 1) = te^{at}$ for polynomial arguments a (`Fabius.bernoulliPolySeries_mul_exp_sub_one`); summing over $a = x + r/q$ and using the finite geometric sum $(\sum_{r < q} e^{rt/q})(e^{t/q} - 1) = e^t - 1$ (`Fabius.sum_pow_mul_sub_one`) gives $G(t)(e^{t/q} - 1) = te^{xt}$ for the left side; rescaling $B(qx;t)$ by $t \mapsto t/q$ (`Fabius.rescale_expSeries`) gives the same identity for the right side; and $e^{t/q} - 1$ is t times a unit, which is cancelled. Equation (29.14) is `Fabius.bernoulli_eval_half`.

29.3 Stirling and Touchard representations

The ordinary Bernoulli numbers form a particularly simple signed Stirling transform. The same identity has a useful integral form in terms of the Touchard polynomial

$$\mathsf{T}_n^{\text{Tot}}(z) := \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} z^k.$$

The integral representation below packages the factor $k!/(k+1)$ as a moment; compare the classical Stirling and exponential-polynomial treatments in [7, 1].

Theorem 29.12 (Bernoulli–Stirling formula and Touchard moments). *For every $n \in \mathbb{N}_0$,*

$$\beta_n = \sum_{k=0}^n (-1)^k \frac{k!}{k+1} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (29.15)$$

Let

$$E_1(x) := \int_1^\infty \frac{e^{-xt}}{t} dt = \int_x^\infty \frac{e^{-v}}{v} dv, \quad x > 0. \quad (29.16)$$

Then

$$\beta_n = \int_0^1 \int_0^\infty e^{-s} \mathsf{T}_n^{\text{Tot}}(-us) ds du, \quad (29.17)$$

$$= \int_0^\infty E_1(x) \mathsf{T}_n^{\text{Tot}}(-x) dx. \quad (29.18)$$

Both integrals are absolutely convergent.

Proof. Put $y = e^t - 1$. Since $t = \log(1 + y)$, formal composition gives

$$\frac{t}{e^t - 1} = \frac{\log(1 + y)}{y} = \sum_{k \geq 0} \frac{(-1)^k}{k+1} y^k.$$

The exponential generating function for Stirling numbers is

$$(e^t - 1)^k = k! \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{t^n}{n!}.$$

Substitution and comparison of $t^n/n!$ prove (29.15). Theorem 29.14 below re-proves it as the value at $x = 0$ of a polynomial identity.

Because $\mathsf{T}_n^{\text{Tot}}$ is a polynomial, the first integral may be evaluated term by term:

$$\begin{aligned} \int_0^1 \int_0^\infty e^{-s} \mathsf{T}_n^{\text{Tot}}(-us) ds du &= \sum_{k=0}^n (-1)^k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \left(\int_0^1 u^k du \right) \left(\int_0^\infty s^k e^{-s} ds \right) \\ &= \sum_{k=0}^n (-1)^k \frac{k!}{k+1} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \end{aligned}$$

Absolute convergence follows from $|\mathsf{T}_n^{\text{Tot}}(-us)| \leq C_n(1 + s^n)$ for $0 \leq u \leq 1$.

For the single integral, Tonelli's theorem gives, for every $k \geq 0$,

$$\int_0^\infty x^k E_1(x) dx = \int_1^\infty \frac{dt}{t} \int_0^\infty x^k e^{-tx} dx = k! \int_1^\infty t^{-k-2} dt = \frac{k!}{k+1}.$$

Termwise integration of the finite Touchard polynomial gives (29.18). The integral is absolutely convergent: for $0 < x \leq 1$,

$$E_1(x) \leq \log(1/x) + \int_1^\infty \frac{e^{-v}}{v} dv,$$

whereas $E_1(x) \leq e^{-x}/x$ for $x \geq 1$.

Remark 29.13 (Formal crosswalk). Equation (29.15) is `Fabius.bernoulli_eq_sum_stirlingSecond` (module `BernoulliStirling`) for Mathlib’s Bernoulli numbers `bernoulli`, whose convention $B_1 = -1/2$ matches the generating function $t/(e^t - 1)$ used here. The Lean proof is the first paragraph of this proof made formal: `Mathlib.bernoulliPowerSeries_mul_exp_sub_one` gives $\mathcal{B}(t)(e^t - 1) = t$, `Fabius.log_subst_exp_sub_one` gives $t = \log(1 + y)$ at $y = e^t - 1$, cancelling the non-zero-divisor $e^t - 1 = t \cdot (e^t - 1)/t$ identifies $\mathcal{B}$ with $\log(1 + y)/y$ composed with $e^t - 1$ (`Fabius.bernoulliPowerSeries_eq_logDivSeries_subst`), and the coefficients are read off by the exponential composition theorem (`Fabius.egfA_subst_bellWeightSeries`) with the Stirling specialization of the partial Bell polynomials. The two integral representations are not formalized. $\square$

Formula (29.15) is the specialization at $x = 0$ of a polynomial identity in x , which is worth recording separately: it expresses the whole Bernoulli polynomial in the finite-difference basis, and it makes the Bernoulli–Stirling formula a corollary of the elementary evaluation $\Delta^k 0^n = k! \binom{n}{k}$.

Theorem 29.14 (Bernoulli polynomials from finite differences). *Let Δ denote the forward difference in x , so that $\Delta^k x^n = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} (x + j)^n$. Then for every $n \in \mathbb{N}_0$,*

$$\beta_n(x) = \sum_{k=0}^n \frac{(-1)^k}{k+1} \Delta^k x^n. \quad (29.19)$$

Proof. The proof of Theorem 29.12 already established, by formal composition of $\log(1 + y)/y$ with $y = e^t - 1 \in t\mathbb{Q}[[t]]$, that

$$\frac{t}{e^t - 1} = \sum_{k \geq 0} \frac{(-1)^k}{k+1} (e^t - 1)^k \quad (29.20)$$

in $\mathbb{Q}[[t]]$; the composition is legitimate, and the sum contributes to each power of t only finitely often, because $(e^t - 1)^k \in t^k \mathbb{Q}[[t]]$. Multiplying (29.20) by e^{xt} in $\mathbb{Q}[x][[t]]$ gives

$$\frac{te^{xt}}{e^t - 1} = \sum_{k \geq 0} \frac{(-1)^k}{k+1} e^{xt} (e^t - 1)^k.$$

Finally $e^{xt}(e^t - 1)^k = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} e^{(x+j)t}$, whose coefficient of $t^n/n!$ is $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} (x + j)^n = \Delta^k x^n$. Comparing coefficients of $t^n/n!$, and noting that $\Delta^k x^n = 0$ for $k > n$ since Δ lowers the degree, yields (29.19). $\square$

Remark 29.15 (Formal crosswalk). `Fabius.bernoulli_eq_sum_fwdDiff` (module `BernoulliNewtonBasis`) is Equation (29.19) as an identity in $\mathbb{Q}[x]$, with $\Delta^k x^n$ written out as $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} (x + j)^n$. The formal proof works in $\mathbb{Q}[x][[t]]$: `Mathlib.Polynomial.bernoulli_generating_function` and `bernoulliPowerSeries_mul_exp_sub_one` give $\sum_n \beta_n(x) t^n/n! = e^{xt} \cdot t/(e^t - 1)$ (`Fabius.bernoulliPolySeries_eq`), the composition Equation (29.20) is used only through its truncation $t/(e^t - 1) \equiv \sum_{k \leq n} \frac{(-1)^k}{k+1} (e^t - 1)^k$ modulo t^{n+1} (`Fabius.X_pow_dvd_bernoulliPowerSeries_sub_gregory`, whose coefficients are compared through the Bernoulli–Stirling formula `Fabius.bernoulli_eq_sum_stirlingSecond` and $(e^t - 1)^k = k! \sum_n S(n, k) t^n/n!$), and $e^{xt}(e^t - 1)^k = \sum_j (-1)^{k-j} \binom{k}{j} e^{(x+j)t}$ is the binomial theorem for `Mathlib.PowerSeries.exp_mul_exp_eq_exp_add`.

Corollary 29.16 (Second proof of the Bernoulli–Stirling formula). *Evaluating (29.19) at $x = 0$ recovers (29.15), by a route independent of the Stirling exponential generating function used in Theorem 29.12.*

Proof. It suffices to evaluate $\Delta^k 0^n$. By the surjection formula Theorem 2.5,

$$\Delta^k 0^n = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\},$$

this being the inclusion–exclusion count of the surjections $[n] \rightarrow [k]$, of which there are $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$. Substituting into (29.19) at $x = 0$ gives

$$\beta_n = \sum_{k=0}^n \frac{(-1)^k}{k+1} k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\},$$

which is (29.15). □

Remark 29.17 (Formal crosswalk). `Fabius.bernoulli_eq_sum_fwdDiff_zero` (module `BernoulliNewtonBasis`) is the evaluation at $x = 0$, $\beta_n = \sum_{k \leq n} \frac{(-1)^k}{k+1} \sum_{j \leq k} (-1)^{k-j} \binom{k}{j} j^n$, and the surjection formula $\Delta^k 0^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ is `Fabius.factorial_mul_stirlingSecond_eq_sum`, so that the two together are `Fabius.bernoulli_eq_sum_stirlingSecond`. (In the formal development the Newton-basis identity itself is derived through the Bernoulli–Stirling formula, so it is not an independent second proof there.)

29.4 Nörlund’s generalized Bernoulli polynomials

The ordinary Bernoulli polynomials sit inside the two-parameter Nörlund family [18]. The order parameter makes convolution visible and turns an isolated exponential diagonal into a polynomial identity.

Definition 29.18 (Generalized Bernoulli polynomials). For $\alpha, x \in \mathbb{C}$, define $\beta_n^{(\alpha)}(x)$ by

$$\left(\frac{t}{e^t - 1} \right)^\alpha e^{xt} = \sum_{n \geq 0} \beta_n^{(\alpha)}(x) \frac{t^n}{n!}. \quad (29.21)$$

Formally, the power means $\exp(\alpha \log(t/(e^t - 1)))$, whose inner logarithm has constant term zero. Analytically, the formula holds for $|t| < 2\pi$, using the unique holomorphic logarithm of $t/(e^t - 1)$ that vanishes at $t = 0$. Thus $\beta_n^{(0)}(x) = x^n$ and $\beta_n^{(1)}(x) = \beta_n(x)$.

Theorem 29.19 (Appell, difference, and convolution calculus). For $\alpha, x, y \in \mathbb{C}$ and $n \geq 1$,

$$\frac{d}{dx} \beta_n^{(\alpha)}(x) = n \beta_{n-1}^{(\alpha)}(x), \quad (29.22)$$

$$\beta_n^{(\alpha)}(x+y) = \sum_{k=0}^n \binom{n}{k} \beta_k^{(\alpha)}(x) y^{n-k}, \quad (29.23)$$

$$\beta_n^{(\alpha)}(x+1) - \beta_n^{(\alpha)}(x) = n \beta_{n-1}^{(\alpha-1)}(x). \quad (29.24)$$

More generally, for $\alpha, \gamma, x, y \in \mathbb{C}$ and $n \geq 0$,

$$\beta_n^{(\alpha+\gamma)}(x+y) = \sum_{k=0}^n \binom{n}{k} \beta_k^{(\alpha)}(x) \beta_{n-k}^{(\gamma)}(y). \quad (29.25)$$

Proof. Differentiating (29.21) with respect to x multiplies its left side by t . Replacing x by $x+y$ multiplies it by e^{yt} . These observations and the exponential Cauchy product prove the first two formulas. For the difference identity,

$$\left(\frac{t}{e^t - 1} \right)^\alpha (e^{(x+1)t} - e^{xt}) = t \left(\frac{t}{e^t - 1} \right)^{\alpha-1} e^{xt}.$$

Coefficient comparison proves the third formula. Multiplying the generating functions of orders α and γ proves the convolution identity. $\square$

Remark 29.20 (Formal crosswalk). Module `NorlundPolynomials` defines $\beta_n^{(a)}(x)$ for natural orders $a \in \mathbb{N}$ by Equation (29.21) with the a -th power of Mathlib’s `bernoulliPowerSeries` (`Fabius.norlund`), so that $\beta_n^{(0)} = x^n$ and $\beta_n^{(1)} = \beta_n$ (`Fabius.norlund_zero`, `Fabius.norlund_one`). For $a, c \in \mathbb{N}$ and rational x, y : Equation (29.22) is `Fabius.derivative_norlund_succ`, proved by differentiating the coefficients of $G(t)e^{xt}$ in x (`Fabius.derivative_coeff_succ_map_C_mul_rescale_exp`); Equation (29.25) is `Fabius.norlund_add_eval_add`, from the product of the two exponential generating functions (`Fabius.egfA_norlund_eval`, `Fabius.egfA_mul`); Equation (29.23) is its case $\gamma = 0$ (`Fabius.norlund_eval_add`); and Equation (29.24) is `Fabius.norlund_succ_eval_add_one_sub`, from $(e^{(x+1)t} - e^{xt})t/(e^t - 1) = te^{xt}$. Complex orders, which need $\exp(\alpha \log(t/(e^t - 1)))$, are not formalized.

Theorem 29.21 (Bell-polynomial construction and the Nörlund diagonal). *Put*

$$c_1(\alpha, x) := x - \frac{\alpha}{2}, \quad c_j(\alpha, x) := -\frac{\alpha \beta_j}{j} \quad (j \geq 2). \quad (29.26)$$

Then $\beta_0^{(\alpha)}(x) = 1$, and for $n \geq 1$,

$$\beta_n^{(\alpha)}(x) = \mathcal{B}_n^{\exp}(c_1(\alpha, x), \dots, c_n(\alpha, x)) \quad (29.27)$$

$$= \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{\prod_{j=1}^n m_j! (j!)^{m_j}} \prod_{j=1}^n c_j(\alpha, x)^{m_j}. \quad (29.28)$$

The diagonal order has the closed form

$$\boxed{\beta_n^{(n+1)}(x) = (x-1)^n}. \quad (29.29)$$

Consequently, for $n \geq 1$,

$$\beta_{n-1}^{(n)}(0) = (-1)^{n-1} (n-1)!, \quad (29.30)$$

$$[t^{n-1}] \left(\frac{t}{e^t - 1} \right)^n = (-1)^{n-1}. \quad (29.31)$$

Proof. The Bernoulli generating function gives

$$\frac{d}{dt} \log \frac{t}{e^t - 1} = -\frac{1}{2} - \sum_{j \geq 2} \beta_j \frac{t^{j-1}}{j!}.$$

The logarithm vanishes at $t = 0$, so

$$\log \frac{t}{e^t - 1} = -\frac{t}{2} - \sum_{j \geq 2} \frac{\beta_j}{j} \frac{t^j}{j!}. \quad (29.32)$$

Thus (29.21) is $\exp(\sum_{j \geq 1} c_j(\alpha, x) t^j / j!)$. The complete Bell-polynomial generating function Theorem 12.4 proves the first formula; expanding the Bell polynomial proves the second.

For the diagonal, set $z = e^t - 1$ in a formal residue:

$$\begin{aligned} \frac{\beta_n^{(n+1)}(x)}{n!} &= \operatorname{Res}_t \frac{e^{xt}}{(e^t - 1)^{n+1}} dt = \operatorname{Res}_z \frac{(1+z)^{x-1}}{z^{n+1}} dz \\ &= [z^n] (1+z)^{x-1} = \binom{x-1}{n}. \end{aligned}$$

The middle equality is Theorem 1.3. Multiplication by $n!$ proves the polynomial diagonal. Replace n by $n - 1$ and set $x = 0$ to obtain the number diagonal; division by $(n - 1)!$ gives the coefficient identity. $\square$

Remark 29.22 (Formal crosswalk). Module `NorlundDiagonal` formalizes the three diagonal displays for natural orders: Equation (29.29) is `Fabius.norlund_diagonal`, in the form $\beta_n^{(n+1)}(x) = (x - 1)^n$ with `Mathlib's descPochhammer`; Equation (29.30) is `Fabius.norlund_eval_zero_diagonal`; and Equation (29.31) is `Fabius.coeff_bernoulliPowerSeries_pow_succ`. The formal proof does not use Theorem 1.3, and needs no residue calculus. It rests instead on the Riccati equation $t \frac{d}{dt} \left(\frac{t}{e^t - 1} \right) = \frac{t}{e^t - 1} - \left(\frac{t}{e^t - 1} \right)^2 - \frac{t^2}{e^t - 1}$ (`Fabius.X_mul_derivative_bernoulliPowerSeries`, obtained by differentiating $B(e^t - 1) = t$), which converts into the order-lowering relation $(a + 1)\beta_{n+1}^{(a+2)}(x + 1) = (a - n)\beta_{n+1}^{(a+1)}(x) + (n + 1)x\beta_n^{(a+1)}(x)$ (`Fabius.norlund_series_step` as generating functions, `Fabius.norlund_step` coefficientwise). Taking $a = n$ annihilates the first term and leaves $\beta_{n+1}^{(n+2)}(x + 1) = x\beta_n^{(n+1)}(x)$ (`Fabius.norlund_diagonal_step`); induction on n against $(x - 1)^{n+1} = (x - 1)(x - 2)^n$ then gives the closed form, and $x = 0$ with $(-1)^n = (-1)^n n!$ (`Fabius.descPochhammer_eval_neg_one`) gives the other two. Complex orders are not formalized, nor is the Bell-polynomial construction Equations (29.27) and (29.28).

29.5 Genocchi numbers

Definition 29.23. The Genocchi numbers G_n^{Gen} are defined by

$$\frac{2t}{e^t + 1} = \sum_{n \geq 1} G_n^{\text{Gen}} \frac{t^n}{n!}. \quad (29.33)$$

Theorem 29.24 (Bernoulli–Euler–Genocchi relations). For $n \geq 1$,

$$G_n^{\text{Gen}} = 2(1 - 2^n)\beta_n = nE_{n-1}(0). \quad (29.34)$$

Thus $G_1^{\text{Gen}} = 1$, $G_{2m+1}^{\text{Gen}} = 0$ for $m \geq 1$, and $G_2^{\text{Gen}} = -1$, $G_4^{\text{Gen}} = 1$, $G_6^{\text{Gen}} = -3$, $G_8^{\text{Gen}} = 17$, $G_{10}^{\text{Gen}} = -155$.

Proof. Use

$$\frac{2t}{e^t + 1} = \frac{2t}{e^t - 1} - \frac{4t}{e^{2t} - 1}$$

and compare Bernoulli coefficients. Multiplying the Euler EGF at $x = 0$ by t gives the second equality. Parity and initial values follow immediately. $\square$

Remark 29.25 (Formal crosswalk). Module `GenocchiNumbers` takes the first equality of Equation (29.34) as the definition, `Fabius.genocchi = 2(1 - 2^n)\beta_n`, and proves the defining generating function Equation (29.33) as `Fabius.egf_genocchi_mul_exp_add_one`, $(\sum_n G_n^{\text{Gen}} t^n / n!)(e^t + 1) = 2t$ over any $\mathbb{Q}$ -algebra, by exactly the displayed partial-fraction identity: the series is $2\mathcal{B}(t) - 2\mathcal{B}(2t)$ (`Fabius.egf_genocchi_eq`), and `Mathlib's bernoulliPowerSeries_mul_exp_sub_one` rescaled to $2t$ gives $(\sum_n G_n^{\text{Gen}} t^n / n!)(e^t + 1)(e^t - 1) = 2t(e^t - 1)$, after which $e^t - 1 = t \cdot (\text{unit})$ is cancelled. The parity statement is `Fabius.genocchi_odd` and the initial values are `Fabius.genocchi_one` and `Fabius.genocchi_two`; the Euler-polynomial equality $G_{n+1}^{\text{Gen}} = (n + 1)E_n(0)$ is `Fabius.genocchi_succ_eq` (module `EulerPolynomials`), from the two generating functions.

29.6 Bernoulli polynomials of the second kind and Cauchy polynomials

The historical names are not uniform. The family below is commonly called the *Bernoulli polynomials of the second kind* and also the *Cauchy polynomials of the first kind*. Some supplied manuscripts call it the Cauchy family “of the second kind,” but that phrase is also used for a distinct reciprocal-log normalization. We therefore use the neutral symbol $b_n(x)$ and regard the generating function, rather than the name, as authoritative; see [18].

Definition 29.26. Define $b_n(x)$ by

$$\frac{t(1+t)^x}{\log(1+t)} = \sum_{n \geq 0} b_n(x) \frac{t^n}{n!}. \quad (29.35)$$

This is a formal identity with $(1+t)^x = \sum_{n \geq 0} (x)^n t^n / n!$. Analytically it holds for $|t| < 1$, with the branch determined near $t = 0$.

Theorem 29.27 (Integral, Stirling, difference, addition, and reflection formulas). *For every $n \geq 0$,*

$$b_n(x) = \int_0^1 (x+u)^n du, \quad (29.36)$$

$$b_n(0) = \sum_{k=0}^n \frac{s(n, k)}{k+1}. \quad (29.37)$$

For $n \geq 1$,

$$b'_n(x) = n(x)^{n-1}, \quad (29.38)$$

$$b_n(x+1) - b_n(x) = nb_{n-1}(x), \quad (29.39)$$

$$b_n(x+y) = \sum_{k=0}^n \binom{n}{k} b_k(x) (y)^{n-k}, \quad (29.40)$$

$$b_n(x) = b_n(0) + \sum_{k=1}^n \frac{n}{k} s(n-1, k-1) x^k. \quad (29.41)$$

Finally, for every $n \geq 0$,

$$\boxed{b_n(n-2-x) = (-1)^n b_n(x)}. \quad (29.42)$$

Thus b_n is even or odd about $(n-2)/2$ according as n is even or odd.

Proof. Coefficientwise integration of the falling-factorial binomial series gives

$$\sum_{n \geq 0} \left(\int_0^1 (x+u)^n du \right) \frac{t^n}{n!} = \int_0^1 (1+t)^{x+u} du = (1+t)^x \frac{t}{\log(1+t)}.$$

This proves the integral representation. At $x = 0$, expand $(u)^n = \sum_k s(n, k) u^k$ and integrate to obtain the Stirling formula.

Differentiating the defining EGF with respect to x cancels its logarithm:

$$\sum_{n \geq 0} b'_n(x) \frac{t^n}{n!} = t(1+t)^x.$$

Coefficient comparison proves the derivative formula. Replacing x by $x+1$ and subtracting multiplies the EGF by t , proving the difference formula. Factoring

$$\frac{t(1+t)^{x+y}}{\log(1+t)} = \frac{t(1+t)^x}{\log(1+t)} (1+t)^y$$

proves addition by the exponential Cauchy product. Expanding $b'_n(x) = n \sum_{j=0}^{n-1} s(n-1, j) x^j$ and integrating from 0 to x proves the explicit polynomial formula.

For reflection, use $(n-1-y)^n = (-1)^n (y)^n$ and put $v = 1-u$ in the integral representation:

$$\begin{aligned} b_n(n-2-x) &= \int_0^1 (n-2-x+u)^n du = \int_0^1 (n-1-(x+v))^n dv \\ &= (-1)^n \int_0^1 (x+v)^n dv = (-1)^n b_n(x). \end{aligned}$$

□

Remark 29.28 (Formal crosswalk). Module `CauchyPolynomials` defines $b_n(x)$ directly by Equation (29.35), as $n!$ times the n -th coefficient of $t(1+t)^x / \log(1+t)$ in $\mathbb{Q}[x][[t]]$ (`Fabius.cauchyPoly`), the factor $t/\log(1+t)$ being the power-series inverse of the Gregory series `Fabius.logDivSeries`. The supporting object is the falling-factorial generating function $(1+t)^u = \sum_n (u)^n t^n / n!$ over any $\mathbb{Q}$ -algebra (`Fabius.fallingSeries`, module `FallingFactorialSeries`), characterised by $(1+t)F' = uF$ with $F(0) = 1$; from that characterisation one gets the product law $(1+t)^u(1+t)^v = (1+t)^{u+v}$ (`Fabius.fallingSeries_mul`), hence the Vandermonde identity for falling factorials (`Fabius.descPochhammer_eval_add`), and, in $\mathbb{Q}[x][[t]]$, the x -derivative $\partial_x(1+t)^x = \log(1+t)(1+t)^x$ (`Fabius.Dx_fallingPoly`, with ∂_x the coefficientwise derivative `Fabius.Dx`). Consequently Equation (29.38) is `Fabius.derivative_cauchyPoly_succ`, Equation (29.39) is `Fabius.cauchyPoly_succ_eval_add_one_sub`, Equation (29.40) is `Fabius.cauchyPoly_eval_add` and Equation (29.41) is `Fabius.cauchyPoly_succ_eq`. Equation (29.37) is `Fabius.cauchyPoly_eval_zero`, proved with the formal integral $\int_0^1 p(u) du = \sum_k p_k / (k+1)$ as a linear functional on $\mathbb{Q}[u]$ (`Fabius.intPoly`), for which $\int_0^1 p' = p(1) - p(0)$ (`Fabius.intPoly_derivative`); applying it coefficientwise to $\partial_x(1+t)^x$ gives $\int_0^1 (1+t)^u du = t/\log(1+t)$ (`Fabius.mapIntPoly_fallingPoly`), which is Equation (29.36) in formal form, although the analytic integral representation itself is not formalized. The reflection Equation (29.42) is not formalized.

Remark 29.29. The first polynomials, useful for checking the normalization, are

$$\begin{aligned} b_0(x) &= 1, & b_1(x) &= x + \frac{1}{2}, & b_2(x) &= x^2 - \frac{1}{6}, \\ b_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{1}{4}, & b_4(x) &= x^4 - 4x^3 + 4x^2 - \frac{19}{30}. \end{aligned}$$

In particular $b_n(0) = 1, 1/2, -1/6, 1/4, -19/30, \dots$, which distinguishes this normalization from the other Cauchy families.

29.7 A modified Bernoulli normalization

Define the even ordinary-series coefficients $\beta_{2m}^\sharp$ by

$$\frac{1}{2} \log \left(\frac{e^{x/2} - e^{-x/2}}{x/2} \right) = \sum_{m \geq 0} \beta_{2m}^\sharp x^{2m}, \quad \beta_0^\sharp = \frac{1}{2} \log 2. \quad (29.43)$$

Proposition 29.30. For even $n \geq 2$,

$$\beta_n^\sharp = \frac{\beta_n}{2n^2 \Gamma(n)}, \quad (29.44)$$

and odd coefficients vanish. In particular, $\beta_2^\sharp = 1/48$, $\beta_4^\sharp = -1/5760$, and $\beta_6^\sharp = 1/362880$.

Proof. Differentiate the defining series:

$$\frac{1}{4} \coth \frac{x}{2} - \frac{1}{2x} = \frac{1}{2x} \left(\frac{x}{2} \coth \frac{x}{2} - 1 \right).$$

The Bernoulli generating function implies

$$\frac{x}{2} \coth \frac{x}{2} = \sum_{m \geq 0} \beta_{2m} \frac{x^{2m}}{(2m)!}.$$

Integrating termwise gives $\beta_{2m}^\# = \beta_{2m}/(4m(2m)!)$, which is the displayed formula. $\square$

29.8 Periodic Bernoulli functions and Fourier expansion

The value of the first periodic Bernoulli function at its jumps must be chosen separately: a Fourier series converges at a jump to the midpoint of the two one-sided limits.

Definition 29.31 (Periodic Bernoulli functions). For $n \geq 2$, put $\tilde{\beta}_n(x) := \beta_n(\{x\})$. For $n = 1$, put

$$\tilde{\beta}_1(x) := \begin{cases} \{x\} - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}. \end{cases} \quad (29.45)$$

Thus $\tilde{\beta}_n$ is continuous and 1-periodic for $n \geq 2$, whereas $\tilde{\beta}_1$ is the midpoint-normalized periodic sawtooth.

Theorem 29.32 (Fourier expansion with exact convergence qualifications). For $n \geq 2$ and every real x ,

$$\tilde{\beta}_n(x) = -n! \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{2\pi i k x}}{(2\pi i k)^n}. \quad (29.46)$$

The series converges absolutely and uniformly. In degree one,

$$\tilde{\beta}_1(x) = - \lim_{N \rightarrow \infty} \sum_{0 < |k| \leq N} \frac{e^{2\pi i k x}}{2\pi i k} = -\frac{1}{\pi} \sum_{k=1}^{\infty} \frac{\sin(2\pi k x)}{k}. \quad (29.47)$$

The symmetric complex series and the paired sine series converge at every real x , and their value at an integer is 0. The complex series is nowhere absolutely convergent, and its symmetric partial sums do not converge uniformly. (The paired sine series is trivially zero at half-integers, so it would be incorrect to call that particular scalar series conditional there.)

Pairing positive and negative frequencies gives, for $m \geq 1$,

$$\tilde{\beta}_{2m}(x) = (-1)^{m+1} \frac{2(2m)!}{(2\pi)^{2m}} \sum_{k=1}^{\infty} \frac{\cos(2\pi k x)}{k^{2m}}, \quad (29.48)$$

$$\tilde{\beta}_{2m+1}(x) = (-1)^{m+1} \frac{2(2m+1)!}{(2\pi)^{2m+1}} \sum_{k=1}^{\infty} \frac{\sin(2\pi k x)}{k^{2m+1}}. \quad (29.49)$$

In particular,

$$\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{2(2m)!} \beta_{2m}. \quad (29.50)$$

For $n \geq 2$,

$$\sup_{x \in \mathbb{R}} |\tilde{\beta}_n(x)| \leq \frac{2n! \zeta(n)}{(2\pi)^n}. \quad (29.51)$$

Proof. For $0 < r < 1$ and $0 < x < 1$, the logarithm series gives

$$\sum_{k=1}^{\infty} \frac{r^k e^{2\pi i k x}}{k} = -\log(1 - r e^{2\pi i x}).$$

Taking imaginary parts and then letting $r \uparrow 1$ yields

$$\sum_{k=1}^{\infty} \frac{\sin(2\pi k x)}{k} = \pi \left(\frac{1}{2} - x \right),$$

because $\arg(1 - e^{2\pi i x}) = \pi x - \pi/2$. Dirichlet's test gives the same ordinary-series limit: $1/k$ decreases to zero and the sine partial sums are bounded for fixed nonintegral x . At an integer every sine term is zero. The complex series is nowhere absolutely convergent because the sum of the moduli of its terms is a constant multiple of $\sum_{k \geq 1} 1/k$. Its symmetric partial sums are continuous, whereas their pointwise limit has jumps at the integers; hence convergence cannot be uniform. This proves the degree-one statement and its convergence claims.

For $n \geq 1$, let

$$c_k^{(n)} = \int_0^1 \beta_n(x) e^{-2\pi i k x} dx.$$

The mean coefficient is zero, because the Appell identity gives

$$\int_0^1 \beta_n(x) dx = \frac{\beta_{n+1}(1) - \beta_{n+1}(0)}{n+1} = 0.$$

For $k \neq 0$ and $n \geq 2$, the endpoint values agree, so integration by parts and $\beta'_n = n\beta_{n-1}$ give $c_k^{(n)} = n c_k^{(n-1)} / (2\pi i k)$. Direct integration gives $c_k^{(1)} = -1/(2\pi i k)$; hence

$$c_k^{(n)} = -\frac{n!}{(2\pi i k)^n} \quad (k \neq 0). \quad (29.52)$$

For $n \geq 2$, the claimed series is absolutely and uniformly convergent by comparison with $2n! \sum_{k \geq 1} k^{-n} / (2\pi)^n$. It defines a continuous 1-periodic function with exactly the coefficients just computed. To make the reconstruction step explicit, let h be its difference from $\tilde{\beta}_n$. Every Fourier coefficient of h is zero, so every Fejér mean is zero:

$$\sigma_N h(x) = \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1}\right) \hat{h}_{2\pi}(k) e^{2\pi i k x} = 0.$$

But $\sigma_N h = h * K_N$, where

$$K_N(y) = \frac{1}{N+1} \left(\frac{\sin \pi(N+1)y}{\sin \pi y} \right)^2.$$

The Fejér kernels are nonnegative, have integral one, and concentrate at zero. Uniform continuity of h therefore gives $\sigma_N h \rightarrow h$ uniformly, so $h = 0$.

Pairing k and $-k$ proves the sine and cosine formulas. Put $x = 0$ in the even formula to obtain the zeta identity. Taking absolute values term by term in the complex formula proves the supremum bound. $\square$

29.9 Euler–Maclaurin summation

Put

$$P_r(x) := \frac{\tilde{\beta}_r(x)}{r!}.$$

The value assigned to P_1 at an integer does not affect the integrals below.

Theorem 29.33 (Euler–Maclaurin with remainder). *Let $m \geq 1$ be an integer, let $a < b$ be integers, and let $f \in C^{2m}([a, b])$. Then*

$$\begin{aligned} \sum_{k=a}^b f(k) &= \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} \\ &\quad + \sum_{r=1}^m \frac{\beta_{2r}}{(2r)!} (f^{(2r-1)}(b) - f^{(2r-1)}(a)) + R_m, \end{aligned} \quad (29.53)$$

where

$$R_m = -\frac{1}{(2m)!} \int_a^b \beta_{2m}(\{x\}) f^{(2m)}(x) dx \quad (29.54)$$

and

$$|R_m| \leq \frac{2\zeta(2m)}{(2\pi)^{2m}} \int_a^b |f^{(2m)}(x)| dx. \quad (29.55)$$

Proof. On $[k, k+1]$, integration by parts gives

$$\int_k^{k+1} \left(x - k - \frac{1}{2}\right) f'(x) dx = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} f(x) dx.$$

Summing yields the trapezoidal term plus $\int_a^b P_1 f'$. Integrate by parts repeatedly, using $P'_r = P_{r-1}$ away from integers and continuity of P_r at integers for $r \geq 2$. Boundary values are $\beta_r/r!$; odd Bernoulli numbers beyond β_1 vanish. The surviving boundary terms and final integral are exactly (29.53) and (29.54).

By (29.51),

$$\sup_{x \in \mathbb{R}} |P_{2m}(x)| \leq \frac{2\zeta(2m)}{(2\pi)^{2m}}.$$

Substitution in (29.54) proves (29.55). $\square$

Both asymptotic applications below — Stirling’s expansion and the harmonic-number expansion — need the same manoeuvre: the lower endpoint contributes nothing that depends on N , and its contribution has to be collected into a single constant that is then identified by a limit. It is worth isolating that step once.

Corollary 29.34 (Constant-of-integration form). *Let $m \geq 1$, let a be an integer, and let $f \in C^{2m}([a, \infty))$ have $f^{(2m)}$ absolutely integrable on $[a, \infty)$. Then there is a constant $C = C(f, a, m)$, independent of N , such that for every integer $N > a$*

$$\sum_{k=a}^N f(k) = \int_a^N f(x) dx + \frac{f(N)}{2} + C + \sum_{r=1}^m \frac{\beta_{2r}}{(2r)!} f^{(2r-1)}(N) + \rho_m(N), \quad (29.56)$$

where

$$\rho_m(N) = \frac{1}{(2m)!} \int_N^\infty \beta_{2m}(\{x\}) f^{(2m)}(x) dx, \quad |\rho_m(N)| \leq \frac{2\zeta(2m)}{(2\pi)^{2m}(2m)!} \int_N^\infty |f^{(2m)}(x)| dx. \quad (29.57)$$

Proof. Take $b = N$ in (29.53) and split the remainder integral as $\int_a^N = \int_a^\infty - \int_N^\infty$, which is legitimate because $\beta_{2m}(\{\cdot\})$ is bounded and $f^{(2m)}$ is absolutely integrable. Every term that involves only a — namely $f(a)/2$, the boundary sum $-\sum_r \beta_{2r} f^{(2r-1)}(a)/(2r)!$, and $-\int_a^\infty \beta_{2m}(\{x\}) f^{(2m)}(x) dx/(2m)!$ — is independent of N , and their sum is the constant C . What survives is (29.56), and the bound on ρ_m is (29.51) applied to the tail. $\square$

Corollary 29.35 (Stirling’s expansion). *For each fixed integer $M \geq 1$, as $n \rightarrow \infty$ through positive integers,*

$$\log(n!) = \left(n + \frac{1}{2}\right) \log n - n + \frac{1}{2} \log(2\pi) + \sum_{r=1}^{M-1} \frac{\beta_{2r}}{2r(2r-1)n^{2r-1}} + \mathcal{O}(n^{-(2M-1)}). \quad (29.58)$$

Proof. Apply Theorem 29.34 to $f(x) = \log x$ with $a = 1$; the hypothesis holds because $|f^{(2M)}(x)| = (2M-1)! x^{-2M}$ is integrable on $[1, \infty)$. Here $\int_1^N \log x dx = N \log N - N + 1$ and $f(N)/2 = \frac{1}{2} \log N$, while $f^{(2r-1)}(x) = (2r-2)! x^{-(2r-1)}$ gives the displayed inverse powers $\beta_{2r}/(2r(2r-1)N^{2r-1})$. The constant C supplied by the corollary is evaluated as $\frac{1}{2} \log(2\pi) - 1$ by Wallis’ product, and the remainder bound one order farther gives the stated error. $\square$

The same machine applied to $f(x) = 1/x$ produces the companion expansion for the harmonic numbers, and identifies the Euler–Mascheroni constant as the associated constant of integration. This is the expansion used silently in several coefficient estimates elsewhere in this work.

Corollary 29.36 (Harmonic-number expansion). *Let $H_N = \sum_{k=1}^N 1/k$. For each fixed integer $M \geq 1$, as $N \rightarrow \infty$,*

$$H_N = \log N + \gamma + \frac{1}{2N} - \sum_{r=1}^{M-1} \frac{\beta_{2r}}{2r N^{2r}} + \mathcal{O}(N^{-2M}), \quad (29.59)$$

so that explicitly $H_N = \log N + \gamma + \frac{1}{2N} - \frac{1}{12N^2} + \frac{1}{120N^4} + \mathcal{O}(N^{-6})$.

Proof. Apply Theorem 29.34 with $m = M$, $f(x) = 1/x$ and $a = 1$; the hypothesis holds because $|f^{(2M)}(x)| = (2M)! x^{-2M-1}$ is integrable on $[1, \infty)$. Here $\int_1^N f = \log N$, $f(N)/2 = 1/(2N)$, and $f^{(2r-1)}(x) = -(2r-1)! x^{-2r}$, so $\beta_{2r} f^{(2r-1)}(N)/(2r)! = -\beta_{2r}/(2r N^{2r})$. Hence

$$H_N = \log N + \frac{1}{2N} + C - \sum_{r=1}^M \frac{\beta_{2r}}{2r N^{2r}} + \rho_M(N). \quad (29.60)$$

By (29.57),

$$|\rho_M(N)| \leq \frac{2\zeta(2M)}{(2\pi)^{2M}(2M)!} \int_N^\infty (2M)! x^{-2M-1} dx = \frac{2\zeta(2M)}{(2M)(2\pi)^{2M}} N^{-2M} = \mathcal{O}(N^{-2M}),$$

so every term of (29.60) except C tends to 0 or is accounted for, and letting $N \rightarrow \infty$ gives $C = \lim_N (H_N - \log N) = \gamma$ — for every M at once, since C is the only N -independent term. Finally the term $r = M$ of the sum is itself $\mathcal{O}(N^{-2M})$ and may be absorbed into the error, which leaves (29.59). The displayed special case uses $\beta_2 = 1/6$ and $\beta_4 = -1/30$. $\square$

Chapter 30

Pochhammer symbols, polygamma functions, and Barnes' G -function

30.1 Pochhammer algebra

Definition 30.1. The rising Pochhammer symbol is

$$(a)_n = (a)^{\overline{n}} = a(a+1) \cdots (a+n-1), \quad (a)_0 = 1. \quad (30.1)$$

Theorem 30.2 (Gamma quotient and binomial series). *Where defined meromorphically,*

$$(a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}, \quad (30.2)$$

and, formally in z and analytically for $|z| < 1$,

$$(1-z)^{-a} = \sum_{n \geq 0} \frac{(a)_n}{n!} z^n. \quad (30.3)$$

Moreover, for all $m, n \in \mathbb{N}_0$,

$$(a)_{m+n} = (a)_m (a+m)_n, \quad (a)^{\overline{m+n}} = (a)^{\overline{m}} (a-m)^{\overline{n}}. \quad (30.4)$$

Proof. Iterate $\Gamma(w+1) = w\Gamma(w)$ to obtain the quotient. For the series, let $F = \sum (a)_n z^n / n!$. The recurrence $(a)_{n+1} = (a+n)(a)_n$ gives $(1-z)F' = aF$ and $F(0) = 1$, whose unique formal solution is $(1-z)^{-a}$. The ratio test gives analytic convergence in the unit disk. The splitting rules (30.4) are immediate from the definition: the product $a(a+1) \cdots (a+m+n-1)$ breaks after its m th factor, and the falling statement is the same identity read in the opposite direction. $\square$

Corollary 30.3 (Chu–Vandermonde).

$$\frac{(a+b)_n}{n!} = \sum_{k=0}^n \frac{(a)_k}{k!} \frac{(b)_{n-k}}{(n-k)!}. \quad (30.5)$$

Proof. Multiply the binomial series for exponents a and b and compare with the series for exponent $a+b$. $\square$

For $r \geq 1$, define shifted generalized harmonic sums

$$H_n^{(r)}(a) = \sum_{j=0}^{n-1} (a+j)^{-r}, \quad H_n(a) = H_n^{(1)}(a). \quad (30.6)$$

Theorem 30.4 (All parameter derivatives). *For $m \geq 1$ away from the apparent poles,*

$$\frac{\partial^m}{\partial a^m}(a)_n = (a)_n \mathcal{B}_m^{\text{exp}} \left(H_n(a), -1!H_n^{(2)}(a), 2!H_n^{(3)}(a), \dots, (-1)^{m-1}(m-1)!H_n^{(m)}(a) \right). \quad (30.7)$$

For $m = 0$ the corresponding statement is the identity $(a)_n = (a)_n \mathcal{B}_0^{\text{exp}}$, with $\mathcal{B}_0^{\text{exp}} = 1$. The right side extends polynomially across all removable singularities.

Proof. Write $L(a) = \log(a)_n$. Then $L^{(r)}(a) = (-1)^{r-1}(r-1)!H_n^{(r)}(a)$. Apply the exponential case of Faà di Bruno to $e^{L(a)}$. Since the left side is a polynomial in a , agreement off the zeros supplies the polynomial continuation. $\square$

The reciprocal needs no separate computation. The complete Bell polynomial $\mathcal{B}_m^{\text{exp}}$ is a sum of monomials $x_1^{j_1} x_2^{j_2} \cdots$ with $\sum_i i j_i = m$, so substituting $x_i \mapsto (-1)^i x_i$ multiplies every monomial by $(-1)^{\sum_i i j_i} = (-1)^m$:

$$\mathcal{B}_m^{\text{exp}}(-x_1, x_2, -x_3, \dots, (-1)^m x_m) = (-1)^m \mathcal{B}_m^{\text{exp}}(x_1, x_2, x_3, \dots, x_m). \quad (30.8)$$

Corollary 30.5 (Derivatives of a reciprocal Pochhammer symbol). *For $m \geq 1$ and $(a)_n \neq 0$,*

$$\frac{\partial^m}{\partial a^m} \frac{1}{(a)_n} = \frac{(-1)^m}{(a)_n} \mathcal{B}_m^{\text{exp}} \left(H_n(a), -1!H_n^{(2)}(a), 2!H_n^{(3)}(a), \dots, (-1)^{m-1}(m-1)!H_n^{(m)}(a) \right), \quad (30.9)$$

with exactly the arguments of (30.7). In particular the two derivative arrays differ only by the alternating sign $(-1)^m$.

Proof. With $L(a) = \log(a)_n$ as above, $1/(a)_n = e^{-L(a)}$, and the exponential Faà di Bruno formula gives

$$\frac{\partial^m}{\partial a^m} e^{-L(a)} = e^{-L(a)} \mathcal{B}_m^{\text{exp}}(-L'(a), -L''(a), \dots, -L^{(m)}(a)).$$

Since $L^{(r)}(a) = (-1)^{r-1}(r-1)!H_n^{(r)}(a)$, the r th argument is $-L^{(r)}(a) = (-1)^r(r-1)!H_n^{(r)}(a)$, which is $(-1)^r$ times the r th argument in (30.7). Now apply (30.8). $\square$

30.2 Euler's limit and the Weierstrass product

The partial-fraction series of the next section are obtained by differentiating Weierstrass' product for $1/\Gamma$. That product is proved here, so that the chapter rests on nothing beyond the integral definition of Γ . The one ingredient is the Beta integral with an integer parameter, which is also the exact statement that converts a Pochhammer symbol into an integral.

Proposition 30.6 (Beta integral with an integer parameter). *For $\Re a > 0$ and every integer $n \geq 0$,*

$$\int_0^1 t^{a-1}(1-t)^n dt = \frac{n!}{(a)_{n+1}}. \quad (30.10)$$

Proof. Induct on n . For $n = 0$ the integral is $1/a = 0!/(a)_1$. For $n \geq 1$, integration by parts with $u = (1-t)^n$ and $dv = t^{a-1} dt$ gives, the boundary terms vanishing because $\Re a > 0$ and $n \geq 1$,

$$\int_0^1 t^{a-1}(1-t)^n dt = \frac{n}{a} \int_0^1 t^a(1-t)^{n-1} dt.$$

The inductive hypothesis at parameter $a + 1$ turns the right side into $\frac{n}{a} \cdot \frac{(n-1)!}{(a+1)_n} = \frac{n!}{a(a+1)_n} = \frac{n!}{(a)_{n+1}}$. $\square$

Theorem 30.7 (Euler's limit and the Weierstrass product). *For $\Re z > 0$,*

$$\Gamma(z) = \lim_{N \rightarrow \infty} \frac{N! N^z}{z(z+1) \cdots (z+N)}, \quad (30.11)$$

locally uniformly on $\{\Re z > 0\}$. Moreover, for every $z \in \mathbb{C}$,

$$\frac{1}{\Gamma(z)} = z e^{\gamma z} \prod_{k \geq 1} \left(1 + \frac{z}{k}\right) e^{-z/k}, \quad (30.12)$$

the product converging locally uniformly, so that $1/\Gamma$ is entire with simple zeros exactly at $0, -1, -2, \dots$

Proof. Put $\Gamma_N(z) = \int_0^N \left(1 - \frac{t}{N}\right)^N t^{z-1} dt$. Substituting $t = Nu$ and applying Theorem 30.6 gives

$$\Gamma_N(z) = N^z \int_0^1 (1-u)^N u^{z-1} du = \frac{N^z N!}{(z)_{N+1}} = \frac{N! N^z}{z(z+1) \cdots (z+N)}. \quad (30.13)$$

For $0 \leq t \leq N$ the elementary inequality $0 \leq (1 - t/N)^N \leq e^{-t}$ holds, and $(1 - t/N)^N \rightarrow e^{-t}$ pointwise. On a compact subset of $\{\Re z > 0\}$ one has $\delta \leq \Re z \leq M$ for some $0 < \delta \leq M$, whence $|t^{z-1}| = t^{\Re z-1} \leq t^{\delta-1} + t^{M-1}$, an integrable majorant against e^{-t} on $(0, \infty)$. Dominated convergence therefore gives $\Gamma_N(z) \rightarrow \int_0^\infty e^{-t} t^{z-1} dt = \Gamma(z)$ for each fixed z with $\Re z > 0$. The convergence is in fact locally uniform, but that does not follow from dominated convergence alone: each Γ_N is holomorphic on $\{\Re z > 0\}$, and the same majorant bounds the whole family uniformly on each compact subset, so Vitali's theorem upgrades pointwise convergence of a locally bounded holomorphic family to local uniform convergence. This is (30.11).

For the product, invert (30.13):

$$\frac{1}{\Gamma_N(z)} = z e^{-z \log N} \prod_{k=1}^N \left(1 + \frac{z}{k}\right) = z e^{\gamma_N z} \prod_{k=1}^N \left(1 + \frac{z}{k}\right) e^{-z/k},$$

where $\gamma_N = \sum_{k=1}^N \frac{1}{k} - \log N$, since the inserted factors $e^{-z/k}$ contribute exactly $e^{-z \sum_{k \leq N} 1/k}$. Now $\gamma_N \rightarrow \gamma$, and the product converges locally uniformly on $\mathbb{C}$: for $|w| \leq \frac{1}{2}$ one has $|\log((1+w)e^{-w})| \leq |w|^2$, so on $|z| \leq \rho$ all factors with $k \geq 2\rho$ satisfy $|\log((1+z/k)e^{-z/k})| \leq \rho^2/k^2$, a convergent bound. Letting $N \rightarrow \infty$ gives (30.12) on $\{\Re z > 0\}$, and both sides are entire, so the identity persists on $\mathbb{C}$ by analytic continuation. The zeros of the right side are visibly simple and located at $0, -1, -2, \dots$ $\square$

30.3 Digamma and polygamma series

Define $\psi = \Gamma'/\Gamma$ and $\psi^{(m)} = D^m \psi$.

Theorem 30.8 (Partial fractions). *Away from nonpositive integers,*

$$\psi(z) = -\gamma + \sum_{k=0}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+z} \right), \quad (30.14)$$

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k=0}^{\infty} (k+z)^{-m-1} \quad (m \geq 1). \quad (30.15)$$

Proof. Logarithmically differentiate the Weierstrass product (30.12), which Theorem 30.7 establishes:

$$-\psi(z) = \frac{1}{z} + \gamma + \sum_{k \geq 1} \left(\frac{1}{k+z} - \frac{1}{k} \right).$$

Combining the term $1/z$ with the sum, and pairing $1/k$ with the index shift, gives (30.14). The local uniform convergence proved in Theorem 30.7 is normal on compact sets away from the poles, so the series may be differentiated termwise any number of times, which gives (30.15). $\square$

Corollary 30.9 (Recurrences and special values).

$$\psi(z+1) = \psi(z) + z^{-1}, \quad (30.16)$$

$$\psi^{(m)}(z+1) = \psi^{(m)}(z) + (-1)^m m! z^{-m-1}, \quad (30.17)$$

$$\psi(n) = H_{n-1} - \gamma, \quad (30.18)$$

$$\psi^{(m)}(1) = (-1)^{m+1} m! \zeta(m+1). \quad (30.19)$$

Proof. Subtract the partial-fraction series at consecutive arguments; the series telescopes. Iterate from $\psi(1) = -\gamma$, and set $z = 1$ in the polygamma series. $\square$

Corollary 30.10 (Hurwitz-zeta form). For $m \geq 1$ and $\Re z > 0$,

$$\psi^{(m)}(z) = (-1)^{m+1} m! \zeta(m+1, z). \quad (30.20)$$

Proof. The Hurwitz zeta function is $\zeta(s, z) = \sum_{k \geq 0} (k+z)^{-s}$ for $\Re s > 1$ and $\Re z > 0$, which is precisely the series in (30.15) at $s = m+1$. $\square$

This is the bridge to the Stirling–Hurwitz sums of Theorem 7.5: the same Hurwitz values occur there as the limits of first-kind Stirling series, and (30.20) identifies them as polygamma values.

Theorem 30.11 (Reflection). For $z \notin \mathbb{Z}$,

$$\psi(1-z) - \psi(z) = \pi \cot(\pi z), \quad (30.21)$$

$$(-1)^m \psi^{(m)}(1-z) - \psi^{(m)}(z) = \pi \frac{d^m}{dz^m} \cot(\pi z). \quad (30.22)$$

Proof. Multiplying the Weierstrass products for $1/\Gamma(1+z)$ and $1/\Gamma(1-z)$ and using Euler's sine product gives $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$. Logarithmic differentiation proves the first formula; differentiating it m more times proves the second. $\square$

Theorem 30.12 (Gauss multiplication). For $q \in \mathbb{N}_{>0}$,

$$\Gamma(qz) = (2\pi)^{(1-q)/2} q^{qz-1/2} \prod_{r=0}^{q-1} \Gamma\left(z + \frac{r}{q}\right). \quad (30.23)$$

Hence

$$\sum_{r=0}^{q-1} \psi\left(z + \frac{r}{q}\right) = q\psi(qz) - q \log q, \quad (30.24)$$

$$\sum_{r=0}^{q-1} \psi^{(m)}\left(z + \frac{r}{q}\right) = q^{m+1} \psi^{(m)}(qz). \quad (30.25)$$

Proof. Define

$$Q(z) = \frac{q^{qz-1/2} \prod_{r=0}^{q-1} \Gamma(z + r/q)}{\Gamma(qz)}.$$

The gamma recurrence shows $Q(z+1/q) = Q(z)$. Stirling's formula in a right half-plane gives $Q(z) \rightarrow (2\pi)^{(q-1)/2}$ as $\Re z \rightarrow +\infty$ in a fixed horizontal strip. Periodicity therefore makes Q equal to that constant; meromorphic continuation extends the identity throughout its domain. Solving for $\Gamma(qz)$ proves (30.23). Logarithmic differentiation gives the digamma identity and further differentiation gives the polygamma identity. $\square$

Corollary 30.13. For every $m \geq 0$,

$$\frac{\partial^{m+1}}{\partial a^{m+1}} \log(a)_n = \psi^{(m)}(a+n) - \psi^{(m)}(a) = (-1)^m m! H_n^{(m+1)}(a). \quad (30.26)$$

Proof. Logarithmically differentiate the gamma quotient and compare with direct differentiation of the finite product. $\square$

30.4 Barnes' function

Theorem 30.14 (Canonical product). *The normally convergent product*

$$\begin{aligned} G_{\text{Barnes}}(z+1) &= (2\pi)^{z/2} \exp\left(-\frac{z(z+1)}{2} - \frac{\gamma z^2}{2}\right) \\ &\quad \times \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right)^k \exp\left(-z + \frac{z^2}{2k}\right) \end{aligned} \quad (30.27)$$

defines an entire function satisfying

$$G_{\text{Barnes}}(1) = 1, \quad G_{\text{Barnes}}(z+1) = \Gamma(z) G_{\text{Barnes}}(z). \quad (30.28)$$

Consequently $G_{\text{Barnes}}(n) = 1!2! \cdots (n-2)!$ for $n \geq 2$.

Proof. The logarithm of the k th factor is $k \log(1 + z/k) - z + z^2/(2k) = z^3/(3k^2) + \mathcal{O}_K(k^{-3})$ on a compact set K , so the product converges normally and defines an entire function with the prescribed integral-multiplicity zeros. At $z = 0$ it equals 1. Divide the finite products at z and $z-1$, rearrange before taking the limit, and use Weierstrass' gamma product; all quadratic convergence factors cancel and the limit is $\Gamma(z)$. Normal convergence justifies passage to the limit, proving the recurrence. Iteration gives the integer values. $\square$

Theorem 30.15 (Alexeiewsky identity). *With compatible logarithm branches,*

$$\log G_{\text{Barnes}}(z+1) = \frac{z}{2} \log(2\pi) - \frac{z(z+1)}{2} + z \log \Gamma(z+1) - \int_0^z \log \Gamma(t+1) dt. \quad (30.29)$$

Proof. Logarithmic differentiation of the product gives

$$\frac{d}{dz} \log G_{\text{Barnes}}(z+1) = \frac{1}{2} \log(2\pi) - z - \frac{1}{2} + z\psi(z+1),$$

where the simplification uses the digamma partial fractions. The derivative of the right side of (30.29) is identical, and both sides vanish at $z = 0$. $\square$

Corollary 30.16. *For $n \geq 1$,*

$$\prod_{k=1}^n k^k = \frac{(n!)^n}{G_{\text{Barnes}}(n+1)}. \quad (30.30)$$

Proof. In $(n!)^n/G_{\text{Barnes}}(n+1)$, the exponent of k is $n - (n-k) = k$. $\square$

Theorem 30.17 (Barnes asymptotics on the positive ray). *The limit*

$$\log A := \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n k \log k - \left(\frac{n^2}{2} + \frac{n}{2} + \frac{1}{12} \right) \log n + \frac{n^2}{4} \right\} \quad (30.31)$$

exists and defines the Glaisher–Kinkelin constant A . For every fixed integer $p \geq 1$, as $x \rightarrow +\infty$ through positive real values,

$$\begin{aligned} \log G_{\text{Barnes}}(x+1) &= \frac{x^2}{2} \log x - \frac{3x^2}{4} + \frac{x}{2} \log(2\pi) - \frac{1}{12} \log x + \frac{1}{12} - \log A \\ &\quad + \sum_{k=1}^{p-1} \frac{\beta_{2k+2}}{2k(2k+2)x^{2k}} + \mathcal{O}_p(x^{-2p}). \end{aligned} \quad (30.32)$$

Equivalently, retaining the exact gamma factor,

$$\begin{aligned} \log G_{\text{Barnes}}(x+1) &= \frac{x^2}{4} + x \log \Gamma(x+1) - \left(\frac{x(x+1)}{2} + \frac{1}{12} \right) \log x - \log A \\ &\quad + \sum_{k=1}^{p-1} \frac{\beta_{2k+2}}{2k(2k+1)(2k+2)x^{2k}} + \mathcal{O}_p(x^{-2p}). \end{aligned} \quad (30.33)$$

An empty sum is understood when $p = 1$.

Proof. The gamma recurrence and Binet's formula (7.20) give, for $t > 0$,

$$\log \Gamma(t+1) = \left(t + \frac{1}{2} \right) \log t - t + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/t)}{e^{2\pi x} - 1} dx.$$

For $u \geq 0$, the exact finite expansion

$$\arctan u = \sum_{r=1}^p \frac{(-1)^{r-1} u^{2r-1}}{2r-1} + (-1)^p \int_0^u \frac{v^{2p}}{1+v^2} dv$$

has remainder of absolute value at most $u^{2p+1}/(2p+1)$. Expanding $(e^{2\pi x} - 1)^{-1}$ as a geometric series and integrating termwise gives

$$\int_0^\infty \frac{x^{2r-1}}{e^{2\pi x} - 1} dx = \frac{(2r-1)! \zeta(2r)}{(2\pi)^{2r}}.$$

Together with (29.50), substitution in Binet's formula yields

$$\log \Gamma(t+1) = \left(t + \frac{1}{2} \right) \log t - t + \frac{1}{2} \log(2\pi) + \sum_{r=1}^p \frac{\beta_{2r}}{2r(2r-1)t^{2r-1}} + R_p(t), \quad (30.34)$$

where $R_p(t) = \mathcal{O}_p(t^{-2p-1})$. Indeed, the arctangent remainder gives the explicit bound

$$|R_p(t)| \leq \frac{2}{(2p+1)t^{2p+1}} \int_0^\infty \frac{x^{2p+1}}{e^{2\pi x} - 1} dx.$$

Split the integral in Alexeiewsky's identity (30.29) at 1, substitute (30.34) on $[1, x]$, and integrate the finite elementary sum. The term $r = 1$ supplies $-\frac{1}{12} \log x$ and a constant. For $r \geq 2$, the contribution of $a_r t^{-(2r-1)}$, where $a_r = \beta_{2r}/(2r(2r-1))$, is

$$x a_r x^{-(2r-1)} - \int_1^x a_r t^{-(2r-1)} dt = \frac{\beta_{2r}}{2r(2r-2)} x^{-2r+2} + C_r.$$

The remainder is equally controlled, but this point is worth making explicit:

$$x R_p(x) - \int_1^x R_p(t) dt = - \int_1^\infty R_p(t) dt + x R_p(x) + \int_x^\infty R_p(t) dt = D_p + \mathcal{O}_p(x^{-2p}).$$

Thus the improper integral contributes only a constant and the asserted next order. Collecting the elementary antiderivatives and putting $k = r - 1$ gives (30.32), initially with a constant C_p in place of $1/12 - \log A$. Comparing the formulas for p and $p+1$ and letting $x \rightarrow \infty$ shows $C_{p+1} = C_p$, since the extra inverse-power term and both remainders tend to zero. Denote the common constant by C .

It remains to identify C without assuming the existence of the limit in (30.31). At a positive integer n , the exact recurrence for G_{Barnes} gives

$$\prod_{k=1}^n k^k = \frac{(n!)^n}{G_{\text{Barnes}}(n+1)}.$$

Subtract the Barnes expansion with $p = 1$ from n times Stirling's expansion. The result is

$$\sum_{k=1}^n k \log k = \left(\frac{n^2}{2} + \frac{n}{2} + \frac{1}{12} \right) \log n - \frac{n^2}{4} + \frac{1}{12} - C + o(1).$$

Hence the limit in (30.31) exists, $\log A = 1/12 - C$, and so $C = 1/12 - \log A$.

Finally expand the exact $x \log \Gamma(x+1)$ in (30.33). At order x^{-2k} the difference between the coefficient in (30.32) and the gamma contribution is

$$\frac{\beta_{2k+2}}{2k(2k+2)} - \frac{\beta_{2k+2}}{(2k+1)(2k+2)} = \frac{\beta_{2k+2}}{2k(2k+1)(2k+2)}.$$

This proves the equivalent gamma-relative form and its remainder. $\square$

Corollary 30.18 (Complete hyperfactorial expansion). *For every fixed integer $p \geq 1$, as $n \rightarrow \infty$ through positive integers,*

$$\begin{aligned} \log \prod_{k=1}^n k^k &= \left(\frac{n^2}{2} + \frac{n}{2} + \frac{1}{12} \right) \log n - \frac{n^2}{4} + \log A \\ &\quad - \sum_{k=1}^{p-1} \frac{\beta_{2k+2}}{2k(2k+1)(2k+2)n^{2k}} + \mathcal{O}_p(n^{-2p}). \end{aligned} \quad (30.35)$$

In particular,

$$\prod_{k=1}^n k^k = A n^{n^2/2 + n/2 + 1/12} e^{-n^2/4} \left(1 + \frac{1}{720n^2} + \mathcal{O}(n^{-4}) \right). \quad (30.36)$$

Proof. Use the exact identity $\prod_{k=1}^n k^k = (n!)^n / G_{\text{Barnes}}(n+1)$, insert Stirling's expansion and (30.32), and subtract. The coefficient at n^{-2k} is $-\beta_{2k+2}/(2k(2k+1)(2k+2))$. Since $\beta_4 = -1/30$, the first logarithmic correction is $1/(720n^2)$; exponentiating changes the error only at order n^{-4} . $\square$

Remark 30.19 (Complex-sector status). The analogous expansion is classical uniformly in closed sectors $|\arg z| \leq \pi - \delta$ [17], but proving that stronger statement requires a sector-uniform complex Stirling remainder and control of the integration path. The positive-real theorem above is the full conclusion of the human-readable proof supplied here; no unproved sector-uniformity is being smuggled into its error term.

Remark 30.20. Polynomial identities such as the Pochhammer derivative formula are formal and require no convergence. The partial-fraction series and Barnes product are convergent analytic identities, while (30.33) is a Poincaré expansion. Keeping these three meanings of “equals” separate is essential in a reliable formula compendium.

Part IX

Catalan Refinements and Singularity Transfer

Chapter 31

Narayana numbers, polygon dissections, associahedra, and the Tamari order

The earlier Lagrange chapter derived the Catalan and Fuss–Catalan formulas. The distinct donor chapters add refinements and geometric models. We fix the convention that $K_{\dim=N-3}^{\text{Assoc}}$ is the associahedron attached to a convex N -gon; hence $\dim K_{\dim=N-3}^{\text{Assoc}} = N - 3$. This eliminates the two-unit index shifts common in sources that index instead by dimension or number of binary-tree leaves.

31.1 Dyck paths and reflection

A Dyck path of semilength n uses n up-steps $U = (1, 1)$ and n down-steps $D = (1, -1)$ and never falls below height 0.

Theorem 31.1 (Catalan formula). *The number of Dyck paths of semilength n is*

$$C_n^{\text{Cat}} = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (31.1)$$

Proof. Among all $\binom{2n}{n}$ balanced paths, reflect the prefix through the first step that reaches height -1 . This is a bijection from bad paths to words with $n+1$ up-steps and $n-1$ down-steps, counted by $\binom{2n}{n-1}$. Subtraction and simplification prove the formula. $\square$

Remark 31.2 (Formal crosswalk). Mathlib defines the Catalan numbers by the first-return recurrence $C_{n+1}^{\text{Cat}} = \sum_{i+j=n} C_i^{\text{Cat}} C_j^{\text{Cat}}$ and proves that they count Dyck words of semilength n (`DyckWord.card_dyckWord_semilength_eq_catalan`, through the bijection with binary trees) and that $C_n^{\text{Cat}} = \binom{2n}{n}/(n+1)$ (`catalan_eq_centralBinom_div`, `succ_mul_catalan_eq_centralBinom`). The difference form of Equation (31.1) is `Fabius.catalan_succ_eq_choose_sub_choose` (module `CatalanGeneratingFunction`), for $n+1$ so that no negative lower index occurs, derived algebraically from $(n+2)C_{n+1}^{\text{Cat}} = \binom{2n+2}{n+1}$ and $\binom{2n+2}{n+1}(n+1) = \binom{2n+2}{n}(n+2)$; the reflection bijection itself is not formalized.

Theorem 31.3 (First return). *The generating function $C(z) = \sum C_n^{\text{Cat}} z^n$ satisfies*

$$C(z) = 1 + zC(z)^2, \quad C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}. \quad (31.2)$$

Proof. Every nonempty path has a unique decomposition $UPDQ$, where D is the first return to height zero and P, Q are Dyck paths. This gives the functional equation. Of the two quadratic roots, only the displayed branch has constant term 1. $\square$

Remark 31.4 (Formal crosswalk). `Fabius.catalanSeries_eq` (module `CatalanGeneratingFunction`) is $C(z) = 1 + zC(z)^2$ in $R[[z]]$ over any commutative ring, read off from `Mathlib`'s recurrence `catalan_succ` by coefficient comparison, and `Fabius.eq_catalanSeries_of_eq_one_add_X_mul_sq` is the uniqueness statement behind the choice of branch: every power series F with $F = 1 + zF^2$ equals $C(z)$, since the equation determines each coefficient from the earlier ones. The closed form with $\sqrt{1 - 4z}$ is not formalized.

31.2 Narayana refinement

Let $N(n, k)$ count Dyck paths with k peaks, where a peak is an occurrence of UD , and let $C(z, u)$ mark semilength by z and peaks by u .

Theorem 31.5 (Narayana formula). *For $1 \leq k \leq n$,*

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}, \quad (31.3)$$

and

$$C(z, u) = 1 + zC(z, u)(C(z, u) + u - 1). \quad (31.4)$$

Consequently $\sum_k N(n, k) = C_n^{\text{Cat}}$ and $N(n, k) = N(n, n+1-k)$.

Proof. Put $H = C - 1$. In the first-return decomposition $UPDQ$, the initial arch is a peak exactly when P is empty. Hence $H = z(uC + (C - 1)C) = z(1 + H)(u + H)$. Lagrange inversion gives

$$[z^n u^k]H = \frac{1}{n} [w^{n-1} u^k] (1 + w)^n (u + w)^n = \frac{1}{n} \binom{n}{k-1} \binom{n}{k}.$$

Setting $u = 1$ gives the Catalan equation, and the closed formula is symmetric under $k \leftrightarrow n+1-k$. $\square$

Remark 31.6 (Formal crosswalk). Module `NarayanaNumbers` defines $N(n, k)$ over $\mathbb{Z}$ by the division-free determinant $\binom{n}{k} \binom{n-1}{k-1} - \binom{n}{k-1} \binom{n-1}{k}$ (`Fabius.narayana`). That shape is chosen so that no side condition is needed: it vanishes identically outside $1 \leq k \leq n$, whereas the quotient $\frac{1}{n} \binom{n}{k} \binom{n}{k-1}$ read with truncated subtraction would give $N(1, 0) = 1$ and poison both the symmetry and the row sum. Equation (31.3) is then `Fabius.narayana_mul` in the cleared form $nN(n, k) = \binom{n}{k} \binom{n}{k-1}$, proved from $k \binom{n}{k} = (n - k + 1) \binom{n}{k-1}$ and $n \binom{n-1}{k} = (n - k) \binom{n}{k}$; the symmetry $N(n, k) = N(n, n+1-k)$ is `Fabius.narayana_symm`, obtained by cancelling n against that cleared form; and $\sum_k N(n, k) = C_n^{\text{Cat}}$ is `Fabius.sum_narayana`, from Vandermonde's identity in the form $\sum_j \binom{n}{j} \binom{n}{j+1} = \binom{2n}{n-1}$ (`Fabius.sum_choose_mul_choose_succ`) together with $nC_n^{\text{Cat}} = \binom{2n}{n-1}$ (`Fabius.succ_mul_catalan_succ`). The peak-counting interpretation and the bivariate generating function Equation (31.4), which the proof above obtains by Lagrange inversion, are not formalized.

Corollary 31.7 (Peak mean and variance). *For a uniformly random Dyck path of semilength $n \geq 2$, the number K_n of peaks satisfies*

$$\mathbb{E}K_n = \frac{n+1}{2}, \quad \text{Var}(K_n) = \frac{n^2 - 1}{4(2n - 1)}. \quad (31.5)$$

Proof. Symmetry gives the mean. Differentiate the Narayana polynomial at $u = 1$ and use Vandermonde's convolution:

$$\begin{aligned} \sum_k k \binom{n}{k} \binom{n}{k-1} &= n \binom{2n-1}{n}, \\ \sum_k k(k-1) \binom{n}{k} \binom{n}{k-1} &= n(n-1) \binom{2n-2}{n-1}. \end{aligned}$$

Divide by nC_n and convert the second factorial moment to the variance. $\square$

31.3 Explicit Catalan bijections

Theorem 31.8 (Binary trees and parenthesizations). *Dyck paths of semilength n , rooted plane full binary trees with n internal vertices, and full parenthesizations of $n + 1$ ordered factors are in mutually explicit bijection.*

Proof. Map the empty path to a leaf. Recursively map $UPDQ$ to an internal root whose left and right subtrees are the images of P, Q . The construction is reversible. A binary tree becomes a parenthesization by labelling its leaves from left to right and interpreting each internal node as the product of its two children; the final multiplication recovers the root split. $\square$

Theorem 31.9 (Triangulations). *Triangulations of a convex $(n + 2)$ -gon are counted by C_n^{Cat} .*

Proof. Fix a root boundary edge. The planar dual of a triangulation has one vertex per triangle and one edge across each internal diagonal. Attach a leaf across every nonroot boundary edge and root at the triangle incident with the root edge. Each triangle has two remaining sides, so the result is a full plane binary tree. Conversely, glue a triangle at each internal vertex along its parent side. The constructions are inverse, and there are n triangles. $\square$

Theorem 31.10 (Noncrossing partitions). *Noncrossing partitions of $[n]$ in circular order are counted by C_n^{Cat} .*

Proof. Let a_n count them. Decompose a partition at the block containing 1. The gaps between successive elements of that block carry independent noncrossing partitions. If the block has $r + 1$ elements, the gap sizes sum to $n - r - 1$. Therefore the OGF $A(z) = \sum a_n z^n$ satisfies

$$A - 1 = \sum_{r \geq 0} z^{r+1} A^{r+1} = \frac{zA}{1 - zA},$$

which rearranges to $A = 1 + zA^2$. The decomposition is reversible, so this is also a recursive bijection. $\square$

31.4 Dissections and the associahedron

Let $D(N, d)$ count sets of d pairwise noncrossing diagonals of a convex N -gon.

Theorem 31.11 (Kirkman–Cayley dissection formula). *For $0 \leq d \leq N - 3$,*

$$D(N, d) = \frac{1}{d+1} \binom{N+d-1}{d} \binom{N-3}{d}. \quad (31.6)$$

Proof. Root one boundary edge and take the planar dual of a dissection. Its $I = d + 1$ regions become internal vertices and its $L = N - 1$ other boundary edges become leaves. This is a rooted plane tree with no unary internal vertices, and the construction is reversible.

In preorder, record outdegree minus one. Internal entries are positive, leaves contribute -1 , and the total is -1 . Choose the I internal positions in $\binom{I+L}{I}$ ways and their positive values, summing to $L - 1$, in $\binom{L-2}{I-1}$ ways. The cycle lemma says exactly one of the $I + L$ cyclic shifts has nonnegative proper partial sums and hence encodes a plane tree. Thus the number is

$$\frac{1}{I+L} \binom{I+L}{I} \binom{L-2}{I-1} = \frac{1}{I} \binom{L+I-1}{I-1} \binom{L-2}{I-1},$$

which becomes (31.6) after substitution. $\square$

Corollary 31.12 (Associahedral face vector). *A j -face of $\mathcal{K}_{\dim=N-3}^{\text{Assoc}}$ corresponds to $N - 3 - j$ fixed noncrossing diagonals, and*

$$f_j(\mathcal{K}_{\dim=N-3}^{\text{Assoc}}) = \frac{1}{N-2-j} \binom{2N-4-j}{N-3-j} \binom{N-3}{j}. \quad (31.7)$$

Thus $f_0 = C_{N-2}^{\text{Cat}}$, $f_{N-4} = N(N-3)/2$, and $f_{N-3} = 1$.

Proof. A face is a partial triangulation, equivalently a set of fixed noncrossing diagonals. Its dimension is the number of diagonals still free, so substitute $d = N - 3 - j$ in (31.6) and simplify. $\square$

31.5 The associahedral h -polynomial

The face vector (31.7) is unwieldy, but its face transform is not: the Narayana numbers of Theorem 31.5 reappear as the h -vector. This is the associahedral counterpart of Theorem 17.2, where the Eulerian numbers played the same role for the permutohedron.

Theorem 31.13 (Associahedral h -polynomial). *For $N \geq 3$,*

$$h_{\mathcal{K}_{\dim=N-3}^{\text{Assoc}}}(t) = \sum_{r=0}^{N-3} N(N-2, r+1) t^r = \sum_{r=0}^{N-3} \frac{1}{N-2} \binom{N-2}{r} \binom{N-2}{r+1} t^r. \quad (31.8)$$

In particular the h -vector is the $(N-2)$ nd Narayana row, its coefficients are palindromic, and $h(1) = C_{N-2}^{\text{Cat}}$ is the number of vertices.

Proof. Write $n = N - 1$, so that $\mathcal{K}_{\dim=N-3}^{\text{Assoc}}$ has dimension $n - 2$, and record (31.7) in the equivalent form

$$f_d = \frac{1}{n} \binom{n-2}{d} \binom{2n-d-2}{n-1}, \quad (31.9)$$

which agrees with (31.7). Indeed, rewriting (31.7) with $n = N - 1$ and taking the complement in the middle binomial coefficient turns it into

$$f_d = \frac{1}{n-1-d} \binom{n-2}{d} \binom{2n-2-d}{n},$$

and the adjacent-coefficient ratio $\binom{M}{k} / \binom{M}{k-1} = (M - k + 1)/k$ with $M = 2n - 2 - d$ and $k = n$ gives

$$\binom{2n-2-d}{n} = \frac{n-1-d}{n} \binom{2n-2-d}{n-1},$$

so the factor $n - 1 - d$ cancels and (31.9) follows. The point of the second form is that its prefactor $1/n$ does not depend on d , which is what makes the alternating sum below collapse. By the face transform of Theorem 17.2,

$$h(t) = \sum_{d=0}^{n-2} f_d (t-1)^d,$$

the sum being over nonempty faces, so

$$[t^r] h = \sum_{d=r}^{n-2} (-1)^{d-r} \binom{d}{r} f_d.$$

Using $\binom{d}{r} \binom{n-2}{d} = \binom{n-2}{r} \binom{n-2-r}{d-r}$ and substituting $j = d - r$, $M = n - 2 - r$,

$$[t^r] h = \frac{1}{n} \binom{n-2}{r} \sum_{j=0}^M (-1)^j \binom{M}{j} \binom{2n-r-j-2}{n-1}.$$

The alternating Vandermonde identity

$$\sum_{j=0}^M (-1)^j \binom{M}{j} \binom{R-j}{S} = \binom{R-M}{S-M} \quad (31.10)$$

follows by extracting $[x^S]$ from

$$(1+x)^R \left(1 - \frac{1}{1+x}\right)^M = x^M (1+x)^{R-M},$$

the left side expanding as $\sum_j (-1)^j \binom{M}{j} (1+x)^{R-j}$. Applying (31.10) with $R = 2n - r - 2$, $S = n - 1$ and $M = n - 2 - r$ gives $\binom{n}{r+1}$, whence

$$[t^r] h = \frac{1}{n} \binom{n-2}{r} \binom{n}{r+1} = \frac{(n-2)!(n-1)!}{r!(r+1)!(n-2-r)!(n-1-r)!} = \frac{1}{n-1} \binom{n-1}{r} \binom{n-1}{r+1},$$

the middle expression being obtained by writing out both binomial coefficients, and the last by reading the same factorial quotient backwards. With $n - 1 = N - 2$ this is $N(N - 2, r + 1)$ by (31.3).

Palindromicity is the Narayana symmetry $N(m, k) = N(m, m + 1 - k)$ recorded after Theorem 31.5, and it is the Dehn–Sommerville relation for this polytope. Finally $h(1) = \sum_k N(N - 2, k) = C_{N-2}^{\text{Cat}}$, which is f_0 as it must be for a simple polytope. $\square$

Example 31.14. For $N = 5$ the pentagon $K_{\dim=2}^{\text{Assoc}}$ has $f = (5, 5, 1)$ and $h(t) = 1 + 3t + t^2$; for $N = 6$ the three-dimensional associahedron has $f = (14, 21, 9, 1)$ and $h(t) = 1 + 6t + 6t^2 + t^3$. Both h -vectors are Narayana rows, and both sum to a Catalan number.

31.6 Loday's realization

The associahedron is not merely a combinatorial object. Loday's coordinates realize it as an explicit lattice polytope inside a hyperplane, which is the form in which it is normally drawn.

Definition 31.15. Let T be a rooted binary tree with n internal vertices, hence $n + 1$ leaves. For the i th internal vertex in infix order, let ℓ_i and r_i be the numbers of leaves in its left and right subtrees, and put

$$M(T) = (\ell_1 r_1, \dots, \ell_n r_n) \in \mathbb{R}^n. \quad (31.11)$$

Proposition 31.16 (Loday's constant hyperplane). *For every rooted binary tree T with n internal vertices,*

$$\sum_{i=1}^n \ell_i r_i = \binom{n+1}{2}. \quad (31.12)$$

Consequently all C_n^{Cat} points $M(T)$ lie in one affine hyperplane of $\mathbb{R}^n$, so their convex hull has dimension at most $n - 1$.

Proof. At the i th internal vertex, the product $\ell_i r_i$ counts the unordered pairs of leaves having one leaf in each of its two child subtrees — that is, exactly the pairs whose lowest common ancestor is that vertex. Every unordered pair of distinct leaves has a unique lowest common ancestor, and is therefore counted at exactly one internal vertex. Summing over i counts every pair of the $n + 1$ leaves once, which is $\binom{n+1}{2}$. $\square$

Remark 31.17. Loday's theorem identifies that convex hull with $K_{\dim=n-1}^{\text{Assoc}}$: the points $M(T)$ are exactly its vertices, and its normal fan is the fan of the Tamari lattice. Only the hyperplane statement is proved here. What (31.12) does supply unaided is the dimension bound, which is what makes the realization plausible: C_n^{Cat} points in $\mathbb{R}^n$ could a priori span the whole space, and the associahedron has dimension $n - 1$.

31.7 The Tamari order

Orient the elementary rotation by

$$((AB)C) \longrightarrow (A(BC)). \quad (31.13)$$

Theorem 31.18 (The rotation relation is a partial order). *Reachability under (31.13) is antisymmetric and therefore defines the Tamari partial order.*

Proof. Label the leaves $1, \dots, N$ from left to right and let $d_T(i)$ be the depth of leaf i . Define the integer potential

$$\Phi(T) = \sum_{i=1}^N 2^i d_T(i).$$

Under a rotation $((AB)C) \rightarrow (A(BC))$, the depths of leaves in A decrease by one, those in C increase by one, and those in B are unchanged. Hence

$$\Phi(A(BC)) - \Phi((AB)C) = \sum_{i \in C} 2^i - \sum_{i \in A} 2^i > 0.$$

Indeed every index in C is larger than every index in A , and the smallest power of two occurring in the first sum exceeds the sum of all preceding powers. Thus the potential strictly increases along each directed rotation, so no directed cycle exists. Reachability is reflexive and transitive by definition, and acyclicity gives antisymmetry. $\square$

Theorem 31.19 (Raney numbers). *If $T = 1 + zT^p$, then for integers $p, r \geq 1$ and $n \geq 0$,*

$$[z^n] T(z)^r = \frac{r}{pn+r} \binom{pn+r}{n}. \quad (31.14)$$

Proof. The case $n = 0$ is immediate. For $n \geq 1$, put $Y = T - 1 = z(1 + Y)^p$. Lagrange–Bürmann applied to $(1 + Y)^r$ gives

$$[z^n] T^r = \frac{r}{n} [w^{n-1}] (1 + w)^{pn+r-1} = \frac{r}{pn+r} \binom{pn+r}{n}.$$

$\square$

Chapter 32

Darboux's finite formula and transfer from singularities

One donor chapter emphasizes an exact finite integration identity, while analytic combinatorics uses Darboux's principle to transfer local singular expansions to coefficient asymptotics. Both are proved here; the latter is stated in a form sufficient for the Catalan and inverse-function examples [8].

32.1 Finite Darboux identity

Theorem 32.1 (Darboux formula). *Let f have $n + 1$ continuous derivatives on the segment from a to z , and let ϕ be a polynomial of degree at most n . Then*

$$\begin{aligned} \phi^{(n)}(0)(f(z) - f(a)) &= \sum_{m=1}^n (-1)^{m-1} (z-a)^m \left(\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a) \right) \\ &\quad + (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}(a + t(z-a)) dt. \end{aligned} \quad (32.1)$$

Proof. Put $d = z - a$, $F_m(t) = f^{(m)}(a + td)$, and

$$S(t) = \sum_{m=1}^n (-1)^{m-1} d^m \phi^{(n-m)}(t) F_m(t).$$

Differentiate and reindex the terms in which the derivative hits F_m . Every interior term cancels, leaving

$$S'(t) = d\phi^{(n)}(0)F_1(t) + (-1)^{n-1} d^{n+1} \phi(t) F_{n+1}(t).$$

Integrate from 0 to 1, move the last integral to the other side, and expand $S(1) - S(0)$. $\square$

Corollary 32.2 (Taylor integral remainder).

$$f(z) = \sum_{m=0}^n \frac{f^{(m)}(a)}{m!} (z-a)^m + \frac{(z-a)^{n+1}}{n!} \int_0^1 (1-t)^n f^{(n+1)}(a + t(z-a)) dt. \quad (32.2)$$

Proof. Take $\phi(t) = (1-t)^n$ in the theorem. All derivatives below order n vanish at $t = 1$, and $\phi^{(n-m)}(0) = (-1)^{n-m} n! / m!$. $\square$

32.2 Algebraic singularities

Lemma 32.3 (Exact algebraic coefficient, including polynomial cases). *Let $R \in \mathbb{C} \setminus \{0\}$, $\alpha \in \mathbb{C}$, and $n \in \mathbb{N}_0$. Then the Taylor germ at the origin satisfies*

$$[z^n] (1 - z/R)^{-\alpha} = R^{-n} \frac{(\alpha)^{\overline{n}}}{n!} = R^{-n} (-1)^n \binom{-\alpha}{n}. \quad (32.3)$$

If $\alpha \notin \{0, -1, -2, \dots\}$, this is also the ordinary gamma quotient

$$R^{-n} \frac{\Gamma(n + \alpha)}{\Gamma(\alpha)\Gamma(n + 1)}. \quad (32.4)$$

If instead $\alpha = -m$ with $m \in \{0, 1, 2, \dots\}$, the germ is the polynomial $(1 - z/R)^m$, and hence

$$[z^n] (1 - z/R)^m = \begin{cases} R^{-n} (-1)^n \binom{m}{n}, & 0 \leq n \leq m, \\ 0, & n > m. \end{cases} \quad (32.5)$$

For fixed $\alpha \notin \{0, -1, -2, \dots\}$,

$$[z^n] (1 - z/R)^{-\alpha} = \frac{R^{-n} n^{\alpha-1}}{\Gamma(\alpha)} \left(1 + \frac{\alpha(\alpha-1)}{2n} + \mathcal{O}(n^{-2}) \right). \quad (32.6)$$

Proof. The generalized binomial theorem gives $(-1)^n \binom{-\alpha}{n} = (\alpha)^{\overline{n}}/n!$; this polynomial identity in α proves the first formula without any excluded parameters. Away from the poles of $\Gamma(\alpha)$, its functional equation identifies the rising factorial with $\Gamma(n + \alpha)/\Gamma(\alpha)$. At $\alpha = -m$ the binomial theorem is the finite expansion of $(1 - z/R)^m$, giving the displayed cases rather than an undefined quotient of gamma poles. Finally, subtract the Stirling expansions of $\log \Gamma(n + \alpha)$ and $\log \Gamma(n + 1)$ and exponentiate to obtain the ratio expansion for the nonexceptional parameters. $\square$

Theorem 32.4 (Darboux transfer with an analytic multiplier). *Let $\rho \in \mathbb{C} \setminus \{0\}$, let h be analytic in $|z| < r$ for some $r > |\rho|$, and fix $J \geq 1$. Then, for every $\alpha \in \mathbb{C}$, as $n \rightarrow \infty$,*

$$\begin{aligned} [z^n] h(z) (1 - z/\rho)^{-\alpha} &= \rho^{-n} \sum_{j=0}^{J-1} \frac{(-\rho)^j h^{(j)}(\rho)}{j!} \frac{(\alpha - j)^{\overline{n}}}{n!} \\ &\quad + O\left(|\rho|^{-n} (n+1)^{\Re \alpha - J - 1}\right). \end{aligned} \quad (32.7)$$

Every summand is literal, including the case in which $\alpha - j$ is a nonpositive integer and therefore contributes only finitely many coefficients. In particular, if $h(\rho) \neq 0$ and $\alpha \notin \{0, -1, -2, \dots\}$,

$$[z^n] h(z) (1 - z/\rho)^{-\alpha} \sim \frac{h(\rho)}{\Gamma(\alpha)} \rho^{-n} n^{\alpha-1}. \quad (32.8)$$

Proof. Taylor-expand $h(z) = \sum_{j < J} h^{(j)}(\rho) (z - \rho)^j / j! + (z - \rho)^J q(z)$, where the removable value at ρ makes q analytic in the same disk. Since $(z - \rho)^j = (-\rho)^j (1 - z/\rho)^j$, the finite terms follow exactly from Lemma 32.3.

Choose s with $|\rho| < s < r$ and write $q(z) = \sum_{k \geq 0} q_k z^k$. Cauchy's estimate gives $q_k = \mathcal{O}(s^{-k})$. For fixed $\beta \in \mathbb{C}$, the exact binomial lemma and its two parameter cases give

$$\left| \frac{(\beta)^{\overline{m}}}{m!} \right| = O\left((m+1)^{\Re \beta - 1}\right);$$

when β is a nonpositive integer the left side simply has finite support. Set $\beta = \alpha - J$ in the convolution for the remainder. For $k \leq n/2$, the factor $(n - k + 1)^{\Re \beta - 1}$ is bounded by a constant

times $(n+1)^{\Re\beta-1}$ and the remaining geometric series has ratio $|\rho|/s < 1$. For $k > n/2$, put $\sigma = \Re\beta - 1$. Since $(n-k+1)^\sigma \leq (n+1)^{\max(\sigma,0)}$, this part of the sum is

$$O\left((|\rho|/s)^{n/2}(n+1)^{\max(\sigma,0)+1}\right) = O((n+1)^\sigma),$$

because exponential decay dominates every positive or negative fixed power of n . This proves the error term. Taking $J = 1$ and applying the nonexceptional gamma-ratio asymptotic proves the final equivalence. $\square$

Theorem 32.5 (Finite-smoothness Darboux subtraction). *Let $R > 0$ and let $N \geq 1$ be an integer. For $1 \leq \nu \leq N$, let*

$$\rho_\nu \in \mathbb{C}, \quad |\rho_\nu| = R, \quad c_\nu, \lambda_\nu \in \mathbb{C}.$$

Let F be analytic in $|z| < R$, and suppose that throughout this disk one has the global decomposition

$$F(z) = \sum_{\nu=1}^N c_\nu \left(1 - \frac{z}{\rho_\nu}\right)^{\lambda_\nu} + E(z), \quad |\rho_\nu| = R, \quad (32.9)$$

where each algebraic factor denotes its Taylor branch at the origin and E is analytic in $|z| < R$. Let

$$\mathcal{N} = \{\nu : \lambda_\nu \notin \{0, 1, 2, \dots\}\}, \quad \mathcal{P} = \{\nu : \lambda_\nu = m_\nu \in \{0, 1, 2, \dots\}\}.$$

Then, for every $n \in \mathbb{N}_0$, the exact coefficient identity is

$$\begin{aligned} [z^n] F(z) &= \sum_{\nu \in \mathcal{N}} c_\nu \rho_\nu^{-n} \frac{\Gamma(n - \lambda_\nu)}{\Gamma(-\lambda_\nu)\Gamma(n+1)} \\ &\quad + \sum_{\substack{\nu \in \mathcal{P} \\ n \leq m_\nu}} c_\nu \rho_\nu^{-n} (-1)^n \binom{m_\nu}{n} + [z^n] E(z). \end{aligned} \quad (32.10)$$

Consequently, the minimal remainder hypothesis

$$[z^n] E(z) = O(R^{-n}(n+1)^{-p}), \quad n \rightarrow \infty, \quad (32.11)$$

for some integer $p \geq 1$ permits the last term in (32.10) to be replaced by that error.

A sufficient, directly checkable condition for (32.11) is the following. Suppose E extends continuously to $|z| \leq R$, and put

$$\varepsilon(\theta) = E(Re^{i\theta}).$$

Assume $\varepsilon \in C^{p-1}([-\pi, \pi])$, its $(p-1)$ st derivative is absolutely continuous, $\varepsilon^{(p)} \in L^1([-\pi, \pi])$, and

$$\varepsilon^{(q)}(-\pi) = \varepsilon^{(q)}(\pi), \quad 0 \leq q < p.$$

Then, for $n \geq 1$,

$$|[z^n] E(z)| \leq \frac{R^{-n}}{2\pi n^p} \left\| \varepsilon^{(p)} \right\|_{L^1([-\pi, \pi])}. \quad (32.12)$$

Proof. Apply Lemma 32.3 with $\alpha = -\lambda_\nu$. The nonpolynomial indices give honest gamma quotients; a polynomial index gives exactly the finite second sum. Linearity of coefficient extraction proves (32.10), and the direct remainder hypothesis proves its asymptotic version.

For the boundary criterion, Cauchy's formula first on $|z| = r < R$ and then uniform convergence as $r \uparrow R$ give

$$[z^n] E(z) = \frac{R^{-n}}{2\pi} \int_{-\pi}^{\pi} \varepsilon(\theta) e^{-in\theta} d\theta. \quad (32.13)$$

Integrate by parts p times. Absolute continuity justifies the last step, and every boundary term vanishes by periodic matching, so

$$[z^n] E(z) = \frac{R^{-n}}{2\pi(\mathrm{i}n)^p} \int_{-\pi}^{\pi} \varepsilon^{(p)}(\theta) e^{-\mathrm{i}n\theta} d\theta.$$

Taking absolute values proves (32.12). $\square$

Remark 32.6. Polynomial exponents must be separated rather than encoded by an undefined ratio of gamma poles; after their finite coefficient support ends they do not affect any tail asymptotic. The word *global* in (32.9) is equally essential for the boundary criterion: a merely local expansion near one singular point does not define a remainder on the whole circle and therefore cannot by itself justify the Fourier integration-by-parts estimate.

Corollary 32.7 (Multiple dominant points). *Let $R > 0$ and $\varepsilon > 0$, and let j range over a finite index set. For every j , let*

$$\rho_j, \alpha_j \in \mathbb{C}, \quad |\rho_j| = R, \quad \alpha_j \notin \{0, -1, -2, \dots\},$$

and let h_j be analytic in $|z| < R + \varepsilon$ with $h_j(\rho_j) \neq 0$. Suppose that, as Taylor germs at the origin,

$$F(z) = \sum_j h_j(z) (1 - z/\rho_j)^{-\alpha_j} + E(z),$$

where E is analytic in $|z| < R + \varepsilon$. Then, as $n \rightarrow \infty$, the dominant contributions add:

$$[z^n] F(z) = \sum_j \frac{h_j(\rho_j)}{\Gamma(\alpha_j)} \rho_j^{-n} n^{\alpha_j-1} (1 + \mathcal{O}(n^{-1})) + \mathcal{O}((R + \varepsilon/2)^{-n}). \quad (32.14)$$

Complex conjugate points therefore produce periodic or oscillatory terms. Any model with $\alpha_j \in \{0, -1, -2, \dots\}$ is analytic in $|z| < R + \varepsilon$ and may therefore be absorbed into E . Its coefficients are exponentially smaller than R^{-n} ; they need not have finite support unless the analytic multiplier h_j is itself a polynomial.

Proof. Apply Theorem 32.4 with $J = 1$ to each model. Cauchy's estimate on $|z| = R + \varepsilon/2$ gives the exponentially smaller bound for E . $\square$

Proposition 32.8 (Logarithmic and meromorphic cases). *For $n \geq 1$,*

$$[z^n] \log \frac{1}{1 - z/R} = \frac{R^{-n}}{n}. \quad (32.15)$$

If a meromorphic function has principal parts $c_{j,m}(1 - z/\rho_j)^{-m}$ at its poles of smallest modulus, then its coefficients equal

$$\sum_{j,m} c_{j,m} \rho_j^{-n} \binom{n+m-1}{m-1} \quad (32.16)$$

up to an exponentially smaller contribution from the analytic remainder.

Proof. The logarithmic formula is its Taylor series. For poles, subtract all principal parts, use the integer binomial coefficient formula, and apply Cauchy's estimate to what remains. $\square$

32.3 Catalan asymptotics

Theorem 32.9. As $n \rightarrow \infty$,

$$C_n^{\text{Cat}} = \frac{4^n}{\sqrt{\pi} n^{3/2}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} + \mathcal{O}(n^{-3}) \right). \quad (32.17)$$

Proof. Use $C_n^{\text{Cat}} = (n+1)^{-1} \binom{2n}{n}$ and

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + \frac{1}{128n^2} + \mathcal{O}(n^{-3}) \right),$$

which follows from Stirling's expansion. Multiply by $(n+1)^{-1} = n^{-1}(1 - n^{-1} + n^{-2} + \mathcal{O}(n^{-3}))$. Alternatively, $C(z) = 2 - 2\sqrt{1-4z} + \mathcal{O}(1-4z)$ near $z = 1/4$, and Darboux transfer with $\alpha = -1/2$ gives the leading term directly. $\square$

Remark 32.10 (Choosing the asymptotic method). Darboux transfer applies to fixed finite singularities. The Bell EGF $\exp(e^z - 1)$ is entire, so its coefficients require the moving saddle point and Lambert W developed earlier. Implicit algebraic functions are normally handled by Lagrange inversion followed by singularity transfer; multivariate systems require the characteristic determinant and often a saddle analysis.

Part X

Further Differential and Reversion Examples

Chapter 33

Higher product rules, quotients, and elementary inverse series

The supplied manuscripts contain several useful consequences of the two master composition theorems that do not belong to a single classical triangle. They are collected here rather than repeated in the Bell-polynomial and inversion chapters.

33.1 Products, reciprocals, and quotients

Theorem 33.1 (Multinomial Leibniz rule). *For functions $f_1, \dots, f_q$ possessing n derivatives,*

$$D^n \left(\prod_{r=1}^q f_r(x) \right) = \sum_{j_1 + \dots + j_q = n} \binom{n}{j_1, \dots, j_q} \prod_{r=1}^q f_r^{(j_r)}(x). \quad (33.1)$$

Proof. Taylor's formula, interpreted either analytically near x or formally in a new variable h , gives

$$\prod_{r=1}^q f_r(x+h) = \prod_{r=1}^q \left(\sum_{j_r \geq 0} f_r^{(j_r)}(x) \frac{h^{j_r}}{j_r!} \right).$$

The coefficient of h^n is the sum over $j_1 + \dots + j_q = n$ of $\prod_r f_r^{(j_r)}(x)/j_r!$. Multiplication by $n!$ proves (33.1). $\square$

Remark 33.2 (Formal crosswalk). The proof above offers two readings, analytic near x and formal in a new variable h ; the formal one is the one that is formalized, and only for two factors. `Fabius.derivative_iterate_mul` (module `IteratedLeibniz`) is $(d/dt)^n(fg) = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$ for formal power series over any commutative ring, proved by induction on n from Pascal's rule in convolution form (`Fabius.sum_pascal_split`, stated for an arbitrary sequence and reusable for any binomial-type two-term recurrence). The general q -factor form Equation (33.1), whose index set is the compositions of n into q parts, is not formalized.

Theorem 33.3 (Higher reciprocal and quotient rules). *If $g(x) \neq 0$, then*

$$\left(\frac{1}{g} \right)^{(n)} = \sum_{k=0}^n (-1)^k k! \frac{\mathcal{B}_{n,k}^{\text{exp}}(g', g'', \dots, g^{(n-k+1)})}{g^{k+1}}. \quad (33.2)$$

Consequently,

$$D^n \left(\frac{f}{g} \right) = \sum_{r=0}^n \binom{n}{r} f^{(n-r)} \sum_{k=0}^r (-1)^k k! \frac{\mathcal{B}_{r,k}^{\text{exp}}(g', g'', \dots, g^{(r-k+1)})}{g^{k+1}}. \quad (33.3)$$

For $n > 0$ the $k = 0$ term in (33.2) vanishes.

Proof. Apply Faà di Bruno's formula to $F(g(x))$ with $F(u) = u^{-1}$. Since $F^{(k)}(u) = (-1)^k k! u^{-k-1}$, the Bell-polynomial form gives (33.2). Formula (33.3) follows by applying the ordinary two-factor Leibniz rule to $f g^{-1}$ and substituting the reciprocal formula. $\square$

Corollary 33.4 (Trigonometric outer functions). *For $n \geq 0$,*

$$D^n \sin(g(x)) = \sum_{k=0}^n \sin\left(g(x) + \frac{k\pi}{2}\right) \mathcal{B}_{n,k}^{\exp}(g', g'', \dots), \quad (33.4)$$

$$D^n \cos(g(x)) = \sum_{k=0}^n \cos\left(g(x) + \frac{k\pi}{2}\right) \mathcal{B}_{n,k}^{\exp}(g', g'', \dots). \quad (33.5)$$

Proof. Use $D^k \sin u = \sin(u + k\pi/2)$ and $D^k \cos u = \cos(u + k\pi/2)$ in Faà di Bruno's formula. $\square$

33.2 Diagonal identities furnished by inverse functions

Theorem 33.5 (Four classical inverse series and their diagonals). *For $|y| < 1$,*

$$\arcsin y = \sum_{m \geq 0} \frac{1}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}, \quad (33.6)$$

$$\arctan y = \sum_{m \geq 0} \frac{(-1)^m}{2m+1} y^{2m+1}, \quad (33.7)$$

$$\operatorname{arsinh} y = \sum_{m \geq 0} \frac{(-1)^m}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}, \quad (33.8)$$

$$\operatorname{artanh} y = \sum_{m \geq 0} \frac{y^{2m+1}}{2m+1}. \quad (33.9)$$

Their inverse pairs give, for every $m \geq 0$,

$$[x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1} = \frac{1}{4^m} \binom{2m}{m}, \quad (33.10)$$

$$[x^{2m}] \left(\frac{x}{\tan x} \right)^{2m+1} = (-1)^m, \quad (33.11)$$

$$[x^{2m}] \left(\frac{x}{\sinh x} \right)^{2m+1} = (-1)^m \frac{1}{4^m} \binom{2m}{m}, \quad (33.12)$$

$$[x^{2m}] \left(\frac{x}{\tanh x} \right)^{2m+1} = 1. \quad (33.13)$$

Proof. The derivatives of the four inverse functions are, in the displayed order,

$$(1 - y^2)^{-1/2}, \quad (1 + y^2)^{-1}, \quad (1 + y^2)^{-1/2}, \quad (1 - y^2)^{-1}.$$

The generalized binomial theorem and the geometric series expand these four functions for $|y| < 1$; termwise integration from 0 to y gives Equations (33.6) to (33.9).

If $g = f^{(-1)}$, $f(0) = 0$, and $f'(0) = 1$, Lagrange inversion gives

$$[y^n] g(y) = \frac{1}{n} [x^{n-1}] \left(\frac{x}{f(x)} \right)^n.$$

Apply this identity with $f = \sin, \tan, \sinh, \tanh$ and $n = 2m + 1$, then compare with the four inverse coefficients just computed. This proves all four diagonal identities. $\square$

Proposition 33.6 (Logarithmic diagonal). *For $n \geq 1$,*

$$[x^{n-1}] \left(\frac{x}{\log(1+x)} \right)^n = \frac{1}{(n-1)!}. \quad (33.14)$$

Proof. The compositional inverse of $\log(1+x)$ is $e^y - 1$, whose coefficient of y^n is $1/n!$. Lagrange inversion therefore gives

$$\frac{1}{n} [x^{n-1}] \left(\frac{x}{\log(1+x)} \right)^n = [y^n] (e^y - 1) = \frac{1}{n!}.$$

Multiplication by n proves (33.14). The parallel exponential diagonal is already the specialization (29.31) of the stronger Nörlund polynomial identity. $\square$

33.3 Expansion about a nonzero equilibrium

Theorem 33.7 (Offset Lagrange–Bürmann formula). *Let $u(t)$ be the unique formal solution of*

$$u = a + t\Phi(u), \quad \Phi(a) \neq 0. \quad (33.15)$$

Then for every formal H and $n \geq 1$,

$$[t^n] H(u(t)) = \frac{1}{n} [z^{n-1}] H'(a+z)\Phi(a+z)^n. \quad (33.16)$$

The same formula holds analytically in a sufficiently small neighbourhood when H and Φ are analytic.

Proof. Put $w = u - a$ and $\phi(w) = \Phi(a+w)$. Then $w = t\phi(w)$, so Theorem 21.1, applied to the outer series $K(w) = H(a+w)$, gives

$$[t^n] K(w(t)) = \frac{1}{n} [z^{n-1}] K'(z)\phi(z)^n,$$

which is (33.16). The analytic statement follows from the analytic inverse-function theorem exactly as in Theorem 21.3. $\square$

33.4 Quadratic critical points and square-root inverses

Theorem 33.8 (Local square-root inverse). *Let f be holomorphic near $a \in \mathbb{C}$ and suppose*

$$f(a) = b, \quad f'(a) = 0, \quad f''(a) \neq 0.$$

Then, after choosing a local square root of $w - b$, the two solutions of $f(z) = w$ near a have convergent Puiseux expansions

$$z = a + \sum_{m \geq 1} c_m (w - b)^{m/2}. \quad (33.17)$$

Passing from one branch to the other changes the sign of every odd-indexed c_m and leaves every even-indexed c_m fixed. In particular,

$$c_1 = \pm \sqrt{\frac{2}{f''(a)}}. \quad (33.18)$$

Proof. Put $z = a + u$. Taylor expansion and the double zero at $u = 0$ give the analytic factorization

$$f(a + u) - b = u^2 h(u), \quad h(0) = \frac{1}{2} f''(a) \neq 0.$$

On a sufficiently small disk, h has an analytic square root. Set $v = u\sqrt{h(u)}$. Its derivative at zero is $\sqrt{h(0)} \neq 0$, so the analytic inverse-function theorem supplies a convergent analytic inverse $u = \chi(v)$. Because $v^2 = w - b$, the two local solutions are $z = a + \chi(\sqrt{w - b})$ and $z = a + \chi(-\sqrt{w - b})$. Expanding χ proves the parity assertion and Equation (33.17). Finally $\chi'(0) = 1/\sqrt{h(0)} = \sqrt{2/f''(a)}$, with the sign determined by the chosen square-root branch, which proves Equation (33.18). $\square$

Corollary 33.9 (Universal coefficient signature). *Let $g(w)$ be analytic at zero and suppose that $\rho \neq 0$ is its unique singularity of smallest modulus. Assume that in a slit neighborhood of ρ one can write*

$$g(w) = H(w) + (1 - w/\rho)^{1/2} K(w),$$

where H and K extend analytically to $|w| < |\rho| + \eta$ for some $\eta > 0$ and $K(\rho) = \lambda \neq 0$. Then

$$[w^n] g(w) \sim -\frac{\lambda}{2\sqrt{\pi}} \rho^{-n} n^{-3/2}. \quad (33.19)$$

Proof. The analytic part H has coefficients $\mathcal{O}((|\rho| + \eta')^{-n})$ for any $0 < \eta' < \eta$. In the remaining term put $w = \rho u$ and apply Theorem 32.4 with $R = 1$, exponent $\alpha = -1/2$, and analytic multiplier $K(\rho u)$. Since $\Gamma(-1/2) = -2\sqrt{\pi}$, its leading term is exactly Equation (33.19). $\square$

Remark 33.10. The theorem explains the ubiquitous exponent $n^{-3/2}$ in inverse-function coefficients. For $f(z) = ze^z$, the critical point $z = -1$ maps to $-e^{-1}$ and produces the first branch point of Lambert W ; the Catalan equation has the analogous quadratic critical value $1/4$. Formal reversion determines coefficients locally, whereas the nearest analytic critical value determines their eventual scale.

Chapter 34

Kepler's equation as a complete reversion example

A classical nonlinear equation with both a formal coefficient expansion and a globally convergent Fourier representation is

$$E = M + e \sin E, \quad |e| < 1. \quad (34.1)$$

Here M is the mean anomaly, E the eccentric anomaly, and e is treated mathematically as a parameter.

Theorem 34.1 (Lagrange and Fourier–Bessel forms). *The unique real solution of (34.1) satisfies*

$$E = M + \sum_{k \geq 1} \frac{e^k}{k!} \left(\frac{d}{dM} \right)^{k-1} \sin^k M, \quad (34.2)$$

$$= M + 2 \sum_{n \geq 1} \frac{J_n(ne)}{n} \sin(nM), \quad (34.3)$$

where, for integral $n \geq 0$,

$$J_n(z) = \frac{1}{2\pi} \int_0^{2\pi} \cos(n\theta - z \sin \theta) d\theta. \quad (34.4)$$

The first formula is an identity of formal series in e ; for every fixed $|e| < 1$, the second series converges absolutely and uniformly in real M . The convergence is locally uniform jointly in (e, M) for e in a compact subset of $(-1, 1)$.

Proof. Formula (34.2) is the shifted reversion formula (23.3) with $x = M$, $y = e$, and $f(x) = \sin x$.

For the Fourier form, the function $M(E) = E - e \sin E$ has derivative $1 - e \cos E \geq 1 - |e| > 0$. Hence it has a smooth inverse satisfying $E(M + 2\pi) = E(M) + 2\pi$, and $E(M) - M$ is a smooth odd 2π -periodic function. Write

$$E(M) - M = \sum_{n \geq 1} c_n \sin(nM).$$

Its sine coefficient, after changing variables from M to E , is

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^{2\pi} (E - M) \sin(nM) dM \\ &= \frac{e}{\pi} \int_0^{2\pi} \sin E \sin(n(E - e \sin E)) (1 - e \cos E) dE. \end{aligned}$$

Since

$$\frac{d}{dE} \cos(n(E - e \sin E)) = -n(1 - e \cos E) \sin(n(E - e \sin E)),$$

integration by parts, with vanishing boundary term, gives

$$c_n = \frac{e}{\pi n} \int_0^{2\pi} \cos E \cos(nE - ne \sin E) dE = \frac{e}{n} (J_{n-1}(ne) + J_{n+1}(ne)).$$

It remains to prove the recurrence used to simplify this expression. The integral over a full period of $\frac{d}{d\theta} \sin(n\theta - z \sin \theta)$ is zero; therefore

$$n \int_0^{2\pi} \cos(n\theta - z \sin \theta) d\theta = z \int_0^{2\pi} \cos \theta \cos(n\theta - z \sin \theta) d\theta.$$

The product-to-sum identity and (34.4) yield $J_{n-1}(z) + J_{n+1}(z) = 2nJ_n(z)/z$, with the value at $z = 0$ understood by continuity. Thus $c_n = 2J_n(ne)/n$, proving (34.3). Finally, the inverse $E(M)$ is smooth for $|e| < 1$. If e ranges over a compact subset of $(-1, 1)$, then $1 - e \cos E$ is uniformly bounded away from zero, so implicit differentiation gives uniform bounds for every fixed M -derivative of $E(M) - M$. Repeated integration by parts therefore makes the Fourier coefficients decay faster than every fixed inverse power of n , uniformly on that compact parameter set. This proves the asserted absolute and local uniform convergence. $\square$

Remark 34.2. The two expansions organize the same solution in different coordinates. The Lagrange series is local in the perturbation parameter e and exposes finite derivative polynomials at every order. The Fourier–Bessel series is global in the angular variable M and diagonalizes the periodic translation symmetry. Their equality is an instructive example of coefficient calculus as a change of basis rather than a collection of unrelated formulas.

Part XI

Algorithms, Normalization, and Reproducible Certificates

Chapter 35

Exact coefficient algorithms and consistency checks

A formula compendium is reliable only when its normalizations can be checked and its coefficients reproduced. This chapter merges the algorithmic and proof-audit appendices of the drafts. Every recurrence is triangular and can therefore be implemented over exact integers, rationals, finite fields, or symbolic coefficient rings.

35.1 Normalization dictionary

Ordinary versus exponential coefficients

If

$$f(z) = \sum_{n \geq 0} \alpha_n z^n = \sum_{n \geq 0} f_n \frac{z^n}{n!},$$

then $f_n = n! \alpha_n$. With the conventions of this monograph,

$$\mathcal{B}_{n,k}^{\text{ord}}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}^{\text{exp}}(1!x_1, 2!x_2, \dots). \quad (35.1)$$

Signed first-kind Stirling numbers are $s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}$; inverse-matrix identities always use the signed array.

Proof of (35.1). The exponential definition gives

$$\frac{1}{k!} \left(\sum_{j \geq 1} x_j t^j \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}^{\text{exp}}(1!x_1, 2!x_2, \dots) \frac{t^n}{n!}.$$

The coefficient on the left is $\mathcal{B}_{n,k}^{\text{ord}}(x)/k!$ in ordinary normalization. Comparison proves the conversion. $\square$

Object	Defining expansion	Primary role
$s(n, k)$	$(x)^{\overline{n}} = \sum_k s(n, k) x^k$	inverse factorial basis
$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$	$(x)^{\overline{n}} = \sum_k \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k$	permutation cycles
$\{n\}_k$	$x^n = \sum_k \{n\}_k (x)^{\overline{k}}$	set partitions
$\mathcal{B}_{n,k}^{\text{exp}}$	exponential partial Bell polynomial	derivatives and labelled composition
$\mathcal{B}_{n,k}^{\text{ord}}$	ordinary partial Bell polynomial	ordinary series composition
$\mathcal{B}_n^{\text{Bell}}$	$\sum_k \{n\}_k$	unweighted partitions
β_n	$t/(e^t - 1) = \sum \beta_n t^n / n!$	power sums and Euler–Maclaurin

35.2 Elementary series operations

Let $A(z) = \sum a_n z^n$, $B(z) = \sum b_n z^n$, and $C(z) = \sum c_n z^n$.

Algorithm 35.1 (Product, reciprocal, and quotient). For $C = AB$,

$$c_n = \sum_{j=0}^n a_j b_{n-j}. \quad (35.2)$$

If a_0 is invertible and $C = A^{-1}$,

$$c_0 = a_0^{-1}, \quad c_n = -a_0^{-1} \sum_{j=1}^n a_j c_{n-j}. \quad (35.3)$$

For $C = B/A$,

$$c_n = a_0^{-1} \left(b_n - \sum_{j=1}^n a_j c_{n-j} \right). \quad (35.4)$$

Proof. The product formula is Cauchy's rule. Equate coefficients in $AC = 1$ or $AC = B$ and isolate the term $a_0 c_n$. $\square$

Algorithm 35.2 (Logarithm, exponential, and powers). Assume $a_0 = 1$. If $L = \log A = \sum_{n \geq 1} \ell_n z^n$, then

$$\ell_n = a_n - \frac{1}{n} \sum_{j=1}^{n-1} j \ell_j a_{n-j}. \quad (35.5)$$

If $A = e^L$, then

$$a_n = \frac{1}{n} \sum_{j=1}^n j \ell_j a_{n-j}. \quad (35.6)$$

If $C = A^\alpha$, then

$$c_0 = 1, \quad c_n = \frac{1}{n} \sum_{j=1}^n ((\alpha + 1)j - n) a_j c_{n-j}. \quad (35.7)$$

Proof. Compare coefficients in $A' = AL'$ to obtain the first two recurrences. For powers, compare degree $n - 1$ in $AC' = \alpha A'C$ and isolate nc_n . $\square$

Algorithm 35.3 (Differentiation and integration). If $C = A'$, then $c_n = (n + 1)a_{n+1}$. If $C(z) = c_0 + \int_0^z A(t) dt$, then $c_{n+1} = a_n/(n + 1)$.

Proof. Formal differentiation gives

$$A'(z) = \sum_{m \geq 1} m a_m z^{m-1} = \sum_{n \geq 0} (n+1) a_{n+1} z^n,$$

so coefficient comparison gives $c_n = (n+1)a_{n+1}$. Over the ambient commutative $\mathbb{Q}$ -algebra, every positive integer is invertible and formal integration is defined coefficientwise. Thus

$$c_0 + \int_0^z A(t) dt = c_0 + \sum_{n \geq 0} \frac{a_n}{n+1} z^{n+1},$$

which gives $c_{n+1} = a_n/(n+1)$. □

35.3 Composition and reversion

Algorithm 35.4 (Composition power table). Assume $B(0) = 0$ and set $p_{k,n} = [z^n] B(z)^k$. Initialize $p_{0,0} = 1, p_{0,n} = 0$ for $n > 0$, and use

$$p_{k,n} = \sum_{j=1}^n b_j p_{k-1,n-j}, \quad [z^n] A(B(z)) = \sum_{k=0}^n a_k p_{k,n}. \quad (35.8)$$

Proof. The first recurrence is $B^k = B B^{k-1}$; the second is $A(B) = \sum_k a_k B^k$. Since $B(0) = 0$, only finitely many powers contribute to each degree. □

Algorithm 35.5 (Lagrange reversion). If $F(z) = a_1 z + a_2 z^2 + \dots$ with invertible a_1 , then its inverse $G(w) = \sum_{n \geq 1} g_n w^n$ has

$$g_n = \frac{1}{n} [z^{n-1}] \left(\frac{z}{F(z)} \right)^n. \quad (35.9)$$

Proof. This is Lagrange–Bürmann with outer function $H(z) = z$. The coefficient g_n depends only on $a_1, \dots, a_n$, so every truncation is exact through its declared order. □

The next proposition is an independent check on Theorem 35.5, on the Bell-polynomial inverse coefficients of Theorem 21.4, and on the derivative recursion of Theorem 27.3: it obtains the same coefficients by undetermined coefficients alone, using no inversion theorem whatever. Two routes to the same fourth nontrivial order is the kind of redundancy this chapter exists to supply.

Proposition 35.6 (Direct reversion through degree five). *Let*

$$F(z) = z + a_2 z^2 + a_3 z^3 + a_4 z^4 + a_5 z^5 + \mathcal{O}(z^6)$$

over a commutative $\mathbb{Q}$ -algebra. Then its compositional inverse is

$$\begin{aligned} F^{(-1) \circ}(z) = & z - a_2 z^2 + (2a_2^2 - a_3) z^3 + (-5a_2^3 + 5a_2 a_3 - a_4) z^4 \\ & + (14a_2^4 - 21a_2^2 a_3 + 3a_3^2 + 6a_2 a_4 - a_5) z^5 + \mathcal{O}(z^6). \end{aligned} \quad (35.10)$$

Proof. Write $G(z) = z + b_2 z^2 + b_3 z^3 + b_4 z^4 + b_5 z^5 + \mathcal{O}(z^6)$ and expand $F(G(z))$. Since $G^k = z^k(1 + b_2 z + b_3 z^2 + \dots)^k$, the coefficients of z^2, z^3, z^4, z^5 in $F(G(z))$ are respectively

$$\begin{aligned} & b_2 + a_2, \\ & b_3 + 2a_2 b_2 + a_3, \\ & b_4 + a_2(2b_3 + b_2^2) + 3a_3 b_2 + a_4, \\ & b_5 + a_2(2b_4 + 2b_2 b_3) + a_3(3b_3 + 3b_2^2) + 4a_4 b_2 + a_5. \end{aligned}$$

Requiring $F(G(z)) = z$ sets each of these to zero. The system is triangular in b_2, b_3, b_4, b_5 and solves successively:

$$\begin{aligned} b_2 &= -a_2, \\ b_3 &= -2a_2b_2 - a_3 = 2a_2^2 - a_3, \\ b_4 &= -a_2(2b_3 + b_2^2) - 3a_3b_2 - a_4 = -5a_2^3 + 5a_2a_3 - a_4, \\ b_5 &= -a_2(2b_4 + 2b_2b_3) - a_3(3b_3 + 3b_2^2) - 4a_4b_2 - a_5 \\ &= 14a_2^4 - 21a_2^2a_3 + 3a_3^2 + 6a_2a_4 - a_5, \end{aligned}$$

which is (35.10). Triangularity is what makes the argument work over any commutative $\mathbb{Q}$ -algebra: no division occurs, because F was normalized to have leading coefficient 1. $\square$

Algorithm 35.7 (Reciprocal Newton iteration). Suppose $A(0)$ is invertible and $AC_m \equiv 1 \pmod{z^m}$. Define

$$C_{2m} = C_m(2 - AC_m) \pmod{z^{2m}}. \quad (35.11)$$

Then $AC_{2m} \equiv 1 \pmod{z^{2m}}$: one Newton step doubles the number of known reciprocal coefficients.

Proof. Write $AC_m = 1 - E$ with $E \in z^m R[[z]]$. Then

$$AC_m(2 - AC_m) = (1 - E)(1 + E) = 1 - E^2.$$

Since $E^2 \in z^{2m} R[[z]]$, reduction modulo z^{2m} proves the claim. Only addition and multiplication are used once the initial inverse of $A(0)$ has been chosen. $\square$

Algorithm 35.8 (Newton reversion). If $F(G_m(w)) \equiv w \pmod{w^m}$, define

$$G_{2m} = G_m - \frac{F(G_m) - w}{F'(G_m)} \pmod{w^{2m}}. \quad (35.12)$$

Then $F(G_{2m}) \equiv w \pmod{w^{2m}}$.

Proof. Put $E = F(G_m) - w = \mathcal{O}(w^m)$ and $\Delta = -E/F'(G_m) = \mathcal{O}(w^m)$. Taylor's formula in the formal ring gives $F(G_m + \Delta) = F(G_m) + F'(G_m)\Delta + \mathcal{O}(\Delta^2) = w + \mathcal{O}(w^{2m})$. The denominator is invertible because its constant term is a_1 . $\square$

Remark 35.9 (Arithmetic complexity). The triangular product, reciprocal, exponential, and logarithm recurrences use $\mathcal{O}(N^2)$ scalar operations through degree N ; a direct composition table uses $\mathcal{O}(N^3)$. Faster multiplication and Newton iteration reduce asymptotic cost, but the triangular forms are preferable for symbolic audits because every coefficient carries a short dependency certificate.

35.4 Small exact cross-checks

$n \setminus k$	0	1	2	3	4	5	6	$n \setminus k$	0	1	2	3	4	5	6
0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0
2	0	1	1	0	0	0	0	2	0	1	1	0	0	0	0
3	0	2	3	1	0	0	0	3	0	1	3	1	0	0	0
4	0	6	11	6	1	0	0	4	0	1	7	6	1	0	0
5	0	24	50	35	10	1	0	5	0	1	15	25	10	1	0
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

(*)

The left table contains the unsigned first-kind Stirling numbers; its n th row sums to $n!$. The right table contains the second-kind Stirling numbers, whose row sums through $n = 6$ are

$$1, 1, 2, 5, 15, 52, 203.$$

After the first-kind array is signed, its product with the second-kind array must be the identity through this order.

Proposition 35.10 (Recurrence uniqueness). *If two triangular arrays have the same boundary data and obey the same recurrence from row $n - 1$ to row n , then they agree identically.*

Proof. Induct on the row number; every new entry is determined by already equal entries in the preceding row. $\square$

35.5 Deterministic and modular certificates

Theorem 35.11 (Polynomial grid certificate). *Let $P, Q \in K[x_1, \dots, x_d]$ and suppose $\deg_{x_i}(P - Q) \leq D_i$. If $P = Q$ on a Cartesian grid $S_1 \times \dots \times S_d$ with $D_i + 1$ distinct values in S_i , then $P = Q$ identically.*

Proof. For one variable, a nonzero degree- D polynomial has at most D roots. Induct on d , viewing $P - Q$ as a polynomial in x_d . At each grid point in the first $d - 1$ coordinates all its coefficient polynomials vanish; the induction hypothesis makes each coefficient identically zero. $\square$

Theorem 35.12 (Chinese-remainder certificate). *If an integer N has a proved bound $|N| < M/2$, where $M = p_1 \cdots p_s$ and the p_i are pairwise coprime, and if $N \equiv 0 \pmod{p_i}$ for every i , then $N = 0$.*

Proof. The congruences give $M \mid N$. The only multiple of M in $(-M/2, M/2)$ is zero. $\square$

Remark 35.13. Random evaluations are excellent error detectors but become proofs only after a degree bound (Schwartz–Zippel or a full deterministic grid) is supplied. Modular computations become exact certificates only after an integer-size bound is recorded. This distinction prevents numerical evidence from being silently promoted to an identity.

35.6 Disposition of the Bell-number decimal table

The arrivals contain a six-decimal table for $R_n = (\mathcal{B}_n^{\text{Bell}})^{1/n} \log(n + 1)/n$ and describe its entries as directed interval bounds. They do not supply the interval endpoints before rounding, the rounding implementation, a replay script, or a checksum-pinned output. Consequently the table is not an exact certificate and is omitted from the canonical mathematical record. The symbolic recurrence for $\mathcal{B}_n^{\text{Bell}}$ remains available for a future reproducible check, while Theorem 11.11 supplies the proved asymptotic bound needed in this monograph. Decimal agreement alone is evidence, not proof.

35.7 Verification protocol

A robust workflow for a new identity is: normalize all signed/unsigned and ordinary/exponential conventions; derive the identity by coefficient extraction, a bijection, recurrence uniqueness, or incidence inversion; expand through at least fourth order, where the Faà di Bruno multiplicities 1, 4, 3, 6, 1 expose most missing factorials; and retain exact or bounded modular certificates for every computational step. The derivation is the proof; the other stages protect it against transcription and implementation errors.

Part XII

Source Concordance, Deduplication Record, and Verification

Chapter 36

How the six sibling manuscripts are unified

The six arrivals are independent sibling syntheses with a common core, not a chronological sequence of revisions. The merger therefore works theorem by theorem. For a repeated result, the canonical text retains one statement with the strongest correct hypotheses and one complete proof; a second proof survives only when it exposes a genuinely different mechanism, such as a bijection rather than coefficient extraction. An absent theorem is imported at the point where its hypotheses have already been established, never as a parallel appendix.

36.1 Source-by-source contribution map

Source	Distinct strengths	Canonical disposition
CCC-2	Broad proof-oriented core; Catalan and Tamari refinements; Darboux transfer; exact coefficient algorithms and certificate principles.	Selected as the mathematical spine at the non-suffixed canonical path. Its common analytic defects were repaired before donor material was added.
CCC	Parameterized and Boolean-lattice inversion; Nörlund and Fourier material; detailed Sheffer connection constants; explicit Lagrange–Good examples.	Its unique results are merged into the corresponding incidence, special-family, Sheffer, and reversion chapters; repeated common-core prose is retired.
CFIT	Pochhammer and binomial transforms; polygon dissections; extended Euler–Maclaurin and Darboux statements; reciprocal Newton iteration.	Stronger hypotheses and complete proof details are imported selectively, with formal and analytic versions kept separate.
UCCC	Elementary inverse series; quadratic critical-point inversion; separated polygamma and Barnes developments.	The square-root inverse theorem and nonduplicate special-function results are retained after branch and remainder review.

Continued on the next page

Source	Distinct strengths	Canonical disposition
UCF	Inverse-trigonometric coefficient evaluations and Loday’s associahedral hyperplane argument.	These concise geometric and inverse-series proofs are inserted beside the canonical Lagrange and associahedral results.
UCFIT	Coupled-exponential Good systems; Abel polynomials; Alexeiewsky, Barnes, and hyperfactorial identities; Fourier–Bessel Kepler analysis.	Only the genuinely stronger examples and proofs are retained; duplicated Bernoulli, Stirling, Bell, and reversion chapters are removed.

The exact source hashes and every topic- and claim-level decision are maintained in the package’s external provenance and disposition ledgers. Git revision 55038f2fc7d20a483a48125f086a6c8488e45530 is the immutable pre-consolidation source pin.

36.2 Closure of the merge

The consolidation is complete: the five donor packages have been retired and this manuscript is the single surviving document. The final pass was driven by a label- and formula-level inventory of all six sources rather than by reading order. Of the donor items with no canonical counterpart, the following were imported, each with a proof written for this text and each checked against an independent computation before insertion.

- *The sharp Bell bound.* All six arrivals asserted $B_n^{\text{Bell}} < (0.792 n / \log(n+1))^n$ for every $n \geq 1$ and proved neither the tail nor the finite range. It is now Theorem 11.13, with the monotonicity of the coefficient majorant supplied by Theorem 11.12 and the range $n \leq 38$ settled by the integer certificate (11.14). The result is due to Berend and Tassa [2].
- *Euler’s limit and the Weierstrass product* (Theorem 30.7) with the Beta integral (Theorem 30.6). This is the one import that repairs an existing argument: the polygamma partial fractions were being deduced from a Weierstrass product that appeared nowhere in the manuscript.
- *Harmonic-number expansion* (Theorem 29.36), *reciprocal Pochhammer derivatives* (Theorem 30.5), the *Pochhammer splitting rules* (30.4), and the *Hurwitz-zeta form* of the polygamma function (Theorem 30.10).
- *The associahedral h -polynomial* (Theorem 31.13) and *Loday’s realization* (Theorem 31.16), which give the Catalan chapter the same face-transform treatment the permutohedron already had.
- *The binomial-type characterization* (Theorem 25.8), the converse the canonical Sheffer section lacked.
- *Bernoulli polynomials in the finite-difference basis* (Theorem 29.14), the *coupled two-series Good system* (Theorem 28.4), the *inverse-derivative operator* (Theorem 27.4), *direct reversion through degree five* (Theorem 35.6), and the *first cumulants* (Theorem 25.6).

Two donor items were audited and deliberately not duplicated, because the canonical text already contained them under different labels: the mixed-partial Faà di Bruno formula, which is Theorem 19.3 verbatim, and the finite-difference lowering of the falling factorial, which is proved inline before Theorem 1.4. The exponential and logarithmic diagonal identities of one donor are likewise already present, and in the stronger Nörlund form (29.31).

Independent of mathematical content, the merge also repaired the manuscript’s notation. Eight calls to catalogue macros whose expansion ends in a script group were followed directly by a prime or an exponent, which is a fatal double-superscript error in $\text{\TeX}$; the source did not compile at all

until they were braced. A further 180 sites still spelled catalogue symbols by hand, including an italic imaginary unit standing next to an upright e inside a single exponent, and are now uniform.

36.3 Deduplication rules

The merger applied the following rules consistently.

1. *Definitions are unique.* Every array or polynomial family has one canonical defining equation. Alternate normalizations appear only in an explicit conversion formula.
2. *A theorem is stated once.* Repeated formulas for Stirling transforms, Bell-polynomial composition, Faà di Bruno, and Lagrange–Bürmann were merged at the theorem level. Independent proofs are retained only when they expose different mechanisms.
3. *Boundary values are part of the theorem.* Empty sums, 0^0 in coefficient formulas, signed versus unsigned first-kind numbers, and the indexing of type- B Eulerian numbers are fixed globally.
4. *Formal and analytic assertions are separated.* Algebraic identities in formal series require no convergence; radii, branches, termwise differentiation, and asymptotic remainders are asserted only after suitable analytic hypotheses are supplied.
5. *Computations carry certificates.* A finite table, primality report, modular period, or numerical constant is not called a symbolic theorem. The recurrence, exact integer data, interval enclosure, or modular witness needed to reproduce it is recorded.

Chapter 37

Status of internal legacy concordances

Each arrival contains an internal concordance that refers to older `articles/`, `Wikipedia/`, or `MathWorld/` folders. None of those folders accompanied the six filed packages. Their alleged contents therefore cannot establish coverage, attribution, or verification in the present repository. The canonical monograph removes the invented file-by-file tables and retains those references only as self-reported historical context.

Encyclopedic sources can be useful discovery aids, but a mathematical result enters this volume only through a proof in the body and, where attribution is material, a citation to an identifiable scholarly source. Likewise, an unshipped notebook or PDF cannot serve as a computational certificate. The immutable Git-backed source closure and the live disposition ledger are the complete evidence boundary for this six-manuscript consolidation.

Chapter 38

Formula-status ledger and proof audit

38.1 Classes of source statements

Every substantive source display was assigned to one of four classes before the merger.

1. *Exact identities.* These are stated in normalized notation and proved by a bijection, coefficient extraction, recurrence uniqueness, finite differences, incidence inversion, formal residues, or analytic estimates.
2. *Equivalent coordinate forms.*
Examples include the partition, frequency-vector, and Bell polynomial forms of Faà di Bruno; the matrix, basis-change, and generating-function forms of Stirling inversion; and the coefficient, derivative, and residue forms of Lagrange inversion. They are presented as one theorem with explicit conversions.
3. *Corrected formulas.* An index shift, omitted range, absent Jacobian, ordinary/exponential factorial mismatch, or unstated branch hypothesis is repaired before proof. The type-*B* Eulerian convention, associahedron indexing, multivariate Good determinant, and delta-function Jacobian are the principal examples.
4. *Schematic or computational claims.* An undefined asymptotic placeholder is not repaired by renaming it: it is either reconstructed with a precise scale and uniform remainder proof or withdrawn from theorem status. A finite search or decimal table is retained as a proved claim only when its exact recurrence, interval witnesses, or size-bounded modular certificate is present and replayable.

38.2 Reusable proof engines

The long list of formulas reduces to a small number of mechanisms.

Change of polynomial basis

Falling and rising factorials diagonalize finite differences; Stirling and Lah numbers are transition matrices. Eulerian numbers provide the parallel transition between powers and shifted binomial coefficients.

Labelled decomposition

Exponential generating functions turn disjoint labelled components into products and sets of components into exponentials. Bell polynomials retain the complete block-size profile.

Incidence inversion

Möbius inversion on the Boolean lattice gives binomial inversion; on the partition lattice it gives moments versus cumulants. Sheffer and Riordan arrays

package the same substitutions as lower-triangular operators.

Formal residues The residue change-of-variables rule proves one-variable Lagrange–Bürmann and, with a Jacobian determinant, Good’s multivariate formula.

Singularity localization

Cauchy’s formula, saddle points, Darboux’s identity, and transfer theorems convert local analytic structure into coefficient asymptotics. The assumptions needed for uniformity and error terms are stated at the theorem where they are used.

38.3 Independent verification protocol

The exact algorithms in Chapter 35 provide a second layer of protection against transcription errors. A new identity should be checked by:

1. converting every series to one declared ordinary or exponential normalization;
2. expanding far enough to encounter nontrivial multiplicities—fourth order already tests the Faà di Bruno pattern 1, 4, 3, 6, 1;
3. verifying triangular recurrences and inverse-matrix products over exact integers or rationals;
4. using a degree-bounded grid or a size-bounded Chinese-remainder certificate when the symbolic expression is too large for direct comparison; and
5. keeping analytic remainder estimates separate from finite numerical checks.

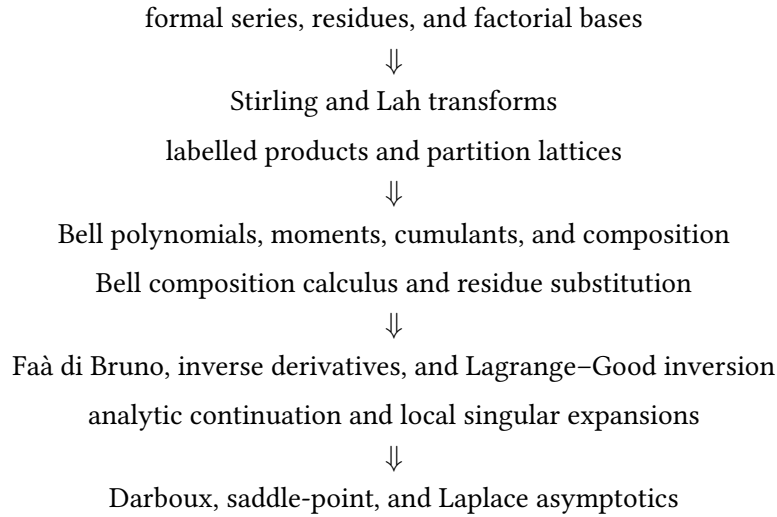
This protocol does not replace derivation. It makes the derivation reproducible and catches the factorial, sign, and index errors most likely to survive visual review.

38.4 References and the proof boundary

The NIST Digital Library of Mathematical Functions [17], Bender and Richmond’s multivariate inversion treatment [3], and Gessel’s proof survey [9] provide attribution and normalization context. A citation does not replace a proof: every theorem asserted by this monograph must either be proved in the body under its stated hypotheses or be removed from theorem status. The external source and claim-disposition ledgers record editorial provenance separately from mathematical validity.

38.5 Dependency map

The mathematical dependence of the unified article is summarized by



Eulerian and Catalan families form two transverse strands: the first translates powers into descent and polyhedral data, while the second translates reversion into trees, paths, polygon dissections, and associahedra. The resulting structure is a coefficient calculus rather than a catalogue of isolated identities.

38.6 Lean formalization register

Every theorem-like environment of this monograph is listed here with the status of its formalization in the Lean development `Analysis/FabiusFunction/Lean/FabiusFunction` (Mathlib v4.32.0). *Lean* means that a public, compiled declaration proves the exact statement or a more general one (the generalization is then noted next to the result); *partial* means that a named part of the statement is proved and the rest is named; *none* means that no Lean counterpart exists yet. A citation, a numerical check, or a plausible derivation does not count. The register is generated from the source and must be regenerated when results are added. Current totals: 48 Lean, 37 partial, 116 none, of 201 results.

Result	Title	Status	Lean declarations (module)
Theorem 1.1	Coefficient rules	partial	the formal rules are Mathlib's: Equation (1.1) is <code>PowerSeries.coeff_mul</code> and Equation (1.2) is <code>PowerSeries.coeff_derivative</code> ; Equation (1.3) is the product with the geometric series (<code>PowerSeries.coeff_mul</code> with <code>PowerSeries.coeff_mk</code>); Cauchy's coefficient formula and bound Equation (1.4) are not formalized in this form
Theorem 1.3	Formal change of variables for residues	none	
Theorem 1.4	Newton expansion	Lean	<code>Fabius.newton_expansion</code> (<code>NewtonExpansion</code>): $p = \sum_{k \leq d} \frac{\Delta^k p(0)}{k!} (x)^k$ in $K[x]$ for every field K of characteristic zero, from the evaluation form <code>Fabius.eval_natCast_eq_sum_choose_fwdDiff</code> ($p(m) = \sum_k \binom{m}{k} \Delta^k p(0)$ over any commutative ring), which packages Mathlib's Gregory–Newton formula <code>shift_eq_sum_fwdDiff_iter</code> with <code>Polynomial.fwdDiff_iter_eq_zero_of_degree_lt</code>

Result	Title	Status	Lean declarations (module)
Theorem 1.6	The exponential formula	Lean	Equation (1.10) is <code>Fabius.bellWeightSeries_pow</code> , Equation (1.11) is <code>Fabius.exp_subst_smul_bellWeightSeries</code> (<code>ExponentialFormula</code>), Equation (1.12) is <code>Fabius.exp_subst_bellWeightSeries</code> , all as substitutions of formal power series over any $\mathbb{Q}$ -algebra
Theorem 1.8	Partition-type count	none	recurrence and boundary values are <code>Mathlib's Nat.stirlingFirst_succ_succ</code> , <code>Nat.stirlingFirst_succ_zero</code> , <code>Nat.stirlingFirst_eq_zero_of_lt</code> ; the defining expansions are <code>Fabius.ascPochhammer_eq_sum_monomial_stirlingFirst</code> and <code>Fabius.descPochhammer_eq_sum_monomial_signedStirlingFirst</code> (<code>StirlingBasisChange</code>); the permutation count itself is not formalized
Theorem 2.2	Permutation interpretation	partial	
Theorem 2.4	Second-kind recurrence	Lean	<code>Nat.stirlingSecond_succ_succ</code> , <code>Nat.stirlingSecond_succ_zero</code> , <code>Nat.stirlingSecond_eq_zero_of_lt</code> (<code>Mathlib</code>); the set-partition count is <code>Mathlib's</code> definition by this recurrence
Theorem 2.5	Surjection formula	Lean	<code>Fabius.factorial_mul_stirlingSecond_eq_sum</code> (over $\mathbb{Z}$) and <code>Fabius.stirlingSecond_eq_sum_div_factorial</code> (over $\mathbb{Q}$) (<code>StirlingBasisChange</code>); proved by binomial inversion of <code>Fabius.pow_eq_sum_stirlingSecond_mul_factorial_mul_choose</code>
Theorem 2.7	Alternative second-kind recurrences	partial	Equation (2.9) is <code>Fabius.stirlingSecond_eq_pow_div_factorial_sub_sum</code> (<code>StirlingTriangularExplicit</code>), for all n, k with the $r = 0$ term included; the two reverse recurrences Equations (2.10) and (2.11) are not formalized
Theorem 2.9	Reverse recurrences of the first kind	Lean	Equation (2.12) is <code>Fabius.first_reverse_row</code> and Equation (2.13) is <code>Fabius.first_reverse_column</code> (module <code>StirlingFirstReverse</code>), the first from the rising factorial, the second by comparing exponential generating functions (<code>Fabius.egfA_first_reverse_column</code>).
Theorem 3.1	Stirling basis transformations	Lean	<code>Fabius.ascPochhammer_eq_sum_monomial_stirlingFirst</code> , <code>Fabius.descPochhammer_eq_sum_monomial_signedStirlingFirst</code> , <code>Fabius.X_pow_eq_sum_stirlingSecond_mul_descPochhammer</code> , <code>Fabius.X_pow_eq_sum_stirlingSecond_mul_ascPochhammer</code> , <code>Fabius.pow_eq_sum_stirlingSecond_mul_factorial_mul_choose</code> (<code>StirlingBasisChange</code>); polynomial identities over every commutative ring
Theorem 3.3	Shifted factorial evaluations	Lean	Equation (3.6) is <code>Fabius.X_pow_eq_sum_stirlingSecond_succ_mul_descPochhammer_comp</code> in $R[x]$ over any commutative ring (cancelling x via <code>Fabius.X_mul_cancel</code>) and Equation (3.7) is <code>Fabius.pow_self_eq_sum_stirlingSecond_mul_descFactorial</code> (<code>StirlingShiftedEvaluations</code>)
Theorem 3.5	Inverse matrices	Lean	<code>Fabius.sum_range_stirlingSecond_mul_signedStirlingFirst</code> , <code>Fabius.sum_range_signedStirlingFirst_mul_stirlingSecond</code> and their <code>Finset.Icc</code> forms (<code>StirlingBasisChange</code>)

Result	Title	Status	Lean declarations (module)
Theorem 3.7	Poisson moments	partial	the moment series $\sum_m m^n \lambda^m / m! = e^\lambda \sum_k S(n, k) \lambda^k$ is <code>Fabius.tsum_pow_mul_pow_div_factorial</code> (<code>BellStirling</code>) for every real λ ; the probabilistic phrasing is not formalized
Theorem 3.8	Moments of the number of fixed points	none	
Theorem 3.10	Conversion between rising and falling factorials	Lean	<code>Fabius.ascPochhammer_eq_sum_lahNumber_mul_descPochhammer</code> , <code>Fabius.descPochhammer_eq_sum_lahNumber_mul_ascPochhammer</code> , <code>Fabius.sum_range_lahNumber_mul_lahNumber</code> , <code>Fabius.lahNumber_eq_sum_stirlingFirst_mul_stirlingSecond</code> (<code>LahNumbers</code>); the closed form of the definition is <code>Fabius.lahNumber_succ_succ_mul_factorial</code>
Theorem 4.1	Column generating functions	partial	<code>Fabius.exp_sub_one_pow</code> , <code>Fabius.egf_stirlingSecond</code> , <code>Fabius.negLogOneSub_pow</code> , <code>Fabius.egf_stirlingFirst</code> , <code>Fabius.log_pow</code> (<code>StirlingGeneratingFunctions</code>), as formal power series over any commutative $\mathbb{Q}$ -algebra; the bivariate generating function Equation (4.4) is not formalized
Theorem 4.3	Fixed- k ordinary generating function	Lean	<code>Fabius.prod_one_sub_mul_X_mul_stirlingColumnOGF</code> and <code>Fabius.stirlingColumnOGF_eq_prod_mk_pow</code> (<code>StirlingOrdinaryGF</code>): the column series $\sum_r \left\{ \begin{smallmatrix} k+r \\ k \end{smallmatrix} \right\} x^r$ times $\prod_{j \leq k} (1 - jx)$ is 1, and equals the product of the geometric series $\sum_r j^r x^r$, in $R[[x]]$ for every commutative ring R ; the complete-homogeneous coefficient formula is the coefficient extraction of the latter and is not stated separately
Theorem 4.5	Stirling transform and its inverse	Lean	the inversion formula is <code>Fabius.stirling_inversion</code> , <code>Fabius.stirling_inversion_symm</code> , <code>Fabius.stirling_inversion_iff</code> (<code>StirlingBasisChange</code>), for sequences in any additive commutative group; the generating-function form is <code>Fabius.egfA_subst_exp_sub_one</code> and <code>Fabius.egfA_subst_log</code> (<code>StirlingTransformEGF</code>): substituting $e^t - 1$, respectively $\log(1 + t)$, into an exponential generating function applies the second-kind, respectively signed first-kind, Stirling transform
Theorem 4.7	Stirling normal-ordering identities	partial	<code>Fabius.iterate_X_mul_derivative</code> (Equation (4.11)) and <code>Fabius.xkDk_eq_sum_signedStirlingFirst</code> (Equation (4.12), by Stirling inversion) in <code>StirlingNormalOrder</code> , as identities of operators applied to an arbitrary polynomial over any commutative ring; the falling-factorial form $(xD)^n$ and the operator series Equations (4.13) and (4.14) are not formalized

Result	Title	Status	Lean declarations (module)
Theorem 5.1	First- and second-kind sum- mations	Lean	<code>StirlingSummations</code> : Equation (5.14) is <code>Fabius.stirlingFirst_succ_succ_eq_sum_choose</code> (module <code>StirlingBasisChange</code>) and <code>Fabius.stirlingFirst_succ_succ_eq_sum_descFactorial</code> , Equation (5.15) is <code>Fabius.stirlingSecond_succ_succ_eq_sum</code> (<code>BellStirling</code>) and <code>Fabius.stirlingSecond_succ_succ_eq_sum_pow</code> , the hockey sticks Equations (5.16) and (5.17) are <code>Fabius.stirlingFirst_add_succ_eq_sum</code> and <code>Fabius.stirlingSecond_add_succ_eq_sum</code> (with lower index k in the first-kind case, correcting a misprint), and the convolutions Equations (5.18) and (5.19) are <code>Fabius.choose_mul_stirlingFirst_add</code> and <code>Fabius.choose_mul_stirlingSecond_add</code>
Theorem 5.3	Explicit first-kind double sum	none	
Theorem 5.4	Near-diagonal partition formula	none	
Theorem 6.1	Rigorous finite-modulus re- placement for the archived spectral form	partial	Equation (6.4) is <code>Fabius.stirlingFirst_cast_eq_coeff_block</code> (module <code>StirlingFirstModH</code>), together with the rising-factorial product form <code>Fabius.ascPochhammer_eq_prod_range</code> ; the eventual vanishing that makes the spectral reading vacuous is <code>Fabius.stirlingFirst_cast_eq_zero_of_lt</code> . The linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and the Jordan decomposition are not formalized.
Theorem 6.4	Parity criterion	partial	Equation (6.5) is <code>Fabius.stirlingSecond_modEq_choose_two</code> (module <code>StirlingParity</code>), for $1 \leq k \leq n$, from the column series modulo 2, <code>Fabius.stirlingColumnOGF_zmod_two</code> ; the bitwise form Equation (6.6) (Lucas) and the power-of-two criterion are not formalized.
Theorem 7.1	Fixed column and fixed di- agonal	none	
Theorem 7.2	Real-rootedness and strict log-concavity	none	
Theorem 7.3	Asymptotic mode and max- imal entry	none	
Theorem 7.4	Uniform Stirling saddle es- timate	none	
Theorem 7.5	Stirling-zeta identities	none	
Theorem 7.6	Logarithmic transfer esti- mate	none	
Theorem 7.7	Convergent Stirling series for $\log \Gamma$ and ψ	none	
Theorem 7.8	Stirling series for the Stielt- jes constants	none	
Theorem 8.1	r -Stirling recurrence	partial	With $\{n\}_r$ defined by the recurrence (<code>Fabius.rStirling</code> , module <code>RStirling</code>): Equation (8.1) is <code>Fabius.rStirling_succ_succ</code> , the explicit formula $\{n+r\}_{k+r} = \sum_j \binom{n}{j} \{j\}_k r^{n-j}$ is <code>Fabius.rStirlingShift_eq_sum</code> and Equation (8.2) is <code>Fabius.egfA_rStirlingPoly</code> ; the distinct-blocks count is not formalized.

Result	Title	Status	Lean declarations (module)
Theorem 8.3	Associated recurrence	partial	With S_r defined as the partial Bell polynomial with weights $[j \geq r]$ (<code>Fabius.associatedStirling</code> , module <code>AssociatedStirling</code>): Equation (8.3) is <code>Fabius.associatedStirling_succ_succ</code> and Equation (8.4) is <code>Fabius.egfA_associatedStirling</code> ; the block-size count is not formalized.
Theorem 8.5 proposition (unlabelled, no. 1)	Reduction formula	none none	
Theorem 9.3	Ordered Bell numbers	partial	<code>Fabius.fubini</code> is defined by Equation (9.1); Equation (9.2) is <code>Fabius.two_sub_exp_mul_egfA_fubini</code> (via <code>Fabius.egfA_fubini</code> : the generating function is $1/(1-u)$ at $u = e^t - 1$) and Equation (9.3) is <code>Fabius.fubini_succ</code> (<code>OrderedBell</code>); the ordered-partition count itself is not formalized
proposition (unlabelled, no. 2)		none	
proposition (unlabelled, no. 3)		none	
theorem (unlabelled, no. 33)	Return-to-order shuffles	none	
theorem (unlabelled, no. 34)		none	
theorem (unlabelled, no. 35)		none	
Theorem 10.1	Binomial recurrence	Lean	<code>Nat.bell_succ</code> (<code>Mathlib</code> , the definition)
Theorem 10.2	Stirling decomposition	Lean	<code>Fabius.bell_eq_sum_stirlingSecond</code> , <code>Fabius.bell_eq_sum_sum_div_factorial</code> (<code>BellStirling</code>)
Theorem 10.4	Spivey's identity	Lean	<code>Fabius.spivey</code> (<code>BellShiftEGF</code>), read off from the shifted generating function <code>Fabius.egfA_bell_add</code> : $\sum_n \mathcal{B}_{n+m}^{\text{Bell}} t^n / n! = \mathcal{T}_m^{\text{Tot}}(e^t) e^{e^t-1}$, proved by induction on m from the Touchard recurrence <code>Fabius.derivative_touchardExp</code> ; the Touchard-polynomial version $\mathcal{T}_{m+n}^{\text{Tot}}(x) = \sum_{j,k} \left\{ \begin{smallmatrix} m \\ j \end{smallmatrix} \right\} x^j \binom{n}{k} \mathcal{T}_k^{\text{Tot}}(x) j^{n-k}$ is <code>Fabius.spivey_touchard</code> (<code>TouchardShiftEGF</code>)
Theorem 10.6	Bell inversion identities	partial	Equation (10.4) is <code>Fabius.bell_eq_sum_neg_one_pow_choose_bell_succ</code> (<code>BellShiftEGF</code>), by <code>Fabius.binomial_inversion_ring</code> applied to <code>Fabius.bell_succ_eq_sum_choose</code> ; Equation (10.5) is not formalized
Theorem 10.8	Weighted Bell shift	Lean	<code>Fabius.weighted_bell_shift</code> is Equation (10.7), <code>Fabius.weighted_bell_shift_one</code> is Equation (10.8) and <code>Fabius.sum_signedStirlingFirst_mul_bell_eq</code> is Equation (10.9), over any commutative ring, from the umbral shift <code>Fabius.bellUmbra_descPochhammer_mul</code> (module <code>BellUmbra</code>).
Theorem 10.10		Lean	<code>Fabius.exp_subst_exp_sub_one</code> (<code>BellGeneratingFunctions</code>): $\exp \circ (e^z - 1) = \sum_n B(n) z^n / n!$ as formal power series over any commutative $\mathbb{Q}$ -algebra; the differential equation is the derivative of this substitution
Theorem 10.12	Dobiński	Lean	<code>Fabius.dobinski</code> (<code>BellStirling</code>), from the general Poisson moment identity <code>Fabius.tsu_m_pow_mul_pow_div_factorial</code>

Result	Title	Status	Lean declarations (module)
corollary (unlabelled, no. 4)	Cauchy integral	none	
Theorem 11.1	Polynomial Touchard congruence	Lean	Equation (11.1) is <code>Fabius.touchardPolynomial_add_prime</code> (<code>TouchardPolyCongruence</code>), an identity in $(\mathbb{Z}/p)[x]$, read off from Spivey's identity for Touchard polynomials (<code>Fabius.touchardPolynomial_add_eq</code>) modulo p with <code>Fabius.stirlingSecond_prime_eq_zero_zmod</code> ($p \mid \{^p_k\}$ for $1 < k < p$); Equation (11.2) is <code>Fabius.bell_add_prime_modEq</code> (<code>TouchardCongruence</code>)
Theorem 11.3	Prime-power shift	Lean	<code>Fabius.bell_add_prime_pow_modEq</code> is Equation (11.3); <code>Fabius.touchardPolynomial_add_prime_pow</code> is Equation (11.4) over $\mathbb{F}_p$; both from the operator identity <code>Fabius.shift_pow_char_pow</code> (module <code>ShiftOperatorCharP</code>).
Theorem 11.5		Lean	<code>Fabius.bell_add_sum_prime_pow_modEq</code> ; N_p is a period modulo p (<code>Fabius.shift_pow_period</code> , via the Fermat product <code>Fabius.prod_range_X_add_C_natCast</code>); <code>Fabius.bell_period_dvd_sum_prime_pow</code> : the least period divides N_p .
Theorem 11.8		none	
Theorem 11.10		none	
Theorem 11.11	Eventual Bell upper bound	none	
Theorem 11.12	The coefficient majorant is decreasing	none	
Theorem 11.13	Uniform numerical Bell bound	none	
Theorem 11.15	Leading Bell asymptotic	none	
Theorem 11.17	de Bruijn logarithmic expansion	none	
Theorem 12.3	Weighted-partition interpretation	none	
Theorem 12.4	Bell-polynomial generating functions	partial	the first identity is <code>Fabius.bellWeightSeries_pow</code> and the third <code>Fabius.exp_subst_bellWeightSeries</code> (<code>BellGeneratingFunctions</code>), as formal power series over any commutative $\mathbb{Q}$ -algebra; the bivariate form is <code>Fabius.exp_subst_smul_bellWeightSeries</code> (<code>ExponentialFormula</code>) and the ordinary Bell polynomials are <code>Fabius.ordPartialBell</code> with <code>Fabius.coeff_pow_eq_ordPartialBell</code> (<code>OrdinaryBellComposition</code>); the ordinary bivariate identity Equation (12.9) is not formalized
Theorem 13.1	Complete and partial recurrences	Lean	<code>Fabius.partialBell</code> is defined by Equation (13.2) (<code>Fabius.partialBell_succ_succ</code> , <code>Fabius.partialBell_succ_succ_eq_binomialConv</code>); Equation (13.1) is <code>Mathlib-free Bell.complete_succ</code> together with <code>Fabius.bell_complete_eq_sum_partialBell</code> (<code>PartialBellPolynomials</code> , <code>BellPolynomialInversion</code>); the boundary values are <code>Fabius.partialBell_zero_succ</code> , <code>Fabius.partialBell_succ_zero</code> , <code>Fabius.partialBell_eq_zero_of_lt</code>
Theorem 13.2	Block-colour convolution	Lean	<code>Fabius.factorial_mul_partialBell_add</code> (<code>BellComposition</code>), in the division-free form $(k_1 + k_2)! B_{n, k_1 + k_2} = k_1! k_2! \sum_i \binom{n}{i} B_{i, k_1} B_{n-i, k_2}$

Result	Title	Status	Lean declarations (module)
Theorem 13.3 Theorem 13.4	Derivative identities	none partial	Equation (13.11) is <code>Fabius.pderiv_partialBell_succ</code> and Equation (13.10) is <code>Fabius.pderiv_bellComplete</code> (module <code>BellDerivative</code>), in $\mathbb{Q}[x_1, x_2, \dots]$ via the coefficientwise derivation <code>Fabius.coeffDerivation</code> ; the chain rule Equation (13.12) is not formalized.
Theorem 13.6 Theorem 14.1		none partial	Equation (14.1) is <code>Fabius.partialBell_factorial_pred</code> , Equation (14.3) is <code>Fabius.partialBell_one</code> , Equation (14.4) is <code>Fabius.bell_complete_one</code> , Equation (14.5) is <code>Fabius.partialBell_factorial</code> (<code>PartialBellPolynomials</code> , <code>BellGeneratingFunctions</code>), and the Touchard form Equation (14.6) is <code>Fabius.bell_complete_const_eq_touchard_eval</code> (<code>BellHomogeneity</code>); the factorial row sum Equation (14.2) is not formalized
Theorem 14.3	Bigraded homogeneity	Lean	<code>Fabius.partialBell_mul_left</code> , <code>Fabius.partialBell_pow_mul</code> , <code>Fabius.partialBell_bihomogeneous</code> (<code>BellHomogeneity</code>), over every commutative semiring
Theorem 14.5	Complete Bell addition formula	Lean	<code>Bell.complete_add</code> (<code>BellPolynomialInversion</code>), over every commutative semiring
Theorem 14.6	Insertion of q leading zeros	Lean	<code>Fabius.partialBell_leadingZeros</code> (<code>BellLeadingZeros</code>), over any $\mathbb{Q}$ -algebra, with the zero-padded weights <code>Fabius.leadingZeros</code> and the rescaled weights <code>Fabius.qScaled</code> ; the proof compares the weight series (<code>Fabius.bellWeightSeries_leadingZeros</code>) exactly as in the text
Theorem 14.8	Bell transform and its inverse	Lean	Equation (14.12) is <code>Fabius.bell_transform_inverse</code> (<code>BellComposition</code>), from $X = \log(1+Y)$ formalized as <code>Fabius.log_subst_exp_sub_one</code> and the composition theorem; the recursive inversion over every commutative ring is <code>Bell.complete_cumulant</code> and <code>Bell.cumulant_complete</code> (<code>BellPolynomialInversion</code>). The more general locally-inverse transform Equation (14.14), which the text states after this theorem rather than inside it, is not formalized
Theorem 14.10 Theorem 14.11 Theorem 15.1 Theorem 15.2 Theorem 15.3	Determinant from traces	none none none none none	
Theorem 15.4		Lean	Equation (15.5) is <code>Fabius.bell_complete_eq_sum_partialBell</code> with <code>Fabius.exp_subst_bellWeightSeries</code> , and Equation (15.6) is <code>Fabius.bell_transform_inverse</code> (<code>BellComposition</code>); the recursive forms are <code>Fabius.completeBellPolynomial_momentCumulant</code> , <code>Fabius.momentCumulant_completeBellPolynomial</code> (<code>MomentCumulantAlgebra</code>) and <code>Bell.complete_cumulant</code>

Result	Title	Status	Lean declarations (module)
Theorem 15.5		partial	with <code>Fabius.binomialTypePoly</code> defined by Equation (15.8) (<code>BinomialType</code>): Equation (15.9) is <code>Fabius.exp_subst_smul_bellWeightSeries_eq_egfA_binomialTypePoly</code> and Equation (15.10) is <code>Fabius.binomialTypePoly_add</code> (from the addition law <code>Bell.complete_add</code>), over any commutative ring; the delta operator Equation (15.11) is not formalized
Theorem 16.2	Eulerian recurrence	partial	<code>Fabius.eulerianNumber</code> is defined by this recurrence (<code>Fabius.eulerianNumber_succ_left</code> , <code>Fabius.eulerianNumber_succ_succ</code> , module <code>EulerianNumbers</code>); the descent count and the polynomial form Equation (16.2) are not formalized
Theorem 16.3	Eulerian EGF	Lean	<code>Fabius.egfA_eulerianPolynomial_mul</code> (<code>EulerianEGF</code>): in $(\mathbb{Q}[t])[x]$, $(\sum_n E_n^{\text{Eul},A}(t)x^n/n!)(t - e^{(t-1)x}) = t - 1$, i.e. Equation (16.5) in multiplicative form; it is the binomial recurrence Theorem 16.7 read coefficient-wise
Theorem 16.5		Lean	<code>Fabius.one_sub_X_pow_mul_succPowSeries</code> (the identity $(1-t)^{n+1} \sum_m (m+1)^n t^m = E_n^{\text{Eul},A}(t)$ in $R[[t]]$) and <code>Fabius.eulerianNumber_eq_sum_int</code> (<code>EulerianGeneratingFunctions</code>); the symmetry Equation (16.3) is <code>Fabius.eulerianNumber_symm</code> and Equation (16.4) is <code>Fabius.eulerianNumber_one_right</code>
Theorem 16.7	A second polynomial recurrence	Lean	<code>Fabius.eulerianPolynomial_binomial_recurrence</code> in $R[t]$ and <code>Fabius.eulerian_binomial_recurrence_series</code> in $R[[t]]$ (<code>EulerianEGF</code>), for $n \geq 1$ over any commutative ring R , proved from the rational generating function Equation (16.6) and the binomial theorem rather than from the EGF
Theorem 16.9	Worpitzky	Lean	<code>Fabius.worpitzky_nat</code> (natural numbers, all n) and <code>Fabius.worpitzky_polynomial</code> (in $\mathbb{Q}[x]$, with <code>Fabius.binomialPoly</code> for $\binom{x+k}{n}$) (<code>EulerianNumbers</code>)
Theorem 16.11	Power sums	Lean	<code>Fabius.sum_range_pow_succ_eq_sum_eulerianNumber</code> (<code>EulerianNumbers</code>), in the form $\sum_{r=0}^m r^{n+1} = \sum_k A(n+1, k) \binom{m+k+1}{n+2}$
Theorem 16.12		Lean	<code>Fabius.sum_eulerianNumber_mul_X_pow_eq_sum_stirlingSecond</code> (<code>EulerianStirling</code>), as an identity in $R[[t]]$ over any commutative ring R , from the rising-factorial expansion of $(m+1)^n$ and $\sum_m \binom{m+k}{k} t^m = (1-t)^{-k-1}$
Theorem 16.14		none	
Theorem 16.15		partial	Equation (16.14) is <code>Fabius.sum_neg_one_pow_mul_eulerianNumber</code> (<code>EulerianAlternating</code>), from the Eulerian EGF at $t = -1$ (<code>Fabius.egfA_eulerianPolynomial_eval_neg_one_mul</code>) and $x \tanh x = x - \mathcal{B}(2x) + \mathcal{B}(4x)$ (<code>Fabius.X_mul_egfA_eulerianPolynomial_eval_neg_one</code>); the two reciprocal-binomial identities are not formalized
theorem (unlabelled, no. 76)		none	
Theorem 17.2	The permutohedral h -polynomial	none	

Result	Title	Status	Lean declarations (module)
Theorem 18.1	Type- B recurrence and Worpitzky series	partial	<code>Fabius.typeBEulerian</code> (<code>TypeBEulerian</code>) is defined by the recurrence Equation (18.1) (<code>Fabius.typeBEulerian_succ_succ</code>); Equation (18.2) is <code>Fabius.one_sub_X_pow_mul_oddPowSeries</code> in $R[[t]]$ (from the type- B Worpitzky identity <code>Fabius.typeB_worpitzky</code>) and Equation (18.3) is <code>Fabius.typeBEulerian_eq_sum_int</code> ; the signed-permutation count is not formalized
Theorem 18.3	The type- B permutohedral h -polynomial	none	
Theorem 18.4	Second-order recurrence	partial	<code>Fabius.secondEulerian</code> (<code>SecondOrderEulerian</code>) is defined by the recurrence Equation (18.4) (<code>Fabius.secondEulerian_succ_succ</code>); the polynomial recurrence Equation (18.5) is <code>Fabius.secondEulerianSeries_succ</code> (as power series) and the row sum Equation (18.6) is <code>Fabius.sum_secondEulerian_eq_doubleFactorial</code> ; the Stirling-permutation count is not formalized
Theorem 18.6	Diagonal Stirling generating function	Lean	<code>Fabius.one_sub_X_pow_mul_diagStirlingSeries</code> (<code>SecondOrderEulerian</code>), in $R[[t]]$ over any commutative ring, by the induction of the text: $(1-t)F_{n+1} = tF'_n$ (<code>Fabius.one_sub_X_mul_diagStirlingSeries_succ</code>) and the polynomial recurrence
Theorem 18.8	Second-order Eulerian diagonal formula	none	
Theorem 19.1	Faà di Bruno, partition form	partial	the analytic statement is not formalized; its formal power-series analogue, the coefficients of $F'(G(t))$ through partial Bell polynomials, is <code>Fabius.egfA_subst_bellWeightSeries</code> (<code>BellComposition</code>)
Theorem 19.2	Faà di Bruno, multiplicity form	partial	the analytic statement is not formalized; the formal power-series analogue <code>Fabius.egfA_subst_bellWeightSeries</code> carries the Bell-polynomial form, with the partial Bell polynomials defined by the pointing recurrence rather than by the multiplicity sum
Theorem 19.3	Multivariate Faà di Bruno for distinct variables	none	
Theorem 20.1	Ordinary composition	partial	Equation (20.3) is <code>Fabius.coeff_subst_eq_sum_ordPartialBell</code> (<code>OrdinaryBellComposition</code>) over any commutative ring, with the ordinary Bell polynomials <code>Fabius.ordPartialBell</code> defined by the composition recurrence $\mathcal{B}_{n,k+1}^{\text{ord}} = \sum_{i \geq 1} b_i \mathcal{B}_{n-i,k}^{\text{ord}}$ (so Equation (20.1) is the definition unrolled) and $[x^n]G^k = \mathcal{B}_{n,k}^{\text{ord}}(b)$ as <code>Fabius.coeff_pow_eq_ordPartialBell</code> ; the reciprocal formula Equation (20.4) is <code>Fabius.coeff_reciprocalSeries</code> ; the multinomial form Equation (20.2) is not formalized
Theorem 20.3	Exponential composition	Lean	<code>Fabius.egfA_subst_bellWeightSeries</code> (<code>BellComposition</code>): substitution of exponential generating functions over any commutative $\mathbb{Q}$ -algebra

Result	Title	Status	Lean declarations (module)
Theorem 21.1	Lagrange–Bürmann	partial	Equation (21.2) is <code>Fabius.Lagrange.coeff_subst_derivative</code> and Equation (21.4) is <code>Fabius.Lagrange.coeff_subst_id</code> (module <code>LagrangeInversion</code>), over any commutative $\mathbb{Q}$ -algebra and in the division-free form $n[z^n]H(g) = [w^{n-1}]H'\phi^n$, from the division-free core <code>Fabius.Lagrange.coeff_subst_mul_derivative</code> . The solution g of Equation (21.1) is constructed, not assumed (<code>Fabius.Lagrange.solution</code> , <code>Fabius.Lagrange.solution_eq</code>), so <code>Fabius.Lagrange.coeff_solution_subst_derivative</code> and <code>Fabius.Lagrange.coeff_solution</code> are unconditional; uniqueness and Equation (21.3) are not formalized.
Theorem 21.3	Analytic Lagrange inversion	none	
Theorem 21.4	Explicit inverse coefficients	none	
Theorem 21.5	Associahedral inversion	none	
Theorem 22.1		partial	Equation (22.2) is <code>Fabius.coeff_fussSolution</code> together with <code>Fabius.coeff_fussSolution_eq_zero</code> (module <code>FussCatalanSeries</code>), for <code>Fabius.fussSolution</code> constructed as the Lagrange solution; the polynomial form of the defining equation is <code>Fabius.fussSolution_sub_pow</code> . The radius Equation (22.3) and the boundary convergence are analytic and are not formalized.
Theorem 22.3		partial	
Theorem 23.1	Lagrange reversion	none	The series Equation (22.4) is <code>Fabius.coeff_lambertW</code> (module <code>LambertWSeries</code>), for <code>Fabius.lambertW</code> constructed as the Lagrange solution of $W = ze^{-W}$; the defining equation in the form $We^W = z$ is <code>Fabius.lambertW_mul_exp_subst</code> . The radius of convergence e^{-1} is analytic and is not formalized.
Theorem 24.1	Laplace–Erdélyi coefficient formula	none	
Theorem 25.1	Binomial inversion	partial	
Theorem 25.3	Möbius function of Π_n	none	Fabius. <code>binomial_inversion_iff</code> (additive commutative groups) and Fabius. <code>binomial_inversion_ring_iff</code> (commutative rings) (<code>BinomialInversion</code>); the kernel orthogonality is <code>Fabius.sum_Icc_neg_one_pow_choose_mu_l_choose</code> ; the EGF form is not formalized
Theorem 25.4	Moment–cumulant inversion	partial	
Theorem 25.8	Generating-function characterization of binomial type	none	In the block-size form: <code>Bell.complete</code> and <code>Bell.cumulant</code> are mutually inverse (<code>Bell.complete_cumulant</code> , <code>Bell.cumulant_complete</code> , module <code>BellPolynomialInversion</code>) and $M = e^K$ is <code>Fabius.exp_subst_bellWeightSeries</code> ; the sums over set partitions and $K = \log M$ are not formalized.

Result	Title	Status	Lean declarations (module)
Theorem 25.9	Addition and lowering	partial	Equation (25.14) is <code>Fabius.shefferPoly_add</code> (<code>BinomialType</code>), for <code>Fabius.shefferPoly = c * p</code> (binomial convolution of the coefficients of g with the binomial-type sequence), with generating function <code>Fabius.egfA_mul_exp_subst_smul_bellWeightSeries</code> ; the lowering law Equation (25.15) is not formalized
Theorem 25.11	Abel polynomials	none	
Theorem 25.13	Riordan action, product, and inverse	Lean	<code>Fabius.expRiordan_action</code> , <code>Fabius.expRiordan_mul</code> , <code>Fabius.expRiordan_mul_inverse</code> (<code>ExponentialRiordan</code>), for exponential Riordan arrays $[g, f]$ over any $\mathbb{Q}$ -algebra, with the inverse law in the form $[g, f][h, \bar{f}] = [1, t]$ whenever $g(h \circ f) = 1$; the Stirling examples are <code>Fabius.expRiordan_one_exp_sub_one</code> and <code>Fabius.expRiordan_one_log</code>
Theorem 26.2	EGF, recurrence, and Dobiński-type series	Lean	<code>Fabius.complementaryBell</code> (<code>ComplementaryBell</code>); Equation (26.2) is <code>Fabius.exp_subst_neg_exp_sub_one</code> , Equation (26.3) is <code>Fabius.complementaryBell_succ</code> , and Equation (26.4) is <code>Fabius.complementaryBell_eq_exp_mul_tsum</code> (the series converges absolutely as the Poisson moment series of <code>Fabius.tsum_pow_mul_pow_div_factorial</code>)
Theorem 26.4	Convolution inverse of the Bell sequence	Lean	<code>Fabius.sum_choose_bell_mul_complementaryBell</code> (<code>ComplementaryBell</code>), from the addition law <code>Bell.complete_add</code> of complete Bell polynomials at weights ± 1
Theorem 27.1	Coordinate-free Faà di Bruno	none	
corollary (unlabelled, no. 9)		none	
Theorem 27.3	Inverse derivative recursion	none	
Theorem 27.4	Inverse-derivative operator	none	
Theorem 28.1	Multivariate residue substitution	none	
Theorem 28.2	Lagrange–Good inversion	none	
Theorem 28.5	Characteristic determinant	none	
Theorem 29.2	Appell, translation, and explicit formulas	Lean	Bernoulli: the derivative identity is <code>Mathlib's Polynomial.derivative_bernoulli</code> , the translation formula is <code>Fabius.bernoulli_eval_add</code> (<code>BernoulliAppell</code>) and the explicit formula Equation (29.6) is the definition <code>Polynomial.bernoulli_def</code> ; Euler: <code>Fabius.eulerPolynomial</code> (<code>EulerPolynomials</code>), defined from $2/(e^t + 1)$ as an Appell sequence), <code>Fabius.derivative_eulerPolynomial</code> and <code>Fabius.eulerPolynomial_eval_add</code> , both translation formulas being instances of the general Appell lemma <code>Fabius.appell_eval_add</code>

Result	Title	Status	Lean declarations (module)
Theorem 29.4	Difference and reflection identities	Lean	Mathlib: the difference identity Equation (29.7) is <code>Polynomial.bernoulli_eval_one_add</code> , the reflection Equation (29.9) is <code>Polynomial.bernoulli_eval_one_sub</code> , and $\beta_{2m+1} = 0$ is <code>bernoulli_eq_zero_of_odd</code> ; the Euler difference identity Equation (29.8) is <code>Fabius.eulerPolynomial_eval_add_one_add</code> (<code>EulerPolynomials</code>), the Euler reflection Equation (29.10) is <code>Fabius.eulerPolynomial_eval_one_sub</code> and $E_{2m+1}(1/2) = 0$ is <code>Fabius.eulerPolynomial_eval_half_odd</code> (<code>EulerReflection</code>)
Theorem 29.6	Faulhaber with an offset	Lean	<code>Fabius.sum_range_add_pow_eq_bernoulli_sub</code> (<code>FaulhaberOffset</code>), telescoping Mathlib's <code>Polynomial.bernoulli_eval_one_add</code> ; the case $x = 0$ is also Mathlib's <code>Polynomial.sum_range_pow_eq_bernoulli_sub</code> (and <code>sum_range_pow</code> , <code>sum_Ico_pow</code> in the Bernoulli-number form)
Theorem 29.8	Alternating power sums	Lean	<code>Fabius.sum_neg_one_pow_mul_pow_eq_eulerPolynomial</code> (<code>EulerPolynomials</code>), in the form $\sum_{j < N} (-1)^j (x + j)^p = (E_p(x) - (-1)^N E_p(x + N))/2$, valid for all $N \geq 0$, by induction from the difference identity
Theorem 29.10	Raabe multiplication	Lean	<code>Fabius.raabe</code> (<code>RaabeMultiplication</code>), in the form $\sum_{r < q} \beta_n(x + r/q) = q(1/q)^n \beta_n(qx)$, proved from the generating functions in $(\mathbb{Q}[x])[t]$ exactly as in the text; Equation (29.14) is <code>Fabius.bernoulli_eval_half</code>
Theorem 29.12	Bernoulli–Stirling formula and Touchard moments	partial	Equation (29.15) is <code>Fabius.bernoulli_eq_sum_stirlingSecond</code> (<code>BernoulliStirling</code>) for Mathlib's <code>bernoulli</code> (with $B_1 = -1/2$), from <code>Fabius.bernoulliPowerSeries_eq_logDivSeries_subst</code> : the Bernoulli generating function is $\log(1 + u)/u$ at $u = e^t - 1$; the two integral representations are not formalized
Theorem 29.14	Bernoulli polynomials from finite differences	Lean	Equation (29.19) is <code>Fabius.bernoulli_eq_sum_fwdDiff</code> (module <code>BernoulliNewtonBasis</code>), an identity in $\mathbb{Q}[x]$ with $\Delta^k x^n$ written out as $\sum_j (-1)^{k-j} \binom{k}{j} (x + j)^n$; the truncated Gregory composition is <code>Fabius.x_pow_dvd_bernoulliPowerSeries_sub_gregory</code> .
Theorem 29.16	Second proof of the Bernoulli–Stirling formula	Lean	<code>Fabius.bernoulli_eq_sum_fwdDiff_zero</code> (module <code>BernoulliNewtonBasis</code>): $\beta_n = \sum_k \frac{(-1)^k}{k+1} \Delta^k 0^n$, and with the surjection formula <code>Fabius.factorial_mul_stirlingSecond_eq_sum</code> this is <code>Fabius.bernoulli_eq_sum_stirlingSecond</code> .
Theorem 29.19	Appell, difference, and convolution calculus	partial	For natural orders $\alpha, \gamma \in \mathbb{N}$ (module <code>NorlundPolynomials</code> , $\beta_n^{(a)}$ defined by <code>Fabius.norlund</code>): Equation (29.22) is <code>Fabius.derivative_norlund_succ</code> , Equation (29.23) is <code>Fabius.norlund_eval_add</code> , Equation (29.24) is <code>Fabius.norlund_succ_eval_add_one_sub</code> and Equation (29.25) is <code>Fabius.norlund_add_eval_add</code> ; complex orders are not formalized.

Result	Title	Status	Lean declarations (module)
Theorem 29.21	Bell-polynomial construction and the Nörlund diagonal	partial	The three diagonal displays are formalized for natural orders (module <code>NorlundDiagonal</code>): Equation (29.29) is <code>Fabius.norlund_diagonal</code> , Equation (29.30) is <code>Fabius.norlund_eval_zero_diagonal</code> and Equation (29.31) is <code>Fabius.coeff_bernoulliPowerSeries_pow_succ</code> ; the Bell-polynomial construction Equations (29.27) and (29.28) is not formalized.
Theorem 29.24	Bernoulli–Euler–Genocchi relations	Lean	with <code>Fabius.genocchi</code> defined by $2(1-2^n)\beta_n$, the generating function Equation (29.33) is <code>Fabius.egf_genocchi_mul_exp_add_one</code> (<code>GenocchiNumbers</code>), i.e. the first equality of Equation (29.34), and the second equality is <code>Fabius.genocchi_succ_eq</code> (<code>EulerPolynomials</code>); <code>Fabius.genocchi_one</code> , <code>Fabius.genocchi_two</code> , <code>Fabius.genocchi_odd</code> give the initial values and the parity
Theorem 29.27	Integral, Stirling, difference, addition, and reflection formulas	partial	Module <code>CauchyPolynomials</code> defines b_n by Equation (29.35) (<code>Fabius.cauchyPoly</code>): Equation (29.38) is <code>Fabius.derivative_cauchyPoly_succ</code> , Equation (29.39) is <code>Fabius.cauchyPoly_succ_eval_add_one_sub</code> , Equation (29.40) is <code>Fabius.cauchyPoly_eval_add</code> , Equation (29.41) is <code>Fabius.cauchyPoly_succ_eq</code> and Equation (29.37) is <code>Fabius.cauchyPoly_eval_zero</code> ; the integral representation Equation (29.36) and the reflection Equation (29.42) are not formalized.
proposition (unlabelled, no. 6)		none	
Theorem 29.32	Fourier expansion with exact convergence qualifications	none	
Theorem 29.33	Euler–Maclaurin with remainder	none	
Theorem 29.34	Constant-of-integration form	none	
corollary (unlabelled, no. 12)	Stirling’s expansion	none	
Theorem 29.36	Harmonic-number expansion	none	
Theorem 30.2	Gamma quotient and binomial series	none	
corollary (unlabelled, no. 14)	Chu–Vandermonde	none	
Theorem 30.4	All parameter derivatives	none	
Theorem 30.5	Derivatives of a reciprocal Pochhammer symbol	none	
Theorem 30.6	Beta integral with an integer parameter	none	
Theorem 30.7	Euler’s limit and the Weierstrass product	none	
Theorem 30.8	Partial fractions	none	
corollary (unlabelled, no. 16)	Recurrences and special values	none	
Theorem 30.10	Hurwitz-zeta form	none	
Theorem 30.11	Reflection	none	
Theorem 30.12	Gauss multiplication	none	
corollary (unlabelled, no. 18)		none	
Theorem 30.14	Canonical product	none	

Result	Title	Status	Lean declarations (module)
Theorem 30.15 corollary (unlabelled, no. 19)	Alexeiewsky identity	none	The Dyck-word count is Mathlib's <code>DyckWord.card_dyckWord_semilength_eq_catalan</code> ; $C_n^{\text{Cat}} = \binom{2n}{n}/(n+1)$ is <code>catalan_eq_centralBinom_div</code> and the difference form is <code>Fabius.catalan_succ_eq_choose_sub_choose</code> (module <code>CatalanGeneratingFunction</code>). $C = 1 + zC^2$ is <code>Fabius.catalanSeries_eq</code> and its uniqueness among power series <code>Fabius.eq_catalanSeries_of_eq_one_add_X_mu1_sq</code> (module <code>CatalanGeneratingFunction</code>); the square-root closed form is not formalized. Module <code>NarayanaNumbers</code> defines $N(n, k)$ by the division-free determinant $\binom{n}{k}\binom{n-1}{k-1} - \binom{n}{k-1}\binom{n-1}{k}$ over $\mathbb{Z}$ (<code>Fabius.narayana</code>), which vanishes outside $1 \leq k \leq n$ without a side condition: Equation (31.3) is <code>Fabius.narayana_mul</code> in the cleared form $nN(n, k) = \binom{n}{k}\binom{n}{k-1}$, the symmetry is <code>Fabius.narayana_symm</code> and the row sum $\sum_k N(n, k) = C_n^{\text{Cat}}$ is <code>Fabius.sum_narayana</code> ; the peak-counting interpretation and the bivariate generating function Equation (31.4) are not formalized.
Theorem 30.17	Barnes asymptotics on the positive ray	none	
Theorem 30.18	Complete hyperfactorial expansion	none	
Theorem 31.1	Catalan formula	Lean	
Theorem 31.3	First return	partial	
Theorem 31.5	Narayana formula	partial	
corollary (unlabelled, no. 21)	Peak mean and variance	none	
Theorem 31.8	Binary trees and parenthesizations	none	
Theorem 31.9	Triangulations	none	
Theorem 31.10	Noncrossing partitions	none	
Theorem 31.11	Kirkman–Cayley dissection formula	none	
Theorem 31.12	Associahedral face vector	none	
Theorem 31.13	Associahedral h -polynomial	none	
Theorem 31.16	Loday's constant hyperplane	none	
Theorem 31.18	The rotation relation is a partial order	none	
Theorem 31.19	Raney numbers	none	
Theorem 32.1	Darboux formula	none	
corollary (unlabelled, no. 23)	Taylor integral remainder	none	
Theorem 32.3	Exact algebraic coefficient, including polynomial cases	none	
Theorem 32.4	Darboux transfer with an analytic multiplier	none	
Theorem 32.5	Finite-smoothness Darboux subtraction	none	
corollary (unlabelled, no. 24)	Multiple dominant points	none	
proposition (unlabelled, no. 9)	Logarithmic and meromorphic cases	none	
theorem (unlabelled, no. 142)		none	

Result	Title	Status	Lean declarations (module)
Theorem 33.1	Multinomial Leibniz rule	partial	The two-factor case, in the formal reading the proof itself sanctions, is <code>Fabius.derivative_iterate_mul</code> (module <code>IteratedLeibniz</code>): $(d/dt)^n(fg) = \sum_k \binom{n}{k} f^{(k)} g^{(n-k)}$ in $R[[t]]$ over any commutative ring; the general q -factor multinomial form is not formalized.
Theorem 33.3	Higher reciprocal and quotient rules	partial	
corollary (unlabelled, no. 25)	Trigonometric outer functions	none	the analytic statement is not formalized; its formal power-series analogue, the coefficients of $1/A$ through ordinary Bell polynomials, is <code>Fabius.coeff_reciprocalSeries</code> (<code>OrdinaryBellComposition</code>)
Theorem 33.5	Four classical inverse series and their diagonals	none	
Theorem 33.6	Logarithmic diagonal	none	
Theorem 33.7	Offset Lagrange–Bürmann formula	none	
Theorem 33.8	Local square-root inverse	none	
Theorem 33.9	Universal coefficient signature	none	
Theorem 34.1	Lagrange and Fourier–Bessel forms	none	
Theorem 35.1	Product, reciprocal, and quotient	none	
Theorem 35.2	Logarithm, exponential, and powers	none	
Theorem 35.3	Differentiation and integration	none	
Theorem 35.4	Composition power table	none	
Theorem 35.5	Lagrange reversion	none	
Theorem 35.6	Direct reversion through degree five	none	
Theorem 35.7	Reciprocal Newton iteration	none	
Theorem 35.8	Newton reversion	none	
proposition (unlabelled, no. 12)	Recurrence uniqueness	none	
Theorem 35.11	Polynomial grid certificate	none	
Theorem 35.12	Chinese-remainder certificate	none	

Selected bibliography and provenance

Bibliography

- [1] E. T. Bell, “Exponential polynomials,” *Annals of Mathematics*, second series, vol. 35 (1934), pp. 258–277.
- [2] D. Berend and T. Tassa, “Improved bounds on Bell numbers and on moments of sums of random variables,” *Probability and Mathematical Statistics*, vol. 30 (2010), pp. 185–205.
- [3] E. A. Bender and L. B. Richmond, “A multivariate Lagrange inversion formula for asymptotic calculations,” *Electronic Journal of Combinatorics*, vol. 5 (1998), Research Paper 33.
- [4] A. Z. Broder, “The r -Stirling numbers,” *Discrete Mathematics*, vol. 49 (1984), pp. 241–259.
- [5] C. A. Charalambides, *Enumerative Combinatorics*, Chapman & Hall/CRC, 2002.
- [6] A. Claesson, “Generalized pattern avoidance,” *European Journal of Combinatorics*, vol. 22 (2001), pp. 961–971.
- [7] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, translated by J. W. Nienhuys, D. Reidel, 1974.
- [8] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [9] I. M. Gessel, “Lagrange inversion,” *Journal of Combinatorial Theory, Series A*, vol. 144 (2016), pp. 212–249.
- [10] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, second edition, Addison–Wesley, 1994.
- [11] I. J. Good, “Generalizations to several variables of Lagrange’s expansion, with applications to stochastic processes,” *Proceedings of the Cambridge Philosophical Society*, vol. 56 (1960), pp. 367–380.
- [12] L. H. Harper, “Stirling behavior is asymptotically normal,” *Annals of Mathematical Statistics*, vol. 38 (1967), pp. 410–414.
- [13] P. Henrici, *Applied and Computational Complex Analysis*, vol. 1, Wiley, 1974.
- [14] L. C. Hsu and P. J.-S. Shiue, “A unified approach to generalized Stirling numbers,” *Advances in Applied Mathematics*, vol. 20 (1998), pp. 366–384.
- [15] W. P. Johnson, “The curious history of Faà di Bruno’s formula,” *American Mathematical Monthly*, vol. 109 (2002), pp. 217–234.
- [16] L. Moser and M. Wyman, “An asymptotic formula for the Bell numbers,” *Transactions of the Royal Society of Canada*, section III, vol. 49 (1955), pp. 49–54.

- [17] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, editors, *NIST Digital Library of Mathematical Functions*, National Institute of Standards and Technology.
- [18] N. E. Nörlund, *Vorlesungen über Differenzenrechnung*, Springer, Berlin, 1924.
- [19] J. Riordan, *An Introduction to Combinatorial Analysis*, Wiley, 1958.
- [20] S. Roman, *The Umbral Calculus*, Academic Press, 1984.
- [21] M. Z. Spivey, “A generalized recurrence for Bell numbers,” *Journal of Integer Sequences*, vol. 11 (2008), Article 08.2.5.
- [22] R. P. Stanley, *Enumerative Combinatorics*, vol. 1, second edition, Cambridge University Press, 2012.
- [23] N. M. Temme, “Asymptotic estimates of Stirling numbers,” *Studies in Applied Mathematics*, vol. 89 (1993), pp. 233–243.
- [24] H. S. Wilf, *generatingfunctionology*, third edition, A K Peters, 2005.

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